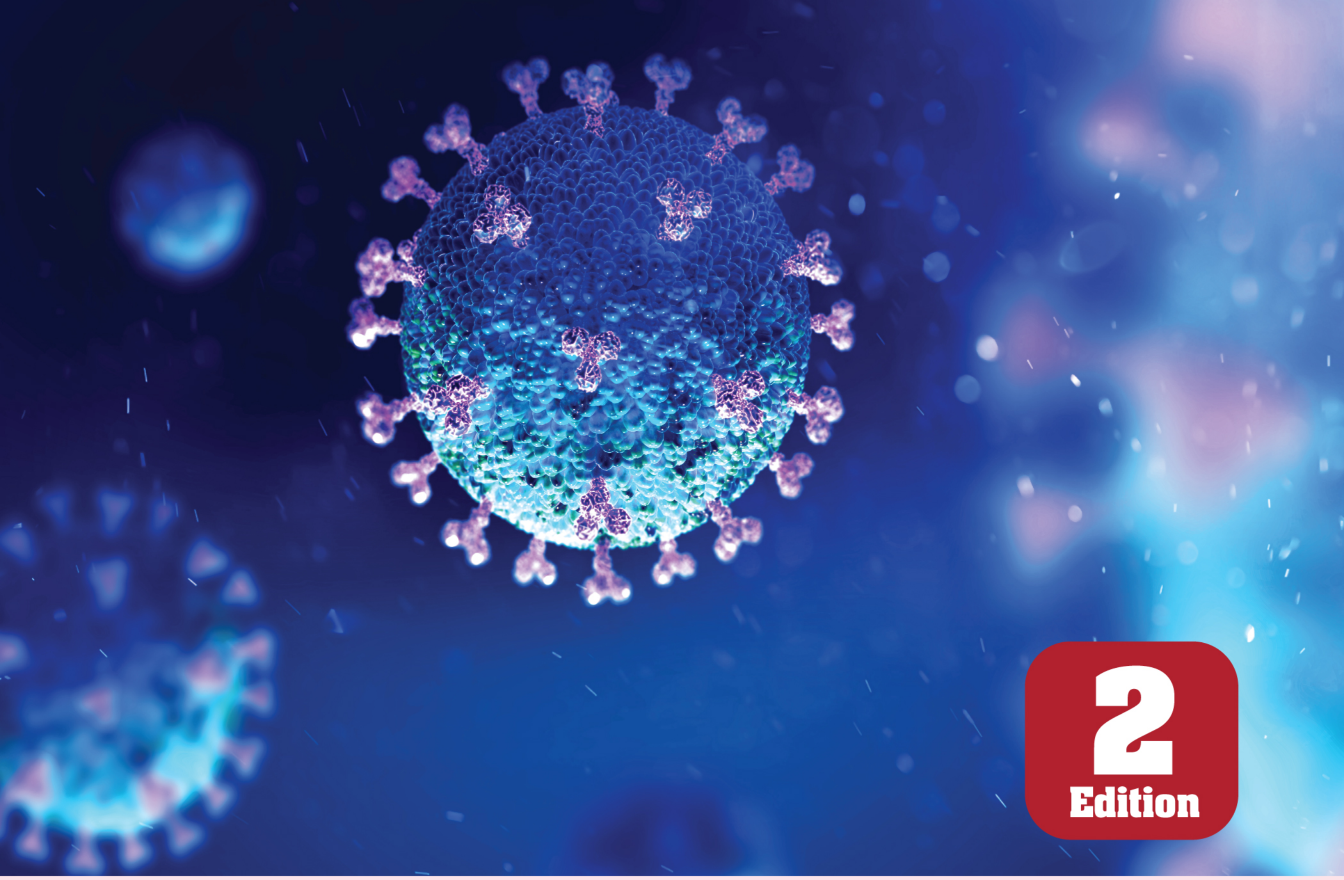

Core Concepts in Biology: Microbiology

Kan Hung Cheng

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Preface

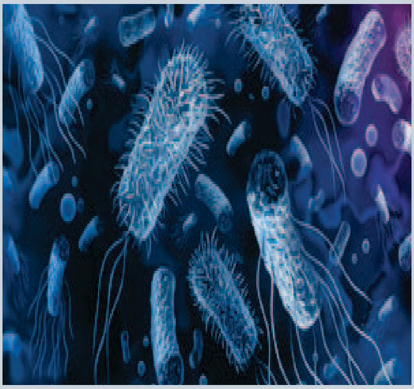
Microbes are found everywhere in significant quantities. Their presence extends beyond our familiar earthly realm, even reaching into the vastness of space. These tiny organisms play a crucial role in the natural cycle of life by efficiently recycling organic matter. Living organisms like bacteria and fungi have the remarkable ability to decompose all naturally occurring substances since they are biodegradable. While many of these microbes positively impact plants, animals, and humans in various ways, it is important to acknowledge that some can also pose a threat.

Microorganisms play a significant role in the natural world by actively participating in various biogeochemical cycling processes and assisting in the decomposition of organic matter in both soil and aquatic ecosystems. These tiny lifeforms engage in symbiotic relationships with plants, leading to mutual benefits for both parties. On the other hand, certain microorganisms can also act as pathogens, causing diseases in plants and animals. This diverse range of microbial activities greatly influences the overall balance and functioning of ecosystems.

Core Concepts in Microbiology is a comprehensive and detailed guide that investigates the captivating realm of microorganisms. This extensive book provides a vast array of information, covering various essential topics, such as the remarkable diversity of microorganisms, their intricate metabolic processes, their interactions with hosts, and their crucial role in both environmental and industrial processes. Whether you are a student pursuing a study in microbiology or simply someone with a curious mind seeking to explore this fascinating field, Core Concepts in Microbiology will undoubtedly enrich your understanding. This invaluable resource offers unparalleled insights and clarity, serving as an indispensable tool for gaining a comprehensive understanding of the captivating world of microbiology.

CHAPTER 1

DEVELOPMENT OF MICROBIOLOGY



Microbiology is the study of microscopic organisms, such as bacteria, viruses, archaea, fungi and protozoa. This discipline includes fundamental research on the biochemistry, physiology, cell biology, ecology, evolution and clinical aspects of microorganisms, including the host response to these agents.

After studying this chapter, you will understand the core concepts of:

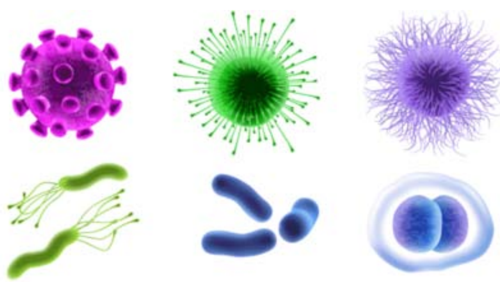
- Microbiology
- Diversity in the Living World
- Spontaneous Generation
- Biogenesis
- Koch's Postulates

INTRODUCTION

Microbiology is the study of a variety of living organisms which are invisible to the naked eye like bacteria and fungi and many other microscopic organisms. Although tiny in size these organisms form the basis for all life on earth. These microbes are also known to produce the soil in which plants grow and fix the atmospheric gases that both plants and animals use. About 3 billion years ago at the time of the formation of the earth, microbes were the only lives on earth. Microorganisms have played a key role in the evolution of the planet earth.

Microorganisms affect animals, the environment, the food supply and also the healthcare industry. There are many different areas of microbiology including environmental, veterinary, food, pharmaceutical and medical microbiology, which is the most prominent.

Microorganisms are very important to the environment, human health and the economy. Few have immense beneficial effects without which we could not exist. Others are really harmful, and our effort to overcome their effects tests our understanding and skills. The uses of microbiology can be beneficial or harmful depending on what we require from them.



CORE CONCEPT: MICROBIOLOGY

Microbiology, study of microorganisms, or microbes, a diverse group of generally minute, simple life-forms that include bacteria, archaea, algae, fungi, protozoa, and viruses. The field is concerned with the structure, function, and classification of such organisms and with ways of both exploiting and controlling their activities.

The 17th-century discovery of living forms existing invisible to the naked eye was a significant milestone in the history of science, for from the 13th century onward it had been postulated that "invisible" entities were responsible for decay and disease. The word *microbe* was coined in the last quarter of the 19th century to describe these organisms, all of which were thought to be related. As microbiology eventually developed into a specialized science, it was found that microbes are a very large group of extremely diverse organisms.

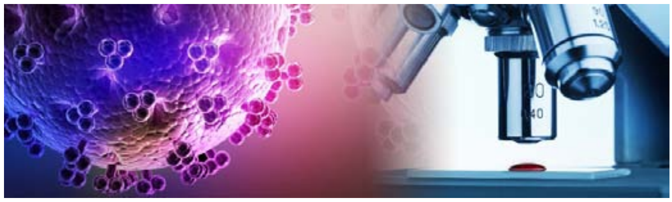
Daily life is interwoven inextricably with microorganisms. In addition to populating both the inner and outer surfaces of the human body, microbes abound in the soil, in the seas, and in the air. Abundant, although usually unnoticed, microorganisms provide ample evidence of their presence—sometimes unfavourably, as when they cause decay of materials or spread diseases, and sometimes favourably, as when they ferment sugar to wine and beer, cause bread to rise, flavour cheeses, and produce valued products such as **antibiotics** and insulin. Microorganisms are of incalculable value to Earth's ecology, disintegrating animal and plant remains and converting them to simpler substances that can be recycled in other organisms.

Antibiotic is a type of antimicrobial drug used in the treatment and prevention of bacterial infections.

Historical Background

Microbiology essentially began with the development of the microscope. Although others may have seen microbes before him, it was Antonie van Leeuwenhoek, a Dutch draper whose hobby was lens grinding and making microscopes, who was the first to provide proper documentation of his observations. His descriptions and drawings included protozoans from the guts of animals and bacteria from teeth scrapings. His records were excellent because he produced magnifying lenses of exceptional quality. Leeuwenhoek conveyed his findings in a series of letters to the British Royal Society during the mid-1670s. Although his observations stimulated much interest, no one made a serious attempt either to repeat or to extend them. Leeuwenhoek's "animalcules," as he called them, thus remained mere oddities of nature to the scientists of his day, and enthusiasm for the study of microbes grew slowly. It was only later, during the 18th-century revival of a long-standing controversy about whether life could develop out of nonliving material, that the significance of microorganisms in the scheme of nature and in the health and welfare of humans became evident.

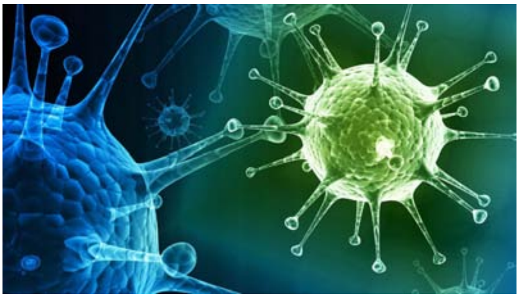
Microscope is an instrument used to see objects that are too small to be seen by the naked eye.



Spontaneous generation versus biotic generation of life

The early Greeks believed that living things could originate from nonliving matter (abiogenesis) and that the goddess Gea could create life from stones. Aristotle discarded this notion, but he still held that animals could arise spontaneously from dissimilar organisms or from soil. His influence regarding this concept of spontaneous generation was still felt as late as the 17th century, but toward the end of that century a chain of observations, experiments, and arguments began that eventually refuted the idea. This advance in understanding was hard fought, involving series of events, with forces of personality and individual will often obscuring the facts.

Although Francesco Redi, an Italian physician, disproved in 1668 that higher forms of life could originate spontaneously, proponents of the concept claimed that microbes were different and did indeed arise in this way. Such illustrious names as John Needham and Lazzaro Spallanzani were adversaries in this debate during the mid-1700s. In the early half of the 1800s, Franz Schulze and Theodor Schwann were major figures in the attempt to disprove theories of abiogenesis until Louis Pasteur finally announced the results of his conclusive experiments in 1864. In a series of masterful experiments, Pasteur proved that only preexisting microbes could give rise to other microbes (biogenesis). Modern and accurate knowledge of the forms of bacteria can be attributed to German botanist Ferdinand Cohn, whose chief results were published between 1853 and 1892. Cohn's classification of bacteria, published in 1872 and extended in 1875, dominated the study of these organisms thereafter.



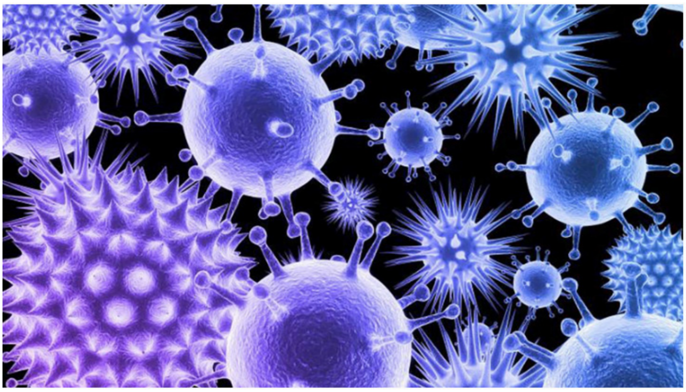
Microbes and disease

Girolamo Fracastoro, an Italian scholar, advanced the notion as early as the mid-1500s that contagion is an infection that passes from one thing to another. A description of precisely what is passed along eluded discovery until the late 1800s, when the work of many scientists, Pasteur foremost among them, determined the role of bacteria in fermentation and disease. Robert Koch, a German physician, defined the procedure (Koch's postulates) for proving that a specific organism causes a specific disease.

The foundation of microbiology was securely laid during the period from about 1880 to 1900. Students of Pasteur, Koch, and others discovered in rapid succession a host of bacteria capable of causing specific diseases (pathogens). They also elaborated an extensive arsenal of techniques and laboratory procedures for revealing the ubiquity, diversity, and abilities of microbes.

Progress in the 20th century

All of these developments occurred in Europe. Not until the early 1900s did microbiology become established in America. Many microbiologists who worked in America at this time had studied either under Koch or at the Pasteur Institute in Paris. Once established in America, microbiology flourished, especially with regard to such related disciplines as biochemistry and genetics. In 1923 American bacteriologist David Bergey established that science's primary reference, updated editions of which continue to be used today.



Since the 1940s microbiology has experienced an extremely productive period during which many disease-causing microbes have been identified and methods to control them developed. Microorganisms have also been effectively utilized in industry; their activities have been channeled to the extent that valuable products are now both vital and commonplace.

Metabolism is the set of life-sustaining chemical transformations within the cells of organisms.

The study of microorganisms has also advanced the knowledge of all living things. Microbes are easy to work with and thus provide a simple vehicle for studying the complex processes of life; as such they have become a powerful tool for studies in genetics and **metabolism** at the molecular level. This intensive probing into the functions of microbes has resulted in numerous and often unexpected dividends. Knowledge of the basic metabolism and nutritional requirements of a pathogen, for example, often leads to a means of controlling disease or infection.

CORE CONCEPT: DIVERSITY IN THE LIVING WORLD

Biology is the science of life forms and living processes. The living world comprises an amazing diversity of living organisms. Early man could easily perceive the difference between inanimate matter and living organisms. Early man deified some of the inanimate matter and some among the animals and plants.

If you look around you will see a large variety of living organisms, be it potted plants, insects, birds, your pets or other animals and plants. There are also several organisms that you cannot see with your naked eye but they are all around you. If you were to increase the area that you make observations in, the range and variety of organisms that you see would increase. Obviously, if you were to visit a dense forest, you would probably see a much greater number and kinds of living organisms in it. Each different kind of plant, animal or organism that you see, represents a species. The number of species that are known and described range between 1.7-1.8 million. This refers to **biodiversity** or the number and types of organisms present on earth. We should remember here that as we explore new areas, and even old ones, new organisms are continuously being identified.

Biodiversity a portmanteau of biological (life) and diversity, generally refers to the variety and variability of life on Earth.

As stated earlier, there are millions of plants and animals in the world; we know the plants and animals in our own area by their local names. These local names would vary from place to place, even within a country. Probably you would recognise the confusion that would be created if we did not find ways and means to talk to each other, to refer to organisms we are talking about.

Hence, there is a need to standardise the naming of living organisms such that a particular organism is known by the same name all over the world. This process is called nomenclature. Obviously, nomenclature or naming is only possible when the organism is described correctly and we know to what organism the name is attached to. This is identification.

In order to facilitate the study, number of scientists have established procedures to assign a scientific name to each known organism. This is acceptable to biologists all over the world. For plants, scientific names are based on agreed principles and criteria, which are provided in International Code for Botanical Nomenclature (ICBN). You may ask, how are animals named?

Animal taxonomists have evolved International Code of Zoological Nomenclature (ICZN). The scientific names ensure that each organism has only one name. Description of any organism should enable the people (in any part of the world) to arrive at the same name. They also ensure that such a name has not been used for any other known organism.

While some fear microbes due to the association of some microbes with various human diseases, many microbes are also responsible for numerous beneficial processes such as industrial fermentation (e.g. the production of alcohol, vinegar and dairy products), antibiotic production and act as molecular vehicles to transfer DNA to complex organisms such as plants and animals.

Biologists follow universally accepted principles to provide scientific names to known organisms. Each name has two components – the Generic name and the specific epithet. This system of providing a name with two components is called Binomial nomenclature. This naming system given by Carolus Linnaeus is being practised by biologists all over the world. This naming system using a two-word format was found convenient. Let us take the example of mango to understand the way of providing scientific names better. The scientific name of mango is written as Mangifera indica. Let us see how it is a binomial name. In this name Mangifera represents the genus while indica, is a particular species, or a specific epithet. Other universal rules of nomenclature are as follows:

1. Biological names are generally in Latin and written in italics. They are Latinised or derived from Latin irrespective of their origin.
2. The first word in a biological name represents the genus while the second component denotes the specific epithet.
3. Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate their Latin origin.
4. The first word denoting the genus starts with a capital letter while the specific epithet starts with a small letter. It can be illustrated with the example of Mangifera indica.

Name of the author appears after the specific epithet, i.e., at the end of the biological name and is written in an abbreviated form, e.g., Mangifera indica Linn. It indicates that this species was first described by Linnaeus.

Since it is nearly impossible to study all the living organisms, it is necessary to devise some means to make this possible. This process is classification. Classification is the process by which anything is grouped into convenient categories based on some easily observable characters. For example, we easily recognise groups such as plants or animals or dogs, cats or insects. The moment we use any of these terms, we associate certain characters with the organism in that group. What image do you see when you think of a dog? Obviously, each one of us will see ‘dogs’ and not ‘cats’. Now, if we were to think of ‘Alsatians’ we know what we are talking about. Similarly, suppose we were to say ‘mammals’, you would, of course, think of animals with external ears and body hair. Likewise, in plants, if we try to talk of ‘Wheat’, the picture in each of our minds will be of wheat plants, not of rice or any other plant. Hence, all these - 'Dogs', 'Cats', 'Mammals', 'Wheat', 'Rice', 'Plants', 'Animals', etc., are convenient categories we use to study organisms. The scientific term for these categories is taxa. Here you must recognise that taxa can indicate categories at very different levels. 'Plants' – also form some taxa. 'Wheat' is also a tax on. Similarly, 'animals', 'mammals', 'dogs' are all taxa – but you know that a dog is a mammal and mammals are animals. Therefore, 'animals', 'mammals' and 'dogs' represent taxa at different levels.

Taxonomy is the science of defining and naming groups of biological organisms on the basis of shared characteristics.

Hence, based on characteristics, all living organisms can be classified into different taxa. This process of classification is taxonomy. External and internal structure, along with the structure of cell, development process and ecological information of organisms are essential and form the basis of modern taxonomic studies.

Hence, characterisation, identification, classification and nomenclature are the processes that are basic to taxonomy. **Taxonomy** is not something new. Human beings have always been interested in knowing more and more about the various kinds of organisms, particularly with reference to their own use. In early days, human beings needed to find sources for their basic needs of food, clothing and shelter. Hence, the earliest classifications were based on the 'uses' of various organisms.

Human beings were, since long, not only interested in knowing more about different kinds of organisms and their diversities, but also the relationships among them. This branch of study was referred to as systematics. The word systematics is derived from the Latin word 'systema' which means systematic arrangement of organisms. Linnaeus used Systema Naturae as the title of his publication. The scope of systematics was later enlarged to include identification, nomenclature and classification. Systematics takes into account evolutionary relationships between organisms.

Taxonomic Categories

Classification is not a single step process but involves hierarchy of steps in which each step represents a rank or category. Since the category is a part of overall taxonomic arrangement, it is called the taxonomic category and all categories together constitute the taxonomic hierarchy. Each category, referred to as a unit of classification, in fact, represents a rank and is commonly termed as taxon (pl.: taxa).

Taxonomic categories and hierarchy can be illustrated by an example. Insects represent a group of organisms sharing common features like three pairs of jointed legs. It means insects are recognisable concrete objects which can be classified, and thus were given a rank or category. Can you name other such groups of organisms? Remember, groups represent category. Category further denotes rank. Each rank or taxon, in fact, represents a unit of classification. These taxonomic groups/ categories are distinct biological entities and not merely morphological aggregates.

Taxonomical studies of all known organisms have led to the development of common categories such as kingdom, phylum or division (for plants), class, order, family, genus and species. All organisms, including those in the plant and animal kingdoms have species as the lowest category. Now the question you may ask is, how to place an organism in various categories? The basic requirement is the knowledge of characters of an individual or group of organisms. This helps in identifying similarities and dissimilarities among the individuals of the same kind of organisms as well as of other kinds of organisms.

Species

Taxonomic studies consider a group of individual organisms with fundamental similarities as a species. One should be able to distinguish one species from the other closely related species based on the distinct morphological differences. Let us consider **Mangifera indica**, Solanum tuberosum (potato) and Panthera leo (lion). All the three names, indica, tuberosum and leo, represent the specific epithets, while the first words Mangifera, Solanum and Panthera are genera and represents another higher level of taxon or category. Each genus may have one or more than one specific epithets representing different organisms, but having morphological similarities. For example, Panthera has another specific epithet called tigris and Solanum includes species like nigrum and melongena. Human beings belong to the species sapiens which is grouped in the genus Homo. The scientific name thus, for human being, is written as Homo sapiens.

Mangifera indica is a species of flowering plant in the sumac and poison ivy family Anacardiaceae.

Genus

Genus comprises a group of related species which has more characters in common in comparison to species of other genera. We can say that genera are aggregates of closely related species. For example, potato and brinjal are two different species but both belong to the genus Solanum. Lion (Panthera leo), leopard (P. pardus) and tiger (P. tigris) with several common features, are all species of the genus **Panther**. This genus differs from another genus Felis which includes cats.

Panthera is a genus within the Felidae family that was named and first described by the German naturalist Oken in 1816.

Family

The next category, Family, has a group of related genera with still less number of similarities as compared to genus and species. Families are characterised on the basis of both vegetative and reproductive features of plant species. Among plants for example, three different genera Solanum, Petunia and Datura are placed in the family Solanaceae. Among animals for example, genus Panthera, comprising lion, tiger, leopard is put along with genus, Felis (cats) in the family Felidae. Similarly, if you observe the features of a cat and a dog, you will find some similarities and some differences as well. They are separated into two different families – Felidae and Canidae, respectively.

Order

You have seen earlier that categories like species, genus and families are based on a number of similar characters. Generally, order and other higher taxonomic categories are identified based on the aggregates of characters. Order being a higher category, is the assemblage of families which exhibit a few similar characters. The similar characters are less in number as compared to different genera included in a family. Plant families like Convolvulaceae, Solanaceae are included in the order Polymoniales mainly based on the floral characters. The animal order, Carnivora, includes families like Felidae and Canidae

Class

This category includes related orders. For example, order Primata comprising monkey, gorilla and gibbon is placed in class Mammalia along with order Carnivora that includes animals like tiger, cat and dog. Class Mammalia has other orders also.

Phylum

Classes comprising animals like fishes, amphibians, reptiles, birds along with mammals constitute the next higher category called Phylum. All these, based on the common features like presence of notochord and dorsal hollow neural system, are included in phylum Chordata. In case of plants, classes with a few similar characters are assigned to a higher category called Division.

Kingdom

All animals belonging to various phyla are assigned to the highest category called Kingdom Animalia in the classification system of animals. The Kingdom Plantae, on the other hand, is distinct, and comprises all plants from various divisions. Henceforth, we will refer to these two groups as animal and plant kingdoms.

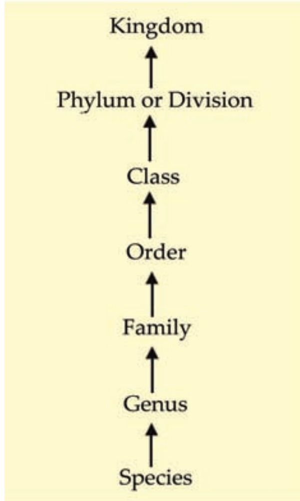


Figure 1: Taxonomic categories showing hierarchial arrangement in ascending order.

The taxonomic categories from species to kingdom have been shown in ascending order starting with species in Figure 1. These are broad categories. However, taxonomists have also developed sub-categories in this hierarchy to facilitate more sound and scientific placement of various taxa.

Look at the hierarchy in Figure 1. Can you recall the basis of arrangement? Say, for example, as we go higher from species to kingdom, the number of common characteristics goes on decreasing. Lower the taxa, more are the characteristics that the members within the taxon share. Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level. Hence, the problem of classification becomes more complex

Table 1 indicates the taxonomic categories to which some common organisms like housefly, man, mango and wheat belong.

Table 1: Organisms with their Taxonomic Categories

Common Name	**Biological Name**	**Genus**	**Family**	**Order**	**Class**	**Phylum/Division**
Man	Homo sapiens	Homo	Hominidae	Primata	Mammalia	Chordata
Housefly	Musca domestica	Musca	Muscidae	Diptera	Insecta	Arthropoda
Mango		Mangifera	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	Triticum aestivum	Triticum	Poaceae	Poales	Monocotyledonae	Angiospermae

Taxonomical Aids

Taxonomic studies of various species of plants, animals and other organisms are useful in agriculture, forestry, industry and in general in knowing our bio-resources and their diversity. These studies would require correct classification and identification of organisms. Identification of organisms requires intensive laboratory and field studies. The collection of actual specimens of plant and animal species is essential and is the prime source of taxonomic studies. These are also fundamental to studies and essential for training in systematics. It is used for classification of an organism, and the information gathered is also stored along with the specimens. In some cases, the specimen is preserved for future studies.

Biologists have established certain procedures and techniques to store and preserve the information as well as the specimens. Some of these are explained to help you understand the usage of these aids

Herbarium

Herbarium is a store house of collected plant specimens that are dried, pressed and preserved on sheets. Further, these sheets are arranged according to a universally accepted system of classification. These specimens, along with their descriptions on herbarium sheets, become a store house or repository for future use (Figure 2). The herbarium sheets also carry a label providing information about date and place of collection, English, local and botanical names, family, collector's name, etc. Herbaria also serve as quick referral systems in taxonomical studies.

Figure 2: Herbarium showing stored specimens.

Botanical

These specialised gardens have collections of living plants for reference. Plant species in these gardens are grown for identification purposes and each plant is labelled indicating its botanical/scientific name and its family.

Museum

Biological museums are generally set up in educational institutes such as schools and colleges. Museums have collections of preserved plant and animal specimens for study and reference. Specimens are preserved in the containers or jars in preservative solutions. Plant and animal specimens may also be preserved as dry specimens. Insects are preserved in insect boxes after collecting, killing and pinning. Larger animals like birds and mammals are usually stuffed and preserved. Museums often have collections of skeletons of animals too.

Zoological Parks

These are the places where wild animals are kept in protected environments under human care and which enable us to learn about their food habits and behaviour. All animals in a zoo are provided, as far as possible, the conditions similar to their natural habitats.

CORE CONCEPT: SPONTANEOUS GENERATION

Spontaneous generation refers to an obsolete body of thought on the ordinary formation of living organisms without descent from similar organisms. The theory of spontaneous generation held that living creatures could arise from non-living matter and that such processes were commonplace and regular. For instance, it was hypothesized that certain forms such as fleas could arise from inanimate matter such as dust, or that maggots could arise from dead flesh.

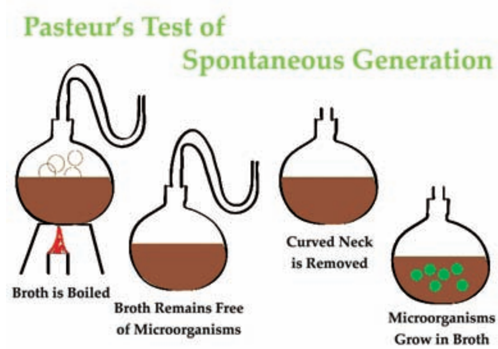
Humans have been asking for millennia: Where does new life come from? Religion, philosophy, and science have all wrestled with this question. One of the oldest explanations was the theory of spontaneous generation, which can be traced back to the ancient Greeks and was widely accepted through the Middle Ages.

Pneuma is an ancient Greek word for "breath", and in a religious context for "spirit" or "soul".

The Theory of Spontaneous Generation

The Greek philosopher Aristotle (384–322 BC) was one of the earliest recorded scholars to articulate the theory of spontaneous generation, the notion that life can arise from nonliving matter. Aristotle proposed that life arose from nonliving material if the material contained ***pneuma*** ("vital heat"). As evidence, he noted several instances of the appearance of animals from environments previously devoid of such animals, such as the seemingly sudden appearance of fish in a new puddle of water.

This theory persisted into the seventeenth century, when scientists undertook additional experimentation to support or disprove it. By this time, the proponents of the theory cited how frogs simply seem to appear along the muddy banks of the Nile River in Egypt during the annual flooding. Others observed that mice simply appeared among grain stored in barns with thatched roofs. When the roof leaked and the grain melded, mice appeared. Jan Baptista van Helmont, a seventeenth century Flemish scientist, proposed that mice could arise from rags and wheat kernels left in an open container for 3 weeks. In reality, such habitats provided ideal food sources and shelter for mouse populations to flourish.



However, one of van Helmont's contemporaries, Italian physician Francesco Redi (1626–1697), performed an experiment in 1668 that was one of the first to refute the idea that maggots (the larvae of flies) spontaneously generate on meat left out in the open air. He predicted that preventing flies from having direct contact with the meat would also prevent the appearance of maggots. Redi left meat in each of six containers (Figure 3). Two were open to the air, two were covered with gauze, and two were tightly sealed. His hypothesis was supported when maggots developed in the uncovered jars, but no maggots appeared in either the gauze-covered or the tightly sealed jars. He concluded that maggots could only form when flies were allowed to lay eggs in the meat, and that the maggots were the offspring of flies, not the product of spontaneous generation.

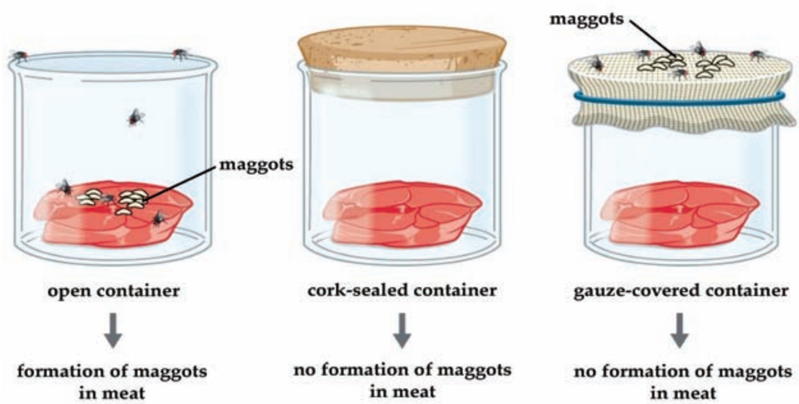


Figure 3: Francesco Redi's experimental setup consisted of an open container, a container sealed with a cork top, and a container covered in mesh that let in air but not flies. Maggots only appeared on the meat in the open container. However, maggots were also found on the gauze of the gauze-covered container.

In 1745, John Needham (1713–1781) published a report of his own experiments, in which he briefly boiled broth infused with plant or animal matter, hoping to kill all preexisting microbes. He then sealed the flasks. After a few days, Needham observed that the broth had become cloudy and a single drop contained numerous microscopic creatures. He argued that the new microbes must have arisen spontaneously. In reality, however, he likely did not boil the broth enough to kill all preexisting microbes.

Lazzaro Spallanzani (1729–1799) did not agree with Needham's conclusions, however, and performed hundreds of carefully executed experiments using heated broth. As in Needham's experiment, broth in sealed jars and unsealed jars was infused with plant and animal matter. Spallanzani's results contradicted the findings of Needham: Heated but sealed flasks remained clear, without any signs of spontaneous growth, unless the flasks were subsequently opened to the air. This suggested that microbes were introduced into these flasks from the air. In response to Spallanzani's findings, Needham argued that life originates from a "life force" that was destroyed during Spallanzani's extended boiling. Any subsequent sealing of the flasks then prevented new life force from entering and causing spontaneous generation (Figure 4).

Figure 4: (a) Francesco Redi, who demonstrated that maggots were the offspring of flies, not products of spontaneous generation. (b) John Needham, who argued that microbes arose spontaneously in broth from a "life force." (c) Lazzaro Spallanzani, whose experiments with broth aimed to disprove those of Needham.

Microorganism is a microscopic organism, which may exist in its single-celled form or in a colony of cells.

Disproving Spontaneous Generation

The debate over spontaneous generation continued well into the nineteenth century, with scientists serving as proponents of both sides. To settle the debate, the Paris Academy of Sciences offered a prize for resolution of the problem. Louis Pasteur, a prominent French chemist who had been studying microbial fermentation and the causes of wine spoilage, accepted the challenge. In 1858, Pasteur filtered air through a gun-cotton filter and, upon microscopic examination of the cotton, found it full of microorganisms, suggesting that the exposure of a broth to air was not introducing a "life force" to the broth but rather airborne **microorganisms**.

Later, Pasteur made a series of flasks with long, twisted necks ("swan-neck" flasks), in which he boiled broth to sterilize it (Figure 5). His design allowed air inside the flasks to be exchanged with air from the outside, but prevented the introduction of any airborne microorganisms, which would get caught in the twists and bends of the flasks' necks. If a life force besides the airborne microorganisms were responsible for microbial growth within the sterilized flasks, it would have access to the broth, whereas the microorganisms would not. He correctly predicted that sterilized broth in his swan-neck flasks would remain sterile as long as the swan necks remained intact. However, should the necks be broken, microorganisms would be introduced, contaminating the flasks and allowing microbial growth within the broth. Pasteur's set of experiments irrefutably disproved the theory of spontaneous generation and earned him the prestigious Alhumbert Prize from the Paris Academy of Sciences in 1862. In a subsequent lecture in 1864, Pasteur articulated "*Omne vivum ex vivo*" ("Life only comes from life"). In this lecture, Pasteur recounted his famous swan-neck flask experiment, stating that "life is a germ and a germ is life. Never will the doctrine of spontaneous generation recover from the mortal blow of this simple experiment. "To Pasteur's credit, it never has.

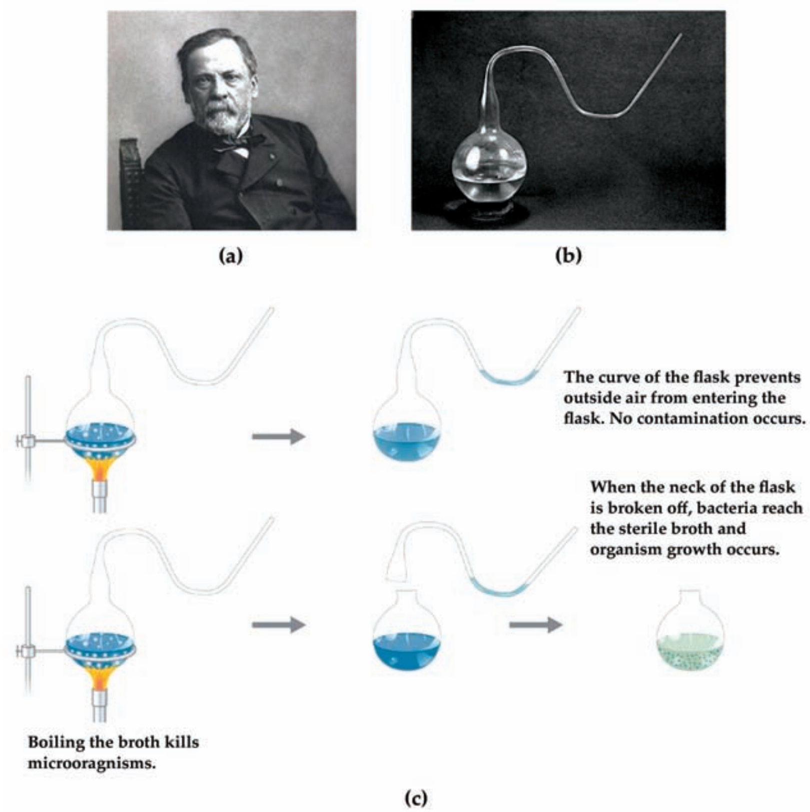


Figure 5: (a) French scientist Louis Pasteur, who definitively refuted the long-disputed theory of spontaneous generation. (b) The unique swan-neck feature of the flasks used in Pasteur's experiment allowed air to enter the flask but prevented the entry of bacterial and fungal spores. (c) Pasteur's experiment consisted of two parts. In the first part, the broth in the flask was boiled to sterilize it. When this broth was cooled, it remained free of contamination. In the second part of the experiment, the flask was boiled and then the neck was broken off. The broth in this flask became contaminated.

CORE CONCEPT: BIOGENESIS

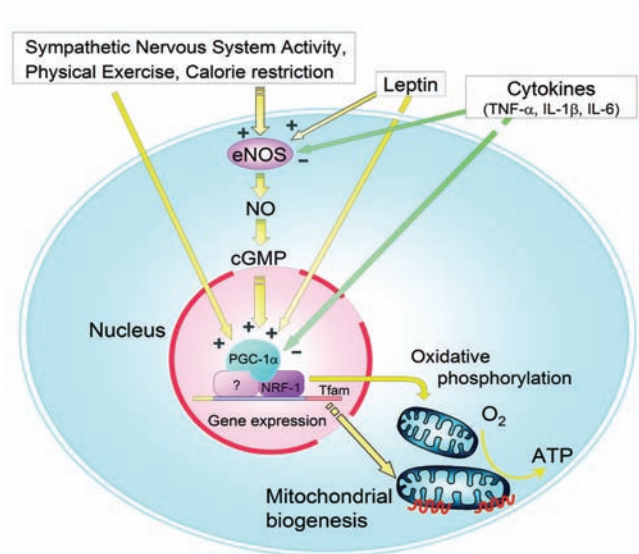
Biogenesis is the production of new living organisms or organelles. Conceptually, biogenesis is primarily attributed to Louis Pasteur and encompasses the belief that complex living things come only from other living things, by means of reproduction. That is, life does not spontaneously arise from non-living material, which was the position held by spontaneous generation. This is summarized in the phrase Omne vivum ex vivo, Latin for “all life [is] from life.”

Biogenesis refers to the process wherein life arises from similar life forms.

Biogenesis is the production of life. In Latin, *bio* means life, and *genesis* means the beginning or origin of. For most of history, mankind thought that biogenesis often occurred through spontaneous generation from earth or vegetable matter, alongside reproduction, which we now know is the only way that biogenesis ever happens. Anaximenes and Anaxagoras, pre-Aristotle Greek natural philosophers, believed that biogenesis could occur from the action of the Sun on *primordial terrestrial slime*, a combination of water and earth. A related idea is *xenogenesis*, which argues that one type of life form can arise from another, completely different life form.

Around 343 BCE, Aristotle wrote the book *History of Animals* which laid out the spontaneous generation theory of biogenesis which would remain dominant for over 2000 years. Besides including lengthy descriptions of countless species of fish, shellfish, and other animals, the book also presented Aristotle’s theory of how animals come to be in the first place. Aristotle believed that different animals could spontaneously arise from different forms of inanimate matter -- clams and scallops in sand, oysters in slime, and the barnacle and the limpet in the hollows of rocks. However, no one seemed to claim that humans could emerge from spontaneous generation, being higher creatures that can apparently only be produced through direct reproduction by other humans.

As early as 1668, the Italian physician Francesco Redi proposed that higher forms of life (microbes) did not arise spontaneously, and the idea got more popular, but the proponents of spontaneous generation still maintained that microbes arose via these means. In 1745, John Needham, an English biologist and Roman Catholic priest, added chicken broth to a sealed flask, boiled it, waited, then observed microbial growth, pointing to this as an example of spontaneous generation. In 1768, Lazzaro Spallanzani repeated this same experiment, but removed all air from the jar, and microbes did not grow within it. This must have been one of the earliest experiments to conclusively disprove spontaneous generation, but the idea that spontaneous generation was false did not spread at the time.



Moving on to 1859, French biologist Louis Pasteur finally disproved spontaneous generation for good. He boiled meat broth in a goose-necked flask. The goose neck allowed in air but not, as the reasoning went, tiny particles from the air. The experiment showed that microbial growth did not occur in the flask until the flask was turned so that particles could fall down the bends, at which point the water quickly became cloudy, showing the presence of microorganisms. After 2000 years, the spontaneous generation theory of biogenesis was finally brought to rest. Today, it has been replaced by cell biology and the biology of reproduction.

> Bacteria can be used for the industrial production of amino acids. Corynebacterium glutamicum is one of the most important bacterial species with an annual production of more than two million tons of amino acids, mainly L-glutamate and L-lysine.

Macroscopic Spontaneous Generation

In 1668, Francesco Redi addressed the question of macroscopic spontaneous generation when he published the results of an experiment in which he placed rotting meat in a container and covered the container's opening with gauze. If the gauze was absent, maggots would grow on the meat. If the gauze was present, maggots would not grow on the meat, but would appear on the gauze. Redi was observing flies depositing eggs as close to a food source as could be reached.

Microscopic Spontaneous Generation

A century later, an experiment performed by Lazzaro Spallanzani in 1768 indicated biogenesis on the microscopic level. Spallanzani wanted to avoid contamination by boiling a meat broth in a sealed container. The problem with this approach was that air in the container could shatter the container upon heating. Therefore, he evacuated the container after sealing it closed. The broth did not subsequently cloud with bacterial growth, supporting the theory of biogenesis.

Critics charged that air is needed for life. The lack of bacterial growth therefore was presumed due to a lack of air, not because bacteria spread through contamination. This criticism stood for almost a century before Pasteur entered the scene and overturned it.

Pasteur's Experimental Equipment

The 1859 experiment performed by Pasteur unequivocally overturned the theory of spontaneous generation at the microscopic level. He boiled a meat broth in a flask that had a long neck that curved downward, then upward, like a goose neck. The bend in the neck prevented contaminating particles from reaching the broth, while still allowing the free diffusion of air. The fact that the flask allowed for the passage of air was a design breakthrough that finally addressed the critics of Spallanzani.

Pasteur's flask remained free of bacterial growth for as long as the flask remained upright. To show where the contaminating elements were located, he tipped the flask enough for the broth to sweep out the bend in the goose neck; the broth would then quickly become clouded with bacterial growth.

A Common Misconception

Some creationists have argued that the law of biogenesis undermines evolutionary theory and the theory that all life originated from inorganic material billions of years ago. However, biogenesis simply invalidates the theory of spontaneous generation--it speaks to what can be accomplished in generational time spans, not over the course of thousands of generations or millions of years.

Theories about the origin of life take into account the lack of predators and the very different chemical makeup of the Earth's atmosphere at the time. They also consider what can be accomplished in millions of years through trial and error. Neither of these is considered in the law of biogenesis. The theory of spontaneous generation speaks of complex life appearing fully formed in days, which theories of the origin of life postulate took millions of years of trial and error to form in conditions that no longer exist on Earth.

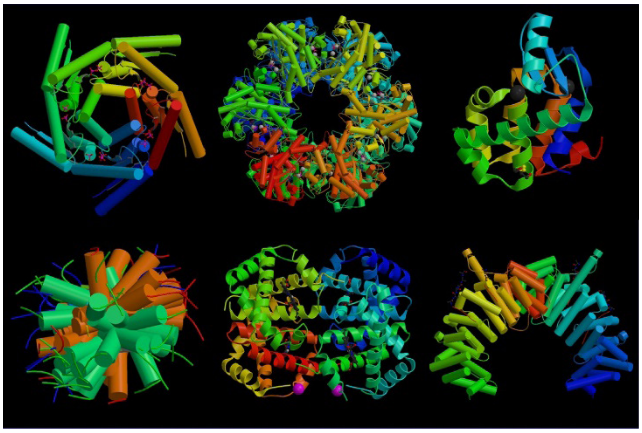
Biogenesis versus Abiogenesis

It took several clever experiments, which appear too simple today, and more than hundred years to resolve the controversy. The abiogenesis for plants and animals was disapproved as a result of the experiments by Italian Physician Fransesco Redi in 1665, who showed that maggots developing in putrefying meat are the larval stages of flies and will never appear if the meat is well protected in a vessel with the fine gauge mesh so that flies cannot lay their eggs on meat.

In 1745, John Needham took hot boiling mutton gravy (meat infusion) in a flask and closed this flask with a cork. He found the spoilage of this infusion and observed animalcules in it. He killed and destroyed the living matter by boiling and thus concluded that animalcules arose spontaneously from the meat. In 1769, Lazaro Spallanzani (1729-1799) an Italian naturalist performed a series of experiments and showed that heating can prevent appearance of animalcules in infusion although duration and level of heating required is variable. He was not satisfied with using cork to plug the flask and sealed it hermetically to prevent contact with the air completely. He found that sealed infusions remained barren for a long time.

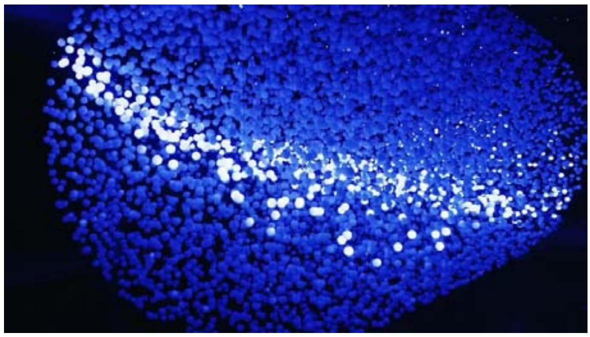
Protozoa is an informal term for single-celled eukaryotes, either free-living or parasitic, which feed on organic matter such as other microorganisms or organic tissues and debris.

A tiny crack in the flask can result in development of animalcules and they will not appear unless new air entered the flask to come in contact with the infusion. Spallanzani infact took a series of flasks and gave heat treatments for different interval of times. He could distinguish animalcules of different types, i.e., Superior or animalcules of higher class which were destroyed by slight heating undoubtedly **protozoa** and animalcules of lower class which were very minute, and much more heat resistant - the bacteria.



Abiogenesis is the natural process by which life arises from non-living matter, such as simple organic compounds.

Although the experiments Spallanzani conducted were very good but faulty experiments continued to be performed and evidence gathered in favor of **abiogenesis**. Moreover, Needham objected to the observations by Spallanzani that there was no growth in the infusions because air which is essential for life had been excluded from his flasks. An interesting practical application of Spallanzani's observation was done by Francois Appert for preservation of foods by enclosing them in airtight containers and then heating the containers called 'Appertization'. He was thereby able to preserve highly perishable food and the process named canning (called later) was widely used much before its scientific principles were understood. He performed an experiment by first making the air free of microorganisms by passing it through red hot tube. This air which still contained 19.4 % oxygen, with and without heating was passed through a set of flasks containing boiled infusions, the former remained unaffected.



Proof of Biogenesis

During the period when these experiments were done, a new figure was emerging in science, in the name of Louis Pasteur (1822-1895) in France (Figure 6). A chemist by training, he emerged as one of the greatest biologists of the 19th century. His contributions are the most significant in the history of science and industry and his work with germs and microorganisms opened new areas of scientific studies. Pasteur was born on 22nd Dec 1822 in the eastern French town of Dole and later became the Dean of the new science faculty at Lille University in 1854. We pay tribute to him as he was a great benefactor of humanity.

Figure 6: Louis Pasteur in his laboratory.

Pasteur first demonstrated through a series of definitive experiments that air contains microscopically observable 'organized bodies'. He aspirated large quantities of air through a tube which contained a plug of cotton to serve as a filter. He removed the cotton plug and suspended it in a solution of alcohol and ether. When he examined the sediments microscopically he observed the presence of small oval shaped bodies. He later confirmed that when heated air is passed through a boiled infusion no microbial development takes place but when cotton plug is suspended in the heated infusion, microbial growth occurs. Pasteur repeated his experiment through swan necked and gooseneck flasks (Figure 7) so that the germs from air cannot ascend into it. He boiled the broth in it to kill all microorganisms in the neck as well as in the flask. The infusion remained sterile in this flask until the neck of the flask was broken resulting in the growth of microorganisms. Thus he established that development of microorganisms in organic infusions bring about chemical changes. Schroeder and Von Dusch (1850) started the technique of cotton plugging because cotton acts as a filter and traps microorganisms.

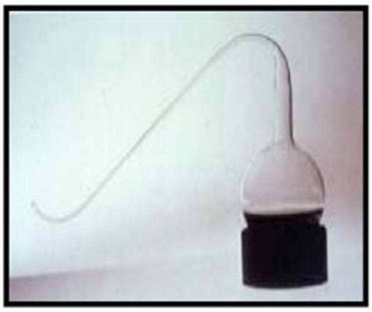


Figure 7: Pasteur's gooseneck flask.

One of the traditional arguments against biogenesis was the claim that heat used to sterilize the air or specimens was destroying an essential "vital force". Those supporting abiogenesis said that, without this force, microorganisms could not spontaneously appear. In response to this argument, an English Physicist John Tyndall (1820-1883, Figure 8) conducted experiments in a specially designed box called 'Tyndall chamber' to prove that dust carries the germs.

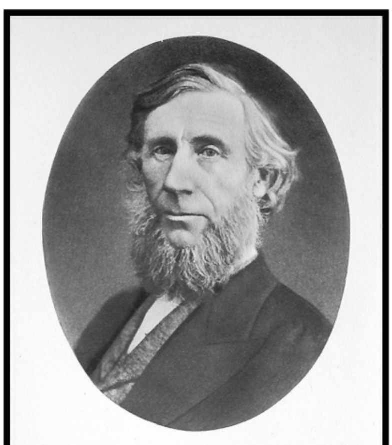


Figure 8: John Tyndall.

He demonstrated that if no dust was present, sterile broth remained sterile for indefinite period. While doing these experiments, Tyndall (1877) also devised a process for complete sterilization by alternate heating and cooling known as 'Tyndallization.' He found that in some cases even boiling the infusion for more than 5 hours was not sufficient to sterilize it and concluded that bacteria have both thermo stable and thermo labile phases. These thermo stable resting bodies were also observed by Ferdinand Cohn in hay bacteria and were called endospores. The experiments by Pasteur and Tyndall finally disapproved the theory of spontaneous generation and promoted the general acceptance of theory of biogenesis.

Xenopus is a genus of highly aquatic frogs native to sub-Saharan Africa.

Biogenesis of Ribosomes

Synthesis of ribosome in eukaryotes is complicated and takes place inside the nucleolus. Nucleoli disappear during mitosis, but at telophase new nucleoli are formed at specific chromosomal sites called nucleolar organizers located in secondary constrictions on the chromosomes.

These nucleolar organizers are known to contain genes for 18S, 28S and 5.8S rRNAs. In **Xenopus**, each nucleolar organizer contains 450 rRNA genes. These genes are tandemly repeated along DNA molecule (i.e., head to tail) and are separated from each other by stretches of spacer DNA which is not transcribed. These rRNA genes are being actively transcribed, and nascent RNA chains are spread perpendicularly to the DNA axis. Each gene is transcribed into a long RNA molecule (varying in size from 40S to 45S according to the species) which will eventually be processed giving rise to 18S, 28S and 5.8S RNA. Nucleolar rRNA genes are transcribed by **RNA polymerase** I and these polymerase molecules (about 100 per gene) can be seen at the origin of each nascent RNA chain.

RNA polymerase is an enzyme that catalyzes the chemical reactions that synthesize RNA from a DNA .template

Genes coding for 5S RNA are not located in nucleolus. The 5S genes are also tandemly repeated along DNA molecule and separated from each other by spacer DNA. The 5S RNA is transcribed by RNA polymerase III on the chromosomes and is then transported to the nucleolus, where it is incorporated into the immature large ribosomal subunits. The 70 ribosomal proteins are synthesized in the cytoplasm. All these components (18S, 5.8S, 28S, 5S and 70S) collect in the nucleolus where they are assembled into ribosomes and transported to the cytoplasm. Ribosome biogenesis is thus a striking example of coordination at the cellular and molecular levels.

CORE CONCEPT: KOCH'S POSTULATES

Koch's postulates are four criteria designed to establish a causative relationship between a microbe and a disease. The postulates were formulated by Robert Koch and Friedrich Loeffler in 1884, based on earlier concepts described by Jakob Henle, and refined and published by Koch in 1890.

Robert Koch, in full Robert Heinrich Hermann Koch, (born Dec. 11, 1843, Clausthal, Hannover—died May 27, 1910, Baden-Baden, Ger.), German physician and one of the founders of bacteriology. He discovered the anthrax disease cycle (1876) and the bacteria responsible for tuberculosis (1882) and cholera (1883). For his discoveries in regard to tuberculosis, he received the Nobel Prize for Physiology or Medicine in 1905.

Early Training

Koch attended the University of Göttingen, where he studied medicine, graduating in 1866. He then became a physician in various provincial towns. After serving briefly as a field surgeon during the Franco-Prussian War of 1870–71, he became district surgeon in Wollstein, where he built a small laboratory. Equipped with a microscope, a microtome (an instrument for cutting thin slices of tissue), and a homemade incubator, he began his study of algae, switching later to pathogenic (disease-causing) organisms.

Anthrax Research

One of Koch's teachers at Göttingen had been the anatomist and histologist Friedrich Gustav Jacob Henle, who in 1840 had published the theory that infectious diseases are caused by living microscopic organisms. In 1850 the French parasitologist Casimir Joseph Davaine was among the first to observe organisms in the blood of diseased animals. In 1863 he reported the transmission of anthrax by the inoculation of healthy sheep with the blood of animals dying of the disease and the finding of microscopic rod-shaped bodies in the blood of both groups of sheep. Inspired by the work of the French microbiologist Louis Pasteur, Davaine showed that it was highly probable that, because the sheep did not become diseased in the absence of these rodlike bodies, anthrax was due to the presence of such organisms in the blood. The natural history of the disease was, nevertheless, far from complete.

It was at that point that Koch began. He cultivated the anthrax organisms in suitable media on microscope slides, demonstrated their growth into long filaments, and discovered the formation within them of oval, translucent bodies—dormant spores. Koch found that the dried spores could remain viable for years, even under exposed conditions. The finding explained the recurrence of the disease in pastures long unused for grazing, for the dormant spores could, under the right conditions, develop into the rod-shaped bacteria (bacilli) that cause anthrax. The anthrax life cycle, which Koch had discovered, was announced and illustrated at Breslau in 1876, on the invitation of Ferdinand Cohn, an eminent botanist. Julius Cohnheim, a famous pathologist, was deeply impressed by Koch's presentation. "It leaves nothing more to be proved," he said.

As we explore new areas, and even old ones, new organisms are continuously being identified.

I regard it as the greatest discovery ever made with bacteria and I believe that this is not the last time that this young Robert Koch will surprise and shame us by the brilliance of his investigations.

Cohn, whose discovery of spores had been published in 1875, was also very much impressed and generously helped to prepare the engraving for Koch's epochal, which he also published. One of Cohn's pupils, Joseph Schroeter, found that chromogenic (colour-forming) bacteria would grow on such solid substrates as potato, coagulated egg white, meat, and bread and that those colonies were capable of forming new colonies of the same colour, consisting of organisms of the same type. That was the starting point of Koch's pure-culture techniques, which he worked out a few years later. That a disease organism might be cultured outside the body was a concept introduced by Louis Pasteur, but the pure-culture techniques for doing so were perfected by Koch, whose precise and ingenious experiments demonstrated the complete life cycle of an important organism. The anthrax work afforded for the first time convincing proof of the definite causal relation of a particular microorganism to a particular disease.

Contributions to General Bacteriology and Pathology

In 1877 Koch published an important on the investigation, preservation, and photographing of bacteria. His work was illustrated by superb photomicrographs. In his he described his method of preparing thin layers of bacteria on glass slides and fixing them by gentle heat. Koch also invented the apparatus and the procedure for the very useful hanging-drop technique, whereby microorganisms could be cultured in a drop of nutrient solution on the underside of a glass slide.

In 1878 Koch summarized his experiments on the **etiology** of wound infection. By inoculating animals with material from various sources, he produced six types of infection, each caused by a specific microorganism. He then transferred these infections by inoculation through several kinds of animals, reproducing the original six types. In that study, he observed differences in pathogenicity for different species of hosts and demonstrated that the animal body is an excellent apparatus for the cultivation of bacteria.

Koch, now recognized as a scientific investigator of the first rank, obtained a position in Berlin in the Imperial Health Office, where he set up a laboratory in bacteriology. With his collaborators, he devised new research methods to isolate pathogenic bacteria. Koch determined guidelines to prove that a disease is caused by a specific organism. These four basic criteria, called Koch's postulates, are:

Etiology is the study of causation, or origination

- A specific microorganism is always associated with a given disease.
- The microorganism can be isolated from the diseased animal and grown in pure culture in the laboratory.
- The cultured microbe will cause disease when transferred to a healthy animal.
- The same type of microorganism can be isolated from the newly infected animal.

Studies of Tuberculosis and Cholera

Koch concentrated his efforts on the study of tuberculosis, with the aim of isolating its cause. Although it was suspected that tuberculosis was caused by an infectious agent, the organism had not yet been isolated and identified. By modifying the method of staining, Koch discovered the tubercle bacillus and established its presence in the tissues of animals and humans suffering from the disease. A fresh difficulty arose when for some time it proved impossible to grow the organism in pure culture. But eventually Koch succeeded in isolating the organism in a succession of media and induced **tuberculosis** in animals by inoculating them with it. Its etiologic role was thereby established. On March 24, 1882, Koch announced before the Physiological Society of Berlin that he had isolated and grown the tubercle bacillus, which he believed to be the cause of all forms of tuberculosis.

Tuberculosis (TB) is an infectious disease usually caused by the bacterium Mycobacterium tuberculosis (MTB).

Meanwhile, Koch's work was interrupted by an outbreak of cholera in Egypt and the danger of its transmission to Europe. As a member of a German government commission, Koch went to Egypt to investigate the disease. Although he soon had reason to suspect a particular comma-shaped bacterium (vibrio) as the cause of cholera, the epidemic ended before he was able to confirm his hypothesis. Nevertheless, he raised awareness of amebic dysentery and differentiated two varieties of Egyptian conjunctivitis. Proceeding where cholera is endemic, he completed his task, identifying both the organism responsible for the disease and its transmission via drinking water, food, and clothing. Resuming his studies of tuberculosis, Koch investigated the effect an injection of dead bacilli had on a person who subsequently received a dose of living bacteria and concluded that he may have discovered a cure for the disease. In his studies he used as the active agent a sterile liquid produced from cultures of the bacillus. However, the liquid, which he named tuberculin (1890), proved disappointing, and sometimes dangerous, as a curative agent. Consequently, its importance as a means of detecting a present or past tubercular state was not immediately recognized (see tuberculin test). Additional work on tuberculosis came later, but, after the seeming debacle of tuberculin, Koch was also occupied with a great variety of investigations into diseases of humans and animals—studies of leprosy, bubonic plague, livestock diseases, and malaria.

In 1901 Koch reported work done on the pathogenicity of the human tubercle bacillus in domestic animals. He believed that infection of human beings by bovine tuberculosis is so rare that it is not necessary to take any measures against it. That conclusion was rejected by commissions of inquiry in Europe and America but extensive and important work was stimulated by Koch. As a result, successful measures of prophylaxis were devised.

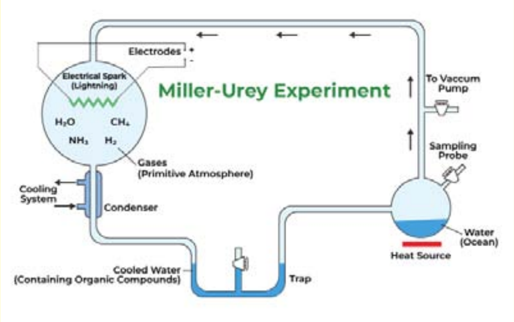
Historical Assessment

Not an eloquent speaker, Koch was nevertheless by example, demonstration, and precept one of the most effective of teachers, and his numerous pupils—from the entire Western world and Asia—were the creators of the new era of bacteriology. His work on trypanosomes was of direct use to the eminent German bacteriologist Paul Ehrlich; that is only one example of Koch’s instigation of epochal work both within and beyond his own immediate sphere. His discoveries and his technical innovations were matched by his fundamental concepts of the etiology of disease. Long before his death, his place in the history of science was universally recognized.

CONCEPT OF ABIOGENESIS THEORY

According to the abiogenesis theory, all life originated from inorganic molecules that undergo various recombinations as a result of energy input. These many forms combined to form a self-replicating molecule, which may have started building the fundamental building blocks of life, such as the cell, using the other molecules generated through abiogenesis. In the same way that populations change over time during the evolution of animals, molecules change over time during the evolution of molecules. According to scientists, RNA molecules are thought to have been the first self-replicating molecules seen in the ribosomes of almost all living things on Earth, some RNA molecules have the known ability to catalyze the production of new RNA molecules. One of these early RNA molecules developed in such a way that it created an identical RNA molecule. This molecule's concentration in the prebiotic soup sharply grew, and it continued to interact with itself and several proteins that also developed around it through abiogenesis. The RNA molecule eventually developed mutations that enabled it to create a protein that would generate further RNA.

Abiogenesis Experiments



In the early 1950s, Stanley Miller, a graduate student from the United States, and Harold Urey, his graduate advisor, decided to recreate an early Earth system to test the **Oparin-Haldane abiogenesis hypothesis**. After combining the theoretical elements and basic compounds in the air, they released sparks from the mixture. Through analysis of the chemical reaction products, they were able to identify the amino acids generated throughout the simulation. This evidence that the first portion of the idea was correct was used to support later studies that tried to establish replicating molecules from amino acids. These tests were unsuccessful. Researchers found that the prebiotic atmosphere of early Earth included more oxygen and fewer other important chemicals than the sample used in the Miller-Urey experiment after this experiment. It led some people to question if the conclusions were still valid. Since then, research has confirmed the initial findings by locating organic compounds, including amino acids, using an adjusted atmospheric composition.

SUMMARY

- Microbiology is the study of a variety of living organisms which are invisible to the naked eye like bacteria and fungi and many other microscopic organisms.
- Microorganisms affect animals, the environment, the food supply and also the healthcare industry. There are many different areas of microbiology including environmental, veterinary, food, pharmaceutical and medical microbiology, which is the most prominent.
- Genus comprises a group of related species which has more characters in common in comparison to species of other genera.
- Herbarium is a store house of collected plant specimens that are dried, pressed and preserved on sheets. Further, these sheets are arranged according to a universally accepted system of classification. These specimens, along with their descriptions on herbarium sheets, become a store house or repository for future use
- Biogenesis is the production of life. In Latin, *bio* means life, and *genesis* means the beginning or origin of. For most of history, mankind thought that biogenesis often occurred through spontaneous generation from earth or vegetable matter, alongside reproduction, which we now know is the only way that biogenesis ever happens.
- Synthesis of ribosome in eukaryotes is complicated and takes place inside the nucleolus. Nucleoli disappear during mitosis, but at telophase new nucleoli are formed at specific chromosomal sites called nucleolar organizers located in secondary constrictions on the chromosomes.

MULTIPLE CHOICE QUESTIONS

1. **What is Microbiology?**
   a. Study of molecules that are visible to human eyes
   b. Study of animals and their family
   c. Study of organisms that are not visible to naked eyes
   d. Study of microscope
2. **Who is known as the father of Microbiology?**
   a. Edwin John Butler
   b. Ferdinand Cohn
   c. Robert Koch
   d. Antoni van Leeuwenhoek
3. **Which microorganism(s) among the following perform photosynthesis by utilizing light?**
   a. Cyanobacteria, Fungi and Viruses
   b. Viruses
   c. Cyanobacteria
   d. Fungi
4. **Which part of the compound microscope helps in gathering and focusing light rays on the specimen to be viewed?**
   a. Condenser lens
   b. Magnifying lens
   c. Objective lens
   d. Eyepiece lens
5. **Which of the following are produced by microorganisms?**
   a. Alcoholic beverages
   b. Fermented dairy products
   c. Breads
   d. All of the mentioned

Answers

1. (c) 2. (d) 3. (c) 4. (a) 5. (d)

REVIEW QUESTIONS

1. Why the classification systems changing every now and then?
2. What different criteria would you choose to classify people that you meet often?
3. What do we learn from identification of individuals and populations?
4. Describe the theory of spontaneous generation and some of the arguments used to support it.
5. Explain how the experiments of Redi and Spallanzani challenged the theory of spontaneous generation.

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CHAPTER 2

MICROBIAL NUTRITION AND GROWTH

Microbiologists use the term growth to indicate an increase in a population of microbes rather than an increase in size. Microbial growth depends on the metabolism of nutrients, and results in the formation of a discrete colony, an aggregation of cells arising from a single parent cell. A nutrient is any chemical required for growth of microbial populations. The most important of these are compounds containing carbon, oxygen, nitrogen, and/or hydrogen.

After studying this chapter, you will understand the core concepts of:

- Sources of Essential Nutrients
- Growth of Bacterial Cultures

INTRODUCTION

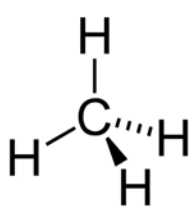
Like all living organisms, microorganisms also require different nutrients for their growth. These nutrients are obtained from the environment or the host in which they inhabit. These nutrients come from the culture medium in which they are grown in the lab. Nutrients are substances used in biosynthesis and energy production, therefore are required for microbial growth.

Growth of a living organism, may be defined as an increase in mass or size in any direction accompanied by synthesis of macromolecules, leading to the production of a newly organized structure. Growth when applied to bacteria normally refers to an increase in number of individual cells and so, is a measure of population density from original inoculums. Bacterial growth can be very rapid for ex. culture of Escherichia coli can double in size in 20 minutes in a rich medium. So, to obtain energy and construct new cellular components, organisms must have a supply of raw materials or nutrients.

CORE CONCEPT: SOURCES OF ESSENTIAL NUTRIENTS

The sources of common essential nutrients are carbon, hydrogen, nitrogen, oxygen, phosphorus and sulfur. There are two categories of essential nutrients; macro-nutrients and micro-nutrients.

Nutrients are materials that are acquired from the environment and are used for growth and metabolism. Microorganisms (or microbes) vary significantly in terms of the source, chemical form, and amount of essential elements they need. Some examples of these essential nutrients are carbon, oxygen, hydrogen, phosphorus, and sulfur. There are two categories of essential nutrients: macro-nutrients (which are needed in large amounts) and micro-nutrients (which are needed in trace or small amounts). Macro-nutrients usually help maintain the cell structure and metabolism. Micro-nutrients help enzyme function and maintain protein structure. Organic nutrients contain some combination of carbon and hydrogen atoms. Inorganic nutrients are elements or simply molecules that are made of elements other than carbon and hydrogen.



The sources of common essential nutrients are carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur. Organisms usually absorb carbon when it is in its organic form. Carbon in its organic form is usually a product of living things. Another essential nutrient, nitrogen, is part of the structure of protein, DNA, RNA, and ATP. Nitrogen is important for heterotroph survival, but it must first be degraded into basic building blocks, such as amino acids, in order to be used. Oxygen is an important component of both organic and inorganic compounds. It is essential to the metabolism of many organisms. Hydrogen has many important jobs including maintaining the pH of solutions and providing free energy in reactions of respiration. Phosphate is an important player in making nucleic acids and cellular energy transfers. Without sufficient phosphate, an **organism** will cease to grow. Lastly, sulfur is found in rocks and sediments and is found widely in mineral form.

Organisms are classified by taxonomy into specified groups such as the multicellular animals, plants, and fungi; or unicellular microorganisms such as a protists, bacteria, and archaea.

Macronutrients or Macro Elements

These are required by microorganisms in relatively large amounts. Carbon, oxygen, hydrogen nitrogen, sulfurs and phosphorous are components of carbohydrates, lipids, proteins and nucleic acids. The remaining four macro elements (K, Ca, Mg and Fe) exist in the cell as cations.

- K^+ - is required for the activity by a number of enzymes, including those involved in protein synthesis.
- Ca^{2+} - contributes to the heat resistance of bacterial endospores. 15% of spore contains dipicolinic acid and calcium.
- Mg^{2+} - serves as a cofactor for many enzymes, complexes with ATP and stabilizes ribosomes and cell membranes.
- Fe^{2+} and Fe^{2+} - part of cytochromes and a cofactor for enzymes and electron-carrying proteins.

Micronutrients or Trace Elements

These are manganese, zinc, cobalt, molybdenum, nickel and copper. These are normally part of enzymes and cofactors, and they aid in the catalysis of reactions and maintenance of protein structure.

- Zn^{2+} - is present at the active site of some enzymes but is also involved in the association of regulatory and catalytic subunits in *E.coli* aspartate carbomoyl transferase.

- Mn^{2+} - aids many enzymes catalyzing the transfer of phosphate groups.
- Mo^{2+} - required for nitrogen fixation.
- Co^{2+} - is a component of Vitamin B12.

Besides macro and micro nutrients, some microorganisms may have particular requirements that reflect the special nature of their morphology or environment. Diatoms need silicic acid to construct their beautiful cell walls of silica. Bacteria growing in saline lakes and oceans depend on the presence of high concentrations of sodium ion. Microorganisms require a balanced mixture of all the above nutrients for proper growth.

Requirements for Carbon, Hydrogen and Oxygen

Carbon is needed for the skeleton or backbone of all organic molecules and molecules serving as carbon sources normally also contribute both oxygen and hydrogen atoms. One important carbon source that does not supply hydrogen or energy is CO_2. Autotrophs – can use CO_2 as their sole or principal source of carbon. Many microorganisms are autotrophic, and most of these carry out photosynthesis and use light as their energy source. Some autotrophs oxidize inorganic molecules and derive energy from electron transfer. Heterotrophs – are organisms that use reduced pre-formed organic molecules as carbon sources. Ex. Glycolytic pathway produces carbon skeleton for use in biosynthesis and also releases energy as ATP and NADH. Actinomycetes will degrade amyl alcohol, paraffin and even rubber. *Burkholderia cepacia* can use over 100 different carbon compounds. Some microorganisms can metabolize even relatively indigestible human-made substances such as pesticides. Indigestible molecules can be oxidized and degraded in the presence of a growth promoting nutrient that is metabolized at the same time, a process called Co-metabolism. The products of this breakdown can then be used as nutrients by other microorganisms.

> In order for it to be an organic molecule or nutrient it has to have both carbon and hydrogen, and it can be a combination of any other element, just as long as it has both.

Nutritional Types of Microorganisms

In addition to Carbon, hydrogen and oxygen all organisms require sources of energy and electrons for growth.

- *Carbon Sources*
    - o Autotrophs - CO_2 sole or principal biosynthetic carbon source
    - o Heterotrophs – reduced, preformed organic molecules from other organisms.
- *Energy Sources*

- o Phototrophs – use light as their energy source.
- o Chemotrophs – obtain energy from the oxidation of chemical compounds (either organic or in organic)

- *Electron Sources*
  - o Lithotrophs – use reduced inorganic substances as their electron source.
  - o Organotrophs – extract electrons from organic compounds.

Four major nutritional classes based on their primary sources of carbon, energy and electrons is known.

- *Phtotolithotrophic Autotrophs or Photoautotrophs or Photolithoautotrophs*

  Source of energy – light energy

  Source of electrons – Inorganic hydrogen/ electron

  Carbon source - CO_2

  Example: Algae, purple and green sulfur bacteria and cyanobacteria.

- *Photoorganotrophic Heterotrophy or Photoorganoheterotrophy*

  Source of energy – light energy

  Source of electrons – organic hydrogen/ electron

  Carbon source – organic carbon sources (CO_2 may also be used)

  Example: Purple and green nonsulfur bacteria (common inhabitants of lakes and streams)

- *Chemolithotrophic Autotrophs or Chemolithoautotrophy*

  Source of energy – Chemical energy source (inorganic)

  Source of electrons – Inorganic hydrogen/ electron donor

  Carbon source - CO_2

  Example: Sulfur-oxidizing bacteria, hydrogen bacteria, nitrifying bacteria, iron-oxidizing bacteria.

- *Chemoorganotrophic Heterotrophs or Chemoorganoheterotrophy*

  Source of energy – Chemical energy source (organic)

Source of electrons – Inorganic hydrogen/ electron donor

Carbon source – organic carbon source

Example: Protozoan, fungi, most non-photosynthetic bacteria (including most pathogens)

The most common nutritional types are photolithoautotrophs and chemoorganoheterotrophs. Bacteria ***Beggiatoa*** rely on inorganic energy sources and organic (or sometimes CO_2) carbon sources. These microbes are sometimes called Mixotrophic because they combine chemolithoautotrophic and heterotrophic metabolic processes.

Beggiatoa is a genus of bacteria in the order Thiotrichales.

Requirements for Nitrogen, Phosphorous and Sulfur

Nitrogen is needed for the synthesis of amino acids, purines, pyramidines, some carbohydrates and lipids, enzyme cofactors and other substances. Most phototrophs and many nonphotosynthetic microorganisms reduce nitrate to ammonia and incorporate the ammonia in assimilatory nitrate reduction. A variety of bacteria like many Cyanobacteria and *Rhizobiium* can reduce and assimilate atmospheric nitrogen using the nitrogenase systems. Phosphorous is present in nucleic acids, phospholipids, ATP, several cofactors, some proteins and other cell components. All microorganisms use inorganic phosphate as their phosphorous source and incorporate it directly. *E.coli* can use both organic and inorganic phosphate. Organophosphates such as hexose 6- phosphate can be taken up directly by transport proteins. Other organophosphates are often hydrolyzed in the periplasm by the enzyme alkaline phosphatase to produce inorganic phosphate which is then transported across the plasma membrane. When inorganic phosphate is outside the bacterium, it crosses the outer membrane by the use of a porin protein channel. Sulfur is needed for the synthesis of substances like the amino acids cysteine and methionine, some carbohydrates biotin and thiamine. Most of them use sulfate as a source of sulfur and reduce it by assimilatory sulfate reduction; a few require a reduced form of sulfur such as cysteine.

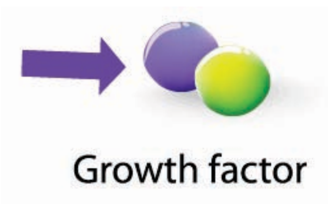
Growth Factors

Many microorganisms have the enzymes and pathways necessary to synthesize all cell components. Many lack one or more enzymes and hence require organic compounds because they are essential cell components or precursors of such components and cannot be synthesized by the organisms are called – growth factors. There are three major classes of growth factors:

- Amino acids – needed for protein synthesis.
- Purines and Pyramidines – for nucleic acid synthesis
- Vitamins – small organic molecules that usually make up all or part of enzyme cofactors, and only very small amounts sustain growth.

Knowledge of the specific growth factor requirements of many microorganisms makes possible quantitative growth response assays for a variety of substances. The observation that many microorganisms can synthesize large quantities of vitamins has led to their use in industry. Several water-soluble and fat-soluble vitamins are produced using industrial fermentations.

- Ribofalvin – *Clostridium, Candida, Ashbya, Eremothecium*
- Coenzyme A – *Brevibacterium*
- Vitamin B_{12} – *Streptomyces, Propionibacterium, Pseudomonas*
- Vitamin C – *Gluconobacter, Erwinia, Corynebacterium*
- β- Carotene – *Dunaliella*
- Vitamin D - *Saccharomyces*



Uptake of Nutrients by the Cell

Uptake of the required nutrients by the microbial cell is important. Since microorganisms live in nutrient poor habitats, they must be able to transport nutrients from dilute solutions into the cell against concentration gradient. Finally, they must pass through a selectively permeable plasma membrane. Microorganisms use different transport mechanisms like facilitated diffusion, active transport and group translocation. Eukaryotic microorganisms do not employ group translocation but take up nutrients by **endocytosis**.

Endocytosis is a form of bulk transport in which a cell transports molecules (such as proteins) into the cell (endo- + cytosis) by engulfing them in an energy-using process.

Movement of materials across the plasmamembrane is mostly done by two processes:

- Passive processes: Substances cross the area from an area of high concentration to an area of low concentration without any expenditure of energy (ATP). Example, simple diffusion, osmosis and facilitated diffusion.
- Active process: The cell must use energy (ATP) to move substances from areas of low concentration to areas of high concentration. Example, Group translocation.

Passive Processes

- *Passive or simple diffusion*: Often called diffusion, is the process in which molecules move from a region of higher concentration to one of lower concentration. The rate is dependent on the size of the concentration gradient between a cell's exterior and its interior. Very small molecules such as water and oxygen and carbon dioxide move across membranes by simple or passive diffusion. Larger molecules, ions, and polar substances do not cross membranes by this method.
- *Osmosis:* Is the net movement of solvent molecules across a selectively permeable membrane from an area in which the solvent molecules are highly concentrated to an area of low concentration until equilibrium is reached. In living systems the chief solvent is water. The three types of solutions which are normally found are isotonic, hypotonic and hypertonic.
- *Facilitated diffusion:* The rate of diffusion across selectively permeable membrane is greatly increased by using carrier proteins, sometimes called permeases which are embedded in the plasma membrane. Because a carrier aids the diffusion process, it is called as facilitated diffusion. Carrier proteins also resemble enzymes in their specificity for the substances to be transported; each carrier is selective and will transport only closely related solutes. Because there is no energy input, molecules will continue to enter only as long as their concentration is greater on the outside. Two widespread major intrinsic protein channels in bacteria are aquaporins that transport water and glycerol facilitators which aid glycerol diffusion. The carrier protein complex spans the membrane. After the solute molecule binds to the outside, the carrier may change conformation and release the molecule on the cell interior. The carrier would subsequently change back to its original shape and be ready to pick up another molecule. The mechanism is driven by concentration gradients and therefore is reversible.

Examples. Glycerol is transported by facilitated diffusion in *E.coli, Salmonella typhimurium*, *Pseudomonas*, *Bacillus* and many other bacteria. This is prominent in eukaryotes where it is used to transport a variety of sugars and amino acids.

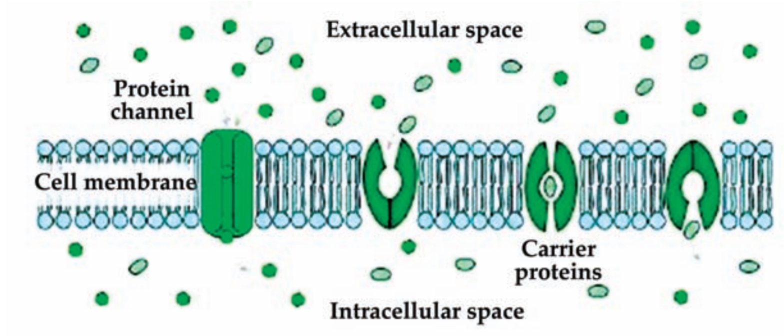


Figure 1. Facilitated diffusion. The carrier proteins aid in the release of solute molecules from extracellular space to the intracellular space.

Bacteria is a type of biological cell. They constitute a large domain of prokaryotic microorganisms.

Active Transport

Active transport is the transport of solute molecules to higher concentrations or against a concentration gradient, with the use of metabolic energy input. It resembles facilitated diffusion in the involvement of protein carrier activity, but differs in its use of metabolic energy and in its ability to concentrate substances. One example of active transport is binding protein transport systems or ATP-binding cassette transporters (ABC transporters) are active in **bacteria,** archaea and eukaryotes.

ABC Transporters

These transporters consist of two hydrophobic membrane spanning domains associated on their cytoplasmic surfaces with two nucleotide-binding domains. The membrane spanning domains from a pore in the membrane and the nucleotide binding domains bind and hydrolyze ATP to drive uptake. ABC transporters employ special substrate binding proteins, which are located in the periplasmic space of gram-negative bacteria or are attached to membrane lipids on the external face of the gram positive plasma membrane. These binding proteins, bind the molecule to be transported and then interact with the membrane transport proteins to move the solute molecule inside the cell. *E. coli* transports a variety of sugars (arabinose, maltose, galactose, and ribose) and amino acids (glutamate, histidine, leucine) by this mechanism.

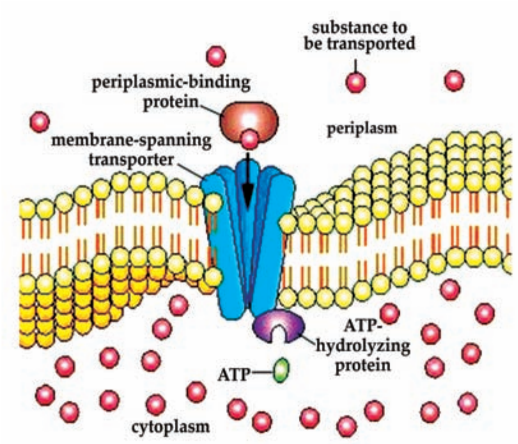


Figure 2. ABC transporter system.

Eukaryotic ABC transporters are sometimes of great medical importance. Some tumor cells pump drugs out using these transporters. Cystic fibrosis results from a mutation that inactivates an ABC transporter that acts as a chloride ion channel in lungs. Bacteria also use Proton gradients generated during electron transport to drive active transport. The lactose permease of *E.coli* transports a lactose molecule inward as a proton simultaneously enters the cell. Such linked transport of two substances in the same direction is called Symport. *E.coli* also uses proton symport to take up amino acids and organic acids like succinate and malate.

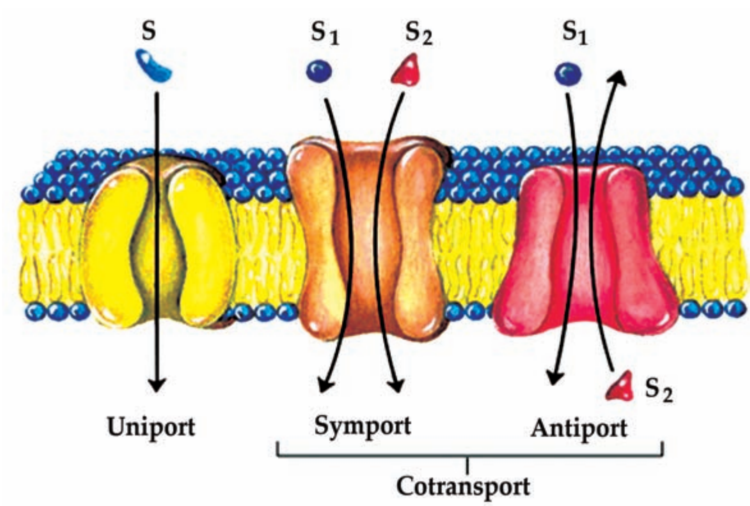


Figure 3. Comparison of the three different types of transport mechanisms.

A proton gradient also can power active transport indirectly, often through the formation of a sodium ion gradient. In *E. coli* , sodium transport system pumps sodium outward in response to the inward movement of protons. Such linked transport in which the transported substances move in opposite directions is called Anitport. The sodium gradient generated by this proton anitport system then drives the uptake of sugars and amino acids. *E.coli* has at least transport systems for the sugar galactose.

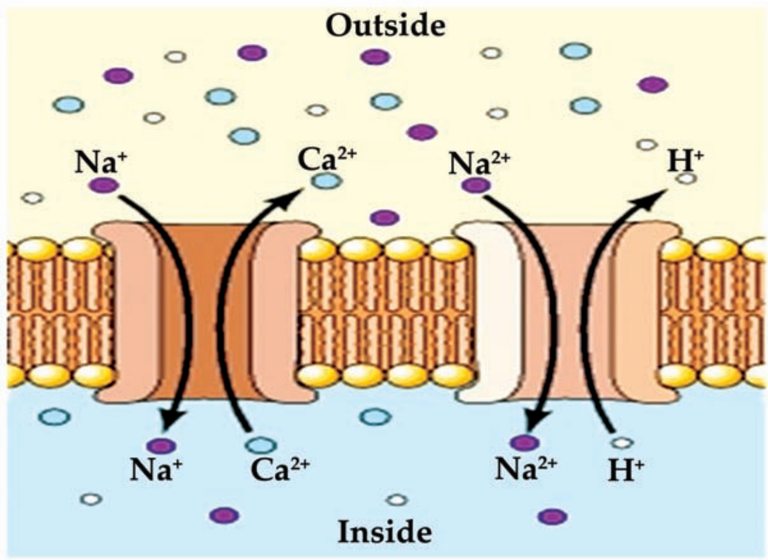


Figure 4. Antiport mechanism - linked transport of two substances in opposite direction.

Enzymes are proteins that help speed up chemical reactions in .our bodies

Group Translocation

In active transport, solute molecules move across a membrane without modification. Many prokaryotes also take up molecules by group translocation, a process in which a molecule is transported into the cell while being chemically altered. For example, Phosphoenolpyruvate: Sugar phosphotransferease system (PTS). It transports a variety of sugars while phosphorylating them using phosphoenolpyruvate (PEP) as the phosphate donor.

PEP + Sugar (outside) → Pyruvate + Sugar-P (inside)

In *E. coli* and *Salmonella typhimurium*, it consists of two enzymes and a low molecular weight heat stable protein (HPr). HPr and enzyme I (EI) are cytoplasmic. **Enzyme** II (EII) is more variable in structure and often composed of three subunits or domains. EIIA is cytoplasmic and soluble. EIIB also is hydrophilic but frequently attached to EIIC, a hydrophobic protein that is embedded in the membrane. A high energy phosphate is transferred from PEP to enzyme II (EII) with the aid of enzyme I (EI) and HPr. Then a sugar molecule is phosphorylated as it is carried across the membrane by enzyme II (EII). Enzyme II (EII) transports only specific sugars and varies with PTS, whereas enzyme I (EI) and HPr are common to all PTS's.

Salmonella is a genus of rod-shaped Gram-negative bacteria of the family .Enterobacteriaceae

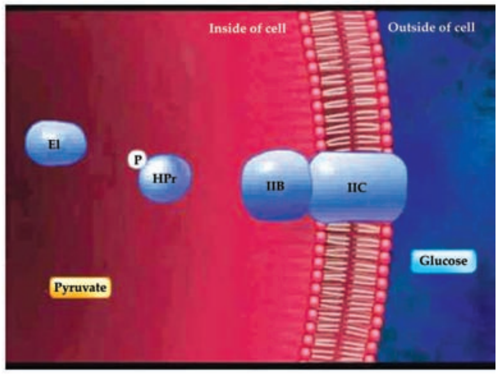


Figure 5. Group translocation: Bacterial PTS transport.

PTS's are widely distributed in prokaryotes. Aerobic bacteria lack PTS's. Genera *Escherichia,* **Salmonella**, *Staphylococcus* and other facultative anaerobic bacteria have phosphotransferase systems; some obligate anaerobic bacteria (*Clostridium*) also have PTS's. Many carbohydrates are transported by these systems. *E. coli* takes up glucose, fructose, mannitol, sucrose, N-acetylglucosamine, cellobiose and other carbohydrates by group translocation.

Iron Uptake

All microorganisms require iron for use in cytochromes and many enzymes. Iron uptake is made difficult by the extreme insolubility of ferric ion (Fe^{3+}) and its derivatives, which leave little free, iron available for transport. Many bacteria and fungi have overcome this difficulty by secreting siderophores. Siderophores – are low molecular weight molecules that are able to complex with ferric ion and supply it to the cell. These are either hydroxamates or phenolates-catecholates. Ferrichrome is a hydroxamate produced by many fungi; enterobactin is the catecholate formed by *E.coli*. Microorganisms secrete sidereophores when little iron is available in the medium. Once the iron-siderophore complex has reached the cell surface, it binds to a siderophore receptor protein. The iron is either released to enter the cell directly or the whole iron-siderophore complex is transported inside by an ABC transporter. In *E.coli*, the siderophore receptor is in the outer membrane of the cell envelope; when the iron reaches the periplasmic space, it moves through the **plasma membrane** with the aid of the transporter. After the iron has entered the cell, it is reduced to the ferrous form (Fe^{2+}). Iron is so crucial to microorganisms that many use more than one route of iron uptake to ensure an adequate supply.

Plasma membrane is an essential part of the cell that encloses the cell's interior components while only allowing certain parts of the outside environment to enter.

Culture Media, Isolation and Cultivation of Pure Cultures

In the laboratory, bacteria are normally grown or cultured in either liquid medium, in flasks, bottles or large culture vessels called fermenters or solid medium in Petri dishes which are round, normally plastic or glass dishes. Introduction of microbes into or onto these media is called inoculation. The nature of the medium depends on the microorganism's natural environment because its nutrient requirements reflect its natural surroundings. The culture medium must contain all the nutrients that the microorganism requires for growth like water, a source of energy, carbon, nitrogen, essential inorganic ions and a number of trace elements. Bacteria that can synthesize all they require from the basic ingredients are called Prototrophs – and most microorganisms that survive in the outside environment can do this. Microbes that have become adapted to life in a situation rich with nutrients such as human body may require other growth factors like vitamins, amino acids or nitrogenous bases to be provided. These are called auxotrophs.

The media that are used in microbiology laboratories to culture bacteria are referred to as artificial media or synthetic media, because they do not occur naturally; rather they are prepared in the laboratory. There are number of ways of categorizing the media, one way is to classify based on whether the exact contents of the media are known or not.

- *Chemically Defined Media*: Is one in which all the ingredients are known; and was prepared in the laboratory by adding a certain number of grams of each of the components (Carbohydrates, amino acids, salts etc.). Particularly photolithoautotrophs such as Cyanobacteria and eukaryotic algae can be grown on relatively simple media containing CO_2 as a carbon source (often added as sodium carbonate or bicarbonate), nitrate or ammonia as a nitrogen source, sulfate, phosphate and a variety of minerals. Chemoorganoheterotrophs can be grown in a defined media with glucose as a carbon source and an ammonium salt as a nitrogen source.
- *Complex Media*: contains undefined ingredients and the exact contents are not known. Complex medium may be sufficiently rich and complete to meet the nutritional requirements of many different microorganisms. They contain undefined components like peptones, meat extract and yeast extract. Peptones – are proteins hydrolysates prepared by partial proteolytic digestion of meat, casein, soyameal, gelatin and other protein sources (Carbon, energy and nitrogen). Beef extract – aqueous extracts from lean beef and contain amino acids, peptides, nucleotides, organic acids and minerals and vitamins.Yeast extract – from brewer's yeast and contain an excellent source of B vitamins, nitrogen and carbon compounds. Three commonly used complex media are
    - o Nutrient broth – Peptone (5.0g/L), Beef Extract (3.0g/L)
    - o Tryptic soya broth – Tryptone (enzyme digest of casein), Peptone (enzyme digest of soybean meal), glucose, NaCl and K_2HPO_4

- MacConkey Agar – Pancreatic digest of gelatin, casein, peptic digest of animal tissue, bile salts mixture, NaCl, neutral red, crystal red and agar.

These media are routinely used for cultivation of bacteria in the laboratory and particularly useful for cultivation of bacteria whose growth requirements have not been defined.

Culture media can also be categorized as liquid or solid. Liquid media (also called broths) are contained in tubes, flasks and fermenters. Solid media are prepared by adding agar to liquid medium and then pouring the media into tubes or Petridishes where the media will solidify. Agar is a complex polysaccharide that is obtained from red marine algae. Other solidifying agents are gelatin, silica. A 1% or 2% agar can be used to solidify, but 1.5% is the most commonly used.

- *Enriched Medium*: Is a broth or solid medium containing a rich supply of special nutrients that promotes the growth of fastidious organisms (that have complex nutrition and environmental requirements). It is usually prepared by adding extra nutrients to a medium called nutrient agar. Blood agar (nutrient agar + 5% sheep red blood cells): It is bright red in color and distinguishes between the hemolytic and non-hemolytic bacteria (*Streptococci* and other pathogens)

Chocolate agar is a nonselective, enriched growth medium used for isolation of pathogenic bacteria.

Chocolate agar (nutrient agar + powdered hemoglobin): It is brown in color and is considered more enriched than blood agar as hemoglobin is more readily accessible in chocolate agar. Pathogens like *Neisseria gonorrhoeae* and *Haemophillus influenza*, which will not grow on blood agar, can be cultured.

Types of Media

The different types of media are:

Selective Media

Are designed to suppress the growth of unwanted bacteria and encourage the growth of the desired microbes.

Examples: MacConkey agar inhibits growth of gram positive bacteria and thus is selective for gram-negative bacteria. Phenylethyl alcohol (PEA) agar and Colistinalidixic acid (CAN) agar are selective for gram positive bacteria.

Manitol salt agar contains a concentration of 7.5% NaCl that is quite inhibitory to most human pathogens. Bile salts, a component of fecus, inhibit most gram positive bacteria while permitting many gram negative rods to grow. This media is used for selecting intestinal pathogens which contain bile salts. Dyes such as methylene blue and crystal violet also inhibit certain gram positive bacteria (Staphylococcus can produce acid from mannitol and turn the phenol red dye to bright yellow.

A medium containing acetate as a carbon source would be selective for organisms that grow on acetate. Sabaraud's dextrose agar is used to isolate fungi and cellulose for cellulose digesting bacteria.

Differential Media

Makes it easier to distinguish colonies of the desired organism from other colonies growing on the same plate.

MacConkey agar is also a differential and selective medium to distinguish between lactose fermenting organisms (e.g, *E.coli*) and other non-fermenters (e.g, *Shigella* sp.). Lactose in the medium is fermented by *E.coli* producing acid which causes an indicator dye to change color to red and colonies that do not ferment lactose are white. Dyes can be used as differential agents because many of them are pH indicators that change color in response to the production of an acid or base. MacConkey agar contains neutral red, a dye that is yellow when neutral and pink or red when acidic.

Mannitol salt agar is used to screen for *Staphylococcus aureus*, and it turns the originally pink medium to yellow due to its ability to ferment mannitol. In a sense, blood agar is also differential because it is used to determine the hemolytic and non-hemolytic bacteria.

Blood agar is both differential and an enriched one. It distinguishes between hemolytic and non-hemolytic bacteria. Hemolytic bacteria (e.g., many *Streptococci* and *Staphylococci*) produce clear zones around their colonies because of red blood cell destruction.

Isolation of Pure Cultures

In natural habitats, microorganisms usually grow in complex, mixed populations containing several species. One needs a pure culture, a population of cells arising from a single cell, to characterize an individual species. Robert Koch introduced pure culture techniques.

Spread Plate and Streak Plate

- *Spread plate* - Mixture of cells is spread out on an agar surface so that every cell grows into a completely separate colony, each colony representing a pure culture. A small volume of around 30 to 300 cells (mixed) is transferred to the center of the plate and spread evenly over the surface with a sterile bent-glass rod.
- *Streak plate* – the microbial mixture is transferred to the edge of an agar plate with an inoculating loop or swab and then streaked out over the surface in one of several patterns. In both these techniques, successful isolation depends on spatial separation of single cells.

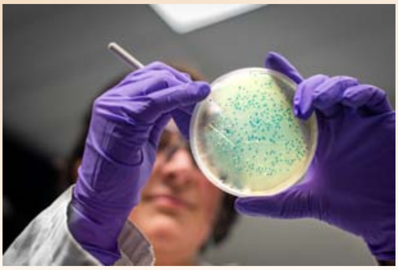
The Pour Plate

Extensively used with bacteria and fungi. The original sample is diluted several times to reduce the microbial population sufficiently to obtain separate colonies when plating. The microbes are mixed with molten agar which has been cooled t 45° Cand then poured into pertidishes. All these techniques require petridishes (special culture dishes) after their inventor Julius Richard Petri (1887). They consist of two round halves, the top half overlapping the bottom. Each bacterium replicates to form a colony that is visible to the naked eye. A colony contains up to 109 copies of the original bacterium.

Chemotaxis is the movement of an organism in response to a chemical stimulus.

Colony Morphology and Growth

Colony development on agar surfaces aids the microbiologist in identifying bacteria because individual species often form colonies of characteristic size and appearance. Structure of bacterial colonies also has been examined with the scanning electron microscope. Generally the most rapid cell growth occurs at the colony edges. Growth is much slower in the center and cell autolysis takes place in the older central portion of some colonies. These differences in growth appear due to gradients of oxygen, nutrients and toxic products within the colony. At the colony edge, oxygen and nutrients are plentiful. Bacteria growing on solid surfaces such as agar can form complex and intricate colony shapes. Nutrient diffusion and availability, bacterial **chemotaxis** and the presence of liquid on the surface all appear to play a role in pattern formation. These patterns vary with nutrient availability and the hardness of the agar surface.



CORE CONCEPT: GROWTH OF BACTERIAL CULTURES

Bacteria grow to a fixed size and then reproduce through binary fission which a form of asexual reproduction. Under optimal conditions, bacteria can grow and divide extremely rapidly.

Growth is defined as an increase in cellular constituents which leads to a rise in cell number. As we are aware, microorganisms reproduce by binary fission or by budding. In order to study growth, normally one follows the changes in the total population number. The cells copy their DNA almost continuously and divide again and again by the process called binary fission. Binary fission which has been described earlier as the form of asexual reproduction in single-celled organisms by which one cell divides into two cells of the same size. Fortunately, few prokaryotic populations can sustain exponential growth for long. Environments are usually limiting in resources such as food and space. Prokaryotes also produce metabolic waste products that may eventually pollute the colony's environment. Still, you can understand why certain bacteria can make you sick so soon after infection or why food can spoil so rapidly. Refrigeration retards food spoilage not because the cold kills the bacteria on food but because most microorganisms reproduce very slowly at such low temperatures.

Binary Fission

In the process of binary fission, the cell elongates and the DNA is replicated. Cell wall and plasma membrane begin to grow inward and cross-wall forms completely around the divided DNA. At the end the cell separates into two individual cells similar to the parent cell and contains all the contents a cell requires for its living including DNA.

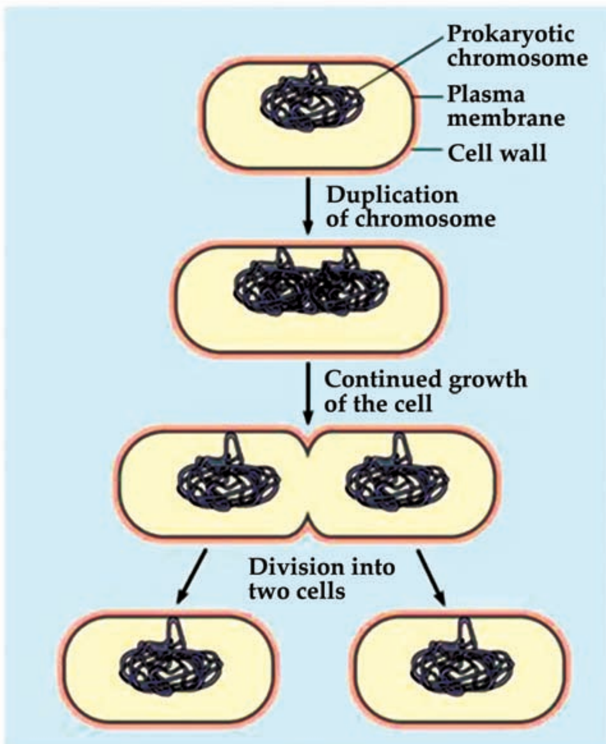


Figure 6. Binary fission in bacteria.

Few bacterial species reproduce by budding. In this method, the cell forms a small initial growth, it enlarges and then it separates from the parent cell. Some filamentous bacteria like actinomycetes reproduce by producing chains of conidiospores and a few bacteria simply fragment.

Growth Curve

The increase in cell number or growth in population is studied by analyzing the growth curve of a **microbial culture**. Bacteria can be grown or cultivated in a liquid medium in a closed system or also called as batch culture. In this method, no fresh medium is added and hence with time, nutrient concentration decreases and an increase in wastes is seen. As bacteria reproduce by binary fission, the growth can be plotted as the logarithm of the number of viable cells verses the time of incubation. The curve plotted shows four basic phases of growth; the lag, log, stationary, and death phase.

Microbial culture is a method of multiplying microbial organisms by letting them reproduce in predetermined culture medium under controlled laboratory conditions.

Lag Phase

As the cells are introduced into the new medium, no immediate increase in cell number occurs. During this phase, the cells are undergoing a period of intense metabolic activity involving synthesis of enzymes and various other molecules required to divide in the coming phase. This phase can vary considerably in length depending on the nature of the medium and the microorganism. The medium may be different from the one the microorganism was growing in previously. The cells may be old and depleted of ATP, essential cofactors and ribosomes; these must be synthesized before growth can begin. So, the microorganism requires time to recover and young, vigorously growing cultures and fresh medium are to be used for the lag phase to be short.

Log Phase

In this phase the cell starts dividing in a logarithmic way and this is also called as exponential phase and the growth is balanced. Cellular reproduction is high during this period and the plot during this phase is a straight line. The cells are most active metabolically during this phase and the population is most uniform during this phase; therefore exponential phase cultures are usually used in biochemical and physiological studies. But during this phase, the microorganisms may be particularly sensitive to adverse conditions. On the whole, in this phase the cells are growing and dividing and increasing in cell number. The rate of exponential growth of a bacterial culture is expressed as generation time, also the doubling time of the bacterial population. Generation time (G) is defined as the time (t) per generation/ n (n = number of generations). Hence, G=t/n is the equation from which calculations of generation time can be derived.

Exponential phase or log phase is balanced growth. That is, all cellular constituents are manufactured at constant rates relative to each other. If nutrient levels or other environmental conditions change, unbalanced growth results. The individual cells may take slightly longer than others to go from lag phase to the log phase, and they do not all divided precisely together. If they divided together and the generation time is same, the number of cells in a culture would increase in a stair – step pattern, exactly doubling every 20 min or a particular time – a hypothetical situation called Synchronous growth. In an actual culture, each cell divides sometime during the 20 min generation time, with about 1/20 cells dividing each minute – a natural situation called nonsynchronous growth or asynchronous growth which appears as a smooth line, not as steps.

Organisms in a tube of culture medium can maintain logarithmic growth for only a limited time. As the number of organisms' increases, nutrients are used up, metabolic wastes accumulate, living space may become limiting factor and aerobes suffer from oxygen depletion

Stationary Phase

Exponential growth cannot be continued forever in a batch culture (e.g. a closed system such as a test tube or flask). Population growth is limited by one of three factors: 1. exhaustion of available nutrients; 2. accumulation of inhibitory metabolites or end products; 3. exhaustion of space, in this case called a lack of "biological space".During the stationary phase, if viable cells are being counted, it cannot be determined whether some cells are dying and an equal number of cells are dividing, or the population of cells has simply stopped growing and dividing. The stationary phase, like the lag phase, is not necessarily a period of quiescence. Bacteria that produce secondary metabolites, such as antibiotics, do so during the stationary phase of the growth cycle (Secondary metabolites are defined as metabolites produced after the active stage of growth). It is during the stationary phase that spore-forming bacteria have to induce or unmask the activity of dozens of genes that may be involved in sporulation process. Starving bacteria frequently produce a variety of starvation proteins, which make the cell much more resistant to damage. They increase peptidoglycan cross-linking and cell wall strength. The Dps (DNA- binding proteins from starved cells) protein protects the DNA. Bacterial pathogens like *Salmonella typhymurium* become more virulent when starved.

Death Phase

Due to the conditions during the stationary phase, the death phase is seen as there is a decline in the number of viable cells. This phase also is like the log phase where the cell number is declining in a logarithmic way. The **cell** is said to be dead if it does not revive itself and reproduce when incubated again in a fresh medium. In this phase, the number of live cells decreases at a logarithmic rate, as indicated by the straight downward sloping diagonal line. The duration of this phase is as highly variable as the duration of log phase. Both depend primarily on the genetic characteristics of the organism.

Cell is the basic structural, functional, and biological unit of all known living organisms.

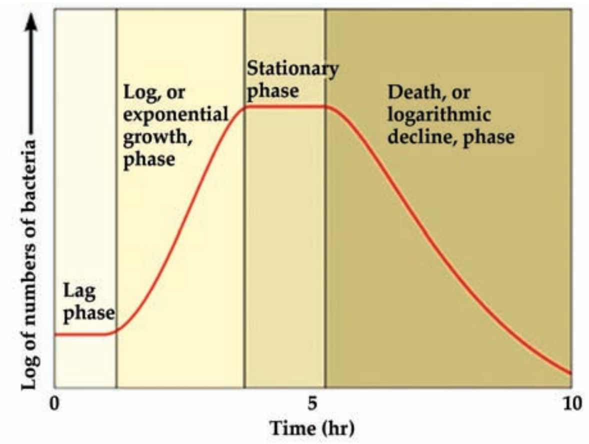


Figure 7. Growth curve of a typical bacterial cell.

Mathematics of Growth

Microbial growth during the exponential phase is very important and of interest to microbiologists and the analysis applies to microorganisms dividing by binary fission. The time required by a cell to divide is called the generation time or doubling time. In the laboratory, under favorable conditions, a growing bacterial population doubles at regular intervals. Growth is by geometric progression: 1, 2, 4, 8, etc. or 2^0, 2^1, 2^2, 2^3.........2^n (where n = the number of generations). This is called exponential growth. In reality, exponential growth is only part of the bacterial life cycle, and not representative of the normal pattern of growth of bacteria in Nature. This might vary from organism to organism depending upon the environmental conditions etc. For example in *E.coli* the generation time is 20 min and hence after 20 generations a single initial cell would increase to over 1 million cells. This would require a little less than 7 hours. The population is doubling every generation; hence the increase in population is always $2^n$where n is the number of generations. The resulting population increase is exponential or logarithmic.

When growing exponentially by binary fission, the increase in a bacterial population is by geometric progression. If we start with one cell, when it divides, there are 2 cells in the first generation, 4 cells in the second generation, 8 cells in the third generation, and so on. The generation time is the time interval required for the cells (or population) to divide.

G (generation time) = (time, in minutes or hours)/n(number of generations)

$G = t/n$

t = time interval in hours or minutes

B = number of bacteria at the beginning of a time interval

b = number of bacteria at the end of the time interval

n = number of generations (number of times the cell population doubles during the time interval)

$b = B \times 2^n$ (This equation is an expression of growth by binary fission)

Solve for n:

$\log b = \log B + n\log 2$

$n=\log b-\log B/\log 2$

$n= \log b-\log B/ .301$

$n = 3.3 \log b/B$

$G = t/n$

Solve for G

$G= t/3.3\log b/B$

Example: What is the generation time of a bacterial population that increases from 10,000 cells to 10,000,000 cells in four hours of growth?

$G= t/3.3\log b/B$

$G= 240\text{minutes}/3.3\log 10^7 /10^4$

$G= 240\text{minutes}/3.3\text{x}3$

$G = 24$ minutes

Measurement of Microbial Growth

A number of techniques are available in order to measure growth of microbial populations. Either population number of mass may be calculated ad growth leads to increase in both.

Direct Measurement of Cell Numbers

Bacteria or microorganisms can be counted directly on the plate and also called as plate counting. Advantage of this method is that it measures the number of viable cells. Disadvantage is that, it is time consuming and expensive as one needs media and other conditions need to be maintained. Bacteria counted on plate counts are referred to as colony forming units as a single cell or a clump of bacterial cells can lead to a colony which contains many cells. The colonies when they are counted in plate count method are to be present sparsely for accurate counting as overcrowding can lead to incorrect counting. To solve this, one has to adapt the serial dilution method in order to get an accurate count.

Serial Dilution and Pour and Spread Plate

Supposing one has to accurately count the number of cells given in a solution, then serial dilution needs to be performed. A 1ml of the sample is taken and transferred to a tube containing 9ml of sterile water and this process can be repeated until we reach a considerable dilution (say 10^6 to 10^7). Once the original inoculum is diluted one needs to perform a pour plate or a spread plate technique in order to count the number of bacteria present in the diluted sample and then the original sample. In pour plate method the diluted sample is poured into the petriplate and then the medium which is at nearly 50^0C is poured over the inoculum and mixed by gentle agitation. With this method, colonies grow within the nutrient agar as well as on the surface of the agar plate. As certain disadvantages are encountered in this method like heat sensitive microorganisms might not grow and also bacteria when they grow within the nutrient medium might not be useful for diagnostic purposes. In order to avoid these problems, spread plate method is mostly used. A 0.1ml of the diluted sample is added to the surface of the nutrient medium and spread uniformly with the help of a glass spreader and after incubation, the colonies can be counted and the concentration of the bacterial cells in the original sample is calculated as follows:

Number of bacteria/ml = Number of colonies on plate x reciprocal of dilution of sample

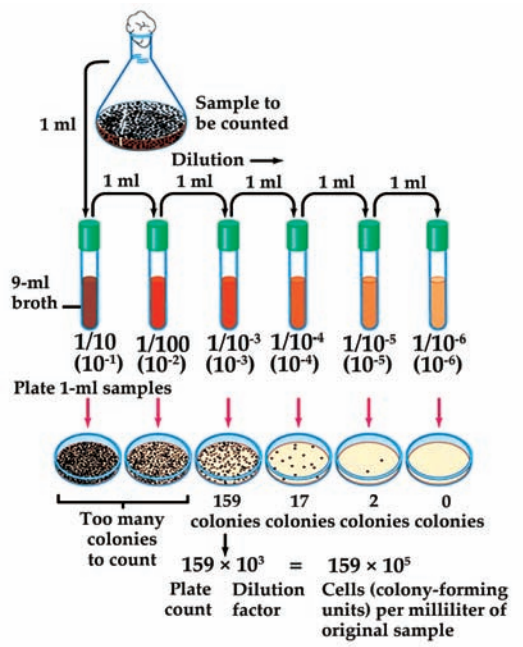


Figure 8. Serial dilution methodology.

Membrane Filtration

This method can be used in order to study if the quantity of the bacteria is very small as in aquatic samples like lakes, streams etc. Membranes with different pore sizes are used to trap different microorganisms. The sample is drawn through these

Selective media are used for the growth of only selected microorganisms.

special membrane filters and placed on an agar medium or on a pad soaked with liquid media. After incubation, the number of colonies can be counted and the number determined in the original sample. **Selective media** or differential media can be used for specific microorganisms. This is mostly used for analyzing aquatic samples.

Microscopic Count

The Petroff-Hausser counting chamber or slide is easy, inexpensive and relatively quick method and also gives information about the size and morphology of the microorganisms. These specially designed slides have chamber of known depth with an etched grid on the chamber bottom.

Bacteria/mm^3 = (bacteria/square) (25 squares)/ (50)

Bacteria can be counted by taking into account the chamber's volume and any sample dilution. The disadvantage encountered in this method is that fairly large volume is required and also it is difficult to distinguish between living and dead cells. Microorganisms of larger sizes can be counted by using electronic counters such as coulter counter; where in the number of cells in a measured volume of liquid is counted. This method gives accurate results with larger cells and is extensively used in hospital laboratories to count red and white blood cells.

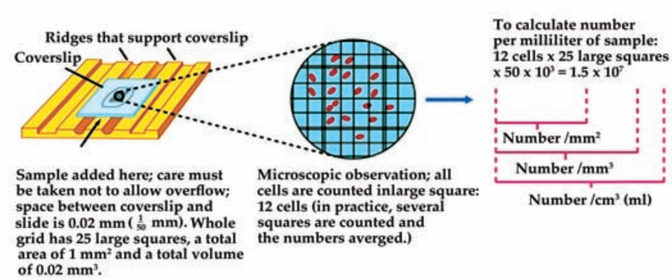


Figure 9. Direct microscopic count of bacterial cells.

Indirect Methods of Measurement of Cell Mass

Population growth leads to increase in the total cell mass, as well as in cell numbers. The following methods can be used.

Turbidity

As bacteria grow/multiply in a liquid medium, the medium becomes turbid. Spectrophotometer is used in order to measure the turbidity. A beam of light is transmitted through a bacterial suspension to a light-sensitive detector. The fact that microbial cells scatter light striking them, the amount of scattering is directly proportional to the biomass of cells present and indirectly related to cell number. The extent of light scattering can be measured and is almost linearly related to bacterial concentration at low absorbance levels.

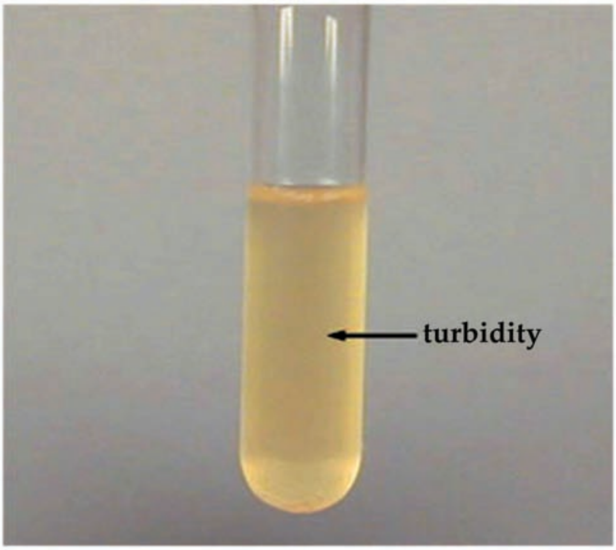


Figure 10. Broth culture showing turbidity.

Dry Weight

This method is mostly used for filamentous bacteria and molds. The microorganism is grown in liquid medium, filtered or centrifuged to remove extraneous material, and dried in an oven and then weighted. It is time consuming and hence not very sensitive.

Continuous Culture of Microorganisms

- *Batch cultures*: Nutrient supplies are not renewed nor wastes removed.
- *Continuous cultures*: continuous provision of nutrients and removal of wastes takes place. The population can be maintained in the exponential phase and at a constant biomass concentration for extended periods.
- These can again be categorized into two types:
- *Chemostat*: Where sterile medium is fed into the culture vessel at the same rate as the media containing microorganisms is removed.
- *Turbidostat*: It has a photocell that measures the absorbance or turbidity of the cell culture in the growth vessel. The flow rate of media through the vessel is automatically regulated to maintain a predetermined turbidity or cell density.

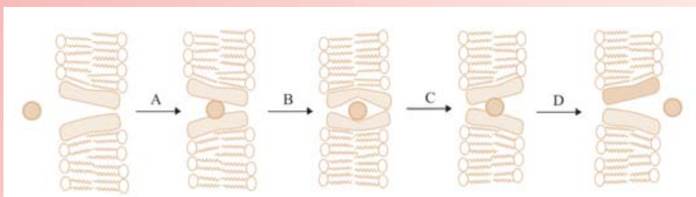
Used in food and industrial microbiology.

CONCEPT OF HOW DO NUTRIENTS GET INTO THE MICROBIAL CELL?

Having found a source of a given nutrient, a microorganism must:

- have some means of taking it up from the environment
- possess the appropriate enzyme systems to utilize it.

The plasma membrane represents a selective barrier, allowing into the cell only those substances it is able to utilized. This selectivity is due in large part to the hydrophobic nature of the lipid bilayer. A substance can be transported across the cell membrane in one of three ways, known as simple diffusion, facilitated diffusion and active transport. In simple diffusion, small molecules move across the membrane in response to a concentration gradient (from high to low), until concentrations on either side of the membrane are in equilibrium. The ability to do this depends on being small (H_2O, Na^+, Cl^-) or soluble in the lipid component of the membrane (non-polar gases such as O_2 and CO_2). Larger polar molecules such as glucose and amino acids are unable to enter the cell unless assisted by membrane-spanning transport proteins by the process of facilitated diffusion. Like enzymes, these proteins are specific for a single/small number of related solutes; another parallel is that they too can become saturated by too much 'substrate'. As with simple diffusion, there is no expenditure of cellular energy, and an inward concentration gradient is required. The transported substance tends to be metabolized rapidly once inside the cell, thus maintaining the concentration gradient from outside to inside.



Diffusion is only an effective method of internalizing substances when their concentrations are greater outside the cell than inside. Generally, however, microorganisms find themselves in very dilute environments; hence the concentration gradient runs in the other direction, and diffusion into the cell is not possible. Active transport enables the cell to overcome this unfavorable gradient. Here, regardless of the direction of the gradient, transport takes place in one direction only, into the cell. Energy, derived from hydrolysis of adenosine triphosphate is required to achieve this, and again specific transmembrane proteins are involved. They bind the solute molecule with high affinity outside of the cell, and then undergo a conformational change that causes them to be released into the interior. Procaryotic cells can carry out a specialized form of active transport called group translocation, whereby the solute is chemically modified as it crosses the membrane, preventing its escape. A well-studied example of this is the phosphorylation of glucose in E. coli by the phosphotransferase system. Glucose present in very low concentrations outside the cell can be concentrated within it by this mechanism. Glucose is unable to pass back across the membrane in its phosphorylated form (glucose-6-phosphate), however it can be utilized in metabolic pathways in this form. Often it may be necessary to employ extracellular enzymes to break down large molecules before any of these mechanisms can be used to transport nutrients into the cell.

SUMMARY

- Nutrients are substances used in biosynthesis and energy production, therefore are required for microbial growth.
- Growth of a living organism, may be defined as an increase in mass or size in any direction accompanied by synthesis of macromolecules, leading to the production of a newly organized structure. Growth when applied to bacteria normally refers to an increase in number of individual cells and so, is a measure of population density from original inoculums.
- Nutrients are materials that are acquired from the environment and are used for growth and metabolism. Microorganisms (or microbes) vary significantly in terms of the source, chemical form, and amount of essential elements they need. Some examples of these essential nutrients are carbon, oxygen, hydrogen, phosphorus, and sulfur.
- There are two categories of essential nutrients: macro-nutrients (which are needed in large amounts) and micro-nutrients (which are needed in trace or small amounts). Macro-nutrients usually help maintain the cell structure and metabolism. Micro-nutrients help enzyme function and maintain protein structure.
- Carbon is needed for the skeleton or backbone of all organic molecules and molecules serving as carbon sources normally also contribute both oxygen and hydrogen atoms.
- Nitrogen is needed for the synthesis of amino acids, purines, pyramidines, some carbohydrates and lipids, enzyme cofactors and other substances. Most phototrophs and many nonphotosynthetic microorganisms reduce nitrate to ammonia and incorporate the ammonia in assimilatory nitrate reduction.
- Active transport is the transport of solute molecules to higher concentrations or against a concentration gradient, with the use of metabolic energy input.
- Growth is defined as an increase in cellular constituents which leads to a rise in cell number. As we are aware, microorganisms reproduce by binary fission or by budding.

MULTIPLE CHOICE QUESTIONS

1. **Which of the following is the nutritional characterization of Escherichia coli?**
   a. Chemotrophic
   b. Organotrophic
   c. Autotrophic
   d. Chemotrophic, Organotrophic, Heterotrophic
2. **Nutrient content and biological structures are considered as**
   a. implicit factor for microbial growth
   b. intrinsic factor for microbial growth
   c. processing factor
   d. none of the above
3. **The temperature that allows for most rapid growth during a short period of time is known as __________**
   a. Minimum Temperature
   b. Maximum Temperature
   c. Optimum Temperature
   d. Growth Temperature
4. **Mesophiles are group of bacteria that grow within the temperature range of?**
   a. 0-20 degree Celsius
   b. 25-40 degree Celsius
   c. 45-60 degree Celsius
   d. more than 60 degree Celsius
5. **The bacterium Staphylococcus aureus is which type of bacteria?**
   a. Psychrophile
   b. Mesophile
   c. Thermophile
   d. Mesophile and psychrophile

Answers

1. (d) 2. (b) 3. (c) 4. (b) 5. (b)

REVIEW QUESTIONS

1. Name the different nutritional types of microorganisms?
2. What are growth factors and how are they important for microbial nutrition?
3. State the role of following micro and macronutrients:
   a. K^{+}
   b. Ca^{2+}
   c. Fe^{2+} and Fe^{2+}
   d. Zn^{2+}
   e. Mo^{2+}
4. Explain the process of Iron uptake.
5. Write short notes on the following:
   a. Chemically defined media
   b. Complex media
   c. Enriched media
   d. Selective media
   e. Differential media
6. Define growth. Explain the different phases in growth curve with proper diagram.
7. Explain generation time or doubling time of the bacterial population mathematically.
8. What are the different ways to measurement of cell numbers?
9. Differentiate between batch culture and continuous culture.

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CHAPTER 3

FUNGI

A fungus is any member of the group of eukaryotic organisms that includes microorganisms such as yeasts and molds, as well as the more familiar mushrooms. These organisms are classified as a kingdom, separately from the other eukaryotic kingdoms, which, by one traditional classification, includes Plantae, Animalia, Protozoa, and Chromista.

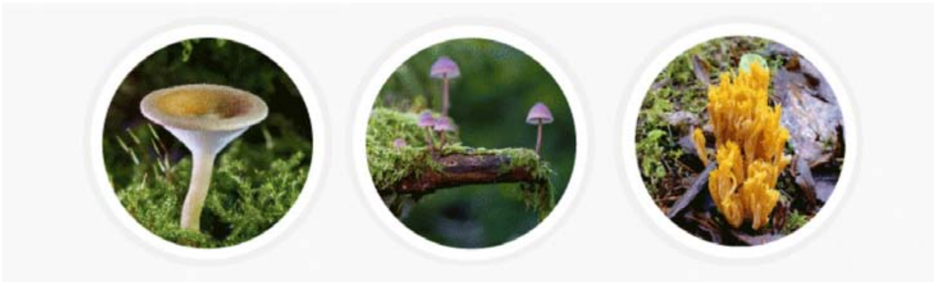
After studying this chapter, you will understand the core concepts of:

- The Science of Fungi
- Fungi Nutrition System
- Classification of Fungi
- Features and Life Cycle of Lichen in Fungus

INTRODUCTION

Fungi are eukaryotic organisms that include microorganisms such as yeasts, molds and mushrooms. These organisms are classified under kingdom fungi.

The organisms found in Kingdom fungi contain a cell wall and are omnipresent. They are classified as heterotrophs among the living organisms.



To name a few – the appearance of black spots on bread left outside for some days, the mushrooms and the yeast cells, which are commonly used for the production of beer and bread are also fungi. They are also found in most skin infections and other fungal diseases.

If we observe carefully, all the examples that we cited involve moist conditions. Thus, we can say that fungi usually grow in places which are moist and warm enough to support them.

CORE CONCEPT: THE SCIENCE OF FUNGI

Fungi are everywhere - beneath your feet, on virtually every surface you rest your eyes upon, even in the air you breathe. Far from this being a disturbing prospect, rest assured that without these strange and fascinating life forms, neither we, nor the inhabitants of our native forests, would survive for long. However, the mushrooms and toadstools with which we are familiar are just a tiny, partial glimpse of the true bodies of fungi.

Fungus, plural fungi, any of about 99,000 known species of organisms of the kingdom Fungi, which includes the yeasts, rusts, smuts, mildews, molds, and mushrooms. There are also many funguslike organisms, including slime molds and oomycetes (water molds), that do not belong to kingdom Fungi but are often called fungi. Many of these funguslike organisms are included in the kingdom Chromista. Fungi are among the most widely distributed organisms on Earth and are of great environmental and medical importance. Many fungi are free-living in soil or water;

others form parasitic or symbiotic relationships with plants or animals.

Fungi are eukaryotic organisms; i.e., their cells contain membrane-bound organelles and clearly defined nuclei. Historically, fungi were included in the plant kingdom; however, because fungi lack chlorophyll and are distinguished by unique structural and physiological features (i.e., components of the cell wall and cell membrane), they have been separated from plants. In addition, fungi are clearly distinguished from all other living organisms, including animals, by their principal modes of vegetative growth and nutrient intake. Fungi grow from the tips of filaments (hyphae) that make up the bodies of the organisms (mycelia), and they digest organic matter externally before absorbing it into their mycelia.

Eukaryotes are organisms whose cells have a nucleus enclosed within membranes, unlike Prokaryotes.

Fungi are **eukaryotic** organisms; i.e., their cells contain membrane-bound organelles and clearly defined nuclei. Historically, fungi were included in the plant kingdom; however, because fungi lack chlorophyll and are distinguished by unique structural and physiological features (i.e., components of the cell wall and cell membrane), they have been separated from plants. In addition, fungi are clearly distinguished from all other living organisms, including animals, by their principal modes of vegetative growth and nutrient intake. Fungi grow from the tips of filaments (hyphae) that make up the bodies of the organisms (mycelia), and they digest organic matter externally before absorbing it into their mycelia.

While mushrooms and toadstools (poisonous mushrooms) are by no means the most numerous or economically significant fungi, they are the most easily recognized. The Latin word for mushroom, fungus (plural fungi), has come to stand for the whole group. Similarly, the study of fungi is known as mycology—a broad application of the Greek word for mushroom, mykēs. Fungi other than mushrooms are sometimes collectively called molds, although this term is better restricted to fungi of the sort represented by bread mold. (For information about slime molds, which exhibit features of both the animal and the fungal worlds, see protist.)

Species of Fungi

There are about 75,000 scientifically identified species of fungi, with scientists believing there may be as many as a million fungal species yet unidentified. As differing species may look the same superficially, classifying them accurately is difficult, and usually requires the application of molecular tools such as DNA sequencing. As DNA sequencing is still relatively expensive, even for fungi with genomes far shorter than mammals, it will likely be many decades before the majority are classified with certainty.

Common types include molds — which grow in strands called hyphae, mushrooms — fruiting bodies of fungal colonies, and yeasts — the name for any single-celled fungi. However, these are broad terms, and molds, yeasts, and mushrooms can be found across several taxonomic categories. Fungal classification at the phyla level is complicated, and is constantly being reshuffled. Fungi were first misclassified as plants, but subsequent investigations found they actually have more in common with animals. Like plants and animals, they are eukaryotes.

Phylogenetically, there are seven phyla of fungi. The first is the Chytridiomycota, or chytrids, the most primitive form, with about 1,000 identified species. These produce spores with flagella (zoospores), and go after amphibians, maize, alfalfa, potatoes, and other vulnerable organisms. These are most representative of the types that lived throughout the Paleozoic era, being primarily aquatic.

Blastocladiomycota is the second phlya, only created as a distinct category in 2007. Like the chytrids, they use zoospores to reproduce, and parasitic of all major eukaryotic groups. The third phyla, Neocallimastigomycota, are anaerobic fungi that primarily occupy the stomachs of ruminants. Their name contains the Greek suffix referring to whips, -mastix, for their numerous flagella. The second and third phyla were both initially misclassified as chytrids.

The fourth phyla are the more familiar Zygomycota, named for the hardy spherical spores they produce. If you see a fungus with tiny dots at the tips of the hyphae (filaments), that's Zygomycota. There are over 600 species of this genus, and it includes black bread mold, one of the most frequently sighted by humans. Another is Pilobolus, which is capable of ejecting spores several meters through the air.

The fifth phyla are the Glomeromycota, known as Arbuscular mycorrhizae (AM) fungi. Basically, that term means “tree fungi.” They can be found in large numbers in the roots of more than 80% of families of vascular plants. This relationship is symbiotic and ancient, extending back at least 460 million years, to the beginning of plant life on land.

The sixth phyla are the Ascomycota, known as sac fungi. These make distinct spherical sacs to hold their spores, and contain the most species out of all the phyla. Examples include Penicillium, morels, truffles, Baker’s yeast, lichens, powdery mildews, and many others. Many of this phyla are plant-pathogenic.

The seventh phyla are the Basidiomycota, or the club fungi. This group contains most common mushrooms. It is distinguished by the presence of a spore-producing structure called the basidium, more commonly known as a cap. Along with Ascomycota, they are known as Higher Fungi.

Importance of Fungi

Humans have been indirectly aware of fungi since the first loaf of leavened bread was baked and the first tub of grape must was turned into wine. Ancient peoples were familiar with the ravages of fungi in agriculture but attributed these diseases to the wrath of the gods. The Romans designated a particular deity, Robigus, as the god of rust and, in an effort to appease him, organized an annual festival, the Robigalia, in his honour.

Fungi are everywhere in very large numbers—in the soil and the air, in lakes, rivers, and seas, on and within plants and animals, in food and clothing, and in the human body. Together with bacteria, fungi are responsible for breaking down organic matter and releasing carbon, oxygen, nitrogen, and phosphorus into the soil and the atmosphere. Fungi are essential to many household and industrial processes, notably the making of bread, wine, beer, and certain cheeses. Fungi are also used as food;

for example, some mushrooms, morels, and truffles are epicurean delicacies, and mycoproteins (fungal proteins), derived from the mycelia of certain species of fungi, are used to make foods that are high in protein.

Ergot is a source of several chemicals used in drugs that induce labour in pregnant women and that control hemorrhage after birth. Ergot is also the source of lysergic acid, the active principle of the psychedelic drug lysergic acid diethylamide (LSD).

Studies of fungi have greatly contributed to the accumulation of fundamental knowledge in biology. For example, studies of ordinary baker's or brewer's yeast (Saccharomyces cerevisiae) led to discoveries of basic cellular biochemistry and metabolism. Some of these pioneering discoveries were made at the end of the 19th century and continued during the first half of the 20th century. From 1920 through the 1940s, geneticists and biochemists who studied mutants of the red bread mold, Neurospora, established the one-gene–one-enzyme theory, thus contributing to the foundation of modern genetics. Fungi continue to be useful for studying cell and molecular biology, genetic engineering, and other basic disciplines of biology.

The medical relevance of fungi was discovered in 1928, when Scottish bacteriologist Alexander Fleming noticed the green mold Penicillium notatum growing in a culture dish of Staphylococcus bacteria. Around the spot of mold was a clear ring in which no bacteria grew. Fleming successfully isolated the substance from the mold that inhibited the growth of bacteria. In 1929 he published a scientific report announcing the discovery of penicillin, the first of a series of antibiotics—many of them derived from fungi—that have revolutionized medical practice.

Another medically important fungus is Claviceps purpurea, which is commonly called ergot and causes a plant disease of the same name. The disease is characterized by a growth that develops on grasses, especially on rye. Other species of fungi contain chemicals that are extracted and used to produce drugs known as statins, which control cholesterol levels and ward off coronary heart disease. Fungi are also used in the production of a number of organic acids, enzymes, and vitamins.

Form and Function of Fungi

Size range

The mushrooms, because of their size, are easily seen in fields and forests and consequently were the only fungi known before the invention of the microscope in the 17th century. The microscope made it possible to recognize and identify the great variety of fungal species living on dead or live organic matter. The part of a fungus that is generally visible is the fruiting body, or sporophore. Sporophores vary greatly in size, shape, colour, and longevity. Some are microscopic and completely invisible to the unaided eye; others are no larger than a pin head; still others are gigantic structures. Among the largest sporophores are those of mushrooms, bracket fungi, and puffballs. Some mushrooms reach a diameter of 20 to 25 cm (8 to 10 inches) and a height of 25 to 30 cm (10 to 12 inches). Bracket, or shelf, fungi can reach 40 cm (16 inches) or more in diameter. A specimen of the bracket fungus Fomitiporia ellipsoidea discovered in 2010 on Hainan Island in southern China had a fruiting body measuring 10.8 metres (35.4 feet) in length and 82–88 cm (2.7–2.9 feet) in width. It may have held some 450 million spores and weighed an estimated 400–500 kg (882–1,102 pounds), at the time making it the largest fungal fruiting body ever documented. Puffballs also can grow to impressive sizes. The largest puffballs on record measured 150 cm (5 feet) in diameter. The number of spores within such giants reaches several trillion.

Distribution and abundance

Fungi are either terrestrial or aquatic, the latter living in freshwater or marine environments. Freshwater species are usually found in clean, cool water because they do not tolerate high degrees of salinity. However, some species are found in slightly brackish water, and a few thrive in highly polluted streams. Soil that is rich in organic matter furnishes an ideal habitat for a large number of species; only a small number of species are found in drier areas or in habitats with little or no organic matter. Fungi are found in all temperate and tropical regions of the world where there is sufficient moisture to enable them to grow. A few species of fungi live in the Arctic and Antarctic regions, although they are rare and are more often found living in symbiosis with algae in the form of lichens (see below Lichens). About 99,000 species of fungi have been identified and described, but mycologists estimate that there may be as many as 5.1 million total species.

Basic morphology

A typical fungus consists of a mass of branched, tubular filaments enclosed by a rigid cell wall. The filaments, called hyphae (singular hypha), branch repeatedly into a complicated, radially expanding network called the mycelium, which makes up the thallus, or undifferentiated body, of the typical fungus. The mycelium grows by utilizing nutrients from the environment and, upon reaching a certain stage of maturity, forms—either directly or in special fruiting bodies—reproductive cells called spores. The spores are released and dispersed by a wide variety of passive or active mechanisms; upon reaching a suitable substrate, the spores germinate and develop hyphae that grow, branch repeatedly, and become the mycelium of the new individual. Fungal growth is mainly confined to the tips of the hyphae, and all fungal structures are therefore made up of hyphae or portions of hyphae.

Flagellum is a lash-like appendage that protrudes from the cell body of certain bacterial and eukaryotic cells. The primary role of the flagellum is locomotion, but it also often has function as a sensory organelle, being sensitive to chemicals and temperatures outside the cell.

Some fungi, notably the yeasts, do not form a mycelium but grow as individual cells that multiply by budding or, in certain species, by fission. In addition, the so-called cryptomycota, a primitive group of microscopic fungi, diverge significantly from the standard body plan of other fungi in that their cell walls lack the rigid polymer known as chitin. These microscopic fungi also possess a whiplike **flagellum**.

Structure of the thallus

In almost all fungi the hyphae that make up the thallus have cell walls. (The thalli of the true slime molds lack cell walls and, for this and other reasons, are classified as protists rather than fungi.) A hypha is a multibranched tubular cell filled with cytoplasm. The tube itself may be either continuous throughout or divided into compartments, or cells, by cross walls called septa (singular septum). In nonseptate (i.e., coenocytic) hyphae the nuclei are scattered throughout the cytoplasm. In septate hyphae each cell may contain one to many nuclei, depending on the type of fungus or the stage of hyphal development. The cells of fungi are similar in structure to those of many other organisms. The minute nucleus, readily seen only in young portions of the hypha, is surrounded by a double membrane and typically contains one nucleolus. In addition to the nucleus, various organelles—such as the endoplasmic reticulum, Golgi apparatus, ribosomes, and liposomes—are scattered throughout the cytoplasm.

Hyphae usually are either nonseptate (generally in the more primitive fungi) or incompletely septate (meaning that the septa are perforated). This permits the movement of cytoplasm (cytoplasmic streaming) from one cell to the next. In fungi with perforated septa, various molecules are able to move rapidly between hyphal cells, but the movement of larger organelles, such as mitochondria and nuclei, is prevented. In the absence of septa, both mitochondria and nuclei can be readily translocated along hyphae. In mating interactions between filamentous Basidiomycota, the nuclei of one parent often invade the hyphae of the other parent, because the septa are degraded ahead of the incoming nuclei to allow their passage through the existing hyphae. Once the incoming nuclei are established, septa are re-formed.

Variations in the structure of septa are numerous in the fungi. Some fungi have sievelike septa called pseudosepta, whereas fungi in other groups have septa with one to few pores that are small enough in size to prevent the movement of nuclei to adjacent cells. Basidiomycota have a septal structure called a dolipore septum that is composed of a pore cap surrounding a septal swelling and septal pore. This organization permits cytoplasm and small organelles to pass through but restricts the movement of nuclei to varying degrees.

The wall of the hypha is complex in both composition and structure. Its exact chemical composition varies in different fungal groups. In some funguslike organisms the wall contains considerable quantities of cellulose, a complex carbohydrate that is the chief constituent of the cell walls of plants. In most fungi, however, two other polymers—chitin and glucan (a polymer of glucose linked at the third carbon and branched at the sixth), which forms an α-glucan layer and a special β-1,3-1,6-glucan layer—form the main structural components of the wall. Among the many other chemical substances in the walls of fungi are some that may thicken or toughen the wall of tissues, thus imparting rigidity and strength. The chemical composition of the wall of a particular fungus may vary at different stages of the organism's growth—a possible indication that the wall plays some part in determining the form of the fungus. In some fungi, carbohydrates are stored in the wall at one stage of development and are removed and utilized at a later stage. In some yeasts, fusion of sexually functioning cells is brought about by the interaction of specific chemical substances on the walls of two compatible mating types.

When the mycelium grows in or on a surface, such as in the soil, on a log, or in culture medium, it appears as a mass of loose, cottony threads. The richer the composition of the growth medium, the more profuse the threads and the more feltlike the mass. On the sugar-rich growth substances used in laboratories, the assimilative (somatic) hyphae are so interwoven as to form a thick, almost leathery colony. On the soil, inside a leaf, in the skin of animals, or in other parasitized plant or animal tissues, the hyphae are usually spread in a loose network. The mycelia of the so-called higher fungi does, however, become organized at times into compact masses of different sizes that serve various functions. Some of these masses, called sclerotia, become extremely hard and serve to carry the fungus over periods of adverse conditions of temperature and moisture. One example of a fungus that forms sclerotia is ergot (Claviceps purpurea), which causes a disease of cereal grasses. The underground sclerotia of Wolfiporia extensa, an edible pore fungus also known as tuckahoe, may reach a diameter of 20 to 25 cm (8 to 10 inches).

If a small piece of mycelium is placed under conditions favorable for growth, it develops into a new thallus, even if no growing tips are included in the severed portion.

Various other tissues are also produced by the interweaving of the assimilative hyphae of some fungi. Stromata (singular stroma) are cushionlike tissues that bear spores in various ways. Rhizomorphs are long strands of parallel hyphae cemented together. Those of the honey mushroom (Armillaria mellea), which are black and resemble shoestrings, are intricately constructed and are differentiated to conduct water and food materials from one part of the thallus to another.

Sporophores and spores

When the mycelium of a fungus reaches a certain stage of growth, it begins to produce spores either directly on the somatic hyphae or, more often, on special sporiferous (spore-producing) hyphae, which may be loosely arranged or grouped into intricate structures called fruiting bodies, or sporophores.

The more primitive fungi produce spores in sporangia, which are saclike sporophores whose entire cytoplasmic contents cleave into spores, called sporangiospores. Thus, they differ from more advanced fungi in that their asexual spores are endogenous. Sporangiospores are both naked and flagellated (zoospores) or walled and nonmotile (aplanospores). The more primitive aquatic and terrestrial fungi tend to produce zoospores. The zoospores of aquatic fungi and funguslike organisms swim in the surrounding water by means of one or two variously located flagella (whiplike organs of locomotion). Zoospores produced by terrestrial fungi are released after a rain from the sporangia in which they are borne and swim for a time in the rainwater between soil particles or on the wet surfaces of plants, where the sporangia are formed by parasitic fungi. After some time, the zoospores lose their flagella, surround themselves with walls, and encyst. Each cyst germinates by producing a germ tube. The germ tube may develop a mycelium or a reproductive structure, depending on the species and on the environmental conditions. The bread molds, which are the most advanced of the primitive fungi, produce only aplanospores (nonmotile spores) in their sporangia.

The more advanced fungi do not produce motile spores of any kind, even though some of them are aquatic in fresh or marine waters. In these fungi, asexually produced spores (usually called conidia) are produced exogenously and are typically formed terminally or laterally on special spore-producing hyphae called conidiophores. Conidiophores may be arranged singly on the hyphae or may be grouped in special asexual fruiting bodies, such as flask-shaped pycnidia, mattresslike acervuli, cushion-shaped sporodochia, or sheaflike synnemata.

Sexually produced spores of the higher fungi result from meiosis and are formed either in saclike structures (asci) typical of the Ascomycota or on the surface of club-shaped structures (basidia) typical of the Basidiomycota. Asci and basidia may be borne naked, directly on the hyphae, or in various types of sporophores, called ascocarps (also known as ascomata) or basidiocarps (also known as basidiomata), depending on whether they bear asci or basidia, respectively. Well-known examples of ascocarps are the morels, the cup fungi, and the truffles. Commonly encountered basidiocarps are mushrooms, brackets, puffballs, stinkhorns, and bird's-nest fungi.

Growth

Under favourable environmental conditions, fungal spores germinate and form hyphae. During this process, the spore absorbs water through its wall, the cytoplasm becomes activated, nuclear division takes place, and more cytoplasm is synthesized. The wall initially grows as a spherical structure. Once polarity is established, a hyphal apex forms, and from the wall of the spore a germ tube bulges out, enveloped by a wall of its own that is formed as the germ tube grows.

The hypha may be roughly divided into three regions: (1) the apical zone about 5–10 micrometres (0.0002–0.0004 inch) in length, (2) the subapical region, extending about 40 micrometres (0.002 inch) back of the apical zone, which is rich in cytoplasmic components, such as nuclei, Golgi apparatus, ribosomes, mitochondria, the endoplasmic reticulum, and vesicles, but is devoid of vacuoles, and (3) the zone of vacuolation, which is characterized by the presence of many vacuoles and the accumulation of lipids.

Growth of hyphae in most fungi takes place almost exclusively in the apical zone (i.e., at the very tip). This is the region where the cell wall extends continuously to produce a long hyphal tube. The cytoplasm within the apical zone is filled with numerous vesicles. These bubblelike structures are usually too small to be seen with an ordinary microscope but are clearly evident under the electron microscope. In higher fungi the apical vesicles can be detected with an ordinary microscope equipped with phase-contrast optics as a round spot with a somewhat diffuse boundary. This body is universally known by its German name, the Spitzenkörper, and its position determines the direction of growth of a hypha.

The growing tip eventually gives rise to a branch. This is the beginning of the branched mycelium. Growing tips that come in contact with neighbouring hyphae often fuse with them to form a hyphal net. In such a vigorously growing system, the cytoplasm is in constant motion, streaming toward the growing tips. Eventually, the older hyphae become highly vacuolated and may be stripped of most of their cytoplasm. All living portions of a thallus are potentially capable of growth. If a small piece of mycelium is placed under conditions favourable for growth, it develops into a new thallus, even if no growing tips are included in the severed portion.

Morphogenesis is the biological process that causes an organism to develop its shape. It is one of three fundamental aspects of developmental biology along with the control of cell growth and cellular differentiation, unified in evolutionary developmental biology (evo-devo).

Growth of a septate mycelium (i.e., with cross walls between adjacent cells) entails the formation of new septa in the young hyphae. Septa are formed by ringlike growth from the wall of the hypha toward the centre until the septa are complete. In the higher fungi the septum stops growing before it is complete; the result is a central pore through which the cytoplasm flows, thus establishing organic connection throughout the thallus. In contrast to plants, in which the position of the septum separating two daughter cells determines the formation of tissues, the fungal septum is always formed at right angles to the axis of growth. As a result, in fungal tissue formation, the creation of parallel hyphae cannot result from longitudinal septum formation but only from outgrowth of a new branch. In fungi, therefore, the mechanism that determines the point of origin and subsequent direction of growth of hyphal branches is the determining factor in developmental **morphogenesis**.

The individual fungus is potentially immortal, because it continues to grow at the hyphal tips as long as conditions remain favourable. It is possible that, in undisturbed places, mycelia exist that have grown continuously for many thousands of years. The older parts of the hyphae die and decompose, releasing nitrogen and other nutrients into the soil.

Some species of endophytic fungi, such as Neotyphodium and Epichloë, which invade the seeds of grasses (e.g., ryegrass and fescue) and grow within the plant, grow not through extension of the hyphal tips but by intercalary growth, in which the hyphae attach to the growing cells of the plant. This type of growth enables the hyphae of the fungus to grow at the same rate that the plant grows. Intercalary growth of endophytic fungi was discovered in 2007, although for many years scientists suspected that these fungi possessed unique adaptations that allow them to grow as if they were natural parts of their hosts.

The underground network of hyphae of a mushroom can grow and spread over a very large area, often several metres (yards) in diameter. The underground hyphae obtain food from organic matter in the substratum and grow outward. The hyphal branches at the edge of the mycelium become organized at intervals into elaborate tissues that develop aboveground into mushrooms. Such a circle of mushrooms is known as a fairy ring, because in the Middle Ages it was believed to represent the path of dancing fairies. The ring marks the periphery of an enormous fungus colony, which, if undisturbed, continues to produce ever wider fairy rings year after year. Fungi can grow into enormous colonies. Some thalli of Armillaria species, which are pathogens of forest trees, are among the largest and oldest organisms on Earth.

Features of Fungi

General features of fungi are as follows:

- Biosynthesis of chitin occurs in fungi.
- Depending on the species and conditions both sexual and asexual spores may be produced.
- During mitosis the nuclear envelope is not dissolved.
- Fungi are eukaryotic organisms.
- Fungi are heterotrophic organisms.
- Fungi exhibit the phenomenon of alteration of generation.
- Fungi store their food as starch.
- Nutrition in fungi - they are saprophytes, or parasites or symbionts.
- Reproduction in fungi is both by sexual and asexual means. Sexual state is referred to as teleomorph, asexual state is referred to as anamorph.
- The nuclei of the fungi is very small.
- The structure of cell wall is similar to plants but chemically the fungi cell wall are composed of chitin.
- The vegetative body of the fungi may be unicellular or composed of microscopic threads called hyphae.
- They are non-vascular organisms.
- They are typically non-motile.
- They fungi digest the food first and then ingest the food, to accomplish this the fungi produce exoenzymes.
- They reproduce by means of spores.

Reproductive Processes of Fungi

Following a period of intensive growth, fungi enter a reproductive phase by forming and releasing vast quantities of spores. Spores are usually single cells produced by fragmentation of the mycelium or within specialized structures (sporangia, gametangia, sporophores, etc.). Spores may be produced either directly by asexual methods or indirectly by sexual reproduction. Sexual reproduction in fungi, as in other living organisms, involves the fusion of two nuclei that are brought together when two sex cells (gametes) unite. Asexual reproduction, which is simpler and more direct, may be accomplished by various methods.

Asexual Reproduction

Typically in asexual reproduction, a single individual gives rise to a genetic duplicate of the progenitor without a genetic contribution from another individual. Perhaps the simplest method of reproduction of fungi is by fragmentation of the thallus, the body of a fungus. Some yeast, which are single-celled fungi, reproduce by simple cell division, or fission, in which one cell undergoes nuclear division and splits into two daughter cells; after some growth, these cells divide, and eventually a population of cells forms. In filamentous fungi the mycelium may fragment into a number of segments, each of which is capable of growing into a new individual. In the laboratory, fungi are commonly propagated on a layer of solid nutrient agarinoculated either with spores or with fragments of mycelium.

Budding, which is another method of asexual reproduction, occurs in most yeasts and in some filamentous fungi. In this process, a bud develops on the surface of either the yeast cell or the hypha, with the cytoplasm of the bud being continuous with that of the parent cell. The nucleus of the parent cell then divides; one of the daughter nuclei migrates into the bud, and the other remains in the parent cell. The parent cell is capable of producing many buds over its surface by continuous synthesis of cytoplasm and repeated nuclear divisions. After a bud develops to a certain point and even before it is severed from the parent cell, it is itself capable of budding by the same process. In this way, a chain of cells may be produced. Eventually, the individual buds pinch off the parent cell and become individual yeast cells. Buds that are pinched off a hypha of a filamentous fungus behave as spores; that is, they germinate each giving rise to a structure called a germ tube, which develops into a new hypha.

Sexual Reproduction

Sexual reproduction, an important source of genetic variability, allows the fungus to adapt to new environments. The process of sexual reproduction among the fungi is in many ways unique. Whereas nuclear division in other eukaryotes, such as animals, plants, and protists, involves the dissolution and re-formation of the nuclear membrane, in fungi the nuclear membrane remains intact throughout the process, although gaps in its integrity are found in some species. Thenucleus of the fungus becomes pinched at its midpoint, and the diploid chromosomes are pulled apart by spindle fibers formed within the intact nucleus. The nucleolus is usually also retained and divided between the daughter cells, although it may be expelled from the nucleus, or it may be dispersed within the nucleus but detectable.

Sexual reproduction in the fungi consists of three sequential stages: plasmogamy, karyogamy, and meiosis. The diploid chromosomes are pulled apart into two daughter cells, each containing a single set of chromosomes (a haploid state).Plasmogamy, the fusion of two protoplasts (the contents of the two cells), brings together two compatible haploid nuclei. At this point, two nuclear types are present in the same cell, but the nuclei have not yet fused. Karyogamy results in the fusion of these haploid nuclei and the formation of a diploid nucleus (i.e., a nucleus containing two sets of chromosomes, one from each parent). The cell formed by karyogamy is called the zygote. In most fungi the zygote is the only cell in the entire life cycle that is diploid. The dikaryotic state that results from plasmogamy is often a prominent condition in fungi and may be prolonged over several generations. In the lower fungi, karyogamy usually follows plasmogamy almost immediately. In the more evolved fungi, however, karyogamy is separated from plasmogamy. Once karyogamy has occurred, meiosis (cell division that reduces the chromosome number to one set per cell) generally follows and restores the haploid phase. The haploid nuclei that result from meiosis are generally incorporated in spores called meiospores.

Fungi employ a variety of methods to bring together two compatible haploid nuclei (plasmogamy). Some produce specialized sex cells (gametes) that are released from differentiated sex organs called gametangia. In other fungi two gametangia come in contact, and nuclei pass from the male gametangium into the female, thus assuming the function of gametes. In still other fungi the gametangia themselves may fuse in order to bring their nuclei together. Finally, some of the most advanced fungi produce no gametangia at all; the somatic (vegetative) hyphae take over the sexual function, come in contact, fuse, and exchange nuclei.

Sexual Incompatibility

Many of the simpler fungi produce differentiated male and female organs on the same thallus but do not undergo self-fertilization because their sex organs are incompatible. Such fungi require the presence of thalli of different mating types in order for sexual fusion to take place. The simplest form of this mechanism occurs in fungi in which there are two mating types, often designated + and − (or *A* and *a*). Gametes produced by one type of thallus are compatible only with gametes produced by the other type. Such fungi are said to be heterothallic. Many fungi, however, are homothallic; i.e., sex organs produced by a single thallus are self-compatible, and a second thallus is unnecessary for sexual reproduction. Some of the most complex fungi (e.g., mushrooms) do not develop differentiated sex organs; rather, the sexual function is carried out by their somatic hyphae, which unite and bring together compatible nuclei in preparation for fusion. Homothallism and heterothallism are encountered in fungi that have not developed differentiated sex organs, as well as in fungi in which sex organs are easily distinguishable. Compatibility therefore refers to a physiological differentiation, and sex refers to a morphological (structural) one; the two phenomena, although related, are not synonymous.

> Fungi in which a single individual bears both male and female gametangia arehermaphroditic fungi. Rarely, gametangia of different sexes are produced by separate individuals, one a male, the other a female. Such species are termed dioecious. Dioecious species usually produce sex organs only in the presence of an individual of the opposite sex.

Sexual Pheromones

The formation of sex organs in fungi is often induced by specific organic substances. Although called sex hormones when first discovered, these organic substances are actually sex pheromones, chemicals produced by one partner to elicit a sexual response in the other. In *Allomyces* (order Blastocladiales) a pheromone named sirenin, secreted by the female gametes, attracts the male gametes, which swim toward the former and fuse with them. In some simple fungi, which may have gametangia that are not differentiated structurally, a complex biochemical interplay between mating types produces trisporic acid, a pheromone that induces the formation of specialized aerial hyphae. Volatile intermediates in the trisporic acid synthetic pathway are interchanged between the tips of opposite mating aerial hyphae, causing the hyphae to grow toward each other and fuse together. In yeasts belonging to the phyla Ascomycota and Basidiomycota, the pheromones are smallpeptides. Several pheromone genes have been identified and characterized in filamentous ascomycetes and basidiomycetes.

Life Cycle of Fungi

In the life cycle of a sexually reproducing fungus, a haploid phase alternates with a diploid phase. The haploid phase ends with nuclear fusion, and the diploid phase begins with the formation of the zygote (the diploid cell resulting from fusion of two haploid sex cells). Meiosis (reduction division) restores the haploid number of chromosomes and initiates the haploid phase, which produces the gametes. In the majority of fungi, all structures are haploid except the zygote. Nuclear fusion takes place at the time of zygote formation, and meiosis follows immediately. Only in *Allomyces* and a few related genera and in some yeasts is alternation of a haploid thallus with a diploid thallus definitely known. In the higher fungi a third condition is interspersed between the haploid and diploid phases of the life cycle. In these fungi, plasmogamy (fusion of the cellular contents of two hyphae but not of the two haploid nuclei) results in dikaryotic hyphae in which each cell contains two haploid nuclei, one from each parent. Eventually, the nuclear pair fuses to form the diploid nucleus and thus the zygote. In the Basidiomycota, binucleate cells divide successively and give rise to a binucleate mycelium, which is the main assimilative phase of the life cycle. It is the binucleate mycelium that eventually forms the basidia—the stalked fruiting bodies in which nuclear fusion and meiosis take place prior to the formation of the basidiospores. Fungi usually reproduce both sexually and asexually. The asexual cycle produces mitospores, and the sexual cycle produces meiospores. Even though both types of spores are produced by the same mycelium, they are very different in form and easily distinguished. The asexual phase usually precedes the sexual phase in the life cycle and may be repeated frequently before the sexual phase appears.

Some fungi differ from others in their lack of one or the other of the reproductive stages.

Ecology of Fungi

Relatively little is known of the effects of the environment on the distribution of fungi that utilize dead organic material as food. The availability of organic food is certainly one of the factors controlling such distribution. A great number of fungi appear able to utilize most types of organic materials, such as lignin, cellulose, or otherpolysaccharides, which have been added to soils or waters by dead vegetation. Most saprotrophic fungi are widely distributed throughout the world, only requiring that their habitats have sufficient organic content to support their growth. However, some saprotrophs are strictly tropical and others are strictly temperate-zone forms; fungi with specific nutritional requirements are even further localized.

Moisture and temperature are two additional ecological factors that are important in determining the distribution of fungi. Laboratory studies have shown that many, perhaps the majority, of fungi are mesophilic, meaning they have an optimum growth temperature of 20–30°C (68–86°F). Thermophilic species are able to grow at 50°C (122°F) or higher but are unable to grow below 30°C. Although the optimum temperature for growth of most fungi lies at or above 20°C, a large number of species are able to grow close to or below 0°C (32°F). The so-called snow molds and the fungi that cause spoilage of refrigerated foods are examples of this group. Obviously, temperature relationships influence the distribution of various species. Certain other effects of temperature are also important factors in determining the habitats of fungi. Many coprophilous (dung-inhabiting) fungi, such as *Pilobolus*, although able to grow at a temperature of 20–30°C, require a short period at 60°C (140°F) for their spores to germinate.

CORE CONCEPT: FUNGI NUTRITION SYSTEM

Fungi are mostly saprobes, organisms that derive nutrients from decaying organic matter. They obtain their nutrients from dead or decomposing organic matter, mainly plant material. Fungal exoenzymes are able to break down insoluble polysaccharides, such as the cellulose and lignin of dead wood, into readily absorbable glucose molecules.

Fungi get their nutrition by absorbing organic compounds from the environment. Fungi are heterotrophic: they rely solely on carbon obtained from other organisms for their metabolism and nutrition. Fungi have evolved in a way that allows many of them to use a large variety of organic substrates for growth, including simple compounds such as nitrate, ammonia, acetate, or ethanol. Their mode of nutrition defines the role of fungi in their environment.

Fungi obtain nutrients in three different ways:

- They decompose dead organic matter. A saprotroph is an organism that obtains its nutrients from non-living organic matter, usually dead and decaying plant or animal matter, by absorbing soluble organic compounds. Saprotrophic fungi play very important roles as recyclers in ecosystem energy flow and biogeochemical cycles. Saprophytic fungi, such as shiitake (Lentinula edodes) and oyster mushrooms (Pleurotus ostreatus), decompose dead plant and animal tissue by releasing enzymes from hyphal tips. In this way they recycle organic materials back into the surrounding environment. Because of these abilities, fungi are the primary decomposers in forests (see Figure below).

- They feed on living hosts. As parasites, fungi live in or on other organisms and get their nutrients from their host. Parasitic fungi use enzymes to break down living tissue, which may causes illness in the host. Disease-causing fungi are parasitic. Recall that parasitism is a type of symbiotic relationship between organisms of different species in which one, the parasite, benefits from a close association with the other, the host, which is harmed.
- They live mutualistically with other organisms. Mutualistic fungi live harmlessly with other living organisms. Recall that mutualism is an interaction between individuals of two different species, in which both individuals benefit.

Both parasitism and mutualism are classified as symbiotic relationships, but they are discussed separately here because of the different effect on the host.

Cellulose is mainly used to produce paperboard and paper. Smaller quantities are converted into a wide variety of derivative products such as cellophane and rayon.

Fungal hyphae are adapted to efficient absorption of nutrients from their environments, because hyphae have high surface area-to-volume ratios. These adaptations are also complemented by the release of hydrolytic enzymes that break down large organic molecules such as polysaccharides, proteins, and lipids into smaller molecules. These molecules are then absorbed as nutrients into the fungal cells. One enzyme that is secreted by fungi is cellulase, which breaks down the polysaccharide cellulose. **Cellulose** is a major component of plant cell walls. In some cases, fungi have developed specialized structures for nutrient uptake from living hosts, which penetrate into the host cells for nutrient uptake by the fungus.

Unlike plants, which use carbon dioxide and light as sources of carbon and energy, respectively, fungi meet these two requirements by assimilating preformed organic matter; carbohydrates are generally the preferred carbon source. Fungi can readily absorb and metabolize a variety of soluble carbohydrates, such as glucose, xylose, sucrose, and fructose. Fungi are also characteristically well equipped to use insoluble carbohydrates such as starches, cellulose, and hemicelluloses, as well as very complex hydrocarbons such as lignin. Many fungi can also use proteins as a source of carbon and nitrogen. To use insoluble carbohydrates and proteins, fungi must first digest these polymers extracellularly. Saprotrophic fungi obtain their food from dead organic material; parasitic fungi do so by feeding on living organisms (usually plants), thus causing disease.

Fungi secure food through the action of enzymes (biological catalysts) secreted into the surface on which they are growing; the enzymes digest the food, which then is absorbed directly through the hyphal walls. Food must be in solution in order to enter the hyphae, and the entire mycelial surface of a fungus is capable of absorbing materials dissolved in water. The rotting of fruits, such as peaches and citrus fruits in storage, demonstrates this phenomenon, in which the infected parts are softened by the action of the fungal enzymes. In brown rot of peaches, the softened area is somewhat larger than the actual area invaded by the hyphae: the periphery of the brown spot has been softened by enzymes that act ahead of the invading mycelium. Cheeses such as Brie and Camembert are matured by enzymes produced by the fungus Penicillium camemberti, which grows on the outer surface of some cheeses. Some fungi produce special rootlike hyphae, called rhizoids, which anchor the thallus to the growth surface and probably also absorb food. Many parasitic fungi are even more specialized in this respect, producing special absorptive organs called haustoria.

Saprotrophism

Together with bacteria, saprotrophic fungi are to a large extent responsible for the decomposition of organic matter. They are also responsible for the decay and decomposition of foodstuffs. Among other destructive saprotrophs are fungi that destroy timber and timber products as their mycelia invade and digest the wood; many of these fungi produce their spores in large, woody, fruiting bodies—e.g., bracket or shelf fungi. Paper, textiles, and leather are often attacked and destroyed by fungi. This is particularly true in tropical regions, where temperature and humidity are often very high.

The nutritional requirements of saprotrophs (and of some parasites that can be cultivated artificially) have been determined by growing fungi experimentally on various synthetic substances of known chemical composition. Fungi usually exhibit the same morphological characteristics in these culture media as they do in nature. Carbon is supplied in the form of sugars or starch; the majority of fungi thrive on such sugars as glucose, fructose, mannose, maltose, and, to a lesser extent, sucrose. Decomposition products of proteins, such as proteoses, peptones, and amino acids, can be used by most fungi as nitrogen sources; ammonium compounds and nitrates also serve as nutrients for many species. It is doubtful, however, that any fungus can combine, or fix, atmospheric nitrogen into usable compounds. Chemical elements such as phosphorus, sulfur, potassium, magnesium, and small quantities of iron, zinc, manganese, and copper are needed by most fungi for vigorous growth; elements such as calcium, molybdenum, and gallium are required by at least some species. Oxygen and hydrogen are absolute requirements; they are supplied in the form of water or are obtained from carbohydrates. Many fungi, deficient in thiamine and biotin, must obtain these vitamins from the environment; most fungi appear able to synthesize all other vitamins necessary for their growth and reproduction.

As a rule, fungi are aerobic organisms, meaning they require free oxygen in order to live. Fermentations, however, take place under anaerobic conditions. Knowledge of the physiology of saprotrophic fungi has enabled industry to use several species for fermentation purposes. One of the most important groups of strictly anaerobic fungi are members of the genera Neocallimastix (phylum Neocallimastigomycota), which form a crucial component of the microbial population of the rumen of herbivorous mammals. These fungi are able to degrade plant cell wall components, such as cellulose and xylans, that the animals cannot otherwise digest.

Parasitism in Plants and Insects

In contrast with the saprotrophic fungi, parasitic fungi attack living organisms, penetrate their outer defenses, invade them, and obtain nourishment from living cytoplasm, thereby causing disease and sometimes death of the host. Most pathogenic (disease-causing) fungi are parasites of plants. Most parasites enter the host through a natural opening, such as a stoma (microscopic air pore) in a leaf, a lenticel (small opening through bark) in a stem, a broken plant hair or a hair socket in a fruit, or a wound in the plant. Among the most common and widespread diseases of plants caused by fungi are the various downy mildews (e.g., of grape, onion, tobacco), the powdery mildews (e.g., of grape, cherry, apple, peach, rose, lilac), the smuts (e.g., of corn, wheat, onion), the rusts (e.g., of wheat, oats, beans, asparagus, snapdragon, hollyhock), apple scab, brown rot of stone fruits, and various leaf spots, blights, and wilts. These diseases cause great damage annually throughout the world, destroying many crops and other sources of food. For example, nearly all the chestnut forests of the United States have been destroyed by the chestnut blight fungus (Cryphonectria parasitica), and the elms in both the United States and Europe have been devastated by Ophiostoma ulmi, the fungus that causes Dutch elm disease.

Infection of a plant takes place when the spores of a pathogenic fungus fall on the leaves or the stem of a susceptible host and germinate, each spore producing a germ tube. The tube grows on the surface of the host until it finds an opening; then the tube enters the host, puts out branches between the cells of the host, and forms a mycelial network within the invaded tissue. The germ tubes of some fungi produce special pressing organs called appressoria, from which a microscopic, needlelike peg presses against and punctures the epidermis of the host; after penetration, a mycelium develops in the usual manner. Many parasitic fungi absorb food from the host cells through the hyphal walls appressed against the cell walls of the host's internal tissues. Others produce haustoria (special absorbing structures) that branch off from the intercellular hyphae and penetrate the cells themselves. Haustoria, which may be short, bulbous protrusions or large branched systems filling the whole cell, are characteristically produced by obligate (i.e., invariably parasitic) parasites; some facultative (i.e., occasionally parasitic) parasites also produce them. Obligate parasites, which require living cytoplasm and have extremely specialized nutritional requirements, are exceptionally difficult, and often impossible, to grow in a culture dish in a laboratory. Examples of obligate parasites are the downy mildews, the powdery mildews, and the rusts.

Certain fungi form highly specialized parasitic relationships with insects. For example, the fungal genus Septobasidium is parasitic on scale insects (order Homoptera) that feed on trees. The mycelium forms elaborate structures over colonies of insects feeding on the bark. Each insect sinks its proboscis (tubular sucking organ) into the bark and remains there the rest of its life, sucking sap. The fungus sinks haustoria into the bodies of some of the insects and feeds on them without killing them. The parasitized insects are, however, rendered sterile. The perpetuation of the insect species and the spread of the fungus are accomplished by the uninfected members of the colony, which are protected from enemies by the fungus body. Newly hatched scale insects crawl over the surface of the fungus, which is at that time sporulating. Fungal spores adhere to the young insects and germinate. As the young insects settle down in a new place on the bark to begin feeding, they establish new fungal colonies. Thus, part of the insect colony is sacrificed to the fungus as food in return for the fungal protection provided for the rest of the insects. The insect is parasitic on the tree and the fungus is parasitic on the insect, but the tree is the ultimate victim.

The sooty molds constitute another interesting ecological group of fungi that are associated with insects. The majority of sooty molds are tropical or subtropical, but some species occur in the temperate zones. All sooty molds are epiphytic (i.e., they grow on the surfaces of other plants), but only in areas where scale insects are present. The fungi parasitize neither the plants nor the insects but rather obtain their nourishment exclusively from the honeydew secretions of the scale insects. Growth of the dark mycelium over the plant leaves, however, is often so dense as to significantly reduce the intensity of the light that reaches the leaf surface; this reduction in turn significantly reduces the rate of photosynthesis. Insect-fungus associations found in the tropical forests of Central and South America include the unique relationship of leafcutter ants (sometimes called parasol ants) with fungi in the family Lepiotaceae (phylum Basidiomycota). The ants cultivate the fungi in their nests as an ongoing food supply and secrete enzymes that stimulate or suppress the growth of the fungi.

Parasitism in Humans

Many pathogenic fungi are parasitic in humans and are known to cause diseases of humans and other animals. In humans, parasitic fungi most commonly enter the body through a wound in the epidermis (skin). Such wounds may be insect punctures or accidentally inflicted scratches, cuts, or bruises. One example of a fungus that causes disease in humans is Claviceps purpurea, the cause of ergotism (also known as St. Anthony’s fire), a disease that was prevalent in northern Europe in the Middle Ages, particularly in regions of high rye-bread consumption. The wind carries the fungal spores of ergot to the flowers of the rye, where the spores germinate, infect and destroy the ovaries of the plant, and replace them with masses of microscopic threads cemented together into a hard fungal structure shaped like a rye kernel but considerably larger and darker. This structure, called an ergot, contains a number of poisonous organic compounds called alkaloids. A mature head of rye may carry several ergots in addition to noninfected kernels. When the grain is harvested, much of the ergot falls to the ground, but some remains on the plants and is mixed with the grain. Although modern grain-cleaning and milling methods have practically eliminated the disease, the contaminated flour may end up in bread and other food products if the ergot is not removed before milling. In addition, the ergot that falls to the ground may be consumed by cattle turned out to graze in rye fields after harvest. Cattle that consume enough ergot may suffer abortion of fetuses or death. In the spring, when the rye is in bloom, the ergot remaining on the ground produces tiny, black, mushroom-shaped bodies that expel large numbers of spores, thus starting a new series of infections.

Other human diseases caused by fungi include athlete’s foot, ringworm, aspergillosis, histoplasmosis, and coccidioidomycosis. The yeast Candida albicans, a normal inhabitant of the human mouth, throat, colon, and reproductive organs, does not cause disease when it is in ecological balance with other microbes of the digestive system. However, disease, age, and hormonal changes can cause C. albicans to grow in a manner that cannot be controlled by the body’s defense systems, resulting in candidiasis (called thrush when affecting the mouth). Candidiasis is characterized by symptoms ranging from irritating inflamed patches on the skin or raised white patches on the tongue to life-threatening invasive infection that damages the lining of the heart or brain. Improved diagnosis and increased international travel, the latter of which has facilitated the spread of tropical pathogenic fungi, have resulted in an increased incidence of fungal disease in humans. In addition, drug therapies used to manage the immune system in transplant and cancer patients weaken the body’s defenses against fungal pathogens. Patients infected with human immunodeficiency virus (HIV), the causative agent of acquired immunodeficiency syndrome (AIDS), have similarly weakened immune defenses against fungi, and many AIDS-related deaths are caused by fungal infections (especially infection with Aspergillus fumigatus).

Mycorrhiza

Among symbiotic fungi, those that enter into mycorrhizal relationships and those that enter into relationships with algae to form lichens (see below Form and function of lichens) are probably the best-known. A large number of fungi infect the roots of plants by forming an association with plants called mycorrhiza (plural mycorrhizas or mycorrhizae). This association differs markedly from ordinary root infection, which is responsible for root rot diseases. Mycorrhiza is a non-disease-producing association in which the fungus invades the root to absorb nutrients. Mycorrhizal fungi establish a mild form of parasitism that is mutualistic, meaning both the plant and the fungus benefit from the association. About 90 percent of land plants rely on mycorrhizal fungi, especially for mineral nutrients (i.e., phosphorus), and in return the fungus receives nutrients formed by the plant. During winter, when day length is shortened and exposure to sunlight is reduced, some plants produce few or no nutrients and thus depend on fungi for sugars, nitrogenous compounds, and other nutrients that the fungi are able to absorb from waste materials in the soil. By sharing the products it absorbs from the soil with its plant host, a fungus can keep its host alive. In some lowland forests, the soil contains an abundance of mycorrhizal fungi, resulting in mycelial networks that connect the trees together. The trees and their seedlings can use the fungal mycelium to exchange nutrients and chemical messages.

There are two main types of mycorrhiza: ectomycorrhizae and endomycorrhizae. Ectomycorrhizae are fungi that are only externally associated with the plant root, whereas endomycorrhizae form their associations within the cells of the host.

Among the mycorrhizal fungi are boletes, whose mycorrhizal relationships with larch trees (Larix) and other conifers have long been known. Other examples include truffles, some of which are believed to form mycorrhizae with oak (Quercus) or beech (Fagus) trees. Many orchids form mycorrhizae with species of Rhizoctonia that provide seedlings of the orchid host with carbohydrate obtained by degradation of organic matter in the soil.

Predation

A number of fungi have developed ingenious mechanisms for trapping microorganisms such as amoebas, roundworms (nematodes), and rotifers. After the prey is captured, the fungus uses hyphae to penetrate and quickly destroy the prey. Many of these fungi secrete adhesive substances over the surface of their hyphae, causing a passing animal that touches any portion of the mycelium to adhere firmly to the hyphae. For example, the mycelia of oyster mushrooms (genus Pleurotus) secrete adhesives onto their hyphae in order to catch nematodes. Once a passing animal is caught, a penetration tube grows out of a hypha and penetrates the host's soft body. This haustorium grows and branches and then secretes enzymes that quickly kill the animal, whose cytoplasm serves as food for the fungus.

Other fungi produce hyphal loops that ensnare small animals, thereby allowing the fungus to use its haustoria to penetrate and kill a trapped animal. Perhaps the most amazing of these fungal traps are the so-called constricting rings of some species of Arthrobotrys, Dactylella, and Dactylaria—soil-inhabiting fungi easily grown under laboratory conditions. In the presence of nematodes, the mycelium produces large numbers of rings through which the average nematode is barely able to pass. When a nematode rubs the inner wall of a ring, which usually consists of three cells with touch-sensitive inner surfaces, the cells of the ring swell rapidly, and the resulting constriction holds the worm tightly. All efforts of the **nematode** to free itself fail, and a hypha, which grows out of one of the swollen ring cells at its point of contact with the worm, penetrates and branches within the animal's body, thereby killing the animal. The dead animal is then used for food by the fungus. In the absence of nematodes, these fungi do not usually produce rings in appreciable quantities. A substance secreted by nematodes stimulates the fungus to form the mycelial rings.

Nematodes are the most numerous multicellular animals on earth. A handful of soil will contain thousands of the microscopic worms, many of them parasites of insects, plants or animals.

CORE CONCEPT: CLASSIFICATION OF FUNGI

The kingdom Fungi contains five major phyla that were established according to their mode of sexual reproduction or using molecular data. Polyphyletic, unrelated fungi that reproduce without a sexual cycle, are placed for convenience in a sixth group called a "form phylum." Not all mycologists agree with this scheme. Rapid advances in molecular biology and the sequencing of 18S rRNA (a part of RNA) continue to show new and different relationships between the various categories of fungi.

Plant and Animal (this was how organisms were classified when was an undergraduate). Fungi, as well as bacteria and algae were classified in the plant kingdom under this system and that is the reason that these organisms are traditionally studied in botany. In the case of fungi, Mycology is that part of botany that studies fungi. Although fungi are no longer classified as plants, there is still good reason to study them in botany. Fungi are most often associated with plants, commonly as decomposers, and pathogens, and as their benefactors, e.g. mycorrhiza, but "What is a fungus?" Based on what your studies on plants, in this course, you know that plants are known to be derived from a single algal ancestor from the algal division: Chlorophyta, i.e. they are monophyletic. Once upon a time, the fungi were also believed to be monophyletic and to be derived from an algal ancestor that lost its ability to photosynthesize. However, over time, with the discovery of new techniques in determining relationships between organisms, it was discovered that the fungi are made up of a polyphyletic group of organisms that, in some cases, are very distantly related to one another.

Distinguishing Taxonomic Features

Taxonomy is the science of classification, identification, and nomenclature. For classification purposes, organisms are usually organized into subspecies, species, genera, families, and higher orders. For eukaryotes, the definition of the species usually stresses the ability of similar organisms to reproduce sexually with the formation of a zygote and to produce fertile offspring. However, bacteria do not undergo sexual reproduction in the eukaryotic sense. Fungi are among the most widely distributed organisms on Earth and are of great environmental and medical importance. Many fungi are free-living in soil or water; others form parasitic or symbiotic relationships with plants or animals. The phylogenetic classification of fungi divides the kingdom into 7 phyla, 10 subphyla, 35 classes, 12 subclasses, and 129 orders.

The true fungi, which make up the monophyletic clade called kingdom Fungi, comprise seven phyla: Chytridiomycota, **Blastocladiomycot**a, Neocallimastigomycota, Microsporidia, Glomeromycota, Ascomycota, and Basidiomycota. When searching for mushrooms in the wild, he said inexperienced foragers should search for mushrooms alongside an experienced and trusted mycologist whenever possible.

Blastocladio-mycota is one of the currently recognized phyla within the king-.dom Fungi

Taxonomic classification is the hierarchical organization of living beings into categories and subcategories that reveal their likenesses. There are eight main levels in the currently used and internationally accepted taxonomic classification system, which has its foundations in the work of Carl Linnaeus.

The Kingdom is the top level of taxonomic classification usually listed – Domain, a higher level, does exist, however. Most biologists recognize five kingdoms: Kingdom Animalia - the multi-cellular animals; Kingdom Plantae — plants except for blue-green algae; Kingdom Fungi — all fungi and yeasts, but not slime molds; Kingdom Protista or Protoctista — protozoa, slime molds, and so on; and Kingdom Monera — bacteria, blue-green algae, and so on. The familiar groups that we use in non-scientific environments to categorize the animals — mammals, birds, amphibians, reptiles, and insects — are the equivalent of the level of classification called Class, where they are referred to as Mammalia, Aves, Amphibia, Reptilia, and Insecta. It seems that there are several ways to handle fish, one being to assign them to four classes.

Several mnemonics have been devised to help recall the order of the taxonomic classification system, some including Variety, and some stopping at Species.

- Kindly place cover on fresh green spring vegetables.
- Krakatoa positively casts off fumes generating sulphurous vapours.
- Kings play cards on fairly good soft velvet.
- Kings play chess on fine grain sand.
- Ken poured coffee on Fran’s good shirt.
- Kangaroos play cellos; orangutans fiddle; gorillas sing.

Annotated Classification

The classification presented here reflects main evolutionary lines. These seem best illustrated at the order level for liverworts and hornworts but at the subclass level for the taxonomically more complex mosses. Those orders that are considered to be most generalized are treated first; and those most specialized, last.

Taxonomists considered fungi to be members of the kingdom Plantae. This early classification was based mainly on similarities in their lifestyles. Both fungi and plants do not move about, they look a little similar, and they grow in similar places. Fungi often grow in soil and, in the case of mushrooms, form fruiting bodies, which may look like plants such as mosses.

The kingdom fungus includes a tremendous variety of organisms that are neither plant nor animal. These unique multicellular eukaryotes include edible examples like mushrooms and organisms such as yeast, which makes our bread rise and ferments our beer and wine.

Kingdom Fungi

Fungi are important organisms that belong to their own kingdom, completely separate from plants and animals. Hugely diverse groups of great economic importance, fungi remain vastly under-studied compared to plants. Eukaryotic acellular (e.g., highly adapted parasites), unicellular (e.g., species adapted to life in small volumes of fluid), or multicellular (filamentous) with hyphae; cell walls composed of chitin, polysaccharides (e.g., glucans), or both; can be individually microscopic in size (i.e., yeasts); at least 99,000 species of fungi have been described.

What do mushrooms, bread, wine, beer, and rotting organisms all have in common? It is obvious that some are edible while others are not. Mushrooms are found in nature, while wine and beer are clearly beverages processed by humans. As for rotting organisms, those typically are not of much interest to most people. However, all of these examples have a common thread: the kingdom Fungi.

Phylum Chytridiomycota

Chytridiomycota, a phylum of fungi (kingdom Fungi) distinguished by having zoospores (motile cells) with a single, posterior, whiplash structure (flagellum). Species are microscopic in size, and most are found in freshwater or wet soils. Most are parasites of algae and animals or live on organic debris (as saprobes). A few species in the order Chytridiales cause plant disease, and one species, Batrachochytrium dendrobatidis, has been shown to cause disease in frogs and amphibians.

Mainly aquatic, some are parasitic or saprotrophic; unicellular or filamentous; chitin and glucan cell wall; primarily asexual reproduction by motile spores (zoospores); mycelia; contains 2 classes.

- Class Chytridiomycetes
- Order Chytridiales

Phylum Neocallimastigomycota

Found in digestive tracts of herbivores; anaerobic; zoospores with one or more posterior flagella; lacks mitochondria but contains hydrogenosomes (hydrogen-producing, membrane-bound organelles that generate energy in the form of adenosine triphosphate, or ATP); contains 1 class.

- Class Neocallimastigomycetes
- Order Neocallimastigales

Phylum Blastocladiomycota

Parasitic on plants and animals, some are saprotrophic; aquatic and terrestrial; flagellated; alternates between haploid and diploid generations (zygotic meiosis); contains 1 class.

- Class Blastocladiomycetes
- Order Blastocladiales

Phylum Microsporidia

Microsporidian, any parasitic fungus of the phylum Microsporidia (kingdom Fungi), found mainly in cells of the gut epithelium of insects and the skin and muscles of fish. They also occur in annelids and some other invertebrates. Infection is characterized by enlargement of the affected tissue.

Microsporidians have minute spores (2 to 20 micrometres, or 0.00008 to 0.0008 inch) that contain a single polar filament and the infective parasite (sporoplasm). When spores are ingested by a new host, the organisms enter the gut epithelium and reach specific tissues through the bloodstream or the body cavity. In the host cells they grow and repeatedly divide asexually. The mature parasites (trophozoites) eventually give rise to sexually produced zygotes that produce new spores. The species Nosema bombycis causes the disease pébrine in silkworms.

Parasitic on animals and protists; unicellular; highly reduced mitochondria; phylum not subdivided due to lack of well-defined phylogenetic relationships within the group.

Phylum Glomeromycota

Forms obligate, mutualistic, symbiotic relationships in which hyphae penetrate into the cells of roots of plants and trees (arbuscular mycorrhizal associations); coenocytic hyphae; reproduces asexually; cell walls composed primarily of chitin.

- Class Archaeosporomycetes
- Order Archaeosporales
- Class Glomeromycetes
- Order Diversisporales
- Order Gigasporales
- Order Glomerales
- Class Paraglomeromycetes
- Order Paraglomerales

Subphylum Mucoromycotina (Incertae Sedis; Not Assigned to Any Phylum)

Parasitic, saprotrophic, or ectomycorrhizal (forms mutual symbiotic associations with plants); asexual or sexual reproduction; branched mycelium; contains 3 orders that represent the traditional Zygomycota.

- Order Mucorales (pin molds)
- Order Endogonales
- Order Mortierellales

Subphylum Entomophthoromycotina (Incertae Sedis)

Pathogenic, saprotrophic, or parasitic; coenocytic or septate mycelium; rhizoids formed by some species; conidiophore branched or unbranched; conidia forcibly discharged; contains 1 order.

- Order Entomophthorales

Subphylum Zoopagomycotina (Incertae Sedis)

Endoparasitic (lives in the body) or ectoparasitic (lives on the body) on nematodes, protozoa, and fungi; thallus branched or unbranched; asexual and sexual reproduction; contains 1 order.

- Order Zoopagales

Subphylum Kickxellomycotina (Incertae Sedis)

Saprotrophic, may be parasitic on fungi, can form symbiotic associations; thallus forms from holdfast on other fungi; mycelium branched or unbranched; asexual and sexual reproduction; contains four orders.

- Order Kickxellales
- Order Dimargaritales
- Order Harpellales
- Order Asellariales

Sac fungi are a phylum of fungi characterized by a saclike structure called the ascus. They are also responsible for causing diseases .like candida

Phylum Ascomycota (Sac Fungi)

Ascomycota, also called sac fungi, a phylum of fungi (kingdom Fungi) characterized by a saclike structure, the ascus, which contains four to eight ascospores in the sexual stage.

The **sac fungi** are separated into subgroups based on whether asci arise singly or are borne in one of several types of fruiting structures, or ascocarps, and on the method of discharge of the ascospores. Many ascomycetes are plant pathogens, some are animal pathogens, a few are edible mushrooms, and many live on dead organic matter (as saprobes). The largest and most commonly known ascomycetes include the morel and the truffle. Other ascomycetes include important plant pathogens, such as those that cause powdery mildew of grape (Uncinula necator), Dutch elm disease (Ophiostoma ulmi), chestnut blight (Cryphonectria parasitica), and apple scab (Venturia inequalis). Perhaps the most indispensable fungus of all is an ascomycete, the common yeast (Saccharomyces cerevisiae), whose varieties leaven the dough in bread making and ferment grain to produce beer or mash for distillation of alcoholic liquors; the strains of S. cerevisiae var. ellipsoideus ferment grape juice to wine.

Neurospora, a genus of widespread species, produces bakery mold, or red bread mold. It has been used extensively in genetic and biochemical investigations. Xylaria contains about 100 species of cosmopolitan fungi. X. polymorpha produces a club-shaped or fingerlike fruiting body (stroma) resembling burned wood and common on decaying wood or injured trees.

Cordyceps, a genus of more than 400 species within the order Hypocreales, are commonly known as vegetable caterpillars, or caterpillar fungi. C. militaris parasitizes insects. It forms a small, 3- or 4-centimetre (about 1.3-inch) mushroomlike fruiting structure with a bright orange head, or cap. A related genus, Claviceps, includes C. purpurea, the cause of ergot of rye and ergotism in humans and domestic animals. Earth tongue is the common name for the more than

80 Geoglossum species of the order Helotiales. They produce black to brown, club-shaped fruiting structures on soil or on decaying wood.

Symbiotic with algae to form lichens, some are parasitic or saprotrophic on plants, animals, or humans; some are unicellular, but most are filamentous; hyphae septate with 1, rarely more, perforation in the septa; cells uninucleate or multinucleate; asexual reproduction by fission, budding, or fragmentation or by conidia that are usually produced on sporiferous (spore-producing) hyphae, the conidiophores, which are borne loosely on somatic (main-body) hyphae or variously assembled in asexual fruiting bodies; sexual reproduction by various means resulting in the production of meiosphores (ascospores) formed by free-cell formation in saclike structures (asci), which are produced naked or, more typically, are assembled in characteristic open or closed bodies (ascocarps, also called ascomata); ascomycota include some cup fungi, saddle fungi, and truffles; this phylum is sometimes included in the subkingdom Dikarya with its sister group, **Basidiomycota**.

Basidiomycota are filamentous fungi composed of hyphae and reproduce sexually via the formation of specialized club-shaped end cells called basidia that normally bear external meiospores.

Subphylum Taphrinomycotina

Pathogenic on some plants; unicellular or filamentous; asci produced on the plant surface; ascocarp absent; contains 4 classes.

- Class Taphrinomycetes
- Order Taphrinales
- Class Neolectomycetes
- Order Neolectales
- Class Pneumocystidomycetes
- Order Pneumocystidales
- Class Schizosaccharomycetes
- Order Schizosaccharomycetales (Fission Yeasts)

Subphylum Saccharomycotina (True Yeasts)

Saprotrophic on plants and animals, including humans, occasionally pathogenic in plants and humans; unicellular; found in short chains; asexual reproduction by budding or fission; contains common yeasts that are relevant to industry (e.g., baking and brewing) and that cause common infections in humans; contains 1 class.

- Class Saccharomycetes
- Order Saccharomycetales (Ascomycete Yeasts)

Phylum Basidiomycota

Parasitic or saprotrophic on plants or insects; filamentous; hyphae septate, with septa typically inflated (dolipore) and centrally perforated; mycelium of two types: primary consisting of uninucleate cells, succeeded by secondary consisting of dikaryotic cells, often bearing bridgelike clamp connections over the septa; asexual reproduction by fragmentation, oidia (thin-walled, free, hyphal cells behaving as spores), or conidia; sexual reproduction by fusion of hyphae (somatogamy), fusion of an oidium with a hypha (oidization), or fusion of a spermatium (a nonmotile male structure that empties its contents into a receptive female structure during plasmogamy) with a specialized receptive hypha (spermatization), resulting in dikaryotic hyphae that eventually give rise to basidia, either singly on the hyphae or in variously shaped basidiocarps(also called basidiomata); meiospores (basidiospores) borne on basidia; in the rusts and smuts, the dikaryotic hyphae produce teleutospores (thick-walled resting spores), which are a part of the basidial apparatus; this is a large phylum of fungi containing the rusts, smuts, jelly fungi, club fungi, coral and shelf fungi, mushrooms, puffballs, stinkhorns, and bird’s-nest fungi; sometimes included in the subkingdom Dikarya with its sister group, Ascomycota.

Subphylum Pucciniomycotina

Pucciniomycotina is a subdivision of fungus within the division Basidiomycota. The subdivision contains 9 classes, 20 orders, and 37 families. Over 8400 species of Pucciniomycotina have been described - more than 8% of all described fungi. The subdivision is considered a sister group to Ustilaginomycotina and Agaricomycotina, which may share the basal lineage of Basidiomycota, although this is uncertain due to low support for placement between the three groups. The group was known as Urediniomycetes until 2006, when it was elevated from a class to a subdivision and named after the largest order in the group, Puccinales.

Subphylum Agaricomycotina

Parasitic or symbiotic on plants, animals, and other fungi, some are saprotrophic ormycorrhizal; basidia may be undivided or have transverse or longitudinal septa; dolipore (inflated) septa and septal pore cap (parenthesomes) present; includes mushrooms, bracket fungi, puffballs; contains 3 classes.

- Class Tremellomycetes
- Class Dacrymycetes
- Class Agaricomycetes

Basidiomycota (Incertae Sedis)

Includes basidiomycota not placed in a subphylum; contains 2 classes.

- Class Wallemiomycetes
- Class Entorrhizomycetes

Kingdom Chromista

Common microorganisms; includes important plant pathogens, such as the cause of potato blight (*Phytophthora*); motile spores swim by means of 2 flagella and grow as hyphae with cellulose-containing walls; includes the majority of the Oomycota; contains a total of approximately 110 genera and 900 species.

Phylum Hyphochytriomycota

Hyphochytriomycota, a phylum of mostly aquatic fungi that contains approximately 23 species and is classified in the kingdom Chromista. The phylum is distinguished by the asexual production of motile cells (zoospores) with a single, anterior, feathery, whiplike structure (flagellum). Sexual reproduction has not been found among these fungi. Microscopic organisms that are parasitic or saprotrophic on algae and fungi in fresh water and in soil; forms a small thallus, often with branched rhizoids; whole of the thallus is eventually converted into a reproductive structure; contains 23 species in 6 genera.

- Order Hyphochytriales

Phylum Labyrinthulomycota

Found in both salt water and fresh water in association with algae and other chromists; feeding stage comprises an ectoplasmic network and spindle-shaped or spherical cells that move within the network by gliding over one another; contains about 45 species in 10 genera.

- Order Labyrinthulales
- Order Thraustochytriales

Phylum Oomycota

Oomycota, phylum of fungi in the kingdom Chromista that is distinguished by its production of asexual reproductive cells, called zoospores. Zoospores move through the use of one or two whiplike swimming structures (flagella). New fungi may germinate from these spores, or mature fungi may reproduce sexually, with the resulting fertilized eggs being converted into nonmobile spores, or oospores, which then also germinate into mature individuals. Oomycetic fungi may occur as saprophytes (living on decayed matter) or as parasites living on higher plants. Among the various aquatic, amphibious, or terrestrial species, one played an important role in modern history: the species Phytophthora infestans destroyed Ireland's potato crop and caused the famine of 1845, which resulted in a mass migration of the Irish to the United States. Some other economically destructive genera include Saprolegnia (water molds), Aphanomyces (the cause of root rot of pea), Plasmopara (downy mildews), and Albugo (white rusts).

Found in fresh water, wet soil, and marine habitats, some are pathogenic (such as *Saprolegnia*and *Phytophthora*); contains about 600 species in 90 genera.

- Order Leptomitales
- Order Myzocytiopsidales
- Order Olpidiopsidales
- Order Peronosporales
- Order Pythiales
- Order Rhipidiales
- Order Salilagenidiales
- Order Saprolegniales (Water Molds)
- Order Sclerosporales
- Order Anisolpidiales
- Order Lagenismatales
- Order Rozellopsidales
- Order Haptoglossales

Characteristics of the Kingdom Fungi

As previously mentioned, the kingdom Fungi encompasses a wide variety of living organisms. Scientists estimate that there are hundreds of thousands of fungus species on Earth. At one time, it was thought that fungi were simply primitive versions of plants. However, further discoveries led to the realization that fungi were different enough to belong to their very own kingdom. Let's take a look at some of the common characteristics of these organisms.

Members of the kingdom Fungi are eukaryotes, meaning they have complex cells with a nucleus and organelles. Most are multicellular, with the exception of single-celled yeast. Structurally, fungi are made up of individual feathery filaments called hyphae. You have certainly observed these filaments, probably with disgust, if you have ever pulled out an old Tupperware of food from the refrigerator.

The hyphae group together to form a conglomerate called the mycelium. You may notice that when you cut into a mushroom, it has a spongy feel. This is because it is actually made up of a mass of very tightly packed hyphae, and thus, is not exactly solid. All fungi are heterotrophic, meaning that they cannot make their own food like plants. Like us, they must gain nutrition from other organisms. So they seek out food sources and eat them, correct? Not exactly. Have you ever seen a mushroom hunting its dinner? Instead, like a parasite inside a body, they grow on organisms and absorb their nutrients.

As an example, let's think about our faithful kitchen friend who we previously mentioned in the Tupperware. One day you have a container of ripe, plump strawberries. After they sit out for several days, you notice the white fuzz of mold begin to appear. If you dare to touch a moldy berry, you are lucky if your finger does not sink right through the mushy skin. But why has it turned so soft?

CORE CONCEPT: FEATURES AND LIFE CYCLE OF LICHEN IN FUNGUS

A lichen is a composite organism that arises from algae or cyanobacteria living among filaments of multiple fungi in a symbiotic relationship. The combined lichen has properties different from those of its component organisms. Lichens come in many colors, sizes, and forms.

When an alga and fungus physically intertwine in a very close symbiotic relationship, they actually form an entirely new growth called lichen. Lichen is a kind of primitive plant species that's nothing more than strands of alga linked with roots and branches of a fungus that together absorb minerals from the ground and conduct photosynthesis. They can grow almost anywhere, from moist bark to recently cooled lava to frozen rocks.

Unlike most parasitic or symbiotic relationships, when a fungus and alga grow together so tightly they can't be separated, they qualify as a different kind of material. Biologists have agreed that lichen straddles the monera and fungi kingdoms of living things since it is part one and part another. Its body, called a thallus, can be made of different types of fungi and blue-green algae, which will determine how much water it needs or to what it can attach.

The alga, called the **phycobiont,** contains chlorophyll, so it can photosynthesize energy that it passes onto the fungus. The mycobiant, the fungus, has roots that leech minerals and water from rocks or plants that it, in turn, passes onto the alga. This allows the new growth to thrive in a greater diversity of climates than either algae or fungi do alone. It can even dry out during drought and reconstitute itself when the rains come.

Phycobiont refers to the algal component of the lichens and mycobiont refers to the fungal component. Both of these are present in symbiotic relationship in which Algae prepare food for Fungi due to presence of chlorophyll whereas the fungus provides shelter to algae and absorbs water and nutrients from the soil.

Three major types of lichen grow into different shapes on various substrates. Crustose grows into unique flat plates that look like mushrooms. Foliose looks leafier, with small, green lobes, and grows in wetter climates beside moss. Lichen that resembles viney clumps hanging from branches is usually fruticose. It can be found in rainforests on partially rotting logs, in arctic regions on rocks, or cliffsides near salty ocean spray.

Lichen is not just a peculiar, uncategorizable kind of living matter; people use it for practical purposes, too. Since it's very sensitive to pollution, especially sulfur dioxide, environmentalists use its presence as a gauge of the cleanliness of the air. Dyes and medicine can be extracted from them as well, and virtually every plant and animal relies on their slow erosion of wood and rock to replenish the topsoil with nutrients.

Different Types of Lichen

Lichens are a type of symbiotic organism made up of a plantlike partner and a fungus. There are three major types of lichen — crustose, foliose, and fructicose — each of which has its own shape, structure, and environmental preferences. Intermediate types include leprose and squamulose lichen, among others. These organisms may also be grouped by the kind of environment in which they prefer to grow.

Every individual lichen is composed of a mycobiont, or fungus, combined with a photobiont or phycobiont in the form of green algae or cyanobacteria. The algae or bacteria photosynthesize, providing nutrients for the fungus, and giving the lichen its characteristic greenish or blueish color. Both parts of the lichen get water and minerals from dust and rain, but some also obtain nutrients from their substrate via the fungal partner.

Not all types of lichen look the same. Crustose are flat and unlobed, with a close attachment to their substrate, and can be difficult to remove from the rock or tree on which they grow. Foliose lichen are leafier-looking, as their name implies, and are made up of two thin sheets of fungus with algae in the middle. They grow in round lobe formations and are easier to pull off of their substrate, since they attach only by small rootlets. Fructicose, or shrubby lichen, have small round branches made of fungus with algae inside, and an unusual pattern of vertical growth that may look beard-like or resemble a small bush.

Other types of lichen include leprose lichen, which form powdery, largely unstructured masses without a smooth surface. Placodioid lichens are lobed or unattached at the edges, and closely attached at the center, making them an intermediate form between crustose and foliose lichens. Another intermediate form, squamulose lichen, has many tiny lobes. Dimorphic lichens have characteristics of both squamulose and fruticose lichens, with small lobes that carry tiny stems or branches.

Environmental grouping divides lichens into seven major categories. Several types of lichen grow on plants, and are called epiphytic. This group includes the coricolous lichens, which prefer to grow on tree trunks, as well as the ramicolous lichens, which inhabit twigs. The musicocolous lichens grow on live moss, and the foliicolous lichens prefer evergreen leaves. Both of those types are ephiphytic, but the legnicolous, saxicolous and terricolous lichens, which inhabit wood, stones, and soil respectively, are not epiphytes.

Life Cycle of Lichen

Lichen, any of about 15,000 species of thallophytic plantlike organisms that consist of a symbiotic association of algae (usually green) or cyanobacteria and fungi (mostly ascomycetes and basidiomycetes). Lichens are found worldwide and occur in a variety of environmental conditions. A diverse group of organisms, they can colonize a wide range of surfaces and are frequently found on tree bark, exposed rock, and as a part of biological soil crust. Lichens have been used by humans as food and as sources of medicine and dye. They also provide two-thirds of the food supply for the caribou and reindeer that roam the far northern ranges.

Lichens were once classified as single organisms—until the advent of microscopy, when the association of fungi with algae or cyanobacteria became evident. Although lichens had been assumed to consist of a single fungus species (usually an ascomycete) and a single photosynthetic partner, research suggests that many macrolichens also feature specific basidiomycete yeasts in the cortex of the organism. There is still some discussion about how to classify lichens, though many taxonomists rely on genetic analyses in addition to traditional morphological data.

Trebouxia is a genus of unicellular green algae in the family Trebouxiaceae. They are common algal symbionts in lichens, but can also be found free-living.

The composite body of a lichen is called a thallus (plural thalli); the body is anchored to its substrate by hairlike growths called rhizines. Lichens that form a crustlike covering that is thin and tightly bound to the substrate are called crustose. Squamulose lichens are small and leafy with loose attachments to the substrate. Foliose lichens are large and leafy, reaching diameters of several feet in some species, and are usually attached to the substrate by their large platelike thalli at the centre. In addition to their morphological forms, lichen thalli are also classified by the ratio of phycobiont cells (i.e., cells of the photosynthetic partner) to mycobiont cells (i.e., cells of the fungus). The homoeomerous type of thallus consists of numerous algal cells distributed among a lesser number of fungal cells, while the heteromerous thallus has a predominance of fungal cells.

Evolutionarily, it is not certain when fungi and algae came together to form lichens for the first time, but it was certainly after the mature development of the separate components. As symbionts, the basis of their relationship is the mutual benefit that they provide each other. The photosynthetic algae or cyanobacteria form simple carbohydrates that, when excreted, are absorbed by fungi cells and transformed into a different carbohydrate. Peltigera polydactyla, the exchange occurs within two minutes. The phycobionts also produce vitamins that the fungi need. Fungi contribute to the symbiosis by absorbing water vapor from the air and by providing much-needed shade for the light-sensitive algae beneath.

Lichens are long-lived and grow relatively slowly, and there is still some question as to how they propagate. Most botanists agree that the most common means of reproduction is vegetative; that is, portions of an existing lichen break off and fall away to begin new growth nearby.

Basic Features of Lichens

A lichen is an association between one or two fungus species and an alga or cyanobacterium (blue-green alga) that results in a form distinct from the symbionts. Although lichens appear to be single plantlike organisms, under a microscope the associations are seen to consist of millions of cells of algae (called the phycobiont) woven into a matrix formed of the filaments of the fungi (called the mycobiont). Many mycobionts are placed in a single group of Ascomycota called the Lecanoromycetes, which are characterized by an open, often button-shaped fruit called an apothecium. Although lichens had long been assumed to consist of a single fungus species and a single phycobiont, research suggests that many macrolichens also feature specific basidiomycete yeasts in the cortex of the organism. There are various types of phycobionts, though half the lichenassociations contain species of *Trebouxia*, a single-celled green alga. There are about 15 species of cyanobacteria that act as the photobiont in lichen associations, including some members of the genera *Calothrix*,*Gloeocapsa*, and *Nostoc*.

Authorities have not been able to establish with any certainty when and how these associations evolved, although lichens must have evolved more recently than their components and probably arose independently from different groups of fungi and algae or fungi and cyanobacteria. It seems, moreover, that the ability to form lichens can spread to new groups of fungi and algae. Lichens are a biological group lacking formal status in the taxonomic framework of living organisms. Although the mycobiont and phycobiont have Latin names, the product of their interaction, a lichen, does not. Earlier names given to lichens as a whole are considered names for the fungus alone, and much of the problem lies in the fact that the taxonomy of lichens was established before their dual nature was recognized; i.e., the association was treated as a single entity. Classification of lichens is difficult and remains controversial, though genetic analyses of the symbionts in a given lichen may serve to clarify the taxonomy of the group.

Approximately 15,000 different kinds of lichens, some of which provide forage for reindeer and products for humans, have been described. Some lichens are leafy and form beautiful rosettes on rocks and tree trunks; others are filamentous and drape the branches of trees, sometimes reaching a length of 2.75 meters (9 feet). At the opposite extreme are those smaller than a pin head and seen only with a magnifying lens. Lichens grow on almost any type of surface and can be found in almost all areas of the world. They are especially prominent in bleak, harsh regions where few plants can survive. They grow farther north and farther south and higher on mountains than most plants.

The thallus of a lichen has one of several characteristic growth forms: crustose, foliose, or fruticose. Crustose thalli, which resemble a crust closely attached to a surface, are drought-resistant and well adapted to dry climates. They prevail in deserts, Arctic and Alpine regions, and ice-free parts of Antarctica. Foliose, or leafy, thalli grow best in areas of frequent rainfall; two foliose lichens, Hydrothyria venosa and Dermatocarpon fluviatile, grow on rocks in freshwater streams of North America. Fruticose (stalked) thalli and filamentous forms prefer to utilize water in vapor form and are prevalent in humid, foggy areas such as seacoasts and mountainous regions of the tropics.

Humans have used lichens as food, as medicine, and in dyes. A versatile lichen of economic importance is *Cetraria* islandica, commonly called Iceland moss and sometimes used either as an appetite stimulant or as a foodstuff in reducing diets; it has also been mixed with bread and has been used to treat diabetes, nephritis, and catarrh. In general, however, lichens have little medical value. One lichen, Lecanora esculenta, is reputed to have been the manna that fell from the skies during the biblical Exodus and has served as a food source for humans and domestic animals.

Lichens are well known as dye sources. Dyes derived from them have an affinity for wool and silk and are formed by decomposition of certain lichen acids and conversion of the products. One of the best-known lichen dyes is orchil, which has a purple or red-violet color. Orchil-producing lichens include species of Ochrolechia, Roccella, and Umbilicaria. Litmus, formed from orchil, is widely used as an acid-base indicator. Synthetic coal tar dyes, however, have replaced lichen dyes in the textile industry, and orchil is limited to use as a food-coloring agent and an acid-base indicator. A few lichens (e.g., Evernia prunastri) are used in the manufacture of perfumes.

Caribou and reindeer depend on lichens for two-thirds of their food supply. In northern Canada an acre of land undisturbed by animals for 120 years or more may contain 250 kg (550 pounds) of lichens; some forage lichens that form extensive mats on the ground are Cladonia alpestris, C. mitis, C. rangiferina, and C. sylvatica. Arboreal lichens such as Alectoria, Evernia, and Usnea also are valuable as forage. An acre of mature black spruce trees in northern Canada, for example, may contain more than 270 kg (595 pounds) of lichens on branches within 3 metres (10 feet) of the ground.

Form and Function of Lichens

Although the fungal symbionts of many lichens have fruiting structures on or within their thalli and may release numerous sporesthat develop into fungi, indirect evidence suggests that natural unions of fungi and algae occur only rarely among some lichen groups, if indeed they occur at all. In addition, free-living potential phycobionts are not widely distributed; for example, despite repeated searches, free-living populations of Trebouxia have not been found. This paradox, an abundance of fungal spores and a lack of algae capable of forming associations, implies that the countless spores produced by lichen fungi are functionless, at least so far as propagation of the association is concerned. Some photobionts, including species of *Nostoc* and *Trentpohlia*, can exist as free-living populations, so that natural reassociations could occur in a few lichens.

Some lichens have solved or bypassed the problem of re-forming the association. In a few lichens (e.g., *Endocarpon*, *Staurothele*) algae grow among the tissues of a fruiting body and are discharged along with fungal spores; such phycobionts are called hymenial algae. When the spores germinate, the algal cells multiply and gradually form lichens with the fungus. Other lichens form structures, especially soredia that are effective in distributing the association. A soredium, consisting of one or several algal cells enveloped by threadlike fungal filaments, or hyphae, may develop into a thallus under suitable conditions. Lichens without soredia may propagate by fragmentation of their thalli. Many lichens develop small thalloid extensions, called isidia that also may serve in asexual propagation if broken off from the thallus.

In addition to these mechanisms for propagation, the individual symbionts have various methods of reproduction. For example, ascolichens (lichens in which the dominant mycobiont is an ascomycete) form fruits called ascocarps that are similar to those of free-living ascomycetes, except that the mycobiont's fruits are capable of producing spores for a longer period of time. The algal symbiont within the lichen thallus reproduces by the same methods as its free-living counterpart.

Most lichen phycobionts are penetrated to varying degrees by specialized fungal structures called haustoria. *Trebouxia* lichens have a pattern in which deeply penetrating haustoria are prevalent in associations lacking a high degree of thalloid organization. On the other hand, superficial haustoria prevail among forms with highly developed thalli. *Lecanora* and *Lecidea*, for example, have individual algal cells with as many as five haustoria that may extend to the cell centre. *Alectoria* and *Cladonia* have haustoria that do not penetrate far beyond the algal cell wall. A few phycobionts, such as *Coccomyxa* and *Stichococcus*, which are not penetrated by haustoria, have thin-walled cells that are pressed close to fungal hyphae.

The flow of nutrients and metabolites between the symbionts is the basic foundation of the symbiotic system. A simplecarbohydrate formed in the algal layer eventually is excreted, taken up by the mycobiont, and transformed into a different carbohydrate. The release of carbohydrate by the phycobiont and its conversion by the mycobiont occur rapidly. Whether the fungus influences the release of carbohydrate by the alga is not known with certainty, but it is known that carbohydrate excretion by the alga decreases rapidly if it is separated from the fungus.

Carbohydrate transfer is only one aspect of the symbiotic interaction in lichens. The alga may provide the fungus with vitamins, especially biotin and thiamine, important because most lichen fungi that are grown in the absence of algae have vitamin deficiencies. The alga also may contribute a substance that causes structural changes in the fungus since it forms the typical lichen thallus only in association with an alga.

One contribution of the fungus to the symbiosis concerns absorption of water vapor from the air; the process is so effective that, at high levels of air humidity, the phycobionts of some lichens photosynthesize at near-maximum rates. The upper region of a thallus provides shade for the underlying algae, some of which are sensitive to strong light. In addition, the upper region may contain pigments or crystals that further reduce light intensity and act as filters, absorbing certain wavelengths of light.

Lichens synthesize a variety of unique organic compounds that tend to accumulate within the thallus; many of these substances are colored and are responsible for the red, yellow, or orange color of lichens.

A lichen thallus or composite body has one of two basic structures. In a homoiomerous thallus, the algal cells, which are distributed throughout the structure, are more numerous than those of the fungus. The more common type of thallus, a heteromerous thallus, has four distinct layers, three of which are formed by the fungus and one by the alga. The fungal layers are called upper cortex, medulla, and lower cortex. The upper cortex consists of either a few layers of tightly packed cells or hyphae that may contain pigments. A cuticle may cover the cortex. The lower cortex, which is similar in structure to the upper cortex, participates in the formation of attachment structures called rhizines. The medulla, located below the algal layer, is the widest layer of a heteromerous thallus. It has a cottony appearance and consists of interlaced hyphae. The loosely structured nature of the medulla provides it with numerous air spaces and allows it to hold large amounts of water. The algal layer, about three times as wide as a cortex, consists of tightly packed algal cells enveloped by fungal hyphae from the medulla.

A heteromerous thallus may have a stalked (fruticose), crustlike (crustose), or leafy (foliose) form; many transitional types exist. It is not known, moreover, which growth form is primitive and which is advanced. Fruticose lichens, which usually arise from a primary thallus of a different growth form (i.e., crustose, foliose), may be shrubby or pendulous or consist of upright stalks. The fruticose form usually consists of two thalloid types: the primary thallus is crustlike or lobed; the secondary thalli, which originate from the crust or lobes of the primary thallus, consist of stalks that may be simple, cup-shaped, intricately branched, and capped with brown or red fruiting bodies called apothecia. Fruticose forms such as *Usnea* may have elongated stalks with a central solid core that provides strength and elasticity to the thallus.

The crustose thallus is in such intimate contact with the surface to which it is attached that it usually cannot be removed intact. Some crustose lichens grow beneath the surface of bark or rock so that only their fruiting structures penetrate the surface. Crustose lichens may have a hypothallus—i.e., an algal-free mat of hyphae extending beyond the margin of the regular thallus. Crustose form varies: granular types such as *Lepraria*, for example, have no organized thalloid structure; but some *Lecanora* species have highly organized thalli, with lobes that resemble foliose lichens lacking a lower cortex.

The foliose forms are flat, leaflike, and loosely attached to a surface. The largest known lichens have a foliose form; species of *Sticta* may attain a diameter of about a metre. Other common foliose genera include *Cetraria*, *Parmelia*, *Peltigera*, and *Physcia*. *Umbilicaria*, called the common rock tripe, differs from other foliose forms in its mode of attachment in that its platelike thallus attaches at the centre to a rock surface.

The complex fruiting bodies (ascocarps) of lichen fungi are of several types. The factors that induce fruiting in lichens have not been established with certainty. Spores of lichen fungi (ascospores) are of extremely varying sizes and shapes; e.g.,*Pertusaria* has one or two large spores in one ascus (saclike bodies containing the ascospores), and *Acarospora* may have several hundred small spores per ascus. Although in most species the ascospore generally has one nucleus, it may be single-celled or multicellular, brown or colorless; the *Pertusaria* spore, however, is a single cell containing 200 nuclei. Another type of fungal spore may be what are sometimes called spermatia (male fungal sex cells) or pycnidiospores; it is not certain that these structures have the ability to germinate and develop into a fungal colony. Few lichen fungi produce conidia, a type of asexual spore common among ascomycetes.

The metabolic activity of lichens is greatly influenced by the water content of the thallus. The rate of photosynthesis may be greatest when the amount of water in the thallus is from 65 to 90 % of the maximum. During drying conditions, the photosynthetic rate decreases; below 30 % it is no longer measurable. Although respiration also decreases rapidly below 80 % water content, it persists at low rates even when the thallus is air-dried. Since lichens have no mechanisms for water retention or uptake from the surface to which they are attached, they very quickly lose the water vapor they absorb from the air. The rapid drying of lichens is a protective device; i.e., a moisture-free lichen is more resistant to temperature and light extremes than is a wet one. Frequent drying and wetting of a thallus is one of the reasons lichens have a slow growth rate.

Maximum photosynthesis in lichens takes place at temperatures of 15–20°C (59–68°F). More light is needed in the spring and summer than in the winter. The photosynthetic apparatus of lichens is remarkably resistant to cold temperatures. Even at temperatures below 0°C (32°F), many lichens can absorb and fix considerable amounts of carbon dioxide. Respiration is much less at low temperatures so that, in nature, the winter months may be the most productive ones for lichens.

CONCEPT OF ERGOT

Members of the genus Claviceps may infect a variety of grains, particularly rye, when they come into flower, giving rise to the condition called ergot. No great damage is caused to the crop, but as the fungus develops in the maturing grain, powerful hallucinatory compounds are produced, which cause ergotism in those who consume bread made from the affected grain. This was relatively common in the Middle Ages, when it was known as St Anthony's Fire. The hallucinatory effects of ergotism have been put forward by some as an explanation for outbreaks of mass hysteria such as witch hunts and also for the cause of the abandonment of the ship, the Mary Celeste. The effects can go beyond the psychological causing convulsions and even death. In small controlled amounts, the drugs derived from ergot can be medically useful in certain situations such as the induction of childbirth and the relief of migraine headaches.

SUMMARY

- Fungus, plural fungi, any of about 99,000 known species of organisms of the kingdom Fungi, which includes the yeasts, rusts, smuts, mildews, molds, and mushrooms. There are also many funguslike organisms, including slime molds and oomycetes (water molds), that do not belong to kingdom Fungi but are often called fungi.
- Fungi are eukaryotic organisms; i.e., their cells contain membrane-bound organelles and clearly defined nuclei. Historically, fungi were included in the plant kingdom; however, because fungi lack chlorophyll and are distinguished by unique structural and physiological features (i.e., components of the cell wall and cell membrane), they have been separated from plants.
- Fungal hyphae are adapted to efficient absorption of nutrients from their environments, because hyphae have high surface area-to-volume ratios. These adaptations are also complemented by the release of hydrolytic enzymes that break down large organic molecules such as polysaccharides, proteins, and lipids into smaller molecules.
- Chytridiomycota, a phylum of fungi (kingdom Fungi) distinguished by having zoospores (motile cells) with a single, posterior, whiplash structure (flagellum). Species are microscopic in size, and most are found in freshwater or wet soils.
- Microsporidian, any parasitic fungus of the phylum Microsporidia (kingdom Fungi), found mainly in cells of the gut epithelium of insects and the skin and muscles of fish. They also occur in annelids and some other invertebrates. Infection is characterized by enlargement of the affected tissue.
- The sac fungi are separated into subgroups based on whether asci arise singly or are borne in one of several types of fruiting structures, or ascocarps, and on the method of discharge of the ascospores.
- Pucciniomycotina is a subdivision of fungus within the division Basidiomycota. The subdivision contains 9 classes, 20 orders, and 37 families. Over 8400 species of Pucciniomycotina have been described - more than 8% of all described fungi.
- Lichens are a type of symbiotic organism made up of a plantlike partner and a fungus. There are three major types of lichen — crustose, foliose, and fructicose — each of which has its own shape, structure, and environmental preferences.

MULTIPLE CHOICE QUESTIONS

1. **What are the cell wall structural components of fungi?**
   a. peptidoglycan
   b. cellulose
   c. chitin
   d. chitin, cellulose, or hemicellulose
2. **The optimum pH for the growth of fungi is ____________**
   a. 7.8-9.0
   b. 12-14
   c. 3.8-5.6
   d. 1.0-2.0
3. **Fungi are sensitive to which of the following antibiotics?**
   a. penicillins
   b. tetracyclins
   c. chloramphenicol
   d. griseofulvin
4. **Pathogenic fungi have a temperature optimum at __________**
   a. 30 to 37 degree celsius
   b. 22 to 30 degree celsius
   c. 37 to 42 degree celsius
   d. 42 to 50 degree celsius
5. **Sabouraud media for the growth of fungi is composed of ____________________**
   a. glucose and ammonia
   b. maltose and peptone
   c. sucrose and peptone
   d. peptone

Answers

1. (d) 2. (c) 3. (d) 4. (a) 5. (b)

REVIEW QUESTIONS

1. How many species of fungi are there?
2. Explain the importance of fungi.
3. Define the 'mycorrhiza'.
4. What are the different types of lichen?
5. Write the form and function of lichens.

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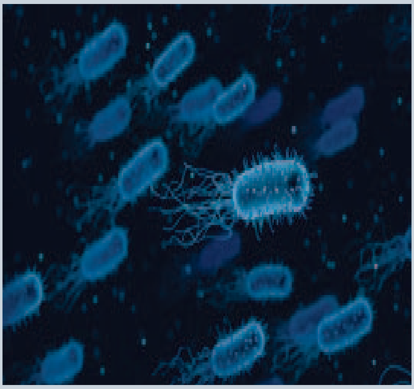
CHAPTER 4

BACTERIOLOGY



Bacteriology is a branch of microbiology that is concerned with the study of bacteria (as well as Archaea) and related aspects. It’s a field in which bacteriologists study and learn more about the various characteristics (structure, genetics, biochemistry and ecology etc.) of bacteria as well as the mechanism through which they cause diseases in humans and animals.

After studying this chapter, you will understand the core concepts of:

- Phenomena of Bacteriology
- Classification of Bacteria

INTRODUCTION

Bacteriology is a scientific discipline. It is the branch of microbiology that focuses on bacteria.

The beginnings of bacteriology paralleled the development of the microscope. The first person to see microorganisms was probably the Dutch naturalist Antonie van Leeuwenhoek, who in 1683 described some animalcules, as they were then called, in water, saliva, and other substances. These had been seen with a simple lens magnifying about 100–150 diameters. The organisms seem to correspond with some of the very large forms of bacteria as now recognized. As late as the mid-19th century, bacteria were known only to a few experts and in a few forms as curiosities of the microscope, chiefly interesting for their minuteness and motility. Modern understanding of the forms of bacteria dates from Ferdinand Cohn's brilliant classifications, the chief results of which were published at various periods between 1853 and 1872. While Cohn and others advanced knowledge of the morphology of bacteria, other researchers, such as Louis Pasteur and Robert Koch, established the connections between bacteria and the processes of fermentation and disease, in the process discarding the theory of spontaneous generation and improving antisepsis in medical treatment.

The modern methods of bacteriological technique had their beginnings in 1870–85 with the introduction of the use of stains and by the discovery of the method of separating mixtures of organisms on plates of nutrient media solidified with gelatin or agar. Important discoveries came in 1880 and 1881, when Pasteur succeeded in immunizing animals against two diseases caused by bacteria. His research led to a study of disease prevention and the treatment of disease by vaccines and immune serums (a branch of medicine now called immunology). Other scientists recognized the importance of bacteria in agriculture and the dairy industry.

Bacteriological study subsequently developed a number of specializations, among which are agricultural, or soil, bacteriology; clinical diagnostic bacteriology; industrial bacteriology; marine bacteriology; public-health bacteriology; sanitary, or hygienic, bacteriology; and systematic bacteriology, which deals with taxonomy.

CORE CONCEPT: PHENOMENA OF BACTERIOLOGY

Bacteriology evolved from physicians needing to apply the germ theory to test the concerns relating to the spoilage of foods and wines in the 18th century. Identification and characterizing of bacteria being associated to diseases led to advances in pathogenic bacteriology.

Bacteriology, branch of microbiology dealing with the study of bacteria. The beginnings of bacteriology paralleled the development of the microscope. The first person to see microorganisms was probably the Dutch naturalist Antonie van Leeuwenhoek, who in 1683 described some animalcules, as they were then called, in water, **saliva**, and other substances. These had been seen with a simple lens magnifying about 100–150 diameters. The organisms seem to correspond with some of the very large forms of bacteria as now recognized.

Saliva is an extracellular fluid produced and secreted by salivary glands in the mouth.

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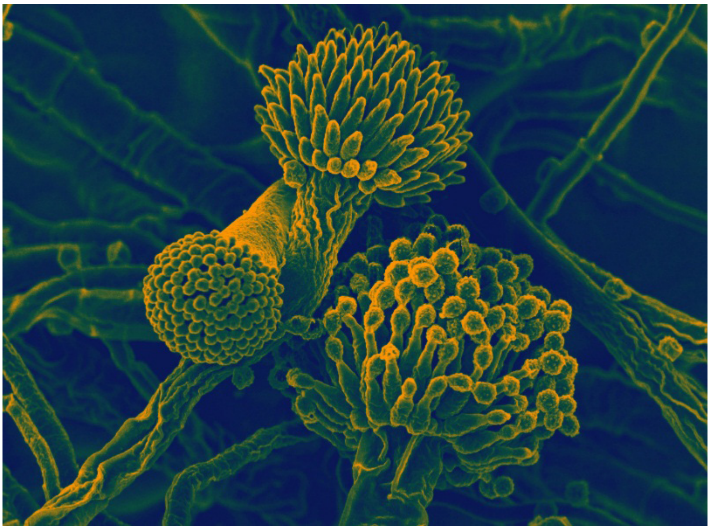

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Structure of Bacteria

The structure of bacteria is simpler than that of other organisms. Bacteria are microorganisms that lack membrane-bound organelles and a well-defined nucleus. They usually adopts one of four shapes, depending on the type of bacteria and have numerous appendages with distinct functions.

Bacteria are unicellular, which means they consist of a single cell only. Everything that the bacteria needs to survive and reproduce, except for food, is contained within this single cell. As such, the bacterial structure has developed in ways that allow the cell to go about its business efficiently.

Bacteria are prokaryotes, meaning they have no organized nucleus. The nucleus is the organelle that houses the deoxyribonucleic acid (DNA) and the genetic materials necessary for replication in more evolved cells. Despite the absence of a nucleus, bacteria still possess DNA, but it floats freely within the cytoplasm.

Peptidoglycan, also known as murein, is a polymer consisting of sugars and amino acids that forms a mesh-like layer outside the plasma membrane of most bacteria, forming the cell wall.

A cell wall surrounds the cytoplasmic region, or interior, of the bacteria and acts as the support structure for the entire organism. The underlying layer of the cell wall is formed from **peptidoglycan,** a molecule that is comprised of amino acids and sugars. This is a feature unique to bacteria and is what makes the cell wall resilient to pressures both internal and external.

The peptidoglycan gives the bacterium its shape. The shape of a given cell is usually distinct to that species and is an important detail in identifying unknown species. Two common shapes for bacteria are the bacillus, which is rod-shaped, and the spirillum, which is spiral-shaped. Other common forms are the coccus, or spherical, and the filamentous, in which the individual cells grow in length but do not separate.

Lining the cell wall is a membrane that selectively allows molecules into and out of the cell. Cytoplasm fills the area inside the cell membrane. Within the cytoplasm are the chromosomes, the DNA of the cell. The cytoplasm also contains the ribosomes that create the proteins used by the cell.

On the exterior of the cell wall, various extensions exist to help bacteria perform actions such as moving and attaching to other objects. Flagella, filaments of protein, are located on the exterior of the organism and move the bacteria through its environment. Bacteria may have one or many flagella depending on the species.

Another surface feature of bacteria are fimbrae. These filaments of protein are used by the bacteria to attach itself to other structures. They exist over the entire surface of the bacterium and are much smaller than flagella. A third component is a thread-like protein extension called a pilus, also found on the exterior of the cell. Pili aid in bacterial conjugation and attachment.

The typical structure of a bacterial cell is shown in figure 1. It consists of the following general structures, found in all bacterial cells; and special structures, found in specific types of bacterial cells:

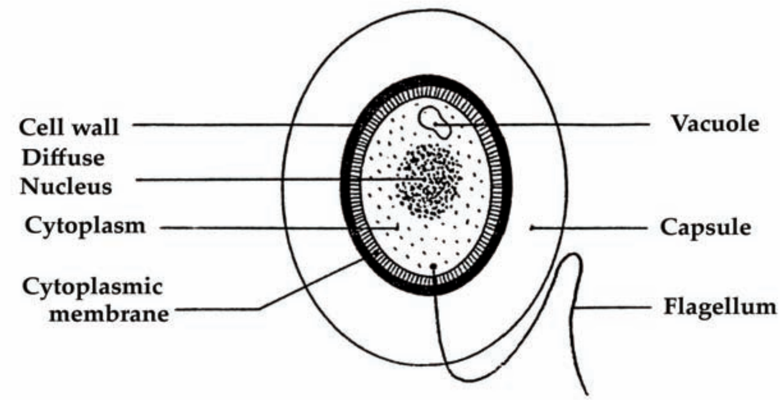


Figure 1: Bacterial cell structure.

General Structures

There are some part of general structures, such as:

Cell wall

A cell wall is a layer of lipid and protein that encloses the protoplasm of the cell giving rigidity to the bacterial shape.

- Cell walls of bacteria are made up of glycoprotein murein.
- The main function of cell wall is it helps in providing support, mechanical strength and rigidity to cell.
- It protects cell from bursting in a hypotonic medium.

Cytoplasmic membrane

A cytoplasmic membrane is a thin semi permeable membrane located directly beneath the cell wall, governing osmotic activity.

- It is also known as cytoplasmic membrane (or) cell membrane.
- It is composed of phospholipids, proteins and carbohydrates, forming a fluid-mosaic.
- It helps in transportation of substances including removal of wastes from the body.
- It helps in providing a mechanical barrier to the cell.
- Plasma membrane acts as a semi permeable membrane, which allows only selected material to move inside and outside of the cell.

Cytoplasm

The cytoplasm is the protoplasmic or vital colloidal material of a cell; the site of metabolic activities.

- Helps in cellular growth, metabolism and replication.
- Cytoplasm is the store houses of all the chemicals and components that are used to sustain the life of a bacterium.

Genetic material

Metabolism is the set of life-sustaining chemical reactions in organisms.

Genetic material is a circular, single-stranded DNA diffused throughout the cytoplasm. It carries the genetic code of heritable traits and is responsible for directing **metabolism** and replication of the cell. The bacterial cell does not have a nuclear membrane or a well-defined nucleus.

Special Structures

There are some part of special structures, such as:

Capsule

A capsule is relatively thick layer of mucoid polysaccharides (slime layer) around a bacterium. A capsule serves as a defense mechanism.

Bacteria secrete extracellular polymers outside of their cell walls. These polymers are usually composed of polysaccharides and sometimes protein. Capsules are relatively impermeable structures that cannot be stained with dyes such as India ink. They are structures that help protect bacteria from phagocytosis and desiccation India ink (or Indian ink in British English), also called Chinese ink since it may have been first developed in either China or India, is a simple black ink once widely used for writing and printing, and now more commonly used for drawing, especially when inking comics and comic strips. Indian ink generally will readily clog the fountain pens if not used for long time, it's then necessary to use water to unclog it. An exception to this is Pelikan Fount India, which does not contain shellac, the substance which can cause clogging.

A capsule is a well-organized but easily washed off layer outside bacteria.eg;A Slimelayer which is a zone of diffuse material tht is unorganized and can be easily removed.The capsules and slime layers forms a network of polysaccharides called as glycocalyx extending from the surface of the bacteria to the other cells.But capsules can be construted of non-glucose substances.eg;Bacillus anthracis has a capsule of poly-D-glutamic acid.Mainly archaebacteria contain glycoprotien made capsules called S layers which promotes the cell adhesion to surfaces.

A capsule made up of polysaccharides (complex carbohydrates). Capsules play a number of roles, but the most important are to keep the bacterium from drying out and to protect it from phagocytosis (engulfing) by larger microorganisms. The capsule is a major virulence factor in the major disease-causing bacteria, such as *Escherichia coli* and *Streptococcus pneumoniae*. Nonencapsulated mutants of these organisms are avirulent, i.e. they do not cause disease.

Most procaryotes contain some sort of a polysaccharide layer outside of the cell wall polymer. In a general sense, this layer is called a capsule. A true capsule is a discrete detectable layer of polysaccharides deposited outside the cell wall. A less discrete structure or matrix which embeds the cells is a called a slime layer or a biofilm. A type of capsule found in bacteria called a glycocalyx is a thin layer of tangled polysaccharide fibers which occurs on surface of cells growing in nature (as opposed to the laboratory). Some microbiologists refer to all capsules as glycocalyx and do not differentiate microcapsules.

Capsules are generally composed of polysaccharide; rarely do they contain amino sugars or peptides (Table 1).

Table 1: Chemical composition of some bacterial capsules

Bacterium	**Capsule composition**	**Structural subunits**
Gram-positive Bacteria		
Bacillus anthracis	polypeptide (polyglutamic acid)	D-glutamic acid
Bacillus megaterium	polypeptide and polysaccharide	D-glutamic acid, amino sugars, sugars
Streptococcus mutans	polysaccharide	(dextran) glucose
Streptococcus pneumoniae	polysaccharides	sugars, amino sugars, uronic acids

Streptococcus pyogenes	polysaccharide (hyaluronic acid)	N-acetyl-glucosamine and glucuronic acid
Gram-negative Bacteria		
Acetobacter xylinum	polysaccharide	(cellulose) glucose
Escherichia coli	polysaccharide (colonic acid)	glucose, galactose, fucose glucuronic acid
Pseudomonas aeruginosa	polysaccharide	mannuronic acid
Azotobacter vinelandii	polysaccharide	glucuronic acid
Agrobacterium tumefaciens	polysaccharide	(glucan) glucose

Capsules have several functions and often have multiple functions in a particular organism. Like fimbriae, capsules, slime layers, and glycocalyx often mediate adherence of cells to surfaces. Capsules also protect bacterial cells from engulfment by predatory protozoa or white blood cells (phagocytes), or from attack by antimicrobial agents of plant or animal origin. Capsules in certain soil bacteria protect cells from perennial effects of drying or desiccation. Capsular materials (e.g. dextrans) may be overproduced when bacteria are fed sugars to becomereserves of carbohydrate for subsequent metabolism.

Some bacteria produce slime materials to adhere and float themselves as colonial masses in their environments. Other bacteria produce slime materials to attach themselves to a surface or substrate. Bacteria may attach to surface, produce slime, divide and produce microcolonies within the slime layer, and construct a biofilm, which becomes an enriched and protected environment for themselves and other bacteria.

Bacterial growth is the asexual reproduction, or cell division, of a bacterium into two daughter cells, in a process called binary fission.

A classic example of biofilm construction in nature is the formation of dental plaque mediated by the oral bacterium, *Streptococcus mutans*. The bacteria adhere specifically to the pellicle of the tooth by means of a protein on the cell surface. The bacteria grow and synthesize a dextran capsule which binds them to the enamel and forms a biofilm some 300-500 cells in thickness. The bacteria are able to cleave sucrose (provided by the animal diet) into glucose plus fructose. The fructose is fermented as an energy source for **bacterial growth**. The glucose is polymerized into an extracellular dextran polymer that cements the bacteria to tooth enamel and becomes the matrix of dental plaque. The dextran slime can be depolymerized to glucose for use as a carbon source, resulting in production of lactic acid within the biofilm (plaque) that decalcifies the enamel and leads to dental caries or bacterial infection of the tooth.

Bacteria that are actively reproducing by fission do not produce spores (sporulate).

Another important characteristic of capsules may be their ability to block some step in the phagocytic process and thereby prevent bacterial cells from being engulfed or destroyed by phagocytes. For example, the primary determinant of virulence of the pathogen *Streptococcus pneumoniae* is its polysaccharide capsule, which prevents ingestion of pneumococci by alveolar macrophages. *Bacillus anthracis* survives phagocytic engulfment because the lysosomal enzymes of the phagocyte cannot initiate an attack on the poly-D-glutamate capsule of the bacterium. Bacteria such as *Pseudomonas aeruginosa*, that construct a biofilm made of extracellular slime when colonizing tissues, are also resistant to phagocytes, which cannot penetrate the biofilm.

Flagellum

Flagella (singular: flagellum) are long, thin, whip-like appendages attached to a bacterial cell that allow for bacterial movement. Bacterial cells are typically between 0.1 micrometers and 50 micrometers in diameter, but average around 2 micrometers. Flagella can be several times longer than the cell, averaging 10 micrometers in length. Some bacteria have a single flagella protruding from one end of the cell, while others have many flagella surrounding the entire cell.

Flagellum, plural flagella, hairlike structure that acts primarily as an organelle of locomotion in the cells of many living organisms. Flagella, characteristic of the protozoan group Mastigophora, also occur on the gametes of algae, fungi, mosses, slime molds, and animals. Flagellar motion causes water currents necessary for respiration and circulation in sponges and coelenterates. Most motile bacteria move by means of flagella.

The structures and pattern of movement of prokaryotic and eukaryotic flagella are different. Eukaryotes have one to many flagella, which move in a characteristic whiplike manner. The flagella closely resemble the cilium in structure. The core is a bundle of nine pairs of microtubules surrounding two central pairs of microtubules (the so-called nine-plus-two arrangement); each microtubule is composed of the protein tubulin. The coordinated sliding of these microtubules confers movement. The base of the flagellum is anchored to the cell by a basal body.

Bacterial flagella are helically shaped structures containing the protein flagellin. The base of the flagellum (the hook) near the cell surface is attached to the basal body enclosed in the cell envelope. The flagellum rotates in a clockwise or counterclockwise direction, in a motion similar to that of a propeller.

A flagellum is a protoplasmic strand of elastic protein originating in the cytoplasmic membrane, and extending from the body of the cell. A flagellum serves as an organ of locomotion. The arrangement of flagella (plural) is peculiar to the species.

Bacteria can be motile or non-motile, which if they are motile means they can freely move in a desired direction, beneficial to that bacterium. To be motile bacteria can have one, two or many flagellum, depending on the type.

A- Monotrichous

B - Lophotrichous

C - Amphitrichous

D - Peritrichous

Flagella are rigid structres about 20nm across and 15-20um long.Flegella has three parts.1- the longest portion is the filament which is a hollow rigid cylinder made of the protien called Flagellin.2-the basal body which is embedded in the cell.3- hook a short,curved segment links the filament to its basal body and acts as a flexible coupling.

Flagella are made from a single type of protein and are attached to the outside of the cell. They can be attached at the end, in the middle, or all over the cell as shown in the diagram above. The flagella is helitical and acts as a propeller. It can turn clockwise or anti clockwise to propel the bacteria in different directions. A protein rings in the cell's membrane that act as bearing to aid rotation. There are motile gram positive and gram negative bacteria.

- This is a rigid rotating tail.
- It helps the cell to move in clockwise and anticlockwise, forward and also helps the cell to spin.
- The rotation is powered by H+ gradient across the cell membrane.

Spore

A spore is a metabolically resistant body formed by a vegetative bacterium, which is in a dormant state, to withstand an unfavorable environment. Only bacilli form spores. The position and size of a spore within a bacillus is peculiar to the species. Bacteria that are actively reproducing by fission do not produce spores (sporulate).

Inclusion bodies

Inclusion bodies are vacuoles of reserve or waste materials contained within the cytoplasm.

Ribosomes

Ribosomes is a site of protein production. Ribosomes consist of a small and large subunits made up of a complex proteins and RNAs. These are the sites of protein synthesis in the cell. The Ribosome's in prokaryotic cells although similar in shape and function to those of eukaryotic cells are in different in the nature of the proteins and RNAs that make up their structure. This have provided to be very useful to the human population as antibiotics that acts by inhibiting protein synthesis in bacteria are but not effective against eukaryotic protein synthesis, thus allowing selective toxicity. Archaebacterial ribosome's are the same as those of the eubacteria but, in some features they are similar to eukaryotic ribosome's in that they are resistant to antibiotics like streptomycin and chloramphenicol, and sensitive to the action of diphtheria toxin.

Ribosomes present in the matrix synthesize protiens to remain within the cell,whereas the ribosomes present in the plasma membrane make protiens for transport ot outside.Procaryotic ribosomes are smaller than eucaryotic ribosomes and are called as 70S ribosomes.They have dimensions of 14-15nm by 20nm and having a molecular weight of 2.7million. 70S ribosomes constitute of two subunits 30S and 50S repectively,but the total sum of the two subunits give 80S ribosome which is present in eucaryotic cells.The S stands for Svedberg units ,which is the sedimentaion coefficient.

DNA is then transferred from the donor to the recipient. Plasmid-mediated conjugation occurs in Bacillus subtilis, Streptococcus lactis, and Enterococcus faecalis but is not found as commonly in the Gram-positive bacteria as compared to the Gram-negative bacteria.

Bacterial Conjugation

Bacterial conjugation is often regarded as the bacterial equivalent of sexual reproduction or mating since it involves the exchange of genetic material. Though it should be noted that this is not sexual reproduction, since no exchange of gamete occurs. During conjugation the donor cell provides a conjugative or mobilizable genetic element that is most often a plasmid or transposon. Most conjugative plasmids have systems ensuring that the recipient cell does not already contain a similar element.

The genetic information transferred is often beneficial to the recipient. Benefits may include antibiotic resistance, xenobiotic tolerance or the ability to use new metabolites. Such beneficial plasmids may be considered bacterial endosymbionts. Other elements, however, may be viewed as bacterial parasites and conjugation as a mechanism evolved by them to allow for their spread.

Each Gram negative F^+ bacterium has 1 to 3 sex pili that bind to a specific outer membrane protein on recipient bacteria to initiate mating. The sex pilus then retracts, bringing the two bacteria in contact and the two cells become bound together at a point of direct envelope-to-envelope contact. In Gram-positive bacteria sticky surface molecules are produced which bring the two bacteria into contact. Gram-positive donor bacteria produce adhesins that cause them to aggregate with recipient cells, but sex pili are not involved.

Resistance plasmid conjugation

Some Gram-negative bacteria harbor plasmids that contain antibiotic resistance genes, such plasmids are called R factors. The R factor has two components, one that codes for self-transfer (like F factor) called RTF (resistance transfer factor) and the other R determinant that contains genes coding for antibiotic resistance. R plasmids may confer resistance to as many as five different antibiotics at once upon the cell and by conjugation; they can be rapidly disseminated through the bacterial population. The difference between F factor and R factor is that the latter has additional genes coding for drug resistance. During conjugation there is transfer of resistance plasmid

(Rplasmid) from a donor bacterium to a recipient. One plasmid strand enters the recipient bacterium while one strand remains in the donor. Each strand then makes a complimentary copy. R-plasmid has genes coding for multiple antibiotic resistance as well as sex pilus formation. The recipient becomes multiple antibiotic resistant and male, and is now able to transfer R-plasmids to other bacteria. When the recipient cells acquire entire R factor, it too expresses antibiotic resistance. Sometimes RTF may disassociate from the R determinant and the two components may exist as separate entities. In such cases though the host cell remains resistant to antibiotics, it cannot transfer this resistance to other cells. Sometimes RTF can have other genes (such as those coding for hemolysin, enterotoxin) apart from R determinants attached to it.

CORE CONCEPT: CLASSIFICATION OF BACTERIA

Most of the bacteria belong to three main shapes: rod (rod shaped bacteria are called bacilli), sphere (sphere shaped bacteria are called cocci) and spiral (spiral shaped bacteria are called spirilla). Some bacteria belong to different shapes, which are more complex than the above mentioned shapes.

Bacteria classification is the process of distinguishing types of bacteria from one another and grouping them according to shared characteristics. Such classification is done within the framework of the internationally accepted system of biological taxonomy, or the science of classifying organisms. A number of things are taken into account during bacteria classification, particularly RNA sequences, but including shape, biochemistry, and characteristics of the external membrane, among others.

While there are differing views on the way organisms should be classified, the current prevailing system divides all life into three domains. Bacteria form one of these domains. They show an extreme degree of diversity and, by number, make up a large majority of the known species and, by mass, outnumber all multicellular organisms on the earth. Bacteria classification can be difficult due to a number of factors, particularly the absence of complex structures found in more advanced organisms, as well as the tendency of bacteria to transfer segments of DNA.

Bacteriologists use a number of techniques in bacteria classification. Shape is the simplest way to tell bacteria apart, and these organisms can exhibit a variety of shapes including rods, spirals, and spheres, among others. Shape is not necessarily an exclusive characteristic, however, and bacteria that are grouped in widely differing classifications can have similar shapes.

Other morphological features, such as size, can vary significantly from species to species, and typical groupings can also aid in classification. Many species tend to form groups of individual cells, which can vary greatly in number. The presence of external structures, such as tiny tentacle-like constructions called flagella, can also help distinguish bacteria species.

Another key tool in bacteria classification is a test known as the gram stain test, named after a 19th century microbiologist. This test quantifies bacteria according to the thickness of their exterior membrane. Gram negative bacteria have a very thin membrane and gram positive bacteria have a thicker membrane.

Factors such as metabolism and other biochemical distinctions are another tool for bacteria classification. Bacteria metabolize a wide variety of different compounds, and the particular compound or compounds used or converted by a specific bacteria can aid in its identification and classification. Analysis of other biochemical data can also help this process.

Advancements in molecular analysis techniques have made it possible for bacteriologists to differentiate bacteria according to differences in RNA, as well as specific gene sequences. Further analysis of the total amount of individual RNA and DNA proteins and the ratios in which they are present provide another means for classification. Using some or all of these techniques and observable characteristics, bacteriologists are able to classify bacteria according to species and to group similar species together.

This classification is done on the basis of shape, cell wall, flagella, nutrition and morphology.

Classification of bacteria on the basis of shape

The three basic bacterial shapes are coccus (spherical), bacillus (rod-shaped), and spiral (twisted), however pleomorphic bacteria can assume several shapes.

Coccus (spherical)

The word "coccus" has a couple of different meanings in biology, but it most commonly refers to a type of bacteria that is spherical or slightly oval in shape. In this application, the plural term used to describe entire bacterial chains or infections is *cocci*, whereas a single coccus bacterium is called a *monococcus*. There are many different strains and species in this broad family. Some benefit life, either by helping keep cell balance or warding off other infections, while others can be quite harmful. In most cases the shape of the bacteria is useful for classification and evolutionary purposes but does not say much about how it will perform or what it does. The bacteria's individual coding and composition more commonly dictate these things.

Physical Characteristics

Cocci are some of the smallest bacteria in existence. Their diameters typically average 0.5 to 1.0 micrometers, and they are usually shaped like a flat, elongated oval. This shape also gives them the largest surface area in relation to their size, which enables them to more efficiently take up nutrients from the environment. In many cases this means that the bacteria are hardier and more likely to survive in harsh environments, but sometimes it can also be a downside. Greater surface area can mean that the bacteria is more susceptible to absorbing harmful substances.

Reproduction

When bacteria are reproducing, monococci are formed only when a new cell splits completely from its parent cell. Some monococci are usually present in *any* species of bacteria with this shape, but it is quite common for them to form groups of two or more cells. It's often the case in these sorts of splits that there is not a complete separation between the parent bacterium and the newly formed one. This can lead to a couple of different formations, though no matter the end result it's important to note that each individual cell retains its original spherical shape. The simplest formation is known as diplococcus, in which two cells remain attached to each other. *Neisseria* is a genus of bacteria that often forms these pairs of cells. Many species of Neisseria are found in healthy human throats, but the bacteria that cause meningitis also belong to this genus.

Staphylococcus is a genus of Gram-positive bacteria. Under the microscope, they appear round (cocci), and form in grape-like clusters.

Chains and Large Groupings

Cocci bacteria can also form larger groups, including groups of four, eight, ten, or more. Groups of four are known as tetrads, while groups of eight are called sarcinae. Many bacteria forming a clump is typically called a staphylococcus, which is also the name of a specific genus of bacteria that most often form in this arrangement. The formal genus *Staphylococcus* includes many species that are commonly found on human skin, as well as some bacteria that reside in the soil. In most cases it also contains **Staphylococcus** aureus, which can cause staph infections; these strains are also often resistant to antibiotics.

Common Species

Multiple chains of the bacteria are generally called streptococci, and many bacteria that take on this shape are found in the genus Streptococcus. A good percentage of these strains are normal throat and nose bacteria, but the species that cause pneumonia, strep throat, and scarlet fever are also in this group. Enterococcus and Lactococcus species produce lactic acid, a substance that is necessary to create fermented dairy products such as yogurt. Lactococci in particular are often used in the dairy industry both as a way to help produce dairy products and to assist consumers in their digestion.

Alternative Definitions

The word "coccus" also has a couple of other meanings in the scientific realm that are unrelated to bacteria. A group of tropical insects that are small, flat, and round carry the name, for instance; these feed primarily on the bark of citrus trees and coffee plants, which makes them a well-known pest to growers and agriculturalists in many parts of the world. In botany the term can also be applied to a single seed that splits apart from a multi-lobed or multi-seeded mass, like a berry or other fruit. It's usually the case that these seeds also have the familiar flattened oval shape.

Types of Coccus (spherical)

There are some types of coccus (spherical), such as:

- *Monococcus*: – they are also called micrococcus and represented by single, discrete round cell. Example: Micrococcus flavus.
- *Diplococcus*: – the cell of the Diplococcus divides ones in a particular plane and after division, the cells remain attached to each other. Example: – Diplococcus pneumonia.
- *Streptococcus*: – here the cells divide repeatedly in one plane to form chain of cells. Example: – Streptococcus pyogenes.
- *Tetracoccus*: – this consists of four round cells, which defied in two planes at a right angles to one another. Example: – Gaffkya tetragena.
- *Staphylococcus:* – here the cells divided into three planes forming a structured like bunches of grapes giving and irregular configuration. Example: – Staphylococcus aureus.
- *Sarcina*: -in this case this cells divide in three planes but they form a cube like configuration consisting of eight or sixteen cells but they have a regular shape. Example: –Sarcina lutea.

Bacillus (rod-shaped)

Bacillus (genus Bacillus), any of a group of rod-shaped, gram-positive, aerobic or (under some conditions) anaerobic bacteria widely found in soil and water. The term bacillus has been applied in a general sense to all cylindrical or rodlike bacteria. The largest known Bacillus species, B. megaterium, is about 1.5 **µm** (micrometres; 1 **µm** = 10^{-6} m) across by 4 **µm** long. Bacillus frequently occur in chains.

In 1877 German botanist Ferdinand Cohn provided an authoritative description of two different forms of hay bacillus (now known as Bacillus subtilis): one that could be killed upon exposure to heat and one that was resistant to heat. He called the heat-resistant forms “spores” (endospores) and discovered that these dormant forms could be converted to a vegetative, or actively growing, state. Today it is known that all Bacillus species can form dormant spores under adverse environmental conditions. These endospores may remain viable for long periods of time. Endospores are resistant to heat, chemicals, and sunlight and are widely distributed in nature, primarily in soil, from which they invade dust particles. Some types of Bacillus bacteria are harmful to humans, plants, or other organisms. For example, B. cereus sometimes causes spoilage in canned foods and food poisoning of short duration. B. subtilis is a common contaminant of laboratory cultures (it plagued Louis Pasteur in many of his experiments) and is often found on human skin. Most strains of Bacillus are not pathogenic for humans but may, as soil organisms, infect humans incidentally. A notable exception is B. anthracis, which causes anthrax in humans and domestic animals. B. thuringiensis produces a toxin (Bt toxin) that causes disease in insects.

Medically useful antibiotics are produced by B. subtilis (bacitracin) and B. polymyxa (polymyxin B). In addition, strains of B. amyloliquefaciens bacteria, which occur in association with certain plants, are known to synthesize several different antibiotic substances, including bacillaene, macrolactin, and difficidin. These substances serve to protect the host plant from infection by fungi or other bacteria and are being studied for their usefulness as biological pest-control agents. A gene encoding an enzyme known as barnase in B. amyloliquefaciens is of interest in the development of genetically modified (GM) plants. Barnase acts to kill plant cells that have become infected by fungal pathogens; this activity limits the spread of disease. The gene controlling production of the Bt toxin in B. thuringiensis has been used in the development of GM crops such as Bt cotton (see genetically modified organism).

Spiral (twisted)

The third bacterial shape is spiral. These bacterial cells are twisted in helices and resemble little corkscrews. Not to gross you out, but if you scrape some gunk off your teeth and look at it under the microscope, you will find many spiral bacteria!

Spiral-shaped bacteria are called spiral bacteria. These types of microorganisms come in three various forms, which include vibrio, spirillum and spirochete.

Bacteria are prokaryotic unicellular microscopic organisms, although some species have been known to grow in sizes that can be perceived by the naked eye. Bacteria are generally classified according to their shapes. The three primary bacterial shapes are spherical, rod-shaped and spiral, or twisted. Spiral bacteria contain one or several twists or waves. Vibrios appear like curved rods, while both spirilla and spirochetes are helical in shape. Spirilla are typically rigid in structure, whereas spirochetes have pliant bodies. Other known bacterial shapes include sheathed, stalked, star-shaped, square, lobed, rectangular and spindle-shaped.

Spiral bacteria have one or more twists.

Vibrios

Vibrio is a bacterial genus which is found in temperate to warm aquatic environments all over the world. The most famous species in this genus is probably V. cholerae, the bacterium which is responsible for causing cholera. Cholera is a disease which causes substantial public health problems in warm areas of the world, and the study of cholera played a critical role in the development of the germ theory of disease, which states that microorganisms are responsible for many diseases.

There are several notable species within the *Vibrio* genus. A few species demonstrate the trait of bioluminescence, and many others can cause a variety of gastrointestinal symptoms similar to those experienced by cholera patients. Many species are also zoonotic, which allows them to hop from species to species to ensure that they are widely distributed. Shellfish, for example, can house *Vibrio* bacteria which may make people sick.

Vibrio vulnificus, another notable species, is endemic to several warm climates. In addition to causing intestinal infections, this bacterium can also infect the skin, and it will take advantage of open wounds to spread into the bloodstream, causing septicemia. People with compromised immune systems are at especially high risk of contracting a dangerous infection from this particular *Vibrio* species.

Many of these species are foodborne, leading some doctors to classify *Vibrio* infection as a foodborne illness. However, because they live in aquatic environments, they can also spread through contaminated water supplies. This can become an especially large issue when sewage spills occur, as untreated sewage can contain bacteria which will enter the water supply and make people sick.

These bacteria often need warm climates to survive, but many can develop dormancy, which allows them to overwinter and appear again in the spring and summer in areas with cold winter. The ability to overwinter is very useful from the point of view of the bacteria, because it ensures that the organisms will survive in a variety of climates. For microbiologists and public health officials, this trait is extremely irritating, as it makes it extremely difficult to eradicate *Vibrio* bacteria.

Spirillum

Spirillum is a type of bacterium that moves in a characteristic spiral, or corkscrew-like, fashion. It typically lives as individual cells, instead of being associated in clusters or chains. There is also a genus of gram-negative bacteria known as Spirillum, most members of which live in water with a large amount of organic matter, such as stagnant pools. A notable exception is Spirillum minus (S. minus), which is one of the two types of bacteria that can cause rat-bite fever.

Most types of bacteria possess flagella, or whip-like appendages, that help them to move through solutions. Bacteria from this genus have unusual types of flagella. It was originally thought that these bacteria had an individual flagellum on each end of their cells, but it is now known that they have a cluster on each end, composed of about 75 individual appendages. It is the motion of these that causes the organisms to move in their characteristic corkscrew fashion.

Spirillum volutans has been studied a great deal, since it is one of the largest species of bacteria. It was observed by early microscopists and identified in 1832. The name of this organism comes from from carbohydrate storage granules inside the organism, known as volutin. It can be difficult to cultivate these bacteria, and *S. volutans* could not be grown in pure culture until recently. It turns out that the organism requires small amounts of oxygen but is poisoned by ambient concentrations. When there were contaminating organisms in the cultures, they used up enough of the oxygen that *S. volutans* could grow, but when it was in pure culture, however, it was unable to grow with the greater concentrations of oxygen. Once this requirement was understood, researchers were able to grow this bacterium in pure culture.

The Spirillaceae family is closely related to the family that includes Spirochaetes. This family of bacteria includes a number of human pathogens, including the bacteria that cause Lyme disease and syphilis. These organisms are also spirilla, and move in a corkscrew manner. They differ from the Spirillaceae in having specialized structures instead of flagella, and having an outer sheath. One species of the genus, *S. minus*, is a human pathogen. It is one of the causal agents of rat-bite fever, although this form of the disease is found less frequently than another bacterial form. It is primarily a problem in Asia, being known as *sodoku* in Japan. This disease is typically contracted from the bites of rats, mice, and other rodents, or through exposure to the urine or secretions of an animal that has been infected.

Spirochete

Spirochete (order Spirochaetales), also spelled spirochaete, any of a group of spiral-shaped bacteria, some of which are serious pathogens for humans, causing diseases such as syphilis, yaws, Lyme disease, and relapsing fever. Examples of genera of spirochetes include *Spirochaeta*, *Treponema*, *Borrelia*, and *Leptospira*.

Spirochetes are gram-negative, motile, spiral bacteria, from 3 to 500 m (1 m = 0.001 mm) long. Spirochetes are unique in that they have endocellular flagella (axial fibrils, or axial filaments), which number between 2 and more than 100 per organism, depending upon the species. Each axial fibril attaches at an opposite end and winds around the cell body, which is enclosed by an envelope. Spirochetes are characteristically found in a liquid environment (e.g., mud and water, blood and lymph). *Treponema* includes the agents of syphilis (*T. pallidum pallidum*) and yaws (*T. pallidum pertenue*). *Borrelia* includes several species transmitted by lice and ticks and causing relapsing fever (*B. recurrentis* and others) and Lyme disease (*B. burgdorferi*) in humans. *Spirochaeta* are free-living nonpathogenic inhabitants of mud and water, typically thriving in anaerobic (oxygen-deprived) environments. Leptospirosis, caused by *Leptospira*, is principally a disease of domestic and wild mammals and is a secondary infection of humans.

Classification of bacteria on the basis of nutrition

On the basis of nutrition bacteria are classified as following:

Autotropic bacteria

An autotroph is an organism able to make its own food. Autotrophic organisms take inorganic substances into their bodies and transform them into organic nourishment. Autotrophs are essential to all life because they are the primary producers at the base of all food chains. There are two categories of autotrophs, distinguished by the energy each uses to synthesize food. Photoautotrophs use light energy; chemoautotrophs use chemical energy.

These bacteria are nonpathogenic, free living, self-sustaining in nature, which prepare their own food by utilisation of solar energy and inorganic components like carbon dioxide, nitrogen etc. They are of two types:

Photoautotrophs

An autotroph is an organism able to make its own food. Photoautotrophs are organisms that carry out photosynthesis. Using energy from sunlight, carbon dioxide and water are converted into organic materials to be used in cellular functions such as biosynthesis and respiration. In an ecological context, they provide nutrition for all other forms of life (besides other autotrophs such as chemotrophs). In terrestrial environments plants are the predominant variety, while aquatic environments include a range of phototrophic organisms such as algae, protists, and bacteria. In photosynthetic bacteria and cyanobacteria that build up carbon dioxide and water into organic cell materials using energy from sunlight, starch is produced as final product. This process is an essential storage form of carbon, which can be used when light conditions are too poor to satisfy the immediate needs of the organism.

These bacteria contain bacterio-chlorophyll and bacterioviridin and can prepare their own food by fixing carbon dioxide the nature by the utilisation of solar energy.

Photoautotrophs are autotrophs that obtain their energy from sunlight. Much like plants, these organisms can turn light energy into chemical energy that will fuel the bacteria's processes. These organisms contain a green pigment called cyanobacteria. Much like the chlorophyll in plants, this substance helps to convert carbon dioxide and sunlight into oxygen and sugar.

They use solar energy for the synthesis of their own food. These bacteria are anaerobic, which could be purple or green. The purple bacteria possess pigment bacteriochlorophyll located in the membranes of thylakoids while green bacteria possess bacteriophaeophytin (chlorobium chlorophyll) located inside small sacs called chlorosomes.

Photoautotrophs carry out anoxygenic photosynthesis in which water is not used as reducing power. Instead sulphur compounds like hydrogen sulphide (H_2S), hydrogen gas (H_2), thiosulphates ($Na_2S_20_3$) or some organic compounds are used to obtain reducing power. Most photoautotrophs live near the bottoms of ponds and lakes where reduced sulphur or other compounds are in plenty and oxygen content is very low.

Chemoautotrophs

Chemoautotrophs are bacteria that use chemical energy. They are able to take in carbon dioxide and water and convert those substances into carbohydrates and sugars. Carbohydrates and sugar are the main energy sources for bacteria. These bacteria are incredibly useful in the atmosphere. Since carbon dioxide is a waste product from other sources, these microorganisms serve as a sort of recycling plants, turning carbon dioxide into a useful source of energy.

Chemical energy is the potential of a chemical substance to undergo a transformation through a chemical reaction to transform other chemical substances.

One interesting fact about chemoautotrophs is that they thrive at higher temperatures. Some can be found in hot springs, which have temperatures much higher than the average bacteria. These bacteria are unique in that they can survive in an environment were almost all other bacteria would die. Halophiles are also a type of chemoautotroph. Halophile is an organism that can live in a salt-abundant environment. Scientists were surprised to discover halophiles in the Dead Sea since the high salt content was sufficient to kill other organisms, but these are able to survive the caustic environment.

Most organisms on the earth need photosynthesis to survive and to produce the energy they need to function. Chemoautotrophic organisms do not need these processes because they are able to oxidize the electron molecules that they take in from their own internal environments in order to survive. They do not live on anything other than molecules and are able to sustain themselves through the use of these electrons.

Chemoautotrophic bacteria are at the bottom of the food chain. They do not eat anything or take anything in other than at a molecular level and are essential to the survival of small sea creatures. Chemoautotrophic bacteria are responsible for methane, hydrogen gas and carbon monoxide that are found in the ocean. These gasses are a result of what happens when the chemoautotrophs convert the electrons to energy. Some chemoautotrophs, like the iron bacteria, are responsible for everyday things like the rust-colored stains that are often found on the inside of a toilet tank.

They do not have photosynthetic pigment and hence utilize **chemical energy** to reduce CO_2 to organic food. The chemical energy is obtained from the oxidation of certain chemicals such as ammonia, nitrites, methane, carbon monoxide, molecular hydrogen, iron salts, sulphur and sulphur compounds (e.g., nitrifying bacteria Nitrosomonas, Nitrobacter, denitrifying bacteria Bacillus denitrificans, sulphur bacteria Thiobacillus thiooxidans, iron bacteria, hydrogen bacteria).

Heterotrophic bacteria

This type of bacteria cannot fix inorganic Carbone but rather depend on external organic Carbone for their nourishment. They also can be classified on the basis of presence and absence of flight and on the basis of the media on which the bacteria are growing.

Heterotrophic cells must ingest biomass to obtain their energy and nutrition. In direct contrast, autotrophs are capable of assimilating diffuse, inorganic energy and materials, and using these to synthesize biochemicals. Green plants, for example, use sunlight and simple inorganic molecules to photosynthesize organic matter. All heterotrophs have an absolute dependence on the biological products of autotrophs for their sustenance—they have no other source of nourishment.

All animals are heterotrophs, as are most microorganisms (the major exceptions being microscopic algae and blue-green bacteria). Heterotrophs can be classified according to the sorts of biomass that they eat. Animals that eat living plants are known as herbivores, while those that eat other animals are known as carnivores. Many animals eat both plants and animals, and these are known as omnivores. Animal parasites are a special type of carnivore that are usually much smaller than their prey, and do not usually kill the animals that they feed upon.

Heterotrophic microorganisms mostly feed upon dead plants and animals, and are known as decomposers. Some animals also specialize on feeding on dead organic matter, and are known as scavengers or detritivores. Even a few vascular plants are heterotrophic, parasitizing the roots of other plants and thereby obtaining their own nourishment. These plants, which often lack chlorophyll, are known as saprophytes.

Heterotrophic bacteria, therefore, are largely responsible for the process of organic matter decomposition. Many pathogenic (disease-causing) bacteria are heterotrophs. However, many species of heterotrophic bacteria are also abundant in the environment and are considered normal flora for human skin. The recycling of minerals in aquatic ecosystems, especially in estuaries, is also made possible by heterotrophic bacteria. Although monitored by health officials, the presence of heterotrophic bacteria in public water supplies is seldom considered a public health threat.

They are most abundant in nature. They do not synthesize their own food but depend on other organisms or on dead organic matter for food. They may be parasites, saprophytes or symbiontes.

Parasites

In saprophytic nutrition, the main classes of matter that are broken down are proteins, fats, and starches. Proteins are digested into amino acids. Fats are broken down into glycerol and fatty acids. Starches are digested into simple sugars. All of the resulting substances are then of a small enough molecular size that they can be transported across the cell membranes.

A parasite is an organism which exploits another organism for the purpose of staying alive. Some parasitic relationships are harmless, while in other cases a parasite can damage or even kill its host. The study of parasitism is an extensive field, because parasites can be found across the biological kingdoms, and many animals host one or more parasites during their lives. A number of organisms also go through a parasitic stage at some point during their lives.

The word is borrowed from the Greek *parasitos*, meaning "one who eats at the table of someone else." In both Greece and Rome, some people made dining at the homes of others a full time occupation, sometimes being called "professional dinner guests." Like biological parasites, these individuals exploited their hosts for food, and they brought nothing to the the table themselves, other than dinner conversation. The existence of parasites has long been known in biology, although the development of high quality microscopes greatly expanded human knowledge of parasites.

In order to be considered a parasite, an organism must depend on another for food, energy, or some other service, such as incubating and raising young. In addition, the parasite must bring nothing to the relationship, creating an arrangement which may be neutral or harmful, but never positive. Numerous organisms team up together to exploit their mutual strengths in a biological process called symbiosis—in this case, the arrangement is mutually beneficial to both creatures and it is not considered parasitism.

Some well known examples of parasites include mites, tapeworms, mistletoe, and fleas. Parasites live in a number of different ways; some, for example, cannot live once their host dies, while others can switch hosts or continue thriving on dead hosts until their nutrients are consumed. There is some dispute over whether bacteria and viruses should be considered parasites; in medical terms, a parasite is usually a eukaryotic organism, meaning that it has a complex cell structure, unlike a bacterium.

Parasites which live inside a host are called endoparasites or internal parasites. Many human diseases are caused by internal parasites, which may infest the intestinal tract, causing symptoms like diarrhea and vomiting. Various treatments are used for parasitic infection, depending on the organism involved. Ectoparasites live outside the host, and they are generally more capable of switching hosts. When a parasite preys on other parasites, it is known as an epiparasite.

They live on other organisms called the host, from which they obtain food. e.g. Streptococcus, Clostridium, Mycobacterium tuberculosis etc. Disease causing parasites are called pathogens.

Saprophytes

Saprophytes are living organisms that feed on dead organic matter. They are considered extremely important in soil biology, as they break down dead and decaying organic matter into simple substances that can be taken up and recycled by plants. The term is usually used to refer to saprophytic fungi or bacteria.

In the strict botanical definition, the term "saprophyte" is something of a misnomer. "Phyte" means a plant, and bacteria and fungi are not classified as plants. Some higher plants such as certain types of orchids and a family of flowering plants called monotropes were once included in this category, because they do not use photosynthesis to make nutrients, so it was believed that they extracted nutrients from dead organic matter. It is now known that these types of plants are actually parasites that obtain their food by growing on living fungi. As such, there are no known true saprophytic plants.

Saprophytes are characterized by their use of a particular kind of digestion mechanism, called extra-cellular digestion. This process involves the secretion of digestive substances into the surrounding environment, where they break down organic matter into simple substances. The resulting nutrients are then absorbed directly through the membranes of the organism's cells, and metabolized.

Suitable conditions are needed for the optimum growth of the common types of saprophytes. There must be sufficient water in the soil or surrounding environment. There must usually be oxygen present, as the majority cannot grow under anaerobic conditions. The acidity of the soil or environment usually needs to be neutral, or slightly acidic, as most of these organisms do not thrive under alkaline conditions.

Some of the most common include certain saprophytic fungus types, such as those in the families of Rhizopus and Mucor. These fungi typically have an extensive network of hyphae, similar to tiny roots, which grow through the soil or through dead wood or other organic matter. They grow in a network called a mycelium. This enables the fungus to thoroughly penetrate the local organic matter, within which the hyphae secrete digestive enzymes and absorb the resulting nutrients.

Symbionts

A symbiont is an organism that is very closely associated with another, usually larger, organism. This larger organism is called a host. A symbiont can live on, in, or sometimes very near its host.

There are two general categories of symbionts: ectosymbionts and endosymbionts. Ectosymbionts live outside of their hosts' cells. For example, the bacteria living on your skin and in your digestive tract are ectosymbionts. These bacteria can be very helpful for fighting off harmful microorganisms.

Endosymbiont, on the other hand, live inside of their hosts' cells. Some plants, for example, have endosymbiotic bacteria living inside their root cells, helping the plants grow.

A symbiotic relationship is the interaction of a symbiont species and a host species. These interacting species can affect each other in positive or negative ways, or sometimes not at all. There are six possible types of symbiotic relationships:

- Parasitism
- Mutualism
- Commensalism
- Amensalism
- Synnecrosis
- Neutralism

(a) *Parasitism and Mutualism:* Parasitism is when the symbionts involved in a symbiotic relationship harm their hosts. These symbionts are called parasites. Just a few examples include lice, fleas, ticks, and tapeworms. These animals are ectosymbionts and benefit by feeding off of their hosts. The negative effects of these parasites are not usually bad enough to cause diseases or death.

Some organisms, called vectors, can cause diseases indirectly by transferring parasites to other organisms. A tick carrying the bacterium that causes Lyme disease could transfer this parasite to its host while drinking blood. In this example, both the tick vector and the bacterium are considered parasites. A mosquito, which is not a parasite, can be a vector if it carries the malaria-causing parasite. This parasite is an endosymbiont because it invades human red blood cells.

When both species in a symbiotic relationship benefit, it's called mutualism. Many species form mutualistic relationships. For example, some ants take care of aphids by protecting both adults and eggs from predators. The aphids provide the ants with sweet-tasting liquid to drink. As another example, many flowers rely on insects, birds, or bats to pollinate them. The pollinators ensure that the flowers can reproduce, while the flowers provide their pollinators with nectar to drink. The ant-protected aphids and flower pollinators are examples of ectosymbionts.

The bacteria that live in plant root cells are mutualistic endosymbionts. These bacteria take unusable nitrogen from the atmosphere and turn it into usable nitrogen. This enables the plant to grow and reproduce. Benefits for the bacteria include protection and nourishment.

(b) *Commensalism and Amensalism:* In commensalism, the symbiont benefits from the host, while the host is unaffected. Some plants, such as orchids and moss, will grow out of large trees to gain easy access to sunlight. They neither help nor harm their host tree. Phoresy is a type of commensal relationship in which the symbiont hitches a ride on its host. For example, some barnacles will attach to the body of a whale. As the whale swims, it brings water and food particles to the barnacles, but is not affected by them.

Sometimes commensalism can be a tricky relationship to confirm. Often on closer examination of the species involved it's found that the host actually does benefit or is in fact harmed. For example, not all small plants growing on trees are commensal. Some, such as mistletoe, will take nutrients and water from their host tree. This makes the relationship parasitic.

When one species kills another species without harm or benefit to itself, it's called amensalism. Humans are one of the most common species found to participate in this relationship. We indiscriminately cut down forests, pollute the environment, and hunt for sport. None of these activities help us to better survive or reproduce, but they do cause harm or death to other species. Another example from the natural world includes red tides. Red tides are huge blooms of algae in the ocean. They are toxic and kill fish and marine mammals. This brings no advantage or detriment to the algae.

Gram-positive bacteria have a thick mesh-like cell wall made of peptidoglycan (50–90% of cell envelope), and as a result are stained purple by crystal violet, whereas gram-negative bacteria have a thinner layer (10% of cell envelope), so do not retain the purple stain and are counter-stained pink by safranin.

(c) *Synnecrosis and Neutralism:* Synnecrosis is a particular case in which the interaction is so mutually detrimental that it results in death, as in the case of some parasitic relationships. It is a rare and necessarily short-lived condition as evolution selects against it. Synnecrosis is a rare type of symbiosis in which the interaction between species is detrimental to both organisms involved.

Neutralism describes the relationship between two species that interact but do not affect each other. It describes interactions where the health of one species has absolutely no effect whatsoever on that of the other. Examples of true neutralism are virtually impossible to prove and most ecologists would agree that this concept does not exist. When dealing with the complex networks of interactions presented by ecosystems, one cannot assert positively that there is absolutely no competition between or benefit to either species. However, the term is often used to describe situations where interactions are negligible or insignificant.

Classification of bacteria on the basis of cell wall

A cell wall is a layer located outside the cell membrane found in plants, fungi, bacteria, algae, and archaea. A peptidoglycan cell wall composed of disaccharides and amino acids gives bacteria structural support. Depending upon the staining reactions by Gram stain bacteria can be classified into two types, those are:

Gram positive

Gram-positive bacteria are bacteria that give a positive result in the Gram stain test. Gram-positive bacteria take up the crystal violet stain used in the test, and then appear to be purple-coloured when seen through a microscope. This is because the thick peptidoglycan layer in the bacterial cell wall retains the stain after it is washed away from the rest of the sample, in the decolorization stage of the test.

If you are not positive you know some examples of gram-positive bacteria and what exactly makes them gram positive, you are about to find out! Gram-positive bacteria are those that can be stained violet using the Gram stain test. The rest are gram negative. That's the easy gist of it, of course. Let's find out a little bit more about this staining technique and why these cells are violet instead of some other cool color. Then let's look at some examples of famous **gram-positive bacteria.**

Gram Stain

A Gram stain refers to a positive or negative test result produced when an iodine wash is introduced to a culture of bacteria in order to identify its species. This test, known as Gram staining, works by detecting the presence of lipopolysaccharides (lipoglycans) and peptidoglycans (mureins) contained within the cell walls of the bacteria sample. Bacteria that have a high level of peptidoglycans are said to be Gram-positive. In contrast, lower levels of peptidoglycans with lipopolysaccharides indicate that the sample is Gram-negative.

First, the bacteria sample is placed on a glass slide and heated only to the point of rendering it innocuous in terms of being infectious to the handler. Next, the bacteria sample is treated with a gentian violet-iodine solution for up to sixty seconds. The slide is then gently rinsed under clean water and the Gram solution is applied, which is a mixture of iodine and potassium iodide diluted in water. This step triggers a reaction to the gentian violet compound.

Initially, the reaction produces a dark blue color. However, a subsequent rinse with ethyl alcohol leads to the color in some bacteria samples to bleed out, but not in others. A final dye solution is applied that uses a contrasting color, usually a variation of red. A sample that accepts this counterstain will appear pink, and is designated as Gram-negative. However, a sample that retains the dark blue color is Gram-positive.

Aside from identification purposes, the significance of the Gram stain test lies in the fact that Gram-negative bacteria produce potent endotoxins that can cause serious diseases, such as cholera and typhoid. Many Gram-negative bacteria are also resistant to antibiotics and it is not possible to manufacture vaccines from them. In addition, not all bacteria produce a positive or negative result. In fact, some species are considered Gram indeterminate or Gram variable. Other species are entirely unaffected by the test simply due to having a wax-like protective layer in their cell walls that the stains cannot permeate.

The Gram stain test was developed in the late 1800s by the noted Danish bacteriologist, Hans Christian Gram. However, the original purpose of the Gram stain test was not to distinguish between different bacterial species at all. In fact, Dr. Gram merely set out to devise a better way to detect the presence of bacteria in sputum samples provided by pneumonia patients. It is also interesting to note that Dr. Gram's discovery, while unintentional, would have a big impact on the study of antibiotic-resistant bacteria half a century later.

Why Are They Called Gram Positive?

To understand why gram-positive bacteria are so called, we need to travel back to more than 100 years ago, back to the late 1800s. Not only that, we need to book a ticket to Denmark to meet the Danish scientist Hans Christian Gram. He is the guy who developed the Gram stain. The Gram stain, simply put, uses the following ingredients:

- A water-soluble stain called crystal violet
- An iodine solution
- A decolorizing agent such as ethyl alcohol or acetone
- A counterstain, typically safranin

In order to stain a bunch of bacteria on a microscope slide, you'd first pour on the crystal violet dye. Then, you'd add iodine in order to form a crystal violet-iodine complex, which is a larger and water-insoluble molecule than the crystal violet alone. The iodine, in short, makes it more difficult for the crystal violet dye to be removed.

Of course, if all the cells are violet, how do we know which ones are gram positive or gram negative? Well, that's where the decolorizer comes in. After pouring on the decolorizer, only the cells with a thick wall called the peptidoglycan layer will retain the crystal violet-iodine complex and remain purple. Cells with a thin peptidoglycan layer, the gram-negative cells, will not. They will lose their color instead.

Since its kind of hard to see colorless cells, we need to stain the gram-negative cells with a counterstain like safranin. This turns the gram-negative cells red. The addition of this lighter counterstain does not affect the violet color of the gram-positive cells with their thick peptidoglycan cell walls. Thus, looking under the microscope, we see red gram-negative cells and purple gram-positive cells.

Examples of Gram-Positive Bacteria

There are many gram-positive bacteria, including the *Staphylococcus* genus (group of bacteria). These typically look like clusters of grapes underneath the microscope. One famous example found in this genus is *Staphylococcus aureus*. This bacterium is well-known for MRSA, or methicillin-resistant *Staphylococcus aureus*. This is a kind of bacteria that's resistant to many antibiotics. As result, MRSA can cause serious skin infections, pneumonia, and blood infections.

Another genus of gram-positive bacteria are the *Streptococci*, which commonly look like chains of bacteria under the microscope. One example from this genus is *Streptococcus pneumoniae*. As you can tell from the name, this bacterium can cause pneumonia. It can also cause ear, sinus, and blood infections, as well as meningitis.

Gram negative

Gram-negative bacteria are bacteria which do not turn purple in the Gram staining process used as a basic step in the identification of bacteria. Most bacteria can be divided into either Gram-positive or Gram-negative bacteria, reflecting key differences in the composition of their cell walls. These differences often have a direct influence on what the bacteria does, with some Gram-negative bacteria being pathogenic in nature.

The Gram stain was developed in 1884 by Hans Christian Gram. In this process, bacteria is fixed on a slide and then bathed in crystal violet, the primary staining solution. All of the cells on the slide turn purple, after which a mordant such as iodine is added to fix the color. Then, a decolorizer is added to the slide. If the bacteria is Gram-negative, the decolorizer will wash the crystal violet away, because the permeable cell wall does not allow the crystal violet to stain the bacteria. Then, a secondary stain is added, turning Gram-negative bacteria a pale pink, but having no effect on the already purple Gram-positive bacteria.

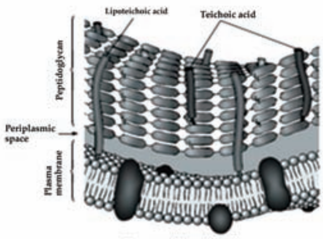
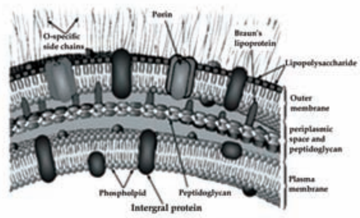
These bacteria have thin cell walls with an outer layer composed of proteins and lypopolysaccharide. This outer layer sometimes reacts with the immune system, causing inflammation and infection. In addition to preventing the bacteria from staining, the outer membrane of the cell also helps the bacteria resist an assortment of drugs, making treatment of infections with Gram-negative bacteria rather challenging.

Some examples of Gram-negative bacteria include Legionella, Salmonella, and E. Coli. Numerous other pathogens are also Gram-negative, including some forms of meningitis, a number of bacterial sources of gastrointestinal distress, and spirochetes. Gram-negative bacteria can be stubborn infectious agents, and many sources of lethal infection are Gram-negative, including the bacteria which contribute to secondary infections in hospitals and clinics.

Gram staining can provide insight into the composition of a bacterium's cell wall, so it is a routine step in examining new bacteria in the laboratory. Once bacteria has been subjected to a Gram stain, additional research will be needed to identify the bacteria, the source, and how infections caused by the bacteria might be treated, but the Gram stain provides a good first step. The stain also has the added benefit of highlighting the key structures of bacteria, including the inner structures of the cell, making them easier to see and understand. Gram staining does not work on all bacteria, however; Gram-indeterminate and Gram-variable bacteria cannot be identified this way.

Differences between Gram positive and Gram negative

There are some differences between gram positive and gram negative, such as:

Gram Positive Bacteria	Gram Negative Bacteria
The bacterial remain coloured with Gram staining even after washing with alcohol or acetone.	The bacteria do not retain the stain when washed with alcohol or acetone.
Outer membrane is absent.	Outer membrane is present.
Cell wall is 20-30 nm thick.	Cell wall is 8-12 nm thick.
The wall is smooth	The wall is wavy and comes in contact with plasma membrane only at a few loci.
The wall contains 70-80% murein.	The wall contains 10-20% murein.
The lipid content in the wall is very low.	The lipid content in the wall is 20-30%
Porin are absent.	Porins or hydrophilic channels occur in outer membrane.
Cell wall contains teichoic acids.  Gram-positive cell wall	Teichoic acids are absent.  Gram-negative cell wall
Basal body of the flagellum contains two rings.	Basal body of the flagellum has four rings.
Mesosomes are quite prominent.	Mesosomes are less prominent.
A few pathogenic bacteria belong to Gram positive group.	Most of the pathogenic bacteria belong to Gram negative group.

Classification on the basis of temperature response

Bacteria can be classified into four major types on the basis of their temperatures response as indicated below: –

Psychrophilic bacteria

Psychrophilic ("cold loving") microorganisms, particularly bacteria , have a preferential temperature for growth at less than 59° Fahrenheit (15° Celsius). Bacteria that can grow at such cold temperatures, but which prefer a high growth temperature, are known as psychrotrophs.

The discovery of psychrophilic microorganisms and the increasing understanding of their functioning has increased the awareness of the diversity of microbial life on Earth. So far, more than 100 varieties of psychrophilic bacteria have been isolated from the deep sea. This environment is very cold and tends not to fluctuate in temperature. Psychrophilic bacteria are abundant in the near-freezing waters of the Arctic and the Antarctic. Indeed, in Antarctica, bacteria have been isolated from permanently ice-covered lakes. Other environments where psychrophilic bacteria have been include high altitude cloud droplets.

Psychrophilic bacteria are truly adapted for life at cold temperatures. The enzymes of the bacteria are structurally unstable and fail to operate properly even at room (or ambient) temperature. Furthermore, the membranes of psychrophilic bacteria contain much more of a certain kind of lipid than is found in other types of bacteria. The lipid tends to be more pliable at lower temperature, much like margarine is more pliable than butter at refrigeration temperatures. The increased fluidity of the membrane makes possible the chemical reactions that would otherwise stop if the membrane were semi-frozen. Some psychrophiles, particularly those from the Antarctic, have been found to contain polyunsaturated fatty acids, which generally do not occur in prokaryotes. At room temperature, the membrane of such bacteria would be so fluid that the bacterium would die.

Aside from their ecological curiosity, psychrophilic bacteria have practical value. Harnessing the enzymes of these organisms allows functions such as the cleaning of clothes in cold water to be performed. Furthermore, in the Arctic and Antarctic ecosystems, the bacteria form an important part of the food chain that supports the lives of more complex creatures. In addition, some species of psychrophiles, including *Listeria monocytogenes* are capable of growth at refrigeration temperatures. Thus, spoilage of contaminated food can occur, which can lead to disease if the food is eaten. Listeriosis, a form of meningitis that occurs in humans, is a serious health threat, especially to those whose immune system is either not mature or is defective due to disease or therapeutic efforts. Other examples of such disease-causing bacteria include *Aeromonas hydrophila*, *Clostridium botulinum*, and *Yersinia enterocolitica*.

These type of bacteria grows just above the freezing temperature, they can cause contamination of food stored in the refrigerator. Example: -Pseudomonas.

Mesophilic bacteria

Mesophiles are microorganisms such as some species of Bacteria , Fungi , and even some Archaea that are best active at median temperatures. For instance, bacterial species involved in biodegradation (i.e., digestion and decomposition of organic matter), which are more active in temperatures ranging from approximately 70° - 90°F (approx. 15°–40°C), are termed mesophilic bacteria. They take part in the web of micro-organic activity that form the humus layer in forests and other fertile soils, by decomposing both vegetable and animal matter.

At the beginning of the decomposition process, another group of bacteria, psychrophylic bacteria, start the process because they are active in lower temperatures up to 55°F (from below zero up to 20°C), and generate heat in the process. When the temperature inside the decomposing layer reaches 50–100°F, it attracts mesophilic bacteria to continue the biodegradation. The peak of reproductive and activity of mesophilic bacteria is reached between 86–99°F (30–37°C), and further increases the temperature in the soil environment. Between 104–170°F (40–85°C, or even higher), another group of bacteria (thermophyllic bacteria) takes up the process that will eventually result in organic soil, or humus. Several species of fungi also take part in each decomposing step.

Mesophilic bacteria are also involved in food contamination and degradation, such as in bread, grains, dairies, and meats. Examples of common mesophilic bacteria are *Listeria monocytogenes, Pesudomonas maltophilia, Thiobacillus novellus, Staphylococcus aureus, Streptococcus pyrogenes, Streptococcus pneumoniae, Escherichia coli, and Clostridium kluyveri*. Bacterial infections in humans are mostly caused by mesophilic bacteria that find their optimum growth temperature around 37°C (98.6°F), the normal human body temperature. Beneficial bacteria found in human intestinal flora are also mesophiles, such as dietary *Lactobacillus acidophilus*.

Thermophilic bacteria

Thermophilic bacteria are microorganisms found in areas that have temperatures too high for other types of bacteria. These bacteria live, reproduce and thrive in temperatures ranging from 113 degrees to more than 212 degrees Fahrenheit.

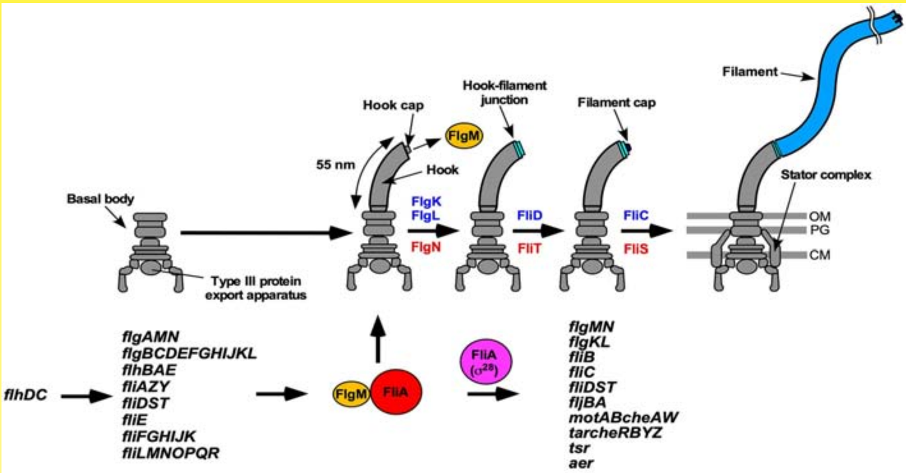
Thermophilic bacteria can be found growing naturally in the hottest places on Earth. Hot springs, volcanoes and geysers are great places for them. They are also found in all types of manure and thrive in compost heaps. It is estimated that 1 trillion bacteria can fit into a single teaspoon of waste material. They are the organisms responsible for the breakdown of organic matter in the compost. Due to the high heat required, and the additional heat they create when eating the materials in the heap, they are the reason for spontaneous compost fires.

Thermophilic bacteria are not only used in agricultural applications, but are finding their way into technology too. The enzyme needed to perform DNA matching is created using Thermus aquaticus and Thermococcus litoralis, two thermophiles.

The thermophiles that reside in the highest temperatures, well above the boiling point of water, are referred to as extreme thermophiles. Some scientists have theorized that thermophiles are the Earth's oldest living organisms. Thermophilic bacteria were first discovered in 1966 in the natural hot springs in Yellowstone National park.

CONCEPT OF FLAGELLAR BIOSYNTHESIS

Flagellar biosynthesis refers to the process by which the flagellum is assembled and constructed. Flagella production is a highly complex and precisely regulated process involving numerous flagellar proteins. It involves the coordinated synthesis and assembly of various flagellar components, including the filament, hook, basal body, and associated proteins.



The specific steps involved in flagellar biosynthesis can vary among different organisms, but the general process includes the following key stages:

- ***Transcription:*** Genes encoding flagellar components are transcribed into mRNA molecules.
- ***Translation:*** The mRNA molecules are translated into protein subunits, such as flagellin proteins and other structural proteins.
- ***Filament Assembly:*** Flagellin proteins are polymerized and arranged in a helical or spiral structure to form the filament.
- ***Hook Formation:*** The hook protein subunits are synthesized and assembled to form the curved region that connects the filament to the basal body.
- ***Basal Body Assembly:*** The various components of the basal body, including the rings, rod, and motor proteins, are synthesized and assembled. The basal body is anchored to the cell membrane.
- ***Export and Secretion:*** Several specialized protein export apparatus, such as the type III secretion system in bacteria, facilitate the export and secretion of flagellar proteins to their appropriate locations.
- ***Integration and Maturation:*** The different flagellar components, including the flagellar filament, hook, and basal bodies, are integrated and undergo maturation processes to become functional flagella.

SUMMARY

- Bacteriology is a scientific discipline. It is the branch of microbiology that focuses on bacteria.
- Genetic material is a circular, single-stranded DNA diffused throughout the cytoplasm. It carries the genetic code of heritable traits and is responsible for directing metabolism and replication of the cell. The bacterial cell does not have a nuclear membrane or a well-defined nucleus.
- A capsule is relatively thick layer of mucoid polysaccharides (slime layer) around a bacterium. A capsule serves as a defense mechanism.
- Ribosomes is a site of protein production. Ribosomes consist of a small and large subunits made up of a complex proteins and RNAs. These are the sites of protein synthesis in the cell.
- Bacterial conjugation is often regarded as the bacterial equivalent of sexual reproduction or mating since it involves the exchange of genetic material.
- Bacillus (genus Bacillus), any of a group of rod-shaped, gram-positive, aerobic or (under some conditions) anaerobic bacteria widely found in soil and water. The term bacillus has been applied in a general sense to all cylindrical or rodlike bacteria. The largest known Bacillus species, B. megaterium, is about 1.5 μm (micrometres; 1 μm = 10^{-6} m) across by 4 μm long. Bacillus frequently occur in chains.
- Spiral-shaped bacteria are called spiral bacteria. These types of microorganisms come in three various forms, which include vibrio, spirillum and spirochete.
- A parasite is an organism which exploits another organism for the purpose of staying alive. Some parasitic relationships are harmless, while in other cases a parasite can damage or even kill its host.
- Saprophytes are living organisms that feed on dead organic matter. They are considered extremely important in soil biology, as they break down dead and decaying organic matter into simple substances that can be taken up and recycled by plants. The term is usually used to refer to saprophytic fungi or bacteria.
- A symbiont is an organism that is very closely associated with another, usually larger, organism. This larger organism is called a host. A symbiont can live on, in, or sometimes very near its host.
- Gram-positive bacteria are bacteria that give a positive result in the Gram stain test. Gram-positive bacteria take up the crystal violet stain used in the test, and then appear to be purple-coloured when seen through a microscope. This is because the thick peptidoglycan layer in the bacterial cell wall retains the stain after it is washed away from the rest of the sample, in the decolorization stage of the test.

MULTIPLE CHOICE QUESTIONS

1. The identification of bacteria by serologic tests is based on the presence of specific antigens. Which of the following bacterial components is least likely to contain useful antigens?

a. Capsule
b. Cell wall
c. Flagella
d. Ribosomes

2. The association of endotoxin in gram-negative bacteria is due to the presence of

a. Steroids
b. Peptidoglycan
c. Lipopolysaccharides
d. Polypeptide

3. Lipopolysaccharide in cell walls is characteristic of?

a. Algae
b. Fungi
c. Gram-negative bacteria
d. Gram-positive bacteria

4. The bacterial genus where sterols are present in the cell membrane is

a. Vibrio
b. Mycoplasma
c. Escherichia
d. Chlamydia

5. The bacterium that infects other gram-negative bacteria is

a. Proteus mirabilis
b. Haemophilus influenza
c. Bdellovibrio
d. Pseudomonas putida

Answers

1. (d) 2. (c) 3. (c) 4. (b) 5. (c)

REVIEW QUESTIONS

1. What does bacteriology mean?
2. Who coined the term bacteriology?
3. Why is bacteriology important?
4. What is the bacteriology of water?
5. What does virulent mean?

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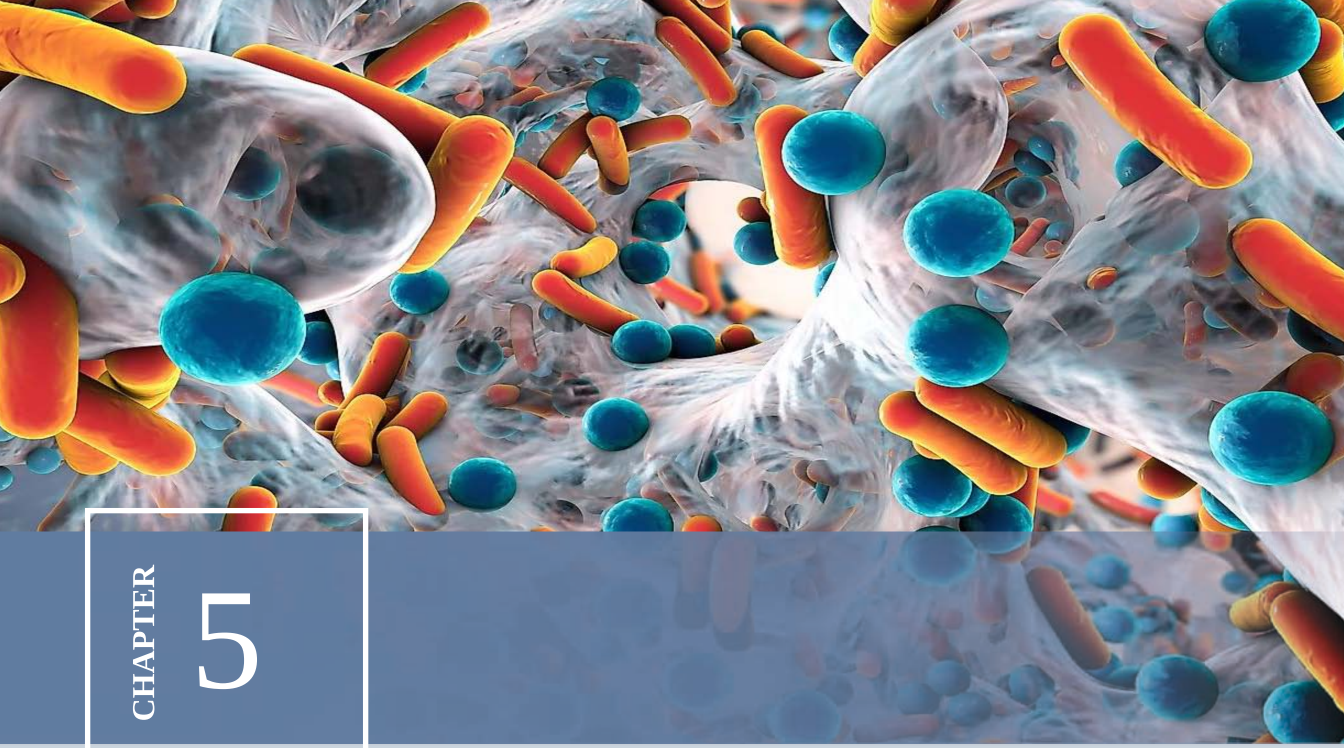
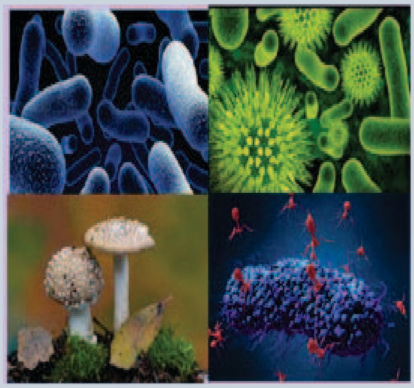


CHAPTER 5

POSITION OF MICROORGANISMS IN BIOLOGICAL WORLD



Microorganisms or microbes are microscopic organisms that exist as unicellular, multicellular, or cell clusters. Microorganims are widespread in nature and are beneficial to life, but some can cause serious harm. They can be divided into six major types: bacteria, archaea, fungi, protozoa, algae, and viruses.

After studying this chapter, you will understand the core concepts of:

- Definition and Types of Microorganisms
- Biological Classification of Microorganism

INTRODUCTION

Microorganisms, or microbes, are a diverse group of minute, simple forms of life that include bacteria, algae, fungi, protozoa, and viruses. Microorganisms are too small to be seen with the naked eye and are normally viewed by means of a microscope. The study of microorganisms is called microbiology.

In addition to populating both the inner and outer surfaces of the human body, microorganisms abound in the soil, in the seas, and in the air. Although usually unnoticed, most microorganisms are essential to life on Earth; for example, different microorganisms cause bread to rise, flavor cheeses, and produce valued products such as antibiotics and insulin. Others, however, are harmful to humans, animals, and plants and can cause diseases, such as chicken pox and rubella.

Common microorganisms include bacteria, which are tiny single-celled organisms that are neither animals nor plants. Bacteria have a variety of shapes, including spheres, rods, and spirals, and they often appear in pairs, chains, groups of four, or clusters.

Viruses are smaller than bacteria. They are not technically living things because they cannot survive on their own; they can multiply only in living cells of animals, plants, or bacteria. Once inside a cell, a virus can reproduce itself, and the copies can leave the original cell and infect other cells.

Fungi, once considered to be plants, are now placed in their own scientific kingdom. This group includes mushrooms, molds, mildews, and yeasts, only some of which are microorganisms. Along with bacteria, fungi are responsible for the decay of organic matter and the release into the atmosphere of carbon, oxygen, nitrogen, and phosphorus. Algae are most commonly found in water, such as oceans, rivers, lakes, and marshes, but some species live in soil or on leaves, wood, and stones. Only some species are so small that they can be seen only through a microscope. Algae are plantlike in that they make their own food through a process called photosynthesis.

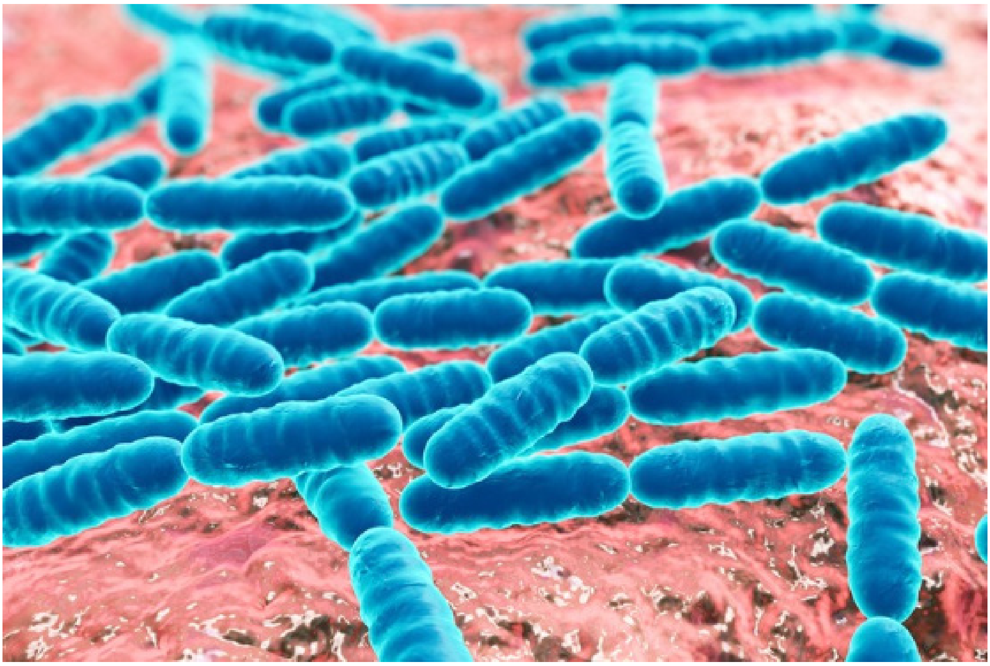
CORE CONCEPT: DEFINITION AND TYPES OF MICROORGANISMS

Microorganisms are, as the name implies, microscopic organisms. They are living things that are too tiny to see with the naked eye. A microorganism can perform all the characteristics that any living thing can perform. It can move, get nutrients from the environment, maintain homeostasis, and evolve.

The vast majority of life on Earth is invisible to the human eye. Microscopic organisms, known as microorganisms or microbes, are found in huge numbers in almost every environment on Earth. Most microbes, such as bacteria and archaea, consist of only a single cell. They study of microorganisms is called microbiology. All branches of life, including animals and plants, contain species of microorganisms. Most commonly however, the study of microscopic organisms focuses on groups such as bacteria, protists and fungi. In recent years, significant effort has focused on the study of a little known group of microbes called Archaea.

Microorganism Definition

A microorganism is a living thing that is too small to be seen with the naked eye. Examples of microorganisms include bacteria, archaea, algae, protozoa, and microscopic animals such as the dust mite. These microorganisms have been often under-appreciated and under-studied. Indeed, until Anton von Leeuwenhoek invented the microscope, we did not know they existed! Until that time, it was thought that phenomena such as illness and food spoilage were caused by “vapors” or “spontaneous generation.”

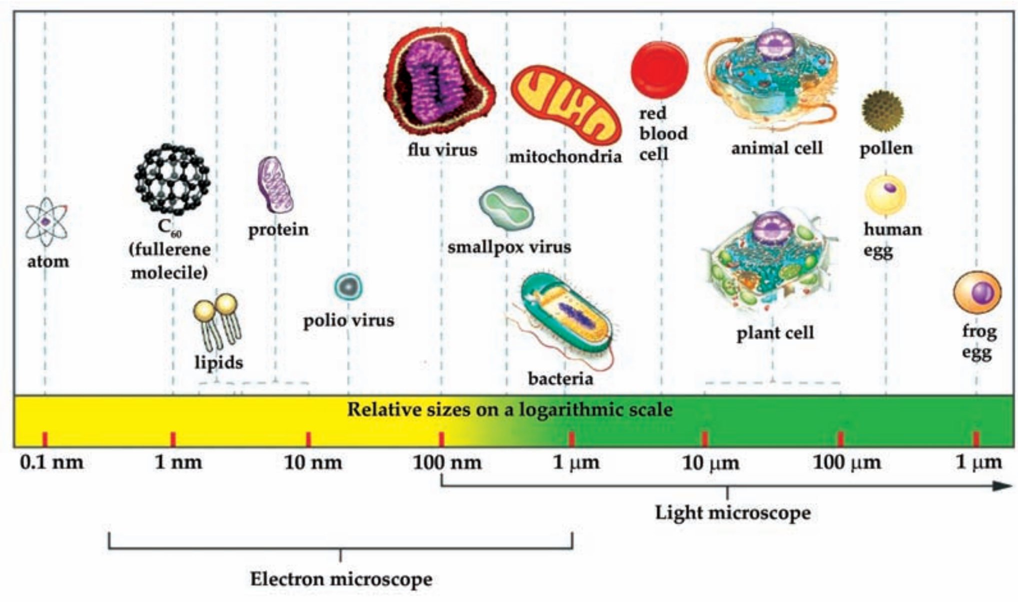


Leeuwenhoek's invention of the microscope soon led Louis Pasteur to realize that many diseases were caused by microorganisms – and to the practice of pasteurization, which kills microorganisms and makes our food products safe to eat today. Now, we know that microorganisms are responsible for many things that happen in the world around us.

- Microorganisms are found virtually everywhere, except for environments that have been made artificially sterile by humans. Even these must be constantly sterilized and carefully protected, lest microorganisms be tracked in from the outside world.
- Microorganisms live in water, in soil, and on the skin and in the digestive tracts of animals. This is why all living things must have immune systems – while many microorganisms can be helpful to them, some can be harmful and cause disease.

Types of Microorganisms

Most microbes are unicellular and small enough that they require artificial magnification to be seen. However, there are some unicellular microbes that are visible to the naked eye, and some multicellular organisms that are microscopic. An object must measure about 100 micrometers (µm) to be visible without a microscope, but most microorganisms are many times smaller than that. For some perspective, consider that a typical animal cell measures roughly 10 µm across but is still microscopic. Bacterial cells are typically about 1 µm, and viruses can be 10 times smaller than bacteria see below figure.



This section will look at different types of microorganisms. It will discuss their cell structure and functions. It will also discuss the position of microbes in food chains and their role in the biosphere. The major groups of microorganisms—namely bacteria, archaea, fungi (yeasts and molds), algae, protozoa, and viruses—are summarized below.

Archaea (bacteria)

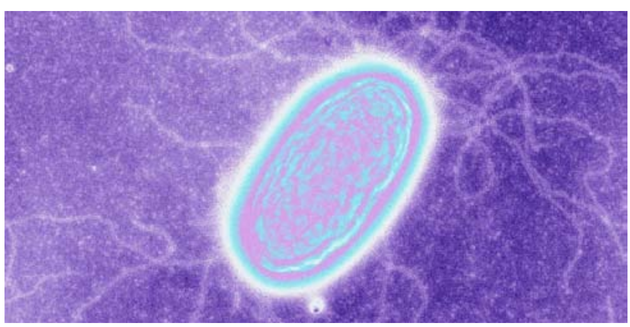
Microbiology came into being largely through studies of bacteria. The experiments of Louis Pasteur in France, Robert Koch in Germany, and others in the late 1800s established the importance of microbes to humans. As stated in the Historical background section, the research of these scientists provided proof for the germ theory of disease and the germ theory of fermentation. It was in their laboratories that techniques were devised for the microscopic examination of specimens, culturing (growing) microbes in the laboratory, isolating pure cultures from mixed-culture populations, and many other laboratory manipulations. These techniques, originally used for studying bacteria, have been modified for the study of all microorganisms—hence the transition from bacteriology to microbiology.

The organisms that constitute the microbial world are characterized as either prokaryotes or eukaryotes; all bacteria are prokaryotic—that is, single-celled organisms without a membrane-bound nucleus. Their DNA (the genetic material of the cell), instead of being contained in the nucleus, exists as a long, folded thread with no specific location within the cell.

Until the late 1970s it was generally accepted that all bacteria are closely related in evolutionary development. This concept was challenged in 1977 by Carl R. Woese and coinvestigators at the University of Illinois, whose research on ribosomal RNA from a broad spectrum of living organisms established that two groups of bacteria evolved by separate pathways from a common and ancient ancestral form. This discovery resulted in the establishment of a new terminology to identify the major distinct groups of microbes—namely, the eubacteria (the traditional or "true" bacteria), the archaea (bacteria that diverged from other bacteria at an early stage of evolution and are distinct from the eubacteria), and the eukarya (the eukaryotes). Today the eubacteria are known simply as the true bacteria (or the bacteria) and form the domain Bacteria. The evolutionary relationships between various members of these three groups, however, have become uncertain, as comparisons between the DNA sequences of various microbes have revealed many puzzling similarities. As a result, the precise ancestry of today's microbes is very difficult to resolve. Even traits thought to be characteristic of distinct taxonomic groups have unexpectedly been observed in other microbes. For example, an anaerobic ammonia-oxidizer—the "missing link" in the global nitrogen cycle—was isolated for the first time in 1999. This **bacterium** (an aberrant member of the order Planctomycetales) was found to have internal structures similar to eukaryotes, a cell wall with archaean traits, and a form of reproduction (budding) similar to that of yeast cells.

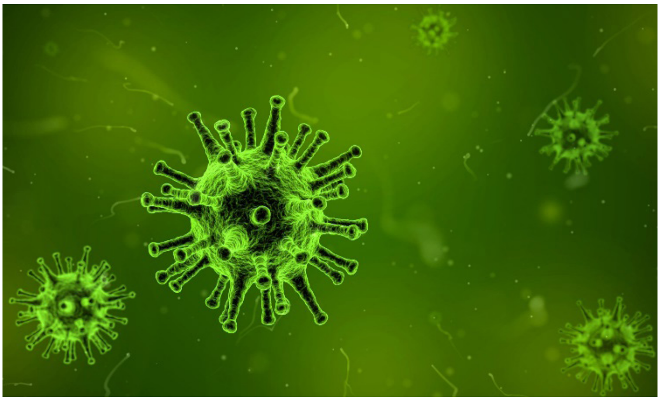
Bacterium is a living cell consisting of a fluid called cytoplasm enclosed by a cell membrane and cell wall. A bacterium contains DNA in the cytoplasm in the form of a chromosome.

Bacteria have a variety of shapes, including spheres, rods, and spirals. Individual cells generally range in width from 0.5 to 5 micrometres (μm; millionths of a metre). Although unicellular, bacteria often appear in pairs, chains, tetrads (groups of four), or clusters. Some have flagella, external whip like structures that propel the organism through liquid media; some have capsule, an external coating of the cell; some produce spores—reproductive bodies that function much as seeds do among plants. One of the major characteristics of bacteria is their reaction to the Gram stain. Depending upon the chemical and structural composition of the cell wall, some bacteria are gram-positive, taking on the stain's purple colour, whereas others are gram-negative.



Through a microscope the archaea look much like bacteria, but there are important differences in their chemical composition, biochemical activities, and environments. The cell walls of all true bacteria contain the chemical substance peptidoglycan, whereas the cell walls of archaeans lack this substance. Many archaeans are noted for their ability to survive unusually harsh surroundings, such as high levels of salt or acid or high temperatures. These microbes, called **extremophiles**, live in such places as salt flats, thermal pools, and deep-sea vents. Some are capable of a unique chemical activity—the production of methane gas from carbon dioxide and hydrogen. Methane-producing archaea live only in environments with no oxygen, such as swamp mud or the intestines of ruminants such as cattle and sheep. Collectively, this group of microorganisms exhibits tremendous diversity in the chemical changes that it brings to its environments.

Extremophile is an organism that thrives in physically or geochemically extreme conditions that are detrimental to most life on Earth.

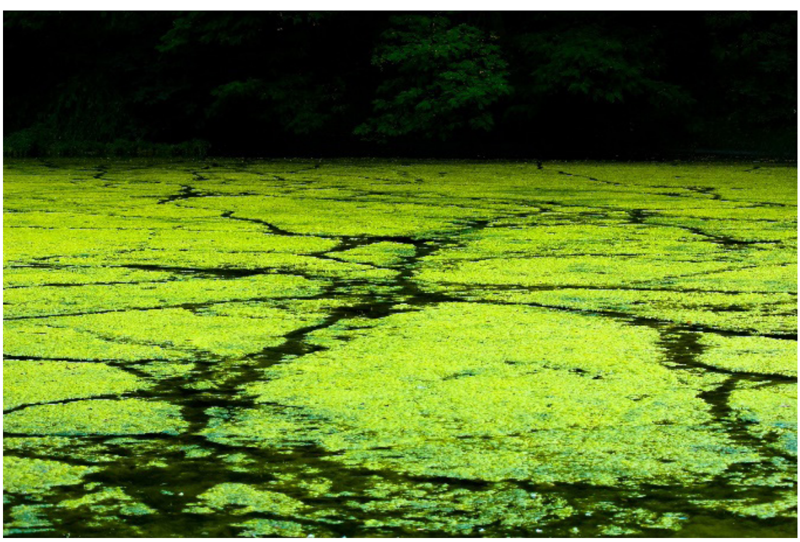


Algae

Algae (singular alga) are a large group of diverse organisms that use photosynthesis to produce food. Although some forms are large and multicellular, they differ from plants in that their cells are not clearly organized into different types of tissue with different functions. This group includes a wide variety of organisms that are not always closely related to one another — the similarities in form are often due to parallel evolution, where different organisms have adapted in similar ways to fill similar niches. They are described as polyphyletic, meaning that not all members of the group share the same common ancestor. By the modern definition, all algae are eukaryotes, which means that the DNA in their cells is contained within a nucleus enclosed by a membrane. Organisms whose cells do not have a nucleus are prokaryotes. The eukaryotes also include plants, fungi and animals. Prokaryotes include bacteria and archaea. The algae can be divided into a number of sub-groups, based largely on the types of pigments they use for photosynthesis.

Green Algae

These forms use the green pigment chlorophyll to photosynthesize, and they are thought to be the ancestors of land plants. Some authorities include them in the plant kingdom, while others prefer to regard them as a separate category of life. They may be single celled or multicellular, and some types live in colonies or form long filaments consisting of many cells. A number of the single-celled types are capable of independent movement using flagellae — long, whip-like structures used by many microorganisms for locomotion. Green algae are found in a wide variety of habitats, including freshwater, the sea, soil, tree trunks and damp walls, but the majority are aquatic. It is thought that land plants evolved from a type of green alga, possibly about 500 million years ago. They contain the same types of chlorophyll and other pigments as land plants. There are further similarities: for example, the chlorophyll is contained in structures called chloroplasts, and many types store sugars in starch granules, as do land plants.



Red Algae

Also known as rhodophyta, these were among the first eukaryotic organisms on the planet, and their signatures have been found in rocks almost 2 billion years old. They are mostly marine organisms, and include many types of seaweed, as well as a number of single-celled species. Their red color comes from the pigments phycoerythrin and phycocyanin, which they use for photosynthesis. These pigments absorb blue light, which reaches further below the ocean surface than the red light trapped by chlorophyll, allowing the rhodophyta to photosynthesize at greater depths. This group also includes coralline algae, which build shells made of calcium carbonate for themselves and can form reefs.

Brown Algae

The proper scientific name for this group is chromista. It is an extremely diverse group, with its members ranging from diatoms — microscopic, single-celled forms with silica shells — to “kelp” seaweeds — large, multicellular organisms that can grow to 164 feet (50 meters) in length. They use a different kind of chlorophyll from that used by plants to photosynthesize and often have additional pigments, such as fucoxanthin, which gives many of these organisms a brown colour. Diatoms are an important part of the phytoplankton, which produce a great deal of the planet’s oxygen through photosynthesis and form the base of many marine food chains. Kelp seaweeds can form extensive seafloor “forests,” which are of great ecological importance.

Prokaryote is a unicellular organism that lacks a membrane-bound nucleus, mitochondria, or any other membrane-bound organelle.

Cyanobacteria

Today, these microorganisms are considered to be bacteria; however, they are still sometimes referred to by their old name, "blue-green algae." They differ mainly in that they are **prokaryotes**, like all bacteria, but they can make their own food by photosynthesis. Cyanobacteria are a very ancient group and may have been the very first organisms to use photosynthesis. Many experts think that, in the distant past, some single-celled, non-photosynthesizing organisms may have incorporated cyanobacteria in a symbiotic relationship, and that these bacteria may have become the chloroplasts that are seen today in algae and plants.

Algal Blooms

From time to time, in certain locations, a species of alga may undergo a population explosion, resulting in what is known as an "algal bloom." These can occur on coastlines and in freshwater lakes. It is not always possible to establish the cause, but often, it seems to be due to agricultural run-off containing fertilizers that stimulate increased growth and multiplication. Algal blooms are often harmful to other aquatic life forms, and occasionally to animals and even humans. The huge numbers of algae can severely reduce the oxygen content of the water, and some species produce toxins that can kill or harm other organisms.

Uses

A number of types of seaweed, especially among the red algae, can be eaten. Seaweeds also provide a number of important food additives, and agar — a type of gel used for culturing microorganisms. Another potential use is in the production of biofuels. The organisms are fast growing and undemanding in terms of conditions and nutritional requirements, and so they can provide a cheap and efficient way of accumulating biomass for fuel.

Fungi

Fungi are a kingdom of eukaryotic (their cells have nuclei) organisms. Other examples of biological kingdoms include Plantae, the plants, and Animalia, the animals. Common fungi include mushrooms, yeasts, and molds. Fungi are essential in decomposing dead organic matter in the soil, and without them, biological refuse would take much longer to degrade, making it difficult for the next generation of organisms to utilize the essential elements therein. Although fungi may look like plants, they are in fact more closely related to animals. The study of mushrooms is known as mycology.

The key characteristic of fungi which sets them apart from other organisms are their chitinous cell walls. This durable material, chitin, also makes up the shells of many insects. Fungi tend to grow in filamentous structures known as mycelium, and reproduce either sexually or asexually via spores. In mushrooms, the spores are visible as black dust underneath the cap.

Fungi have a long history of being utilized by humans. Yeasts are used to give bread the puffy consistency with which we are familiar. Many mushrooms are integrated into dishes; the Portobello mushroom is one of the more popular species. Some fungal species are critical to fermentation, the process underlying the production of alcoholic beverages. Cheeses possess their characteristic odd smell, and sometimes, colour, due to carefully introduced fungi. Psilocybin mushrooms have long been consumed for their hallucinogenic properties. Some species of mushrooms with creative names such as "destroying angel" and "death cap" are very poisonous, however, and can cause death within hours of consumption. In an effort to cut down on the use of chemically polluting pesticides, some agricultural scientists have developed fungi for use as bio pesticides – beneficial fungi which produce alkaloids toxic to a wide range of insects and other pests.

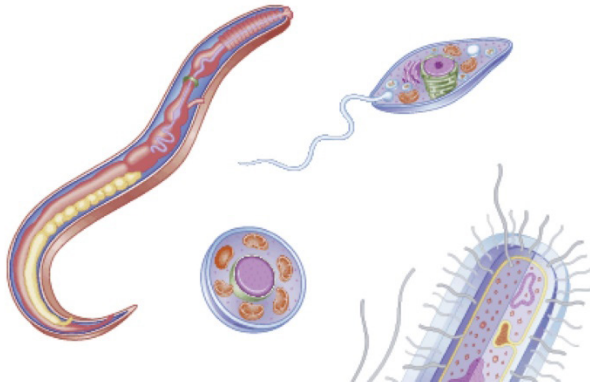
Environment is everything that is around us. It can be living or non-living things. It includes physical, chemical and other natural forces. Living things live in their environment.

Fungi can be found in every **environment** on Earth, even the dry valleys of Antarctica, where small microbial populations exist during the summer. Fungi exist in pretty much every square meter of the Earth's surface that is not permanently frozen or thoroughly sterilized. The only way to restrict fungal growth is by using a refrigerator or freezer.

Protozoa

Protozoa, or protozoans, are single-celled, eukaryotic microorganisms. Some protozoa are oval or spherical, others elongated. Still others have different shapes at different stages of the life cycle. Cells can be as small as 1 μm in diameter and as large as 2,000 μm, or 2 mm (visible without magnification). Like animal cells, protozoa lack cell walls, are able to move at some stage of their life cycle, and ingest particles of food; however, some phytoflagellate protozoa are plantlike, obtaining their energy via photosynthesis. Protozoan cells contain the typical internal structures of an animal cell. Some can swim through water by the beating action of short, hairlike appendages (cilia) or flagella. Their rapid, darting movement in a drop of pond water is evident when viewed through a microscope.

When a fungus reproduces sexually, two haploid hyphae of compatible mating types come together and fuse.

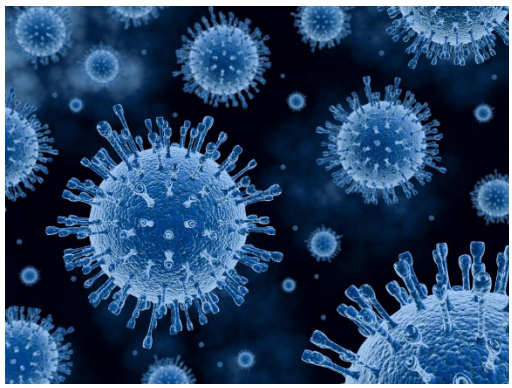


The amoebas (also amoebae) do not swim, but they can creep along surfaces by extending a portion of themselves as a pseudopod and then allowing the rest of the cell to flow into this extension. This form of locomotion is called amoeboid movement. The sporozoans (phylum Apicomplexa) are so named because they form dormant bodies called spores during one phase of their life cycle. Protozoa occur widely in nature, particularly in aquatic environments.

Viruses

Viruses, agents considered on the borderline of living organisms, are also included in the science of microbiology, come in several shapes, and are widely distributed in nature, infecting animal cells, plant cells, and microorganisms. The field of study in which they are investigated is called virology. All viruses are obligate parasites; that is, they lack metabolic machinery of their own to generate energy or to synthesize proteins, so they depend on host cells to carry out these vital functions. Once inside a cell, viruses have genes for usurping the cell's energy-generating and protein-synthesizing systems. In addition to their intracellular form, viruses have an extracellular form that carries the viral nucleic acid from one host cell to another. In this infectious form, viruses are simply a central core of nucleic acid surrounded by a protein coat called a capsid. The capsid protects the genes outside the host cell; it also serves as a vehicle for entry into another host cell because it binds to receptors on cell surfaces. The structurally mature, infectious viral particle is called a virion.

With the electron microscope it is possible to determine the morphological characteristics of viruses. Virions generally range in size from 20 to 300 nanometres (nm; billionths of a metre). Since most viruses measure less than 150 nm, they are beyond the limit of resolution of the light microscope and are visible only by electron microscopy. By using materials of known size for comparison, microscopists can determine the size and structure of individual virions.



Prions

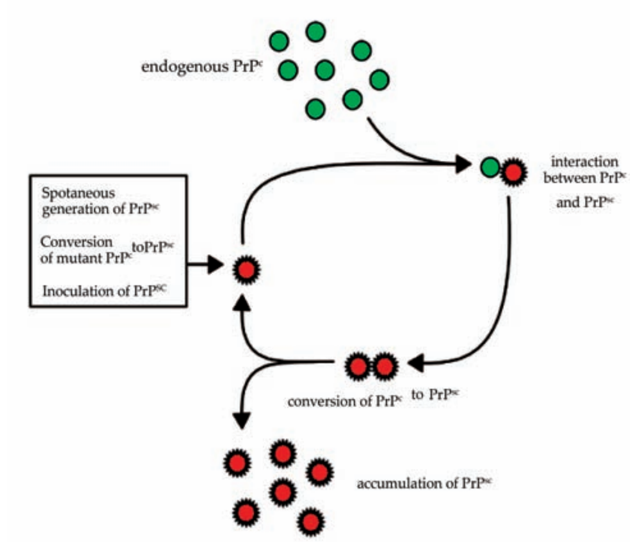
Prions are unique proteins which appear to be responsible for a family of illnesses collectively referred to as Transmissible Spongiform Encephalopathies (TSEs), including scrapie in sheep, kuru in humans, and "mad cow disease" in cows. These prion diseases are all caused by the breakdown of the nervous system, which is readily apparent after autopsy or necropsy. Looking at the tissue of the nervous system, it becomes clear that it has literally been eaten away by these rogue proteins, causing the formation of distinctive holes.

Radiation is energy that comes from a source and travels through space and may be able to penetrate various materials.

The existence of prions was hypothesized as early as the 1960s, when researchers first formulated the so-called "protein only" hypothesis to explain TSEs. These researchers noted that irradiation of potentially infective tissue did not seem to kill the organism responsible, suggesting that it did not contain nucleic acid, which is susceptible to **radiation**. The theory met with a great deal of opposition until the 1980s, when researcher

Stanley Prusiner coined the term “prion” to describe the highly unusual proteins he had discovered; these proteins did not contain nucleic acid, the backbone of all living organisms on Earth.

The word is taken from the longer term “proteinaceous infectious particle.” Despite the fact that prions lack nucleic acid and therefore in theory cannot reproduce, these proteins do, by exhibiting strange folding patterns which subvert otherwise healthy proteins. The protein they are made of is found in many animals, explaining why these diseases strike a wide range of species. Animals and humans can get TSE by ingesting prions in things like infected tissue and through contaminated blood products.



These little proteins are quite stubborn. They resist many techniques used to decontaminate food, raising concerns about the safety of the food supply. Prions cannot be irradiated, and very high temperatures and pressure are required to remove those— temperatures far higher than those consumers would normally use to cook a steak. Fortunately, they appear to lie primarily in nervous system tissue, although consuming meat from an animal with TSE could potentially be dangerous.

These proteins made the news in the 1990s, when numerous cases of Creutzfeldt-Jakob disease (CJD) began to be linked with prions and the consumption of contaminated tissue from cattle and sheep in the United Kingdom. Cases of TSE have clearly been documented in food animals in Europe, and researchers suspect that TSE may also be an issue in the United States as well, perhaps explaining the rising rate of so-called “downer cows,” cows which for some reason are unable to stand up for slaughter. Such cattle are not supposed to enter the food supply out of concerns about prion diseases and other conditions which could pose a threat to human health.

Lichens

Lichens represent a form of symbiosis, namely, an association of two different organisms wherein each benefits. A lichen consists of a photosynthetic microbe (an alga or a cyanobacterium) growing in an intimate association with a fungus. A simple lichen is made up of a top layer consisting of a tightly woven fungal mycelium, a middle layer where the photosynthetic microbe lives, and a bottom layer of mycelium. In this mutualistic association, the photosynthetic microbes synthesize nutrients for the fungus, and in return the fungus provides protective cover for the algae or cyanobacteria. Lichens play an important role ecologically; among other activities they are capable of transforming rock to soil.

Slime molds

Slime mold is a broad categorization of fungi-like, slimy, amoeboid protists (unicellular eukaryotes, i.e., non-bacterial unicellular organisms) that feed on wood, flowers, fruits, mulch, and any other type of dead plant material, bacteria, yeast, and fungal spores. Individual slime mold organisms are usually microscopic, but some species form coenocyte colonies that are usually a few centimetres across. A coenocyte is a large cell with many nuclei which occurs when nuclear division is not accompanied by cellular division. In some instances, these coenocytes can grow in size up to 30 square meters, making them the largest individual cells in the animal world, much larger than the ostrich egg.

Slime molds come in a variety of shapes, including irregular, spherical, bulbous, cylindrical, and colours, including bright pink, orange, yellow, white, gray, and brown. In some ways, their colonial behaviour is reminiscent of the behaviour of multicellular fungi, though they are technically colonies of unicellular organisms. Upon maturity, a colony of slime mold forms a sporangium, a spore-filled bulb on a stalk, that releases spores into the environment. These spores hatch both amoeboid and flagellated gametes which combine to create a **zygote**, which subsequently develops into a slime mold colony, depending on available nutrients.

Zygote is a eukaryotic cell formed by a fertilization event between two gametes.

The group of organisms called "slime molds" is polyphyletic -- that is, they do not descend from a common ancestor. Though slime molds were initially mistakenly categorized as fungi, scientists now consider them non-fungal protists that belong to four distinct families: The Mycetozoa (part of kingdom Amoebozoa), Acrasiomycota (part of kingdom Excavata), Labyrinthulomycota ("slime nets", part of kingdom Chromalveolata), and Plasmodiophorids (plant parasites, part of kingdom Rhizaria). It is interesting to note that the four groups of "slime molds" are actually part of entirely different protist kingdoms, despite their shared qualities, which emerged as a result of convergent evolution.

The different groups of slime molds have different characteristic adaptive strategies and structures. For instance, the "slime nets" are so known because they construct networks of membrane-covered tubes and filaments which are used as tracks to guide cells, as well as absorbing nutrients for them. This structure is known as an "ectoplasmic net". A unique organelle called a bothrosome constructs these filaments. Slime nets are most distantly related to the other slime molds, with some scientists considering them their own group. Another group of slime molds, the Acrasiomycota, are known for releasing pheromones to aggregate amoebal cells in preparation for colonial movement.

Benefits of Microorganisms for the Human Being

The Benefits of Microorganisms For the human being are multiple. From applications in the food industry, to the processes of solid waste degradation or the development of vaccines and medical advances. Microbes or microorganisms are small microscopic entities that can be classified into different groups, such as bacteria, fungi, protozoa, microalgae and viruses. They live on soil, water, food and the intestines of animals, among other means.

Human body is the entire structure of a human being. It is composed of many different types of cells that together create tissues and subsequently organ systems.

Humans have used microorganisms in different industries, such as food or agriculture, where fermented beer, Yogurt and cheese, or microorganisms can be used to release nitrogen from the soil that plants need to grow. Not all microorganisms are beneficial to human life, there are some organisms that limit the production of food or stay in animals and plants causing diseases. In the human body, different microorganisms are responsible for contributing to different processes, such as digestion and defense of other invasive organisms in a complex process that is reflected in the natural course of a disease. Microorganisms are beneficial in different industries and contribute to multiple biological processes taking place within the **human body**.

Food Industry

Microorganisms are used in the production of fermented foods and beverages. Fungi like yeast or bacteria like Lactobacilli Are essential in the food industry. The fermentation process leading to the production of alcoholic beverages or acid based dairy products takes place when microorganisms get energy from food cells without having to take oxygen. In other words, the fermentation process allows the decomposition of complex organic substances.

Foods like cheese, olives, sausages, chocolate, bread, wine, beer and soy sauce are made with the help of different types of bacteria and yeasts. In most of these products, bacteria play a key role. They are in charge of producing lactic acid, a substance that allows the preservation of food.

Medicine and Science

Microorganisms also have significant potential in the field of medicine and science. They are generally used industrially for the production of antibiotics, vaccines and insulin. As well as to make the diagnosis of certain diseases.

In medicine bacteria are used to produce thousands of antibiotics. Species of bacteria as Streptomyces are responsible for the production of more than 500 different antibiotics. Similarly, there are antibiotics produced from fungi and other types of bacteria. The antibiotic name means “against life.” This name is because the main role of these compounds is to attack bacteria and other unicellular organisms that may be pathogenic to humans. The majority of antibiotics used today were discovered from observation on the propagation of fungi on animals in a state of decomposition.

Waste treatment

Microorganisms play a vital role in the handling and disposal of domestic and industrial waste. They are responsible for cleaning the waste through a biological process of decomposition or stabilization of organic matter. This decomposition process is as old as life on planet Earth. The process of controlled biological decomposition is known as composting. The final product thrown by this process is called compost. It can be classified as anaerobic compost when organic matter is decomposed from the use of fungi, bacteria and protozoa. Microorganisms are responsible for breaking down matter by raising its temperature and producing carbon dioxide. In this way, a substance called humus is generated that has a similar appearance to the soil to grow.

Microflora

There are billions of bacteria that inhabit the digestive tract of humans. It is estimated that one kilogram of body weight of each person is composed of bacteria known as microflora. These bacteria are responsible for breaking down food remains that have not been previously processed and digested. The microflora is also responsible for defending the body from fungi and bacteria harmful to human health. It produces vitamin K, which is necessary to regulate blood clotting processes. The human body can house 400 different types of bacterial species, some of which are only beneficial and others are potentially harmful. It is essential that there is a balance between these two types of microorganisms to ensure the sustainability of life. The Beneficial bacteria which live in our gut are known as probiotics and can be obtained commercially when the body fails to preserve them.

Air

The air is composed mainly of gases, dust particles and water vapor. However, it also contains microorganisms in the form of vegetative cells, spores, fungi, algae, viruses and protozoan cysts. Air is not a medium in which microorganisms can grow, but it is the one in charge of transporting them along with the particulate material. However, the amount of microorganisms in the air is considerably lower than can be found on land or water. Microbes in the air are responsible for the degradation of dead cells that are released from the skin of humans. If these microorganisms did not exist, the world would be full of mountains of dead skin.

Biotechnology

Biotechnology is the branch of science that deals with the manipulation of living organisms through genetic engineering. It has multiple applications in the biological sciences and depends directly on microorganisms. Microbial biotechnology is responsible for the study of genomes, which allows to improve vaccines and develop better tools for the diagnosis of diseases.

Advances in microbial biotechnology have allowed the control of pests in animals and plants, from the development of agents that catalyze pathogens and fermentation organisms. All this has allowed the bioreparation of soils and water contaminated by the agricultural processes mainly. In general, microorganisms, together with biotechnology, have allowed the development of alternative energy sources, Biofuels, Bioalcohol and research for the field of agriculture.

Agriculture

The microorganisms that live in the soil allow to improve the Agricultural productivity. Humans naturally use organisms to develop fertilizers and biopesticides. The objective of the development of these substances is to contribute to plant growth and control pests, weed growth and other diseases.

These microorganisms present in the soil allow plants to absorb more nutrients as energy sources needed to live. The plants, in turn, deliver their waste to the microorganisms so that they feed on them and generate biofertilizers. The agricultural industry has used microorganisms during the last hundred years for the generation of biofertilizers and biopesticides. In this way, plant foods can be grown in a controlled and safe manner, blocking potential threats to the environment and contributing to the acceleration of natural processes such as the release of nitrogen from the soil.

Evolution

Life as it is known today exists thanks to the evolution of millions of microorganisms that changed the structure of the world and gave way to complex life forms. These microorganisms are known as Cyanobacteria and were responsible for the development of aerobic conditions in the primitive soil, allowing the process of photosynthesis to be possible. This change in conditions led to the development of life and its evolution over millions of years. Bacteria are unicellular organisms that developed millions of years ago. Some theories suggest that, thanks to the global cooling process, a series of complex chemical reactions took place in the water. For millions of years these chemical reactions allowed bacteria to develop nucleic acid, and protein, taking the form of more complex particles. Eventually, these new primitive particles joined and gave way to the formation of cells that later became new forms of life.

Environment

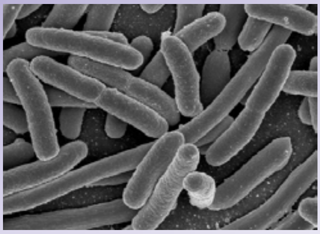
Microorganisms are present anywhere in the body. Biosphere and their presence affects the environment in which they coexist. These effects of microorganisms in the environment can be beneficial, harmful or neutral according to the standards imposed by human observation.

The benefits derived from the action of microorganisms take place thanks to their metabolic activities in the medium. Activities they perform in relation to plants and animals, from which they take their energy to carry out biological processes. In this way, there is the concept of bioreparation, consisting of the elimination of materials toxic to the environment, such as oil spills in water or land. The processes of **biofiltration** and transformation of toxic materials are only possible by the action of microorganisms, since the majority of particles that pollute the environment can be decomposed by different types of bacteria.

Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants.

Body Balance

The more complex communities of microorganisms located in the human body have the power to balance or unbalance it. For this reason, compounds such as probiotics have been developed to administer necessary doses of beneficial bacteria that allow the regulation of internal processes of the body. There are biological therapies in which intestine material is inserted from one patient into another in order to regulate the number of bacteria contained in the intestine. This balances the number of microorganisms needed to perform vital body processes.



CORE CONCEPT: BIOLOGICAL CLASSIFICATION OF MICROORGANISM

The term microbe is short for microorganism, which means small organism. To help people understand the different types of microbes, they are grouped or classified in various ways. Microbes are very diverse and represent all the great kingdoms of life. In fact, in terms of numbers, most of the diversity of life on Earth is represented by microbes.

Microscopic organisms, commonly known as microorganisms or microbes, are found all around us and even inside our bodies. The category 'Microbes' includes a massive range of organisms including bacteria, fungi, viruses, algae, archaea and protozoa. Some of these, such as bacteria and fungi, are well known, but others such as archaea much less so. Since the dawn of civilization, there have been many attempts to classify

Microorganisms used include viruses, bacteria, fungi and protozoans. These may cause disease, or may compete with or otherwise limit the target organism.

living organisms. It was done instinctively not using criteria that were scientific but borne out of a need to use organisms for our own use – for food, shelter and clothing. Aristotle was the earliest to attempt a more scientific basis for classification. He used simple morphological characters to classify plants into trees, shrubs and herbs. He also divided animals into two groups, those which had red blood and those that did not.

In Linnaeus' time a Two Kingdom system of classification with Plantae and Animalia kingdoms was developed that included all plants and animals respectively. This system was used till very recently. This system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organisms. Classification of organisms into plants and animals was easily done and was easy to understand, but, a large number of organisms did not fall into either category. Hence the two kingdom classification used for a long time was found inadequate. A need was also felt for including, besides gross morphology, other characteristics like cell structure, nature of wall, mode of nutrition, habitat, methods of reproduction, evolutionary relationships, etc. Classification systems for the living organisms have hence, undergone several changes over time. Though plant and animal kingdoms have been a constant under all different systems, the understanding of what groups/organisms be included under these kingdoms have been changing; the number and nature of other kingdoms have also been understood differently by different scientists over time.

Table 1: Characteristics of the Five Kingdoms

Characters	Five Kingdoms				
	Monera	**Protista**	**Fungi**	**Plantae**	**Animalia**
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellulosic (Polysaccharide + amino acid)	Present in some	Present (without cellulose)	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation.	Cellular	Cellular	Multiceullar/ loose tissue	Tissue / organ	Tissue /organ / organ system
Mode of nutrition	Autotrophic (chemosyn-thetic and photosynthetic) and Hetero-trophic (sapro-phytic / para-sitic)	Autotrophic (Photosyn-thetic) and Hetero-trophic	Heterotrophic (Saprophytic/ Parastitic)	Autotrophic (Photosyn-thetic)	Heterotrophic (Holozoic / Saprophytic etc.)

R.H. Whittaker (1969) proposed a Five Kingdom Classification. The kingdoms defined by him were named Monera, Protista, Fungi, Plantae and Animalia. The main criteria for classification used by him include cell structure, thallus organization, and mode of nutrition, reproduction and phylogenetic relationships. Table 1 gives a comparative account of different characteristics of the five kingdoms. Let us look at this five kingdom classification to understand the issues and considerations that influenced the classification system. Earlier classification systems included bacteria, blue green algae, fungi, mosses, ferns, gymnosperms and the angiosperms under 'Plants'. The character that unified this whole kingdom was that all the organisms included had a cell wall in their cells. This placed together groups which widely differed in other characteristics. It brought together the prokaryotic bacteria and the blue green algae with other groups which were eukaryotic. It also grouped together the unicellular organisms and the multicellular ones, say, for example, Chlamydomonas and Spirogyra were placed together under algae. The classification did not differentiate between the heterotrophic group – fungi, and the autotrophic green plants, though they also showed a characteristic difference in their walls composition – the fungi had chitin in their walls while the green plants had a cellulosic **cell wall.**

Cell wall is a structural layer surrounding some types of cells, just outside the cell membrane. It can be tough, flexible, and sometimes rigid.

When such characteristics were considered, the fungi were placed in a separate kingdom – Kingdom Fungi. All prokaryotic organisms were grouped together under Kingdom Monera and the unicellular eukaryotic organisms were placed in Kingdom Protista. Kingdom Protista has brought together Chlamydomonas, Chlorella (earlier placed in Algae within Plants and both having cell walls) with Paramoecium and Amoeba (which were earlier placed in the animal kingdom which lack cell wall). It has put together organisms which, in earlier classifications, were placed in different kingdoms. This happened because the criteria for classification changed. This kind of changes will take place in future too depending on the improvement in our understanding of characteristics and evolutionary relationships. Over time, an attempt has been made to evolve a classification system which reflects not only the morphological, physiological and reproductive similarities, but is also phylogenetic, i.e., is based on evolutionary relationships. In this section we will study characteristics of Kingdoms Monera, Protista and Fungi of the Whittaker system of classification.

Kingdom Monera

Bacteria are the sole members of the Kingdom Monera. They are the most abundant micro-organisms. Bacteria occur almost everywhere. Hundreds of bacteria are present in a handful of soil. They also live in extreme habitats such as hot springs, deserts, snow and deep oceans where very few other life forms can survive. Many of them live in or on other organisms as parasites. Bacteria are grouped under four categories based on their shape: the spherical Coccus (pl.: cocci), the rod-shaped Bacillus (pl.: bacilli), the comma-shaped Vibrium (pl.: vibrio) and the spiral Spirillum (pl.: spirilla) (Figure 1).

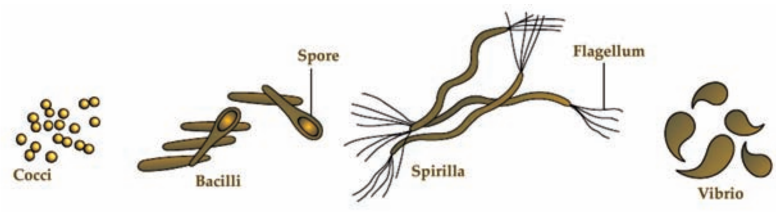


Figure 1: Bacteria of different shapes.

Though the bacterial structure is very simple, they are very complex in behavior. Compared to many other organisms, bacteria as a group show the most extensive metabolic diversity. Some of the bacteria are autotrophic, i.e., they synthesize their own food from inorganic substrates. They may be photosynthetic autotrophic or chemosynthetic autotrophic. The vast majority of bacteria are heterotrophs, i.e., they do not synthesize their own food but depend on other organisms or on dead organic matter for food.

Archaebacteria

These bacteria are special since they live in some of the harshest habitats such as extreme salty areas (halophiles), hot springs (thermoacidophiles) and marshy areas (methanogens). Archaebacteria differ from other bacteria in having a different cell wall structure and this feature is responsible for their survival in extreme conditions. Methanogens are present in the gut of several ruminant animals such as cows and buffaloes and they are responsible for the production of methane (**biogas**) from the dung of these animals.

Biogas typically refers to a mixture of different gases produced by the breakdown of organic matter in the absence of oxygen.

Eubacteria

There are thousands of different eubacteria or ‘true bacteria’. They are characterized by the presence of a rigid cell wall, and if motile, a flagellum. The cyanobacteria (also referred to as blue-green algae) have chlorophyll a similar to green plants and are photosynthetic autotrophs (Figure 2). The cyanobacteria are unicellular, colonial or filamentous, freshwater/marine or terrestrial algae. The colonies are generally surrounded by gelatinous sheath. They often form blooms in polluted water bodies. Some of these organisms can fix atmospheric nitrogen in specialized cells called heterocysts, e.g., Nostoc and Anabaena. Chemosynthetic autotrophic bacteria oxidise various inorganic substances such as nitrates, nitrites and ammonia and use the released energy for their ATP production. They play a great role in recycling nutrients like nitrogen, phosphorous, iron and sulphur.

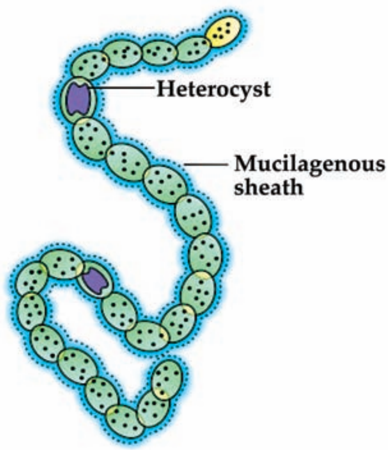


Figure 2: A filamentous blue-green algae – Nostoc.

Heterotrophic bacteria are the most abundant in nature. The majority are important decomposers. Many of them have a significant impact on human affairs. They are helpful in making curd from milk, production of antibiotics, fixing nitrogen in legume roots, etc. Some are pathogens causing damage to human beings, crops, farm animals and pets. Cholera, typhoid, tetanus, citrus canker are well known diseases caused by different bacteria. Bacteria reproduce mainly by fission (Figure 3). Sometimes, under unfavourable conditions, they produce spores. They also reproduce by a sort of sexual reproduction by adopting a primitive type of DNA transfer from one bacterium to the other. The Mycoplasma are organisms that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen. Many mycoplasma are pathogenic in animals and plants.

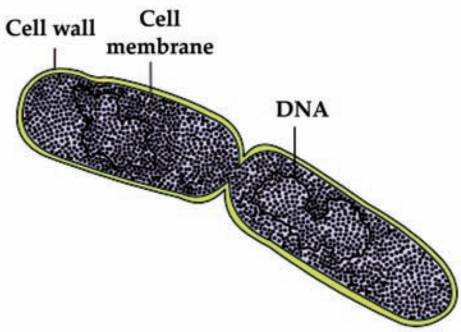


Figure 3: A dividing bacterium.

Difference between Archaebacteria and Eubacteria

According to scientists, there are six differentiated kingdoms into which living things can be divided. The eubacteria and archaebacteria are probably the least known of this categorization. Eubacteria and archaebacteria are two very different kinds of bacteria, each with their own identities and use in our daily lives.

Bacteria do much more than help keep our environment clean. Bacteria also help produce many of the foods we eat every day. They even help make important medicines.

Archaebacteria are one of the oldest of organisms found on planet earth. They are composed of a single cell and are called prokaryotes. Interestingly, archaebacteria are usually found under extremes of conditions. This is not surprising considering the fact that they were one of the first organisms on earth- at a time when earth was a planet with poisonous gases and unbearable heat. The archaebacteria was one of the only organisms that could survive in that unfriendly condition.

The Eubacteria are the common ones we refer to when we are generally talking about bacteria. They are complex in structure and are found under neutral conditions. You can find Eubacteria in a variety of conditions, for instance, you can find them in the human body, in some foods (yikes!) and practically everywhere around us. The following table represents the basic structural and characteristic differences with relation to archaebacteria and eubacteria.

Points of Distinction	**Archaebacteria**	**Eubacteria**
RNA Polymerase Composition	Core architecture of ten subunits	Core architecture of four subunits
Pathogenicity	Absent	Some species are pathogenic
Relationship with Other Organisms	Mutual, and commensal	Mutual, predatory, and pathogenic
Cellular Structure	Peptidoglycan is absent, and ether-liked lipids are in the form of branched chains	Peptidoglycan is present, and ester-linked lipids are in the form of a straight chain
Role in Nature	Unknown	Significant role in nutrient recycling

Although initially believed to have belonged to the same group of unicellular prokaryotes as Eubacteria, the Archaea kingdom was later discovered, and its constituent members were identified as a separate group of microorganisms altogether, based upon their rRNA gene sequences. After much research, it was finally concluded that both eubacteria and archaebacteria belong to two separate microbiological domains, which probably descended from a common unicellular ancestor.

Kingdom Protista

All single-celled eukaryotes are placed under Protista, but the boundaries of this kingdom are not well defined. What may be ‘a photosynthetic protistan’ to one biologist may be 'a plant' to another. In this book we include Chrysophytes, Dinoflagellates, Euglenoids, Slime moulds and Protozoans under Protista. Members of Protista are primarily aquatic. This kingdom forms a link with the others dealing with plants, animals and fungi. Being eukaryotes, the protistan cell body contains a well-defined nucleus and other membrane-bound organelles. Some have flagella or cilia. Protists reproduce asexually and sexually by a process involving **cell fusion** and zygote formation.

Cell fusion is an important cellular process in which several uninuclear cells combine to form a multinuclear cell, known as a syncytium.

Chrysophytes

This group includes diatoms and golden algae (desmids). They are found in fresh water as well as in marine environments. They are microscopic and float passively in water currents (plankton). Most of them are photosynthetic. In diatoms the cell walls form two thin overlapping shells, which fit together as in a soap box. The walls are embedded with silica and thus the walls are indestructible. Thus, diatoms have left behind large amount of cell wall deposits in their habitat; this accumulation over billions of years is referred to as 'diatomaceous earth'. Being gritty this soil is used in polishing, filtration of oils and syrups. Diatoms are the chief 'producers' in the oceans.

Dinoflagellates

These organisms are mostly marine and photosynthetic. They appear yellow, green, brown, blue or red depending on the main pigments present in their cells. The cell wall has stiff cellulose plates on the outer surface. Most of them have two flagella; one lies longitudinally and the other transversely in a furrow between the wall plates. Very often, red dinoflagellates (Example: Gonyaulax) undergo such rapid multiplication that they make the sea appear red (red tides). Toxins released by such large numbers may even kill other marine animals such as fishes.

Euglenoids

Majority of them are fresh water organisms found in stagnant water. Instead of a cell wall, they have a protein rich layer called pellicle which makes their body flexible. They have two flagella, a short and a long one. Though they are photosynthetic in the presence of sunlight, when deprived of sunlight they behave like heterotrophs by predating on other smaller organisms. Interestingly, the pigments of euglenoids are identical to those present in higher plants. Example: Euglena (Figure 4a).

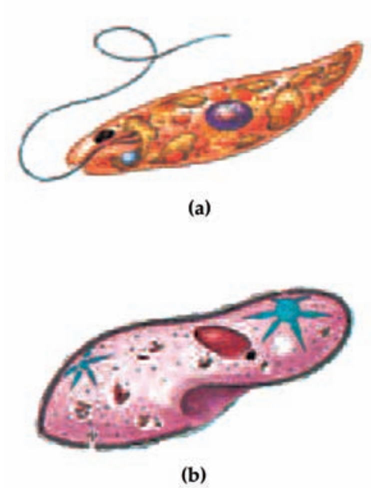


Figure 4: (a) Euglena (b) Paramoecium.

Slime Moulds

Slime moulds are saprophytic protists. The body moves along decaying twigs and leaves engulfing organic material. Under suitable conditions, they form an aggregation called plasmodium which may grow and spread over several feet. During unfavourable conditions, the plasmodium differentiates and forms fruiting bodies bearing spores at their tips. The spores possess true walls. They are extremely resistant and survive for many years, even under adverse conditions. The spores are dispersed by air currents.

Protozoans

All protozoans are heterotrophs and live as predators or parasites. They are believed to be primitive relatives of animals. There are four major groups of protozoans.

- Amoeboid protozoans: These organisms live in fresh water, sea water or moist soil. They move and capture their prey by putting out pseudopodia (false feet) as in Amoeba. Marine forms have silica shells on their surface. Some of them such as Entamoeba are parasites.
- Flagellated protozoans: The members of this group are either free-living or parasitic. They have flagella. The parasitic forms cause diseases such as sleeping sickness. Example: Trypanosoma.
- Ciliated protozoans: These are aquatic, actively moving organisms because of the presence of thousands of cilia. They have a cavity (gullet) that opens to the outside of the cell surface. The coordinated movement of rows of cilia causes the water laden with food to be steered into the gullet. Example: Paramoecium (Figure 4b).
- Sporozoans: This includes diverse organisms that have an infectious spore-like stage in their life cycle. The most notorious is Plasmodium (malarial parasite) which causes malaria, a disease which has a staggering effect on human population.

Kingdom Fungi

The fungi constitute a unique kingdom of heterotrophic organisms. They show a great diversity in morphology and habitat. When your bread develops a mould or your orange rots it is because of fungi. The common mushroom you eat and toadstools are also fungi. White spots seen on mustard leaves are due to a parasitic fungus. Some unicellular fungi, e.g., yeast are used to make bread and beer. Other fungi cause diseases in plants and animals; wheat rust-causing Puccinia is an important example. Some are the source of antibiotics, e.g., Penicillium. Fungi are cosmopolitan and occur in air, water, soil and on animals and plants. They prefer to grow in warm and humid places. Have you ever wondered why we keep food in the refrigerator? Yes, it is to prevent food from going bad due to bacterial or fungal infections.

With the exception of yeasts which are unicellular, fungi are filamentous. Their bodies consist of long, slender thread-like structures called hyphae. The network of hyphae is known as mycelium. Some hyphae are continuous tubes filled with multinucleated cytoplasm – these are called coenocytic hyphae. Others have septae or cross walls in their hyphae. The cell walls of fungi are composed of chitin and polysaccharides. Most fungi are heterotrophic and absorb soluble organic matter from dead substrates and hence are called saprophytes. Those that depend on living plants and animals are called parasites. They can also live as symbionts – in association with algae as lichens and with roots of higher plants as mycorrhiza.

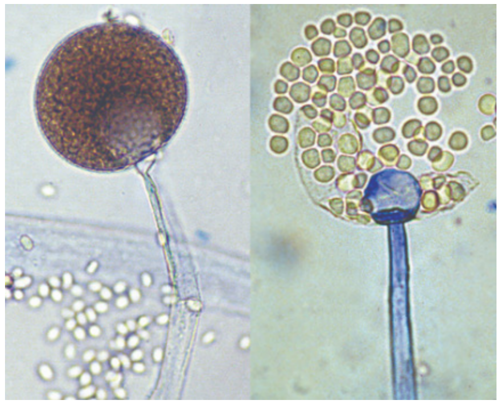
Reproduction in fungi can take place by vegetative means – fragmentation, fission and budding. Asexual reproduction is by spores called conidia or sporangiospores or zoospores, and sexual reproduction is by oospores, ascospores and basidiospores. The various spores are produced in distinct structures called fruiting bodies. The sexual cycle involves the following three steps:

- Fusion of protoplasms between two motile or non-motile gametes called plasmogamy.
- Fusion of two nuclei called karyogamy.
- Meiosis in zygote resulting in haploid spores.

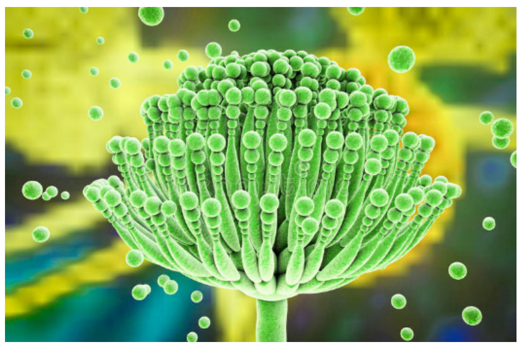
When a fungus reproduces sexually, two haploid hyphae of compatible mating types come together and fuse. In some fungi the fusion of two haploid cells immediately results in diploid cells (2n). However, in other fungi (ascomycetes

Mycelium is the vegetative part of a fungus or fungus-like bacterial colony, consisting of a mass of branching, thread-like hyphae.

and basidiomycetes), an intervening dikaryotic stage (n + n, i.e., two nuclei per cell) occurs; such a condition is called a dikaryon and the phase is called dikaryophase of fungus. Later, the parental nuclei fuse and the cells become diploid. The fungi form fruiting bodies in which reduction division occurs, leading to formation of haploid spores. The morphology of the **mycelium,** mode of spore formation and fruiting bodies form the basis for the division of the kingdom into various classes.



(a)



(b)

(c)

Figure 5: Fungi: (a) Mucor (b) Aspergillus (c) Agaricus.

Phycomycetes

Members of phycomycetes are found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites on plants. The mycelium is aseptate and coenocytic. Asexual reproduction takes place by zoospores (motile) or by aplanospores (non-motile). These spores are endogenously produced in sporangium. A zygospore is formed by fusion of two gametes. These gametes are similar in morphology (isogamous) or dissimilar (anisogamous or oogamous). Some common examples are Mucor (Figure 5a), Rhizopus (the bread mould mentioned earlier) and Albugo (the parasitic fungi on mustard).

Neurospora is a genus of Ascomycete fungi. The genus name, meaning "nerve spore" refers to the characteristic striations on the spores that .resemble axons

Ascomycetes

Commonly known as sac-fungi, the as comycetes are mostly multicellular, e.g., Penicillium, or rarely unicellular, e.g., yeast (Saccharomyces). They are saprophytic, decomposers, parasitic or coprophilous (growing on dung). Mycelium is branched and septate. The asexual spores are conidia produced exogenously on the special mycelium called conidiophores. Conidia on germination produce mycelium. Sexual spores are called ascospores which are produced endogenously in sac like asci (singular ascus). These asci are arranged in different types of fruiting bodies called ascocarps. Some examples are Aspergillus (Figure 5b), Claviceps and **Neurospora**. Neurospora is used extensively in biochemical and genetic work. Many members like morels and truffles are edible and are considered delicacies.

Basidiomycetes

Commonly known forms of basidiomycetes are mushrooms, bracket fungi or puffballs. They grow in soil, on logs and tree stumps and in living plant bodies as parasites, e.g., rusts and smuts. The mycelium is branched and septate. The asexual spores are generally not found, but vegetative reproduction by fragmentation is common. The sex organs are absent, but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes. The resultant structure is dikaryotic which ultimately gives rise to basidium. Karyogamy and meiosis take place in the basidium producing four basidiospores. The basidiospores are exogenously produced on the basidium (pl.: basidia). The basidia are arranged in fruiting bodies called basidiocarps. Some common members are Agaricus (mushroom) (Figure 5c), Ustilago (smut) and Puccinia (rust fungus).

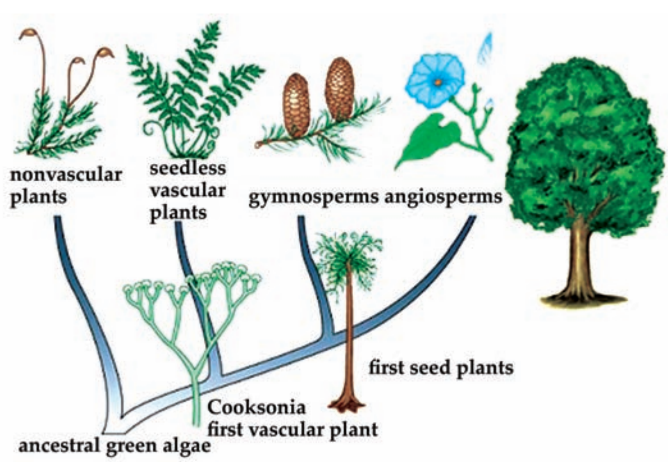
Deuteromycetes

Insectivorous plants are plants that derive some of their nutrients from trapping and consuming animals .or protozoan

Commonly known as imperfect fungi because only the asexual or vegetative phases of these fungi are known. When the sexual forms of these fungi were discovered they were moved into classes they rightly belong to. It is also possible that the asexual and vegetative stage have been given one name (and placed under deuteromycetes) and the sexual stage another (and placed under another class). Later when the linkages were established, the fungi were correctly identified and moved out of deuteromycetes. Once perfect (sexual) stages of members of dueteromycetes were discovered they were often moved to ascomycetes and basidiomycetes. The deuteromycetes reproduce only by asexual spores known as conidia. The mycelium is septate and branched. Some members are saprophytes or parasites while a large number of them are decomposers of litter and help in mineral cycling. Some examples are Alternaria, Colletotrichum and Trichoderma.

Kingdom Plantae

Kingdom Plantae includes all eukaryotic chlorophyll-containing organisms commonly called plants. A few members are partially heterotrophic such as the **insectivorous plants** or parasites. Bladderwort and Venus fly trap are examples of insectivorous plants and Cuscuta is a parasite. The plant cells have an eukaryotic structure with prominent chloroplasts and cell wall mainly made of cellulose. Plantae includes algae, bryophytes, pteridophytes, gymnosperms and angiosperms.



Life cycle of plants has two distinct phases – the diploid sporophytic and the haploid gametophytic – that alternate with each other. The lengths of the haploid and diploid phases, and whether these phases are free– living or dependent on others, vary among different groups in plants. This phenomenon is called alternation of generation.

Kingdom Animalia

This kingdom is characterized by heterotrophic eukaryotic organisms that are multicellular and their cells lack cell walls. They directly or indirectly depend on plants for food. They digest their food in an internal cavity and store food reserves as glycogen or fat. Their mode of nutrition is holozoic – by ingestion of food. They follow a definite growth pattern and grow into adults that have a definite shape and size. Higher forms show elaborate sensory and neuromotor mechanism. Most of them are capable of locomotion. The sexual reproduction is by **copulation** of male and female followed by embryological development.

Copulation is animal sexual behavior in which a male introduces sperm into the female's body, especially directly into her reproductive tract.

Viruses, Viroids and Lichens

In the five kingdom classification of Whittaker there is no mention of some a cellular organisms like viruses and viroids, and lichens. These are briefly introduced here. All of us who have suffered the ill effects of common cold or 'flu' know what effects viruses can have on us, even if we do not associate it with our condition. Viruses did not find a place in classification since they are not truly 'living', if we understand living as those organisms that have a cell structure. The viruses are non-cellular organisms that are characterized by having an inert crystalline structure outside the living cell. Once they infect a cell they take over the machinery of the host cell to replicate themselves, killing the host. Would you call viruses living or non-living?

The name virus that means venom or poisonous fluid was given by Pasteur. D.J. Ivanowsky (1892) recognized certain microbes as causal organism of the mosaic disease of tobacco (Figure 6a). These were found to be smaller than bacteria because they passed through bacteria-proof filters. M.W. Beijerinek (1898) demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants and called the fluid as Contagium vivum fluidum (infectious living fluid). W.M. Stanley (1935) showed that viruses could be crystallized and crystals consist largely of proteins. They are inert outside their specific host cell. Viruses are obligate parasites.

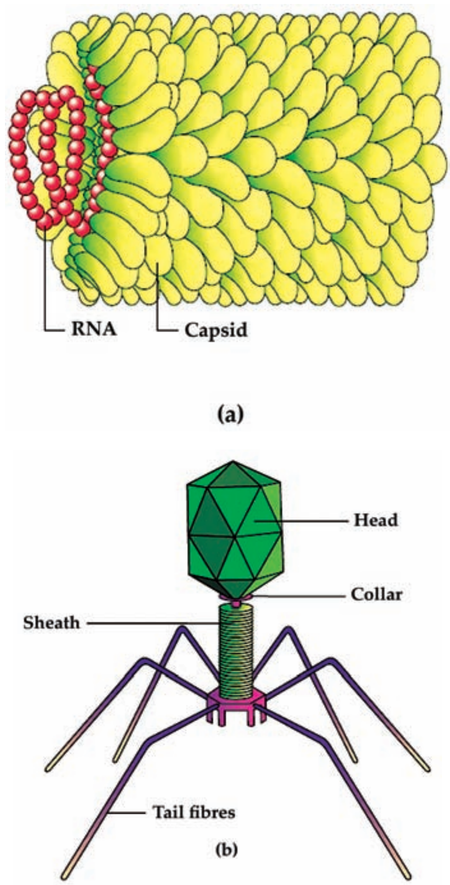


Figure 6: (a) Tobacco Mosaic Virus (TMV) (b) Bacteriophage.

In addition to proteins, viruses also contain genetic material that could be either RNA or DNA. No virus contains both RNA and DNA. A virus is a nucleoprotein and the genetic material is infectious. In general, viruses that infect plants have single stranded RNA and viruses that infect animals have either single or double stranded RNA or double stranded DNA. Bacterial viruses or bacteriophages (viruses that infect the bacteria) are usually double stranded DNA viruses (Figure 6b). The protein coat called capsid made of small subunits called capsomeres, protects the nucleic acid. These capsomeres are arranged in helical or polyhedral geometric forms. Viruses cause diseases like mumps, small pox, herpes and influenza. AIDS in humans is also caused by a virus. In plants, the symptoms can be mosaic formation, leaf rolling and curling, yellowing and vein clearing, dwarfing and stunted growth.

- Viroids: In 1971, T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease. It was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid. The RNA of the viroid was of low molecular weight.

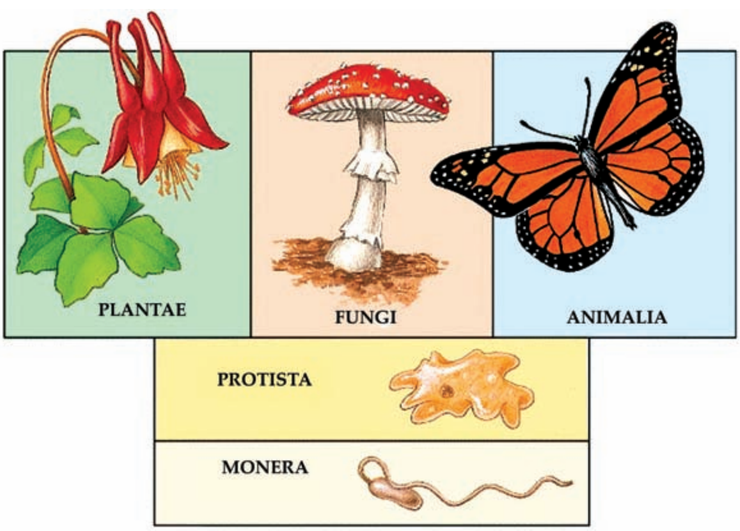
- Lichens: Lichens are symbiotic associations i.e. mutually useful associations, between algae and fungi. The algal component is known as phycobiont and fungal component as mycobiont, which are autotrophic and heterotrophic, respectively. Algae prepare food for fungi and fungi provide shelter and absorb mineral nutrients and water for its partner. So close is their association that if one saw a lichen in nature one would never imagine that they had two different organisms within them. Lichens are very good pollution indicators – they do not grow in polluted areas.

Merits and Demerits of Five Kingdom Classification

The merits and demerits of five kingdom classification are given below.

Merits of R.H Whittaker 5 Kingdom Classification:

- This system of classification is more scientific and natural.
- It is the most accepted system of modern classification as the different groups of animals are placed phylogenetically.
- The prokaryotes are placed in a separate kingdom as they differ from all other organisms in their organization.
- As the unicellular organisms are placed under the kingdom Protista, it has solved many problem related to the position of organisms like euglena.
- The fungi totally differ from other primitive eukarytotes hence, placing the group fungi in a status of kingdom is justifiable.
- The kingdom Plantae and Animalia shows the phylogeny of different life styles, in the five kingdom classification, they are more homogeneous group than the two kingdom classification.
- This system of classification clearly indicates cellular organization and modes of nutrition, the character which have appeared very early in the evolution of life.

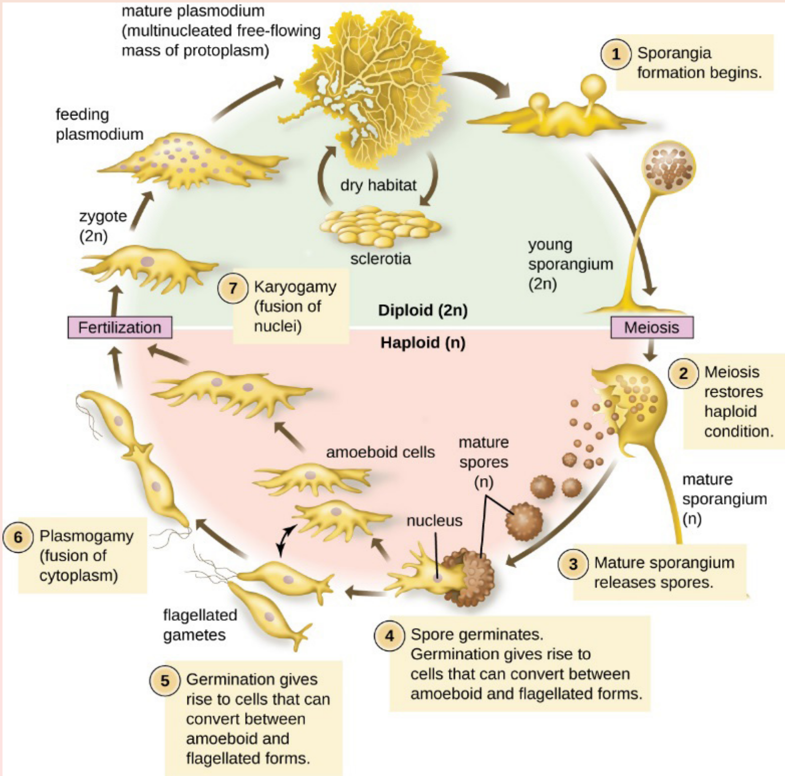


Demerits of R.H. Whittaker 5 Kingdom Classification:

- This system of classification has drawbacks with reference to the lower forms of life.
- The Kingdom of Monerans and the Protists include diverse, heterogenous forms of life. In both the kingdoms there are autoptrophic and hetertrophic organisms. They also include organisms which have cells with cell wall and cells without cell wall.
- All the organisms of these three kingdoms do not originate from a single ancestor.
- Multicellular organisms have originated from protists several times.
- Organisms like the unicellular green algae like volvox and chlamydomonas have not been included under the Kingdom Protista because of their resemblance to other green algae.
- The general organization of the slime moulds are completely different from the members of protists.
- In this system of classification viruses have not been given proper place.

CONCEPT OF LIFE CYCLE OF PLASMODIAL SLIME MOLDS

The life cycle of plasmodial slime molds is best classified as diplontic: the "multicellular" (actually just multinucleate) phase is diploid. Haploid cells that germinate from spores (amoebae or biflagellate swarm cells) do not grow until after they have fused with another haploid cell. In some myxomycetes, amoebae or swarm cells produced from the same parent plasmodium can fuse together to form a new plasmodium. This is called homothallism (homo- meaning same, thallus). In other myxomycetes, these gametes must be from different individuals (heterothallism, hetero- meaning other). The discovery of different mating types in myxomycetes, as well as the genes that determine mating type, was made by O'Neil Ray Collins.



SUMMARY

- Microorganisms, or microbes, are a diverse group of minute, simple forms of life that include bacteria, algae, fungi, protozoa, and viruses. Microorganisms are too small to be seen with the naked eye and are normally viewed by means of a microscope. The study of microorganisms is called microbiology.
- Common microorganisms include bacteria, which are tiny single-celled organisms that are neither animals nor plants. Bacteria have a variety of shapes, including spheres, rods, and spirals, and they often appear in pairs, chains, groups of four, or clusters.
- Viruses are smaller than bacteria. They are not technically living things because they cannot survive on their own; they can multiply only in living cells of animals, plants, or bacteria. Once inside a cell, a virus can reproduce itself, and the copies can leave the original cell and infect other cells.
- Fungi, once considered to be plants, are now placed in their own scientific kingdom. This group includes mushrooms, molds, mildews, and yeasts, only some of which are microorganisms. Along with bacteria, fungi are responsible for the decay of organic matter and the release into the atmosphere of carbon, oxygen, nitrogen, and phosphorus.
- Algae are most commonly found in water, such as oceans, rivers, lakes, and marshes, but some species live in soil or on leaves, wood, and stones. Only some species are so small that they can be seen only through a microscope. Algae are plantlike in that they make their own food through a process called photosynthesis.
- Protozoa, or protozoans, are single-celled, eukaryotic microorganisms. Some protozoa are oval or spherical, others elongated. Still others have different shapes at different stages of the life cycle.
- Prions are unique proteins which appear to be responsible for a family of illnesses collectively referred to as Transmissible Spongiform Encephalopathies (TSEs), including scrapie in sheep, kuru in humans, and "mad cow disease" in cows.
- Lichens represent a form of symbiosis, namely, an association of two different organisms wherein each benefits. A lichen consists of a photosynthetic microbe (an alga or a cyanobacterium) growing in an intimate association with a fungus.
- Archaebacteria are one of the oldest of organisms found on planet earth. They are composed of a single cell and are called prokaryotes.
- The Eubacteria are the common ones we refer to when we are generally talking about bacteria. They are complex in structure and are found under neutral conditions.

MULTIPLE CHOICE QUESTIONS

1. Which organisms are microscopic and dependent on host organisms for reproduction?

a. Algae
b. Protozoa
c. Viruses
d. Bacteria

2. Bdellovibrios come under which group of microorganisms?

a. Viruses
b. Bacteria
c. Fungi
d. Algae

3. Growth of microbes in a solid media is identified by the formation of?

a. pellicle at the top of media
b. colonies
c. sediment at the bottom
d. turbidity

4. Which among the following are produced by microorganisms?

a. Fermented dairy products
b. Breads
c. Alcoholic beverages
d. Fermented dairy products, breads and alcoholic beverages

5. The DNA molecule of microorganisms is made up of base pairs of ________________

a. guanine-cytosine
b. adenine-thymine
c. adenine-cytosine
d. guanine-cytosine and adenine-thymine

Answers

1. (c) 2. (b) 3. (b) 4. (d) 5. (d)

REVIEW QUESTIONS

1. Write the definition of microorganism.
2. Explain the benefits of microorganisms for the human being.
3. State two economically important uses of:
   a. Heterotrophic bacteria
   b. Archaebacteria
4. What is the nature of cell-walls in diatoms?
5. Differentiate between archaebacteria and eubacteria.
6. What are the characteristic features of Euglenoids?
7. How are viroids different from viruses?
8. Write the four major groups of Protozoa.

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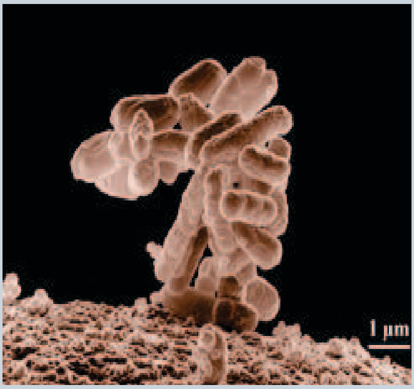
CHAPTER 6

BACTERIAL MORPHOLOGY AND CELLULAR STRUCTURES



All organisms have morphology. Morphology refers to the structural features that have evolved to help the organism interact favorably with the environment. Bacterial morphology includes the shape, arrangement, and size of the cells.

After studying this chapter, you will understand the core concepts of:

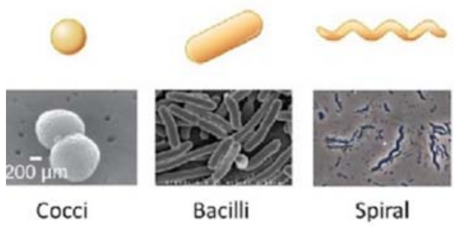
- Definition of Bacteria
- Structure of Bacteria

INTRODUCTION

Members of the domain Bacteria are single-celled prokaryotes that, for 3.5 billion years, have adapted to changes in the environment by evolving a variety of structural characteristics for better survival and reproduction. These physical characteristics are used to identify bacterial species for the purposes of scientific classification and treatment of bacterial infection and disease.

The term morphology comes from the Greek for *form* and means the form and structure of living organisms. The morphology of bacteria describes the external appearance of bacterial cells including shape, arrangement, and size.

The shape of bacterial cells is inherited and subject to the cell wall and cytoskeleton structure. The bacterial shape plays a role in how well bacteria perform life functions such as obtaining nutrition and avoiding predators. In pathogenic bacteria, the shape contributes to the ability to cause disease. The most common shapes of bacteria are cocci, bacilli, and spiral.



Bacteria multiply via binary fission, a reproductive cell division that results in two identical daughter cells. These daughter cells usually separate completely and sometimes remain attached. The type of attachment after binary fission is a morphological feature called arrangement. There are many types of arrangement in bacterial cells, each characterized by the number of cells and pattern in the grouping. The most common types of arrangement are single, diplo arrangement, tetrad, sarcina, staphylo- arrangement, and strepto-arrangement.

Arrangement and shape are often combined in bacterial species descriptions of cell morphology. For example, cocci that are arranged in pairs are identified as diplococci. Cocci that are attached in clusters or chains are identified as streptococci and staphylococci respectively.

The size of bacteria is determined by the need for a high surface-to-volume ratio which maximizes the transport of nutrients and waste products across the cell membrane. Escherichia coli is a well-studied bacterial species and has a cell volume range of 0.4-3 fL and dimensions measuring 0.2-3 um in diameter and 2-8 um in length. One of the smallest known bacteria, Candidatus actinomarina, is 100 times smaller than E. coli. The largest known bacteria, Thiomargarita namibiensis is 100 times larger than the size of an E. coli cell. The largest T. namibiensis cells are large enough to be seen with the naked eye.

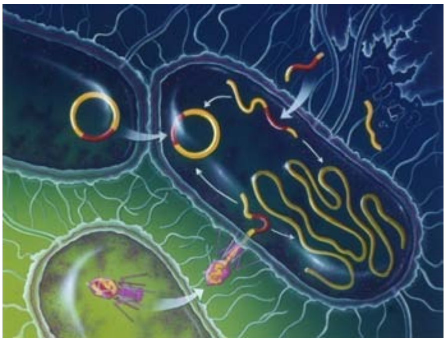
CORE CONCEPT: DEFINITION OF BACTERIA

Bacteria is a type of biological cell. They constitute a large domain of prokaryotic microorganisms. Typically a few micrometres in length, bacteria have a number of shapes, ranging from spheres to rods and spirals. Bacteria were among the first life forms to appear on Earth, and are present in most of its habitats.

Bacteria are single-celled microorganisms with prokaryotic cells, which are single cells that do not have organelles or a true nucleus and are less complex than eukaryotic cells. Bacteria with a capital B refers to the domain Bacteria, one of the three domains of life. The other two domains of life are Archaea, members of which are also single-celled organisms with prokaryotic cells, and Eukaryota. Bacteria are extremely numerous, and the total biomass of bacteria on Earth is more than all plants and animals combined.

Eukaryotes are organisms whose cells have a nucleus enclosed within membranes, unlike Prokaryotes. Eukaryotes belong to the domain Eukaryota or Eukarya.

Bacteria first arose on Earth approximately 4 billion years ago, and they were the first forms of life on Earth. For 3 billion years, bacteria and archaea were the most prevalent kinds of organisms on Earth. Multicellular eukaryotes did not appear until around 1.6-2 billion years ago. Eukaryotic cells, which make up all protists, fungi, animals, and plants, also contain what was once bacteria; it is thought that the mitochondria in eukaryotes, which produce energy through cellular respiration, and chloroplasts in plants and algae, which produce energy through photosynthesis, both evolved from bacteria that got taken up into cells in an endosymbiotic (mutually benefiting) relationship that became permanent over time.



Types of Bacteria

The cell wall also makes Gram staining possible. Gram staining is a method of staining bacteria involving crystal violet dye, iodine, and the counterstain safranin. Many bacteria can be classified into one of two types: gram-positive, which show the stain and appear violet in color under a microscope, and gram-negative, which only show the counterstain, and appear red. Gram-positive bacteria appear violet because they have thick cell walls that trap the crystal violet-iodine complex. The thin cell walls of gram-negative bacteria cannot hold the violet-iodine complex, but they can hold safranin. This makes gram-negative bacteria appear red under Gram staining. Gram staining is used for general identification of bacteria or to detect the presence of certain bacteria; it cannot be used to identify bacteria in any specific way, such as at a species level. Examples of gram-positive bacteria include the genera Listeria, Streptococcus, and Bacillus, while gram-negative bacteria include Proteobacteria, green sulfur bacteria, and cyanobacteria.

Characteristics of Bacteria

The major characteristics of Bacteria are based on their size, shape and arrangements

Size

The unit of measurement used in bacteriology is the micron (micrometer)

1 micron (μ) or micrometer (μm)	one thousandth of a millimeter
1 millimicron (mμ) or nanometer (nm)	one thousandth of a micron or one millionth of a millimeter
1 Angstrom unit (Å)	one tenth of a nanometer

The limit of resolution with the unaided eye is about 200 microns. Bacteria are smaller which can be visualized only under magnification. Bacteria of medical importance generally measure 0.2 – 1.5 μm in diameter and about 3-5 μm in length.

Microscopy

The morphological study of bacteria requires the use of microscopes. Microscopy has come a long way since Leeuwenhoek first observed bacteria using hand-ground lenses.

The types of microscope are

- Light or optical microscope
- Phase contrast microscope
- Dark field/ Dark ground microscope
- Electron microscope

Light or Optical Microscope

They are of two types namely Simple and compound microscope

- Simple Microscope consists of a single lens. A hand lens is an example of a simple Microscope.
- Compound Microscope consists of two or more lenses in series. The image formed by the first lens is further magnified by another lens.

Bacteria may be examined under the compound microscope, either in the living state or after fixation and staining. Examination of wet films or hanging drops indicates the shape, arrangements, motility and approximately size of the cells. But due to lack of contrast details cannot be appreciated.

Phase Contrast Microscope

This imposes the contrast and makes evident the structure within the cells that differ in thickness or refractive index. The difference in the refractive index between bacteria cells and the surrounding medium makes them clearly visible. Retardation, by a fraction of a wavelength, of the rays of light that pass through the object, compared to the rays passing through the surrounding medium, produces phase difference between the two types of rays.

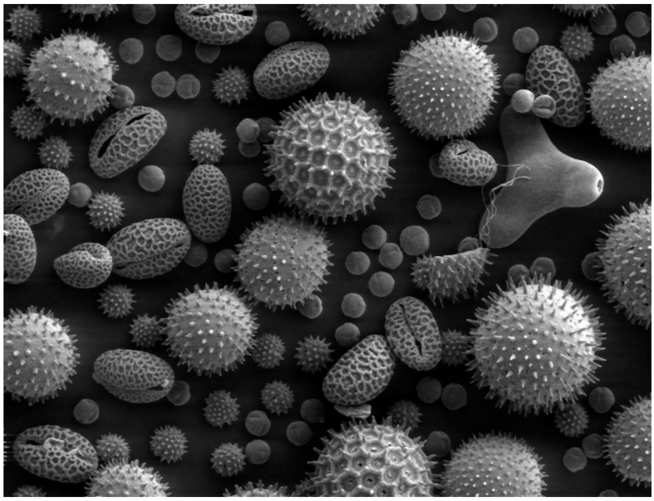
Dark field / Dark ground microscope

Another method of improving the contrast is the dark field microscope in which reflected light is used instead of the transmitted light used in the ordinal microscope. The contrast gives an illusion of increased resolution, so that very slender organisms such as **spirochete**, not visible under ordinary illumination, can be clearly seen under the dark field microscope.

Spirochete is a member of the phylum Spirochaetes, which contains distinctive diderm bacteria, most of which have long, helically coiled cells.

Electron Microscope

Beams of electron are used instead of beam of light, used in light microscope. The object which is held in the path of beam scatters the electrons and produces an image which is focused on a fluorescent viewing screen. Gas molecules scatter electron, therefore it is necessary to examine the object in a vacuum.



Stained Preparations

Live bacteria do not show the structural under the light microscope due to lack of contrast. Hence staining techniques are used to produce color contrast. Routine methods of staining of bacteria involve dying and fixing smears procedures that kill them. Bacteria have an affinity to basic dyes due to acidic nature of their protoplasm.

The commonly used staining techniques are:

Simple Stains

Dyes such as methylene blue or basic fuchsin are used for simple staining. They provide color contrast, but impart the same color to all bacteria.

Negative Staining

Bacteria are mixed with dyes such as Indian ink or nigrosin that provide a uniformly colored background against which the unstained bacteria stand out in contrast. Very slender bacteria like spirochetes that cannot be demonstrated by simple staining methods can be viewed by negative staining.

Impregnation Methods

Cells and structures too thin to be seen under ordinary microscope may be rendered visible if they are impregnated with silver on the surface. These are used for demonstration of spirochetes and bacterial flagella.

Differential Stains

These stains impart different colors to different bacteria or bacterial structures, the two most widely used differential stains are the Gram stain and Acid fast stain. The gram stain was devised by histologist Christian Gram as a method of staining bacteria in tissues.

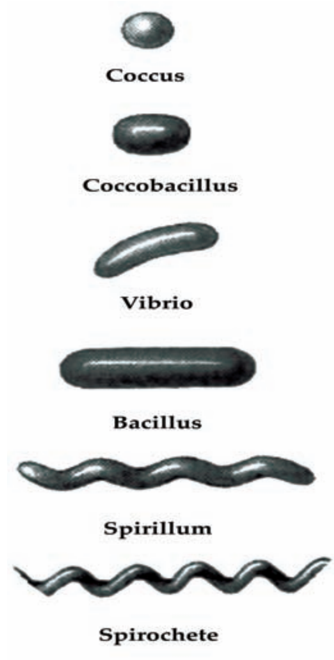
Gram positive cells are simpler chemical structure with a acidic protoplasm. It has a thick peptidoglycan layer. Teichoic acids are intertwined among the peptidoglycan and the teichoic acids are the major surface antigen determinants Gram negative cells are more complex, they are rich in lipids. The membrane is bilayered as phospholipids, proteins and lipopolysaccharide. **Lipopolysaccharides (LPS)** are also known as endotoxin. Gram negative cells have a peptidoglycan layer which is thin and formed by just one or two molecules. No Teichoic acids are found in the cell wall of Gram negative bacteria. The Outer membrane has Lipopolysaccharide channels with porins which transfer the solutes across. Lipoprotein cross link outer membrane and peptidoglycan layer Gram reaction may be related to the permeability of the bacterial cell wall and cytoplasmic membrane to the dye-iodine complex, the Gram-negative, but not the Gram-positive cells, permitting the outflow of the complex during decolourisation. Gram staining is an essential procedure used in the identification of bacteria and is frequently the only method required for studying their morphology. The acid fast stain was discovered by Ehrlich, who found that after staining with aniline dyes, tubercle bacilli resist decolourisation with acids.

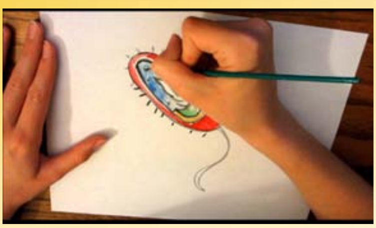
Lipopolysaccharides (LPS) are important outer membrane components of gram-negative bacteria.

Shape of the Bacteria

Depending on their shape, bacteria are classified into several varieties:

- Cocci (from kokkos meaning berry) are spherical or oval cells
- Bacilli (from baculus meaning rod) are rod shaped cells
- Vibrios are comma shaped curved rods and derive their name from their characteristics vibratory motility.
- Spirilla are rigid spiral forms.
- Spirochetes (from speira meaning coil and chaite meaning hair) are flexuous spiral forms
- Actinomycetes are branching filamentous bacteria, so called because of a fancied resemblance to the radiating rays of the sun when seen in tissue lesions (from actis meaning ray and mykes meaning fungus)
- Mycoplasmas are bacteria that are cell wall deficient and hence do not possess a stable morphology. They occur as round or oval bodies and as interlacing filaments.





CORE CONCEPT: STRUCTURE OF BACTERIA

It is a gel-like matrix composed of water, enzymes, nutrients, wastes, and gases and contains cell structures such as ribosomes, a chromosome, and plasmids. The cell envelope encases the cytoplasm and all its components. Unlike the eukaryotic (true) cells, bacteria do not have a membrane enclosed nucleus.

Bacteria are pokaryotic in their organization. The bacterial cell is surrounded by a capsule which encloses the nuclear structure or incipient nucleus. They do not have membrane-bounded organelles or a well organized nucleus and their cell wall composition is entirely different. Beneath the capsule is a cell wall which gives shape to the cell. Cell wall is very rigid and plays a vital role in bacterial biology. It consists of lipids, carbohydrates, proteins, phosphorus and other inorganic substances. Here the rigidity is due to a unique polymer, the mucopeptide, which is based upon a backbone of alternating molecules of N-acetyl glucosamine and N-acetyl-muramic acid. Attached to the muramic acid are short chain molecules of aminoacids which are cross linked to other muramic acid molecules in adjacent chains. This cross linking enables the complex to form a rigid network. At times the slime of cell wall becomes thick, gummy and mucilaginous this is known as capsule. The capsule is a resistant covering against the unfavorable conditions. Capsules and slime layers of certain bacteria are made up of polysoccharides, a material composed of dextrans, levans, etc. and appear like plant gums. Capsules or slime of certain other bacteria are composed of polypeptides and peptides.

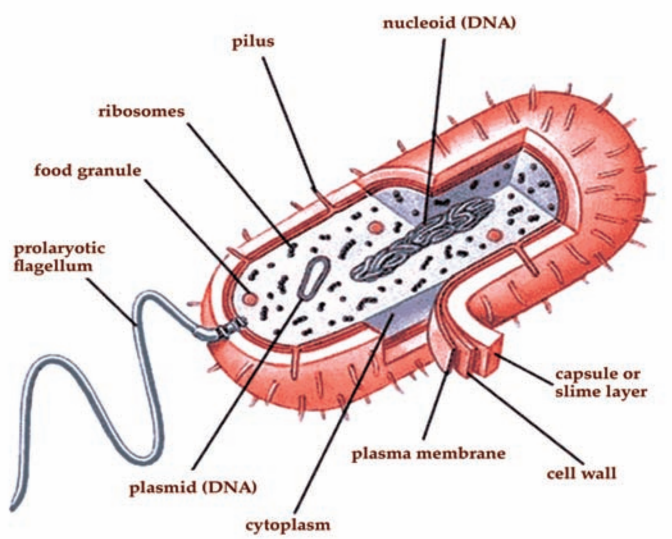
The cytoplasm is granular, viscous and lies between the nuclear material and **plasma membrane**. The protoplasm of bacterial cell is similar to that of other living organisms. The moisture content is 70-85%, and is colloidal in nature. The cytoplasm is densely packed with ribosomes of 70S. The definite components of nucleus like chromonemata, nucleolus and nuclear membrane are not found in the bacterial nucleus.

Plasma membrane is a biological membrane that separates the interior of all cells from the outside environment. It consists of a lipid bilayer with embedded proteins.

The difficulty of demonstrating the nuclear material in bacterial cells is due to the extreme basophily of the cells because of the presence of RNA in the cytoplasm. Most of the bacteriologists now believe that the nucleus in bacteria is present in the cell as a discrete body. The chromatinic structures of bacteria can be distinguished from the cytoplasm of the cell. These structures increase in size and usually give rise to short rods which multiply by splitting in a lengthwise manner. More recently the bacteriologists have demonstrated the staining of a body which takes the stain of DNA.

Structure	Function
Cell Wall	Protects the cell and gives shape
Outer Membrane	Outer membrane is found only in Gram-negative bacteria, it functions as an initial barrier to the environment and is composed of lipopolysaccharide (LPS) and phospholipids.
Cell Membrane	Regulates movement of materials into and out of the cell; contains enzymes important to cellular respiration.
Cytoplasm	Contains DNA, ribosomes, and organic compounds required to carry out life processes
Chromosome	Carries genetic information inherited from past generations.
Plasmid	Contains some genes obtain through genetic recombination
Capsule, and slime layer	Protects the cell and assist in attaching the cell to other surfaces.
Endospore	Protects the cell against harsh environmental conditions, such as heat or drought.
Pilus (Pili)	Assist the cell in attaching to other surfaces, ‘which is important for genetic recombination.
Flagellum	Moves the cell

Bacterial cell has a single chromosome, which consists of two chains of DNA wound round each other. The chromosome is ring shaped and it is embedded in the cytoplasm. This whole body usually rests in the center of the cell. The total length of the DNA strand is proportional to its molecular weight and is often much longer than the length of the cell. Other intracellular organelles like golgibodies, nucleolus, endoplasmic reticulum, lysosomes and mitochondria are absent in bacteria. Microtubules have been reported in L-form of certain bacteria. On the other hand, extra chromosomal genetic elements such as plasmids and episomes are characteristic organelles in many bacteria.



Capsules

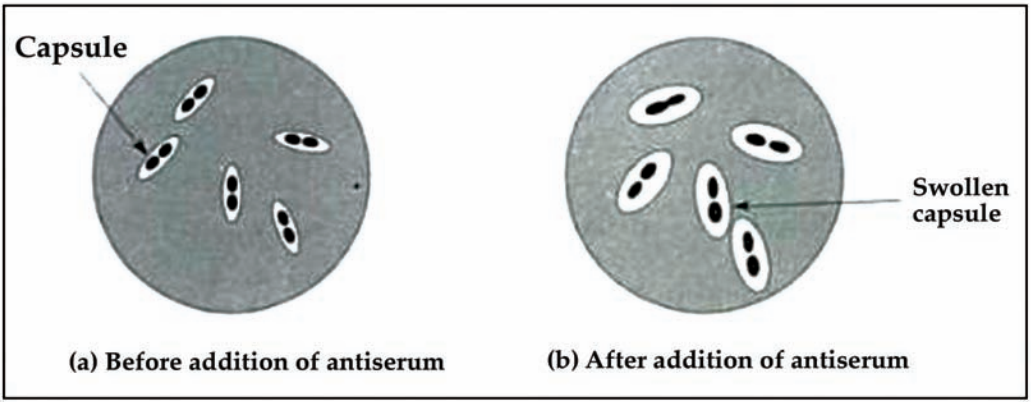
The capsule is not essential for the survival of bacteria under favorable growth conditions. However, it protects the bacteria against adverse conditions. The capsule may also function in the storage of food substances, and as a storage site for the disposal of waste material.

Capsules may be divided into two categories:

- *Macrocapsules*: They are at least 0.2 Earl thick and can be seen under the light microscope.
- *Microcapsules*: They cannot be seen under the light microscope, but can be demonstrated immunologically.

The capsule is secreted by the bacterial cell and is usually composed of polysaccharides or disaccharides, but is sometimes made up of polypepiides. Most bacterial polysaccharides contain more than one type of sugar, i.e. they are heteropolysaccharide, Bacterial capsules are species—specific, and can therefore be used for immunological distinction of closely related species. The daxtrans produced by Leuconostoc mesenteroides are used as plasma expanders in hospitals. They are also used as laboratory gels for filtration. In Acetobacter xylinum the polysaccharide is a homopolysaccharide made up of glucose units linked p1-4. Pseudomonas aeruginosa secretes a heteropolysaccharide consists of D-glucose, D-galactose, D-mannose, D-glucuronic acid and L-rhamnose residues. The capsule of pneumococci, is made up of hexoses, uronic acids and amino sugars. The protein capsule consists of L-amino acids in the streptococci. Polypeptides in the capsule of Bacillus anthracis contain D-glutamic acid. Some bacteria are enclosed within a capsule. This protects the bacterium, even within phagocytes, helping to prevent the cell from being killed.

Encapsulated bacteria grow as ‘smooth’ colonies, whereas colonies of bacteria that have lost their capsules appear rough. Rough colonies do not generally cause disease. Encapsulated bacteria do not succumb to intracellular killing as easily as bacteria that lack capsules. Strains of Streptococcus pnuemoniae that lack capsules do not cause disease. All the bacteria that cause meningitis are encapsulated. Suspending bacteria in India ink is an easy way of demonstrating capsules. Ink particles cannot penetrate the capsular material and encapsulated cells appear to have a halo around them. This is the Quellung reaction.



In the quellung reaction, bacterial cells are resuspended in antiserum that carries antibodies raised against the capsule. This causes the capsule to swell, and this can be easily visualised by suspension in India ink. The ink particles cannot penetrate the capsule, which this appears as a halo around the bacterial cells. Some bacteria produce slime to help them to stick to, surfaces. Slime is produced by several types of pathogenic microbes, and is usually made up from polysaccharides. The slime produced by Streptococcus mutans enables it to stick to the surface of teeth, where it helps to form plaque, leading eventually to dental caries. ‘Coagulase-negative’ staphylococci live on the skin, and some strains produce a slime that enables them to stick to plastics. These bacteria cause infections associated with implanted plastic medical devices.

Cytoplasmic Membrane

The plasma membrane, also called the cytoplasmic membrane, is the most dynamic structure of a procaryotic cell. Its main function is as a selective permeability barrier that regulates the passage of substances into and out of the cell. The plasma membrane is the definitive structure of a cell since it sequesters the molecules of life in a unit, separating it from the environment. The bacterial membrane allows passage of water and uncharged molecules up to mw of about 100 daltons, but does not allow passage of larger molecules or any charged substances except by means

Some prokaryotic cells use one or more flagella to move through water. Peritrichous bacteria, which have numerous flagella, use runs and tumbles to move purposefully in the direction of a chemical attractant.

special membrane transport processes and transport systems. Bacterial membranes are composed of 40 percent phospholipid and 60 percent protein. The phospholipids are amphoteric molecules with a polar hydrophilic glycerol "head" attached via an ester bond to two nonpolar hydrophobic fatty acid tails, which naturally form a bilayer in aqueous environments. Dispersed within the bilayer are various structural and enzymatic proteins which carry out most membrane functions. At one time, it was thought that the proteins were neatly organized along the inner and outer faces of the membrane and that this accounted for the double track appearance of the membrane in electron micrographs. However, it is now known that while some membrane proteins are located and function on one side or another of the membrane, most proteins are partly inserted into the membrane, or possibly even traverse the membrane as channels from the outside to the inside. It is possible that proteins can move laterally along a surface of the membrane, but it is thermodynamically unlikely that proteins can be rotated within a membrane, which discounts early theories of how transport systems might work. The arrangement of proteins and lipids to form a membrane is called the fluid mosaic model, and is illustrated in Figure 1

Archaea constitute a domain of single-celled microorganisms. These microbes are prokaryotes, meaning they have no cell nucleus. Archaea were initially classified as bacteria, receiving the name archaebacteria, but this classification is outdated.

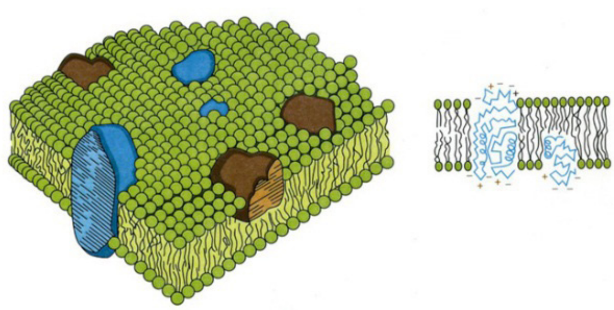


Figure 1: Fluid mosaic model of a biological membrane.

The membranes of Bacteria are structurally similar to the cell membranes of eucaryotes, except that bacterial membranes consist of saturated or monounsaturated fatty acids (rarely, polyunsaturated fatty acids) and do not normally contain sterols. The membranes of **Archaea** form bilayers functionally equivalent to bacterial membranes, but archaeal lipids are saturated, branched, repeating isoprenoid subunits that attach to glycerol via an ether linkage as opposed to the ester linkage found in glycerides of eukaryotic and bacterial membrane lipids (Figure 2). The structure of archaeal membranes is thought to be an adaptation to their existence and survival in extreme environments.

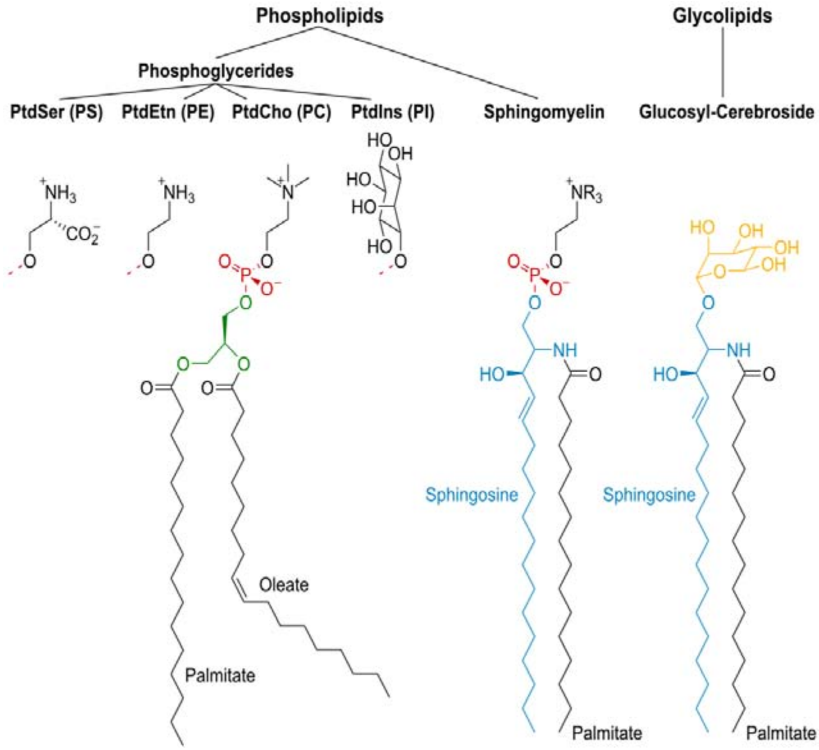


Figure 2: Generalized structure of a membrane lipids.

If polypeptide is partially folded it binds to DnaJ and DnaK, and repeats the same process again.

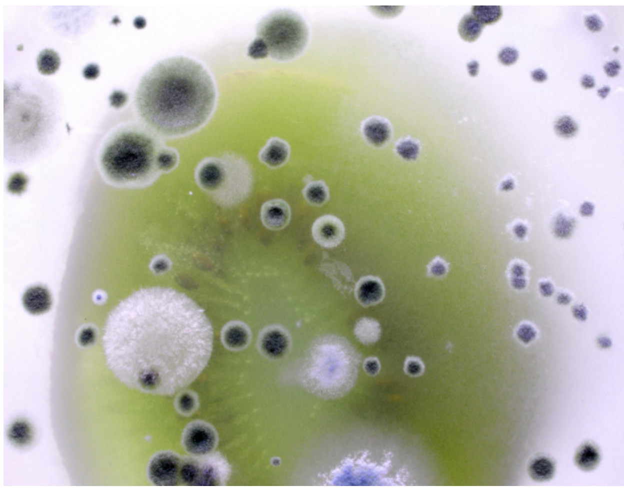
The plasma membrane, also known as the cell membrane, is an essential part of the cell that encloses the cell's interior components while only allowing certain parts of the outside environment to enter. This membrane is one of the few parts that prokaryotic, eukaryotic, plant, and animal cells all have in common. The plasma membrane is far more than a simple barrier; it controls what moves into and out from the cell, and it governs many of the interactions that occur between a cell and its environment. The membrane is composed of many different molecules and proteins that move somewhat fluidly, resulting in the "fluid mosaic" description of the plasma membrane.

The most abundant molecules in the plasma membrane are phospholipids, which are made up of a hydrophobic, "water-fearing" tail and a hydrophilic, "water loving" head. Two layers of phospholipids arranged with the hydrophobic tails on the inside form a phospholipid bilayer that provides the primary structure of the membrane. This bilayer prevents large substances or particularly polar substances from passively diffusing across the cellular membrane.

Many proteins that allow for the transportation of large or polar substances across the membrane are embedded in the phospholipid bilayer. Some allow for the passive diffusion of substances in and out of the cell; this requires no energy. Others actively carry substances from one side of the membrane to the other. This process, generally referred to as active transportation, does require a small expenditure of energy. Not all substances can move in and out of the plasma membrane at all times, so it is said to be "selectively-permeable."

The plasma membrane also plays an important role in positioning, anchoring, and shaping the cell while connecting neighboring cells. Extracellular structural components, which compose the extracellular matrix, connect to a cell at its cellular membrane. Cell walls, which provide rigidity to plant cells and to some bacteria and other small organisms, also tend to connect to a cell's plasma membrane.

Cellular communication is another important function of the plasma membrane. Proteins and protein receptors embedded in the membrane can send and receive chemical signals. Some of these signals prompt cells to some form of action, such as absorbing or expelling particular substances. Other chemical signals serve as identification mechanisms and allow cells to recognize each other. This is particularly important in the immune system so the body's immune response only targets harmful cells and does not harm the body's normal cells.



Functions of the Cytoplasmic Membrane

Since procaryotes lack any intracellular organelles for processes such as respiration or photosynthesis or secretion, the plasma membrane subsumes these processes for the cell and consequently has a variety of functions in energy generation, and biosynthesis. For example, the electron transport system that couples aerobic respiration and ATP synthesis is found in the procaryotic membrane. The photosynthetic chromophores that harvest light energy for conversion into chemical energy are located in the membrane. Hence, the plasma membrane is the site of oxidative phosphorylation and photophosphorylation in procaryotes, analogous to the functions of mitochondria and chloroplasts in eukaryotic cells. Besides transport proteinsthat selectively mediate the passage of substances into and out of the cell, procaryotic membranes may contain sensing proteins that measure concentrations of molecules in the environment or binding proteins that translocate signals to genetic and metabolic machinery in the cytoplasm. Membranes also contain enzymes involved in many metabolic processes such as cell wall synthesis, septum formation, membrane synthesis, DNA replication,

CO_2 fixation and ammonia oxidation. The predominant functions of procaryotic membranes are:

- Osmotic or permeability barrier
- Location of transport systems for specific solutes (nutrients and ions)
- Energy generating functions, involving respiratory and photosynthetic electron transport systems, establishment of proton motive force, and transmembranous, ATP-synthesizing ATPase
- Synthesis of membrane lipids (including lipopolysaccharide in Gram-negative cells)
- Synthesis of murein (cell wall peptidoglycan)
- Assembly and secretion of extracytoplasmic proteins
- Coordination of DNA replication and segregation with septum formation and cell division
- Chemotaxis (both motility per se and sensing functions)
- Location of specialized enzyme system

Some molecules can move across the bacterial membrane by simple diffusion, but most large molecules must be actively transported through membrane structures using cellular energy.

Symposium on Bacterial Structure and Activity

Although in most sections the cytoplasmic membrane appears to be firmly attached to, or associated with the cytoplasm, and to separate from the cell wall, this is not always the case. Even in Gram-negative organisms where plasmolysis is more readily demonstrated, the membrane may often remain partly attached to the cell wall and thus detached from the cytoplasm.

It is true that cell-wall preparations prepared by prolonged shaking with glass beads are devoid of the enzymic activities and lipoprotein associated with the cytoplasmic membrane. However, when prepared by other methods such as the Hughes press or French press fractions containing both cell wall and the bulk of the membrane are readily obtained both from Gram-negative and Gram-positive organisms, essentially free from cytoplasmic contamination. Such preparations when further fragmented, for instance by ultrasonics, yield particulate fractions in which the carbohydrate of the cell wall is still associated with lipoprotein (Hughes, unpublished), suggesting that the membrane may in fact be more firmly associated with the cell wall than the separation during **plasmolysis** suggests.

Plasmolysis is the process in which cells lose water in a hypertonic solution. The reverse process, cytolysis, can occur if the cell is in a hypotonic solution resulting in a lower external osmotic pressure and a net flow of water into the cell.

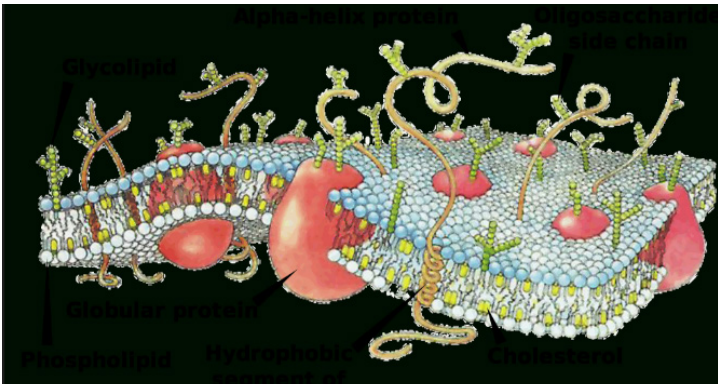
Preparation of Isolated Membranes

Lipoprotein-rich fragments of membranes are commonly prepared by osmotically shocking or bursting by other means sucrosestabilized protoplasts of organisms such as Micrococcus lysodeikticus and Bacillus megaterium. In these organisms, lysozyme removes the bulk of the cell wall. The isolated fragments (so-called 'ghosts ') appear in metal-shadowed specimens to be approximately 500 mp in cross section, about 100 A. thick, and to be free from adhering cell wall and cytoplasm; they nevertheless may contain up to 20% of their dry weight as carbohydrate, the rest beingmainly lipoprotein. Fractions rich in lipoprotein but containing various amounts of cell wall may also be prepared by similar methods from spheroplasts prepared by lysozyme treatment in the presence of versene.

The action of autocatalytic enzymes, unbalanced growth produced by the action of penicillin or other antibiotics or the omission of an essential cell-wall constituent such as lysine or diaminopimelic acid have also been used to prepare spheroplasts from which membrane-enriched fractions may be isolated. Protoplast preparations of certain yeasts have been prepared by the action of carbohydrases from Helix pomatia. In addition to these chemical and enzymic methods for isolating cytoplasmic membranes, cell fractions can be prepared by mechanical rupture of the cell wall followed by procedures to remove the bulk of the cytoplasm; these contain the bulk of the cell wall and cytoplasmic membrane. Such cell-wall membranes prepared from Pseudomonas Jluorescens were made by rupture of the cells in the Hughes press, treatment with deoxyribonuclease, subsequent emptying by shaking with glass beads and extraction with phosphate buffer.

The present author has used this procedure more recently with a wide range of micro-organisms to yield preparations which typically contain the bulk of the cell wall and have an enzymic constitution similar to membrane preparations from protoplasts. Similar preparations may be obtained from cells ruptured in the 'x' press or the French press (Milner et al. 1950) although these presses appear to give greater fragmentation of the wall (Hughes, unpublished). In some cases varying amounts of cell-wall material may be removed from cell-wall membrane preparations by subsequent treatment with lysozyme or other carbohydrases (Hughes, unpublished) ; usually this leads to fragmentation and yields preparations which appear similar to ' ghosts ' by electron microscopy but which still consist of particles which contain both lipoprotein and cell-wall carbohydrates.

Although it is obviously of advantage especially for chemical analysis to prepare the cytoplasmic membrane free from cell wall, it is often also advantageous to be able to isolate cell-wall membranes. The membrane in these appears to be structurally intact and easily freed from supernatant fractions containing cytoplasm particles such as ribosomes. These in turn are relatively free from enzymes and other material associated with particles, for instance so-called 'oxidosomes', which are now generally assumed to be comminuted cytoplasmic membrane.



Chemical Analysis of Cytoplasmic Membrane

The most constant feature of cytoplasmic membrane and cell-wall membrane fractions is their high concentration of lipid, up to 30 % of dry weight, which is mostly present as phospholipid. Carbohydrate is also generally present but this depends on the proportion of adhering cell wall. Where the cell wall is absent, amino sugars are absent and the carbohydrate is probably not derived from the cell wall. For instance, in Micrococcus lysodeikticus, mannose in contrast to glucose is found in the membrane ghosts. The carbohydrate of cell-wall membrane preparations appears indistinguishable from that of purified cell walls (Hughes, unpublished). The amino acid composition of purified membranes appears similar to cell protein and is unlike that of wall polypeptide in that it contains all the common L-amino acids and no D-amino acids. The lipid composition of membranes from M. lysodeikticus has recently been analysedmore fully by McFarlane. About 80% of the total lipids is phospholipid, the remaining 20% being neutral lipid, mainly a diglyceride. Of the phospholipid, about 80% is diphosphatidyl glycerol, the remainder consisting of phosphatidyl inositol and a complex which yields glycerol, mannitol and fatty acids on hydrolysis. The fatty acids in all these fractions were mainly q6 anteiso and iso-acids, an observation which fits well with the 50-60 A. spacing seen in the electron micrographs. The lipid content of these preparations represented approximately 80 % of that of the intact cell. Cell-wall membranes from Pseudomonas jluorescens and Lactobacillus arabinosus contain at least 90-95 % of the total cell lipid. The main fatty acids from these organisms are similar to those from M. Zysodeikticus, and staining reactions suggest they are also present mainly as phospholipid. In L. arabinosuslactobacillic acid can be detected in the membrane extracts and in P. pseudomow an unidentified C, branched acid was detected (Hughes, unpublished).

The Enzymatic Function of the Cytoplasmic Membrane

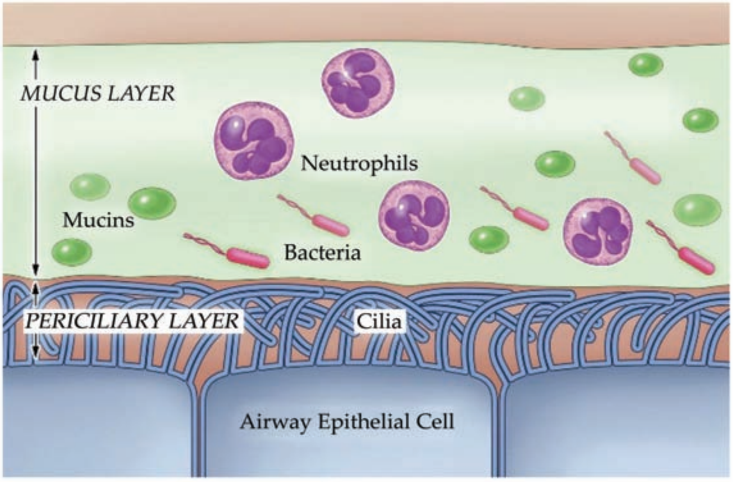
The cytoplasmic membrane was first established as the site of respiratory enzymes and cytochrome-linked electron transport by studies on highly purified ‘ ghost ‘ preparations from Bacillus rmguterium. Since then a similar localization of enzymes concerned with respiration has been found in many other aerobic organisms. In addition to these enzymes in membranes, numerous particulate fractions which have a similar

Lipoprotein is a biochemical assembly that contains both proteins and lipids, bound to the proteins , which allow fats to move through the water inside and outside cells.

enzymic constitution to the membranes have been isolated from mechanically broken cells. These particulate enzyme fractions usually consist mainly of lipoprotein granules, are often heterogeneous in size and contain ribonucleoprotein, which, when it can be separated, is usually without enzymic activity. The enzymic activity associated with the **lipoprotein** particles and membranes is often difficult to solubilize without destroying their structure. However, by treating them with detergents or other surfaceactive reagents such as deoxycholic acid, which disassociate lipid and protein, the enzymes sometimes may be brought into solution and further purified. Granules which reduce tetrazolium in whole cells have been regarded as equivalent to animal and plant mitochondria.These have been identified with theisolated lipoprotein granules as 'bacterial mitochondria'.

There seems little doubt now, however, that most if not all of such preparations have been derived by comminution of the cytoplasmic membrane from which they can be readily prepared by, for instance, treatment with ultra-sound. If analogies for bacterial structures are to be sought in the higher cells, it would seem more correct to regard the cytoplasmic membrane as equivalent to the 'christaemitochondriales ' of animal or plant mitochondria. It would be expected in this case that intact membrane preparations or cellwall membranes would carry out oxidative phosphorylation. So far this has only been demonstrated in particulate fractions from micro-organisms. It is possible that this failure to phosphorylate oxidatively is due to either the lack of essential factors present in supernatant fractions. Crude supernatants from disrupted bacteria cannot always be added to the particulate fractions because they contain material and enzymes such as adenosine triphosphates (ATPase) which interfere with the measurement of oxidative phosphorylation. In the case of cell-wall membranes, the effect of freezing and thawing may also be deleterious.

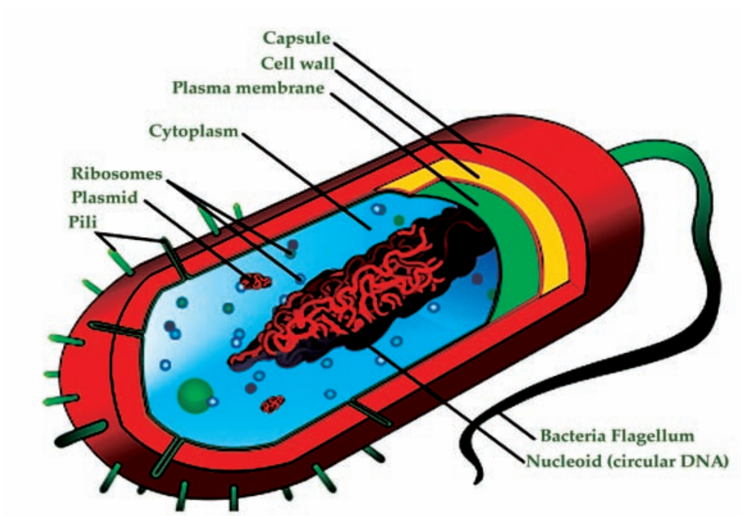
In addition to enzymes associated with energy production many others have been shown to occur in the cytoplasmic membrane or in particles derived from it. These include enzymes such as formic hydrogenlyase, nicotinic acid and other aromatic hydroxylases and many polyol-dehydrogenases which occur exclusively in microorganism.



Most of the work on membrane and particulate-bound enzymes has so far been done with aerobic organisms, and apart from the demonstration of particulate enzymes such as hydrogenase in Clostridiurnpasteurianzcm, little work has been reported on the membrane of facultative or obligate anaerobes. Recently cell-wall membranes have been prepared from a number of lactobacilli. These organisms, although obtaining the bulk of their energy by anaerobic glycolysis, nevertheless can utilize oxygen to oxidize substrates such as pyruvate and lactate at low rates. This oxidation is mediated through riboflavin electron-transport systems; cytochromes are normally completely absent. Neither the glycolytic enzymes, with the exception of some weak hexokinase activity, nor the flavoenzyme systems were found in the cell-wall membranes. The only activity so far found consistently is ATPase. The ATPase in these cell-wall membranes has an unusually low pH optimum, is Mg^{2+}-activated but is not activated by o-dinitrophenol. It has not so far been found possible to free it from the cell-wallmembrane nor to show definitely that it is associated with the lipoprotein portion of this fraction. However, similar ATPase activity has been demonstrated in membranes. The enzyme from all these organisms does not liberate on orthophosphate from a wide range of organic phosphates including AMP; a low rate of hydrolysis is found with other nucleoside triphosphates and ADP. The consistent occurrence of an ATPase in the bacteria is of interest in connexion with the suggested role of this enzyme in ion transport and permeability. The possible role of the cytoplasmic membrane in protein synthesis is discussed by Hunter in this symposium. It is of significance in this connexion that most purified membrane preparations contain about 2 % ribonucleic acid, and that this also forms a. constant feature of cell-wall membrane preparations. It is difficult to remove this without chemical treatment or disruption of the membrane. It seems possible that the cytoplasmic membrane is the site of synthesis as well as of activity of permeases and other inducible membrane-bound enzymes such as the nicotinic hydroxylase system. Also of interest in this connexion is the apparent lability of respiratory cytochromes which appear to be firmly bound and almost exclusively in the membrane but yet alter concentration rapidly when oxygen tension in the growth medium is changed. Such problems, which are connected with the synthesis of the membrane itself, are open to an experimental approach now that methods for isolating the membrane from a wide range of microorganisms are available.

Bacterial Cell Wall

Bacterial cells lack a membrane bound nucleus. Their genetic material is naked within the cytoplasm. Ribosomes are their only type of organelle. The term “nucleoid” refers to the region of the cytoplasm where chromosomal DNA is located, usually a singular, circular chromosome. Bacteria are usually single-celled, except when they exist in colonies. These ancestral cells reproduce by means of binary fission, duplicating their genetic material and then essentially splitting to form two daughter cells identical to the parent. A wall located outside the cell membrane provides the cell support, and protection against mechanical stress or damage from osmotic rupture and lysis. The major component of the bacterial cell wall is peptidoglycan or murein. This rigid structure of peptidoglycan, specific only to prokaryotes, gives the cell shape and surrounds the cytoplasmic membrane. Peptidoglycan is a huge polymer of disaccharides (glycan) cross-linked by short chains of identical amino acids (peptides) monomers. The backbone of the peptidoglycan molecule is composed of two derivatives of glucose: N-acetylglucosamine (NAG) and N-acetlymuramic acid (NAM) with a pentapeptide coming off NAM and varying slightly among bacteria. The NAG and NAM strands are synthesized in the cytosol of the bacteria. They are connected by inter-peptide bridges. They are transported across the cytoplasmic membrane by a carrier molecule called bactoprenol. From the peptidoglycan inwards all bacterial cells are very similar. Going further out, the bacterial world divides into two major classes: Gram positive (Gram +) and Gram negative (Gram -). The cell wall provides important ligands for adherence and receptor sites for viruses or antibiotics.



Bacteria are single-celled organisms that have a prokaryotic cell structure. While bacterial cells vary in some structural elements, such as size and shape, they all share the common traits of prokaryotes. Prokaryotic cells are distinctive in that they do not have nuclei or other organelles bound by membranes.

The bacterial cell is protected and contained by a cell wall, which is made from peptidoglycan, a sugar and protein polymer. Bacteria with thick cell walls are referred to as gram-positive, while those with thin cell walls surrounded by a lipid membrane are called gram-negative. The cell wall protects the cell from the effects of turgor pressure, which results from the higher concentration of solute inside the cell compared to the cell's surrounding environment.

Some bacterial cells have external structures. Flagella, which are long, flexible structures made from the protein flagellin, extend out from the cell wall and give the bacterial cell added motility. Pili and fimbrae are short tubes of protein that are found among **Proteobacteria** and allow the bacterial cell to latch on to a substrate or another bacterial cell.

Proteobacteria is a major phylum of gram-negative bacteria. They include a wide variety of pathogens, such as Escherichia, Salmonella, Vibrio, Helicobacter, Yersinia, Legionellales, and many other notable genera.

Separating the cytoplasm, or internal fluid of the cell, from the cell wall is the cell membrane. This membrane acts as a mediator in the transport of material into and out of the cell. The bacterial cell membrane is a phospholipid bilayer consisting of fatty acids and is permeable only to certain ions and molecules.

The interior of the bacterial cell is fairly simple, since prokaryotic cells do not contain many internal structures. The genetic information of a bacterial cell is encoded in a supercoiled structure of deoxyribonucleic acid (DNA) suspended in the region known as the nucleoid. The bacterial chromosome is usually circular in shape. Other small pieces of DNA known as plasmids float independently in the cytoplasm apart from the main chromosome. These fragments code for nonessential traits and may be exchanged between bacteria. The lack of a membrane-bound nucleus allows the DNA in a bacterial cell to interact more directly with ribosomes, which are responsible for the process of translation, or the transfer of genetic data.

Ribosomes and the bacterial chromosome are the most basic intracellular structures found in the cytoplasm of a bacterial cell, although some types of bacteria include more complicated structures. For example, some types of bacterial plankton have gas vesicles in their cells, which allow them to adjust their buoyancy in water. Structural filaments comprising a cytoskeleton have also been observed in bacterial cells.

Bacterial Cell Cytoplasm

Bacteria are one-celled organisms that can cause disease in humans and yet are also essential to our good health because they play an important role in our digestion. Bacteria are prokaryotic cells; they don't have a nucleus enclosed by a membrane. Instead of having DNA in chromosomes, bacterial genetic information is contained in a loop of cellular material called a plasmid. The plasmid, nuclear materials, water, enzymes, nutrients, waste materials and ribosomes all circulate within the bacterium in a thick liquid called cytoplasm.

Bacteria have a simple internal organization that consists of specialized parts rather than distinct organs surrounded by membranes. These specialized parts are called organelles. The cytoplasm is where the organelles carry out the processes necessary for the life of the bacterium. The components of the cytoplasm are responsible for cell growth, metabolism, elimination of waste and replication (reproduction) of the cell.

Components of Cytoplasm of Bacteria

The following points highlight the six main components of cytoplasm of bacteria. The components are:

- Ribosomes
- Molecular Chaperones
- Nucleoids
- Plasmids
- Cytoplasmic Inclusions
- Spore and Cysts.

Ribosomes

All living cells contain ribosomes which act as a site of protein synthesis. High number of ribosomes represents high rate of protein synthesis and vice versa. Cytoplasm of a prokaryotic cell contains about 10,000 ribosomes which account upto 30% of total dry weight of the cell. Presence of ribosomes in high number gives the cytoplasm a granular appearance.

The eukaryotic ribosomes are found attached to cell membrane, whereas the prokaryotic mesosomes are free in cytoplasm. Prokaryotic ribosomes are smaller and less dense than eukaryotic ribosomes.

Ribosomes of prokaryotes are often called 70S ribosomes and that of eukaryotes as SOS ribosomes. The letter 'S' refers to Svedberg unit which indicates the relative rate of sedimentation during ultracentrifugation. Sedimentation rate depends on size, shape and weight of particles.

Subunits: In general the ultrastructure of ribosomes reveals that these are composed of two subunits, a larger 508 subunit and a smaller 30S subunit. Each subunit is composed of protein and ribosomal RNA (rRNA).

Their association and dissociation depend on the concentration of Mg++ ions. The structure of a ribosome is very complex. The proteins and RNA are inter-wined. James A. Lake (1981) presented the structure and function of ribosome.

According to him the smaller subunit of ribosome consists of a head, a base and a platform. With the help of a cleft the platform and head are separated from the base. The larger subunit comprises of a ridge, a central protuberance and a stalk; the former two are separated by a valley (Figure 3).

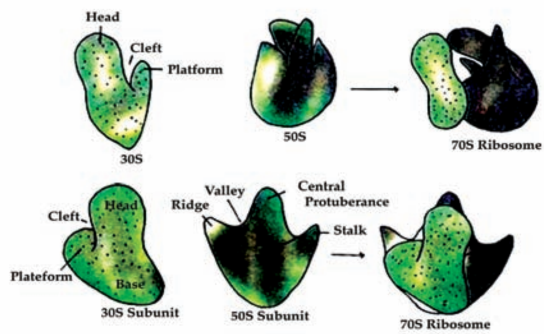


Figure 3: Three dimensional model of E.coli ribosome shown in two different orientation (A and B).

Chemical Composition: The ribosomes of E. coli consist of three types of RNAs, 5S, 16S and 23S, and 53 proteins. The SOS subunit consists of 5S and 23S RNA, and 34 proteins; 30S subunit consists of 16S RNA and 21 proteins. The 5S RNA is 120 nucleotides long, 16S RNA is about 1,600 nucleotides long and 23S RNA is about 3,200 nucleotides long.

Base sequence of 5S RNA has been strongly conserved throughout the evolution and that of 16S form double stranded hairpin loops. Only 30-35% bases of 16S form single stranded loops. Interactions between rRNA and cellular RNA (mRNA and tRNA) occur (Figure 4).

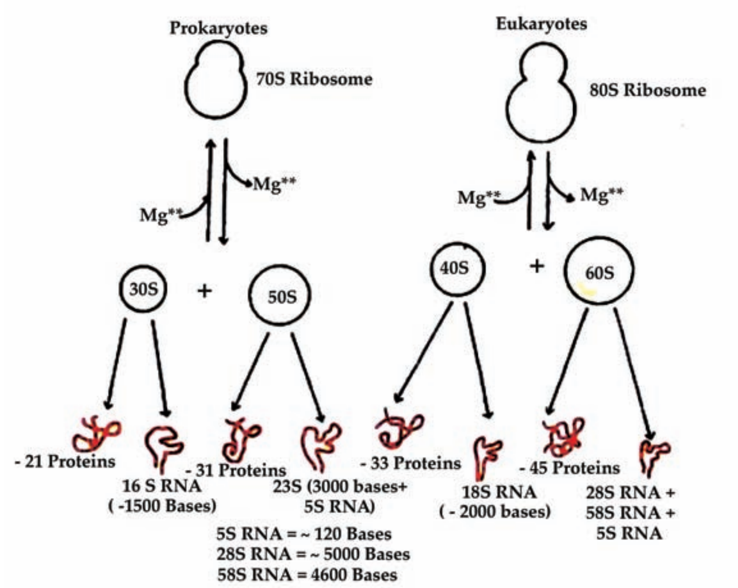


Figure 4: The rRNA and protein of prokaryotic (A) and eukaryotiv (B) ribosomes.

From 30S RNA of E. coli, 21 proteins have been isolated that are designated as SI to S21. Similarly from larger subunit (50S), 31 proteins (LI to L34) have been isolated. Protein map of ribosome showing their sites on two subunits is presented in Figure 5.

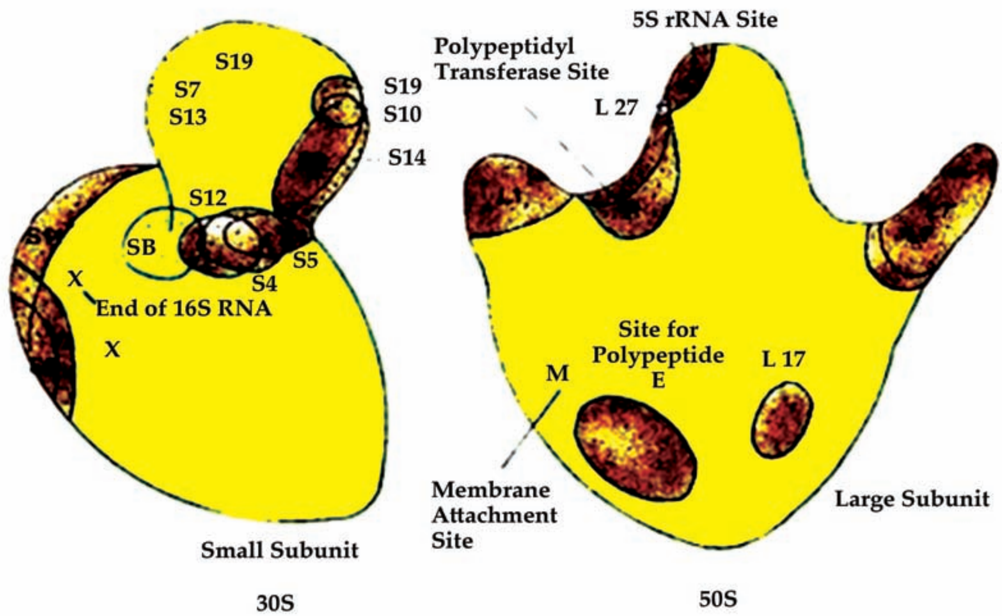


Figure 5: Proteine map showing their sites on small subunit and large subunit of rivosomes.

Molecular Chaperones

It was thought for many years that polypeptides after synthesis fold into native stage and this folding is not determined by its amino acids. It is now clear that there are certain helper proteins called molecular chaperones or chaperones which recognize the newly formed polypeptides and fold to its proper shape.

Proteins fold rapidly into the secondary structure. This unusually open and flexible conformation is called as molten globule. It is the starting points for slow process which results in correct tertiary structure.

There are several chaperones involved in proper protein folding in bacteria. Chaperones were first identified in E. coli mutant that did not allow to replicate, phage lambda. In E. coli at least four chaperones viz., DnaK, DnaJ, GroEL and GroES, and stress protein GrpE are involved in folding process.

They play an important role because after protein synthesis the cytoplasmic matrix is filled with nascent polypeptides and proteins. It is possible that these polypeptides become folded and form a nonfunctional complex. The chaperones check wrong in-folding and promote correct folding. The chaperones are found both in the cells of prokaryotes and eukaryotes.

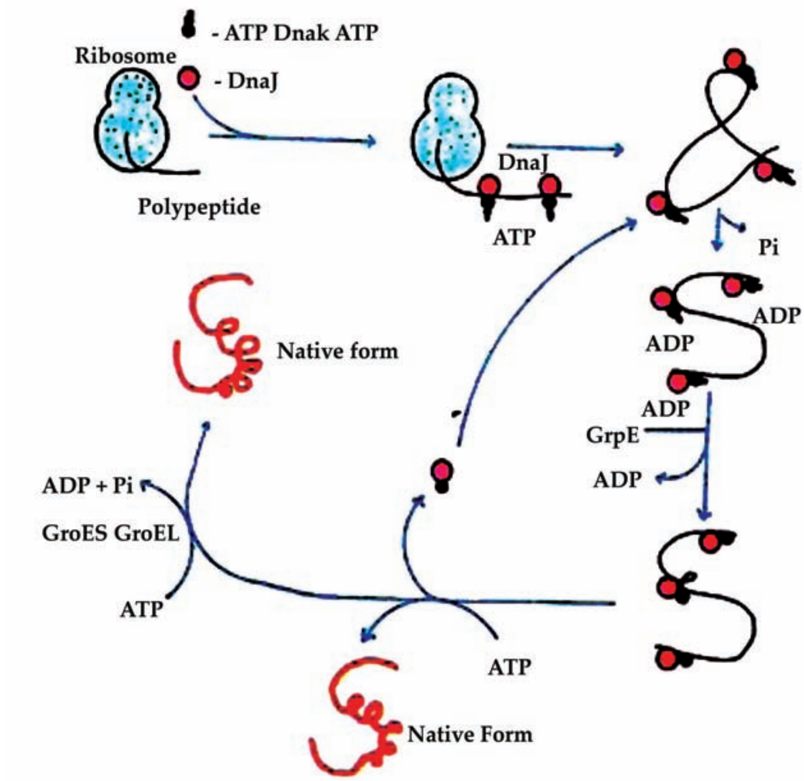


Figure 6: Mechanism of polypeptide folding by chaperones.

After synthesis of a sufficient length of polypeptide from the ribosome, DnaJ binds to unfolded chain (Figure 6). DnaK complexed with ATP attaches to polypeptide. These two chaperones check the polypeptide from folding. After binding of DnaK to polypeptide ATP is hydrolysed to ADP which increases the ability of DnaK to bind with unfolded peptide.

When polypeptide synthesis is completed, GrpE protein binds to DnaK-polypeptide complex and causes the DnaK to release ADP. Thereafter, ATP binds to DnaK, and DnaK and DnaJ are released from the polypeptide. During these events, polypeptide is folded and reaches to its final native conformation. At this stage if polypeptide is partially folded it binds to DnaJ and DnaK, and repeats the same process again.

Mostly DnaK and DnaJ transfer the polypeptide to GroEL and GroES where final foldings occurs. GroEL is a long hollow barrel shaped complex of 14 subunits which are stacked in two rings, whereas GroES contains four subunits arranged in one ring and can combine to both ends of GroEL. ATP binds to GroEL and changes the ability of the later for polypeptide binding. GroES binds to GroEL and helps in binding and release of refolding peptide.

Heat Shock Proteins: When E. coli cells are exposed to high temperature, metabolic poisons and other stressful conditions, the concentrations of chaperones increase. In E. coli cultures, at temperature between 30 and 40°C, 20 different chaperones often called heat shock proteins are produced within 5 minutes.

These protect the cell from thermal damage and stress, and promote the proper folding of polypeptides. In hypo-thermophiles (e.g. Pyrodictum occultum) that grow at about 110°C, a large amount of chaperones are present.

Mutant E. coli resistant to phage X produces slightly two changed chaperones like heat shock proteins 60 and 70 (hsp60 and hsp70). The eukaryotic cells have families of hsp60 and hsp70 proteins, and different family members function in different organelles.

The mitochondria contain their own hsp60 and hsp70 molecules which are different from those functioning in the cytosol. A special hsp70 helps to fold proteins in the endoplasmic reticulum. The other function of chaperones is the transport of proteins across the membrane.

Nucleoids (Bacterial Chromosome)

As in eukaryotes, in prokaryotes too the basic dye stains the nuclear material and reveals as dense and centrally located bodies of irregular outline. Upon observation with electron microscope it was found that this central region is not separated from the cytoplasm by a membrane and consists of nuclear structure besides the DNA fibrils. The eukaryotes contain a well organized nucleus in which the genetic material is enclosed by a nuclear membrane.

Therefore, the DNA material is not enclosed by any covering. Hence the bacterial chromosome is known as chromatin bodies or nucleoids. The nucleoid is a single long circular double stranded DNA molecule devoid of highly conserved histone protein. The histone is present in eukaryotes, therefore, results the eukaryotic DNA into the beaded structures i.e. nucleosomes.

By the turn of 1960s, the nuclear material was studied in detail. In 1963, J. Cairns succeeded in extracting E. coli DNA under conditions that minimize its shearing. Auto-radiographic studies showed that the DNA was extremely long threads measuring about 1mm in length.

A few threads were circular. Hobot (1985) found that the fully extended E. coli DNA was 1 mm long ($4 \times 10^6 \times 3.4$ Å) having molecular weight 3×10^9 Dalton. Through electron microscope they also observed the compaction of chromosome into irregular shaped nucleoids.

The nucleoid is observed as a coralline (coral like) shaped structure the branches of which spread far into the cytoplasm and over the entire area of the cell. By using serial sections it has been possible to reconstruct the ribosome free area of the nucleoid.

Two types of nuclear bodies can be observed, an envelope associated nucleoid and an envelope free nucleoid. Associated with the first type a large amount of RNA, proteins, lipids and peptidoglycan are found, whereas the second type contains less amount of it.

Generally, the number of nucleoid per bacterial cell is one, but in some bacteria the number may go even to four or more. The DNA molecule appears to be present in 10 to 80 super coils. Worcel and Burg proposed the structure of the folded chromosome of E. coli and showed as seven loops, each twisting into a super helix. These loops are held together by a core of DNA (Figure 7).

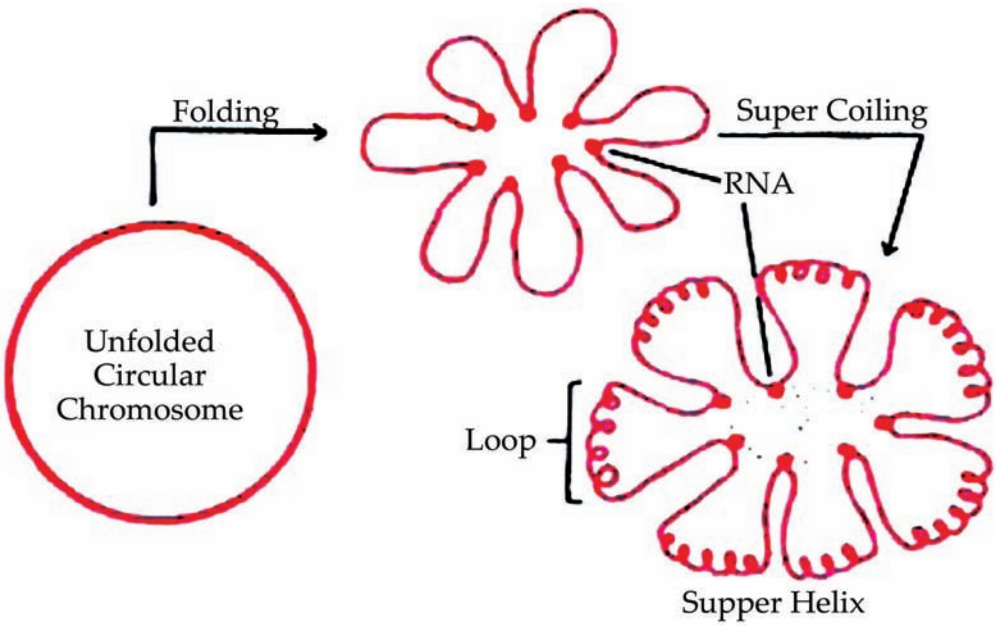


Figure 7: A model representing the process of folding and super coiling of bacterial chromosome**.**

Super coiling may be induced enzymatically. Possibly it may be a factor for the formation of nucleoids. The folded structure was found to be attached to a fragment of cell membrane. This shows that the bacterial chromosome remains associated to a point of cell membrane. This helps in separation of newly replicated DNA molecules.

Plasmids

Plasmids are small genetic structures found in many bacteria that contain strands of coiled DNA. Plasmid DNA is not used in reproduction. Instead, plasmids carry out other specific and important functions. When bacteria reproduce, plasmids carry along special properties such as antibiotic drug resistance, resistance to heavy metals and factors necessary for infection of animals or plants. These properties confer certain advantages and protections to the bacteria.

A plasmid contains 5-100 genes that determine several biological functions. Under certain circumstances they provide special characteristics to the bacterial cell and help them in survivability. They may even lose without harming the bacterial cell.

Plasmids are the circular DNA molecule but in resting stage helix twists in right hand direction at every 400-600 base pairs and forms supercoils. The twisted form is called covalently closed circular- DNA. After cleaving the twists this form is converted into an open circular form of double stranded DNA molecule (Figure 8).

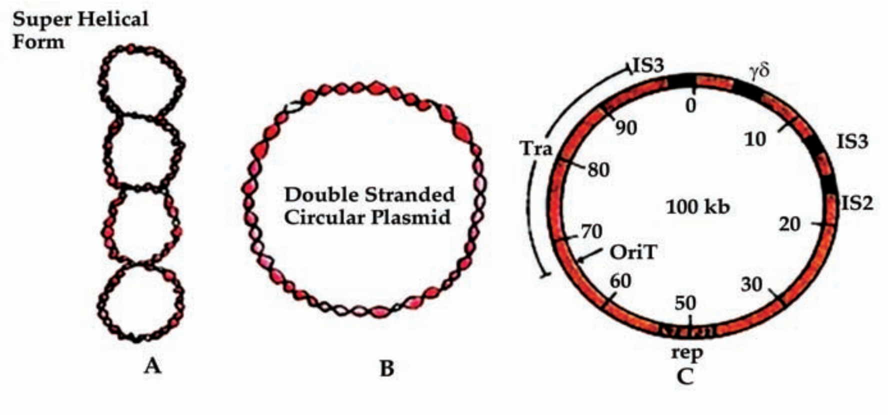


Figure 8: Plasmid, A: super helical form, B: Double stranded circular form; C: F-plasmid.

Cytoplasmic Inclusions

The cytoplasm of prokaryotic and eukaryotic cells contains several reserve deposits which are called inclusions. The Inclusion bodies are aggregates of polymers produced when there is excess of nutrients in the environment and they are the storage reserve for granules, phosphates and other substances. Volutin granules are polymetaphosphates which are reserves of energy and phosphate for cell metabolism and they are also known as metachromatic granules.

The inclusion bodies are of two types (a) free inclusion bodies (e.g. polyphosphate granules and cyanophycean granules), and (b) single-layered non-unit membrane enclosed inclusion bodies (such as poly P-hydroxybutyrate granules, glycogen granules, sulphur granules, carboxysomes and gas vaculoes). The membrane of inclusion bodies is made up of proteins or lipids.

Some of the inclusions are:

(a) *Volutin Granules*: The volutin granules are also known as polyphosphate granules or metachromatic granules because after staining the bacteria with blue dye (e.g. methylene blue) these granules take stain and appear reddish purple in color. Polyphosphate is a linear polymer of orthophosphates joined by ester bonds.

These granules are found in algae, fungi, protozoa and bacteria. These are present in high amount in Corynebacterium diphtheriae, hence it can easily be diagnosed. The volutin granules are composed of polyphosphates i.e. inorganic phosphates which are used in synthesis of ATP.

The phosphate is incorporated into nucleic acid. These are generally formed when the cells grow in phosphate rich environment. Growing cells of Aerobacter aero-genes contain traces of inorganic phosphate when there is nutrient deficiency; a large amount of phosphate is accumulated when synthesis of nucleic acid gets ceased.

(b) *Polysaccharide Granules:* The polysaccharide granules are found in protozoa, yeasts, fungi and algae. These can be identified by using

iodine solution. After reacting with iodine, glycogen turns into reddish brown and starch into blue color.

(c) *Lipid Inclusions:* Lipids are found in high amount in several species of Bacillus, Azotobacter, Mycobacterium, Spirillum, etc. These are present as storage material and are polymer of poly β-hydroxybutyric acid. It is formed by the condensation of acetyl CoA.

In Bacillus megaterium poly β-hydroxybutyrate (PHB) accounts for 60% of total dry weight when the bacterium has grown on acetate or butyrate (PHB). The monomers of poly β-hydroxybutyrate are linked by ester linkage forming the long poly β-hydroxybutyrate polymer.

The chemical structure of poly β-hydroxybutyrate is given in Figure 9. The poly p- hydroxybutyrate polymer accumulates as granules of 0.2-0.7 Hm diameter.

The collective term poly β-hydroxybutyrate represents the all classes of carbon storage reservoir polymer acting as a source of energy and biosynthesis. These are found naturally both in Archaea but not in Eukarya. The presence of lipid inclusions can be demonstrated by using fat soluble dyes, for example sudan dye.

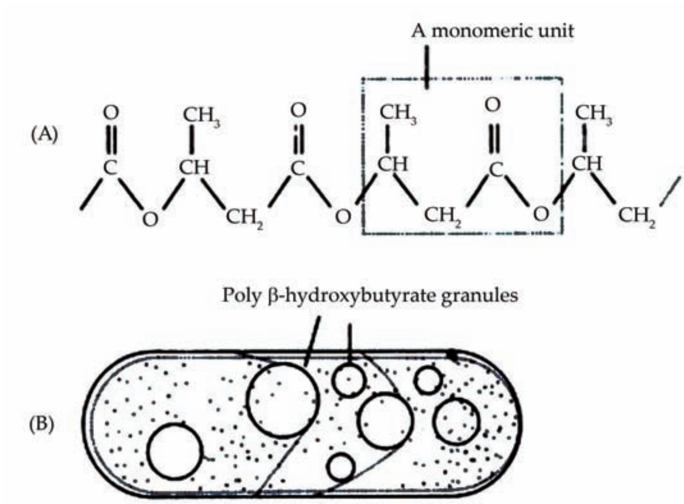


Figure 9: (A) Chemical structure of poly β-hydroxybutyrate (PHB). (B) Rhodospillium sodomense vontaining granules of poly β-hydroxybutyrate.

(d) *Glycogen:* Glycogen is another storage product of prokaryotes. It is polymer of glucose sub-units looking like starch. The glucose units are linked by α(1→4) glycosidic bonds. The branching chains are connected by α(1→6) glycosidic bonds. Glycogen is also a storage reservoir for carbon and energy.

(e) *Carboxysomes (Polyhedral Bodies):* Carboxysomes are found in photosynthetic bacteria, nitrifying bacteria, cyanobacteria and thermo bacilli. These are polyhedral or hexagonal inclusions containing ribulose-1, 5-bi-phosphate carboxylase. This enzyme is required in carbon dioxide fixation during photosynthesis.

(f) *Sulphur Granules:* Sulphur granules are also temporarily stored by some bacteria which are also called a second type of inorganic inclusion body. It is exemplified by photosynthetic purple sulphur bacteria (e.g. Thiobacillus, Thiospirillum, Thiocapsa, and Chromatium) which are found in anaerobic, sulphide-rich zones of lake. They oxidize H_2O to S and internally deposit as sulphur granules within invaginated structures of plasma membrane. Sulphur is oxidised into sulphate as the reduced sulphur source becomes limiting.

(g) *Maghetosomes:* Magnetosomes are the intracellular chains of 40-50 magnetite (Fe_3O_4) particles of about 40-100 nm diameter found in magneto tactic bacteria. These bacteria are motile, highly microaerophilic spirilla isolated from fresh water habitats such as Aquaspirillum magnetotacticum. In addition, some bacteria isolated from sulphide habitats have also been found to possess magnetosomes containing greigite (Fc_3S_4) and pyrite (FeS_2).

Magnetosomes are surrounded by membrane containing phospholipids, proteins and glycoproteins. The membrane proteins of magneto-some precipitate Fe^{3+} as Fe_3O4_3 in the magneto-some. The shape of magnetosomes varies from square to rectangular or spike-shaped.

Each iron particle is a tiny magnet. Hence, the bacteria employ their magnetosomes to determine northward and downward directions. With the help of magneto-some they swim to nutrient- rich sediments or locate the optimum depth in fresh water and marine habitats. Besides, magnetosomes are also found in some eukaryotic algae, heads of birds, dolphins, green turtles and other animals to aid navigation.

(h) *Gas Vesicles:* There are many prokaryotic microorganisms found in floating forms in lakes and sea that possess gas vesicles. Gas vesicles are gas-filled structures which contain the same gas in which the organisms are suspended. Gas vesicles provide buoyancy and keep the cells in floating form.

The most interesting ‘floating and sinking’ phenomenon is found in those cyanobacteria which cause water blooms in lakes such as Microcystis aquaticus and Anabaena flosaquae. Besides, gas vesicles are also found in certain purple and green phototrophic bacteria and some archaebacteria such as halo bacteria e.g. Thiopedia and Amoebobacter.

Gas vesicles look like spindles of varying dimension (300-700×60-110 nm) and numbers (a few to hundreds per cell). They contain a rigid membrane and internally a hollow structure. The membrane consists of only proteins called GvpA and GvpC. They are 2 nm thick and impermeable to water and solutes; but they are permeable to most of gases.

Gas vesicles cannot resists high hydrostatic pressure and can be collapsed resulting in loss of buoyancy. The collapsed vesicles cannot resume the normal shape. Gas vesicles attain a density of about 5-20% of the cell. Presence of gas vesicles in photoautotrophic microorganisms provides a mechanism to adjust in a water column to move in a region to rescue from high or low light intensity.

(i) *Chlorobium Vesicles (Chlorosomes):* In Heliobacteria (strict anaerobic, green coloured phototrophs) bacteriochlorophylls are associated with the cytoplasmic membrane. But in green bacteria photosynthetic apparatus is different. These bacteria consist of a series of cylindrical, flat sheets (lamellae) or ellipsoidal vesicles called chlorosomes or chlorobium vesicles. Structure of chlorosomes varies in different bacteria.

Chlorosomes are attached to cytoplasmic membrane. They lack bilayered thin membrane called non-unit membrane. Depending upon bacterial species bacteriochlorophylls a, c, d or e is present in chlorosomes. But most of the bacteriochlorophyll a (and component of electron transport chain) are found in the cytoplasmic membrane. Chlorosomes are the site of photosynthesis.

Spores and Cysts

Certain species of bacteria produce metabolically dormant structures known as spores. The spores may be endospores (i.e. produced inside the cell) or exospores (i.e. produced external to the cell). After return of suitable conditions spores undergo germination and produce vegetative cells.

In addition, some bacteria such as Azotobacter produce thick-walled dormant, and resistant spores which are known as cysts. Cysts develop after differentiation of a vegetative cell. To some extent cysts resemble endospores that differ in structure and chemical composition.

(a) *Endospores:* The endospore formation is restricted to six bacterial genera including two Gram-positive bacteria such as Bacillus (e.g. B. megaterium, B. spharericus) and Clostridium (C. tetani). They form endospores when there is lack of water or depletion of essential nutrients in the environment.

Endospores, therefore, are highly durable and dehydrated resting bodies produced inside plasma membrane of the cells. These have additional layers of thick walls which are made up of **peptidoglycan** (Figure 10).

Peptidoglycan, also known as murein, is a polymer consisting of sugars and amino acids that forms a mesh-like layer outside the plasma membrane of most bacteria, forming the cell wall.

Frankein and Bradlay reported the spores in a majority of species of Bacillus and Clostridium which differed by surface layer. The surface may be smooth or ribbed. Van den Hooff and Aninga found that the spore coat consists of an outer and an inner layer separated by a space. The outer layer is called exine and the inner layer intine. The central core is separated from the intine by a regular non- osmophilic space or cortex.

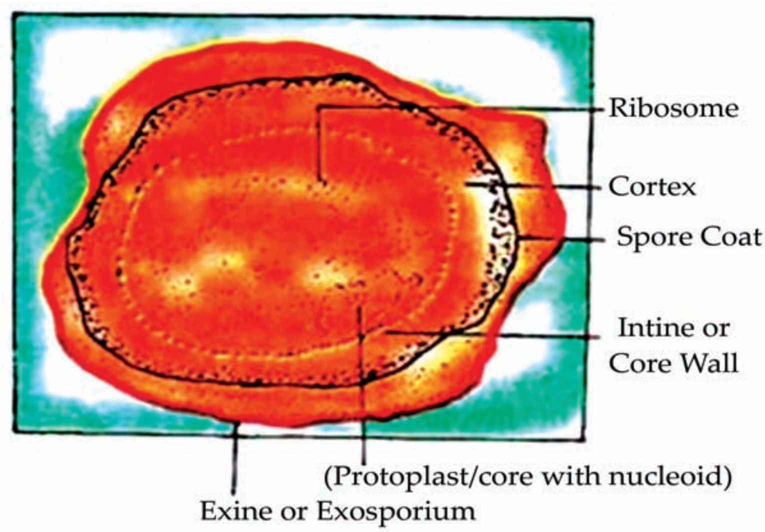


Figure 10: Structure of an endospore of bacillus anthracis.

Some of the endospores can remain viable over 100 years. Several species of endospore forming bacteria are the dangerous pathogens. The endospores are of much practical importance in food, industrial and medical microbiology. Several physicochemical processes are adopted to kill the vegetative cells.

If the endospores are produced the process of killing by chemicals is useless. In the processed food the endospores are adapted and produce toxins when suitable growth conditions prevail. In such conditions special methods are adopted. Generally the endospores are difficult to stain for deletion. Therefore, specially prepared stain i.e. the Schaeffer-Fulton endospore stain is commonly used for detection purpose.

Sporulation: The process of endospore formation is called sporulation or sporogenesis. It occurs normally when growth of bacterium ceases due to lack of nutrients. Sporulation is a complex process and occurs in several stages (Figure 11). Generally sporulation process requires 10 hours to get accomplished. Formation of endospores begins with the development of ingrowth of plasma membrane known as spore septum which becomes double layered.

The spore septum surrounds the cytoplasm and chromosome. A structure entirely enclosed within cell is formed which is known as fore-spore. Between the two membranes a thick layer of peptidoglycan is laid down.

A thick coat of protein is formed around this structure that provides resistance to endospores. All the endospores contain DNA, small amount of RNA and large amount of organic acid (dipicolinic acid). These cellular components are important for resuming metabolism in later stage.

Spore Germination: The endospores may be produced terminally, sub-terminally or centrally depending upon species. After maturity wall layer dissolved and endospores are set free (Figure 11). This process is called spore germination. Germination occurs only in favorable environmental conditions which are accomplished in three stages: activation, germination and outgrowth.

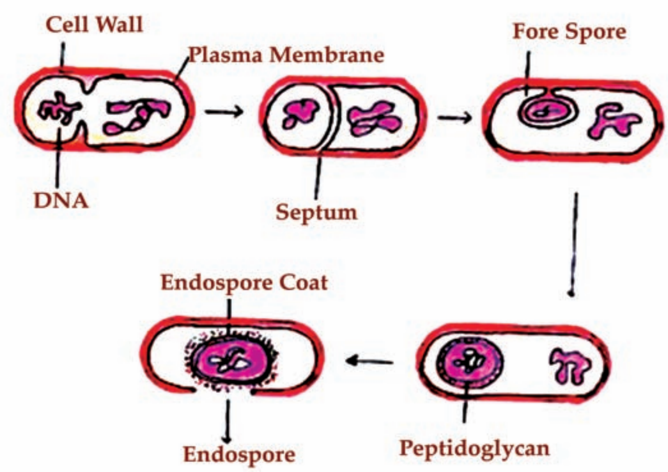


Figure 11: The Process of endospore formation i.e. sporulation in a bacterial cell.

(b) *Exospores:* Cells of Methylosinus (methane oxidizing bacterium) produce exospores through budding at one end of the cell. The exospores do not contain dipicolinic acid. They can resist desiccation and heat.

Following are some of the conditions that stimulate spore germination:

- Heat shock at sub-lethal temperature for varying period of time from 1 minute to 1 hour or more (time of heating varies with species, age of spores and conditions of storage).
- Incubation of spores at 30°C for 5 minutes in water-ethanol mixtures.
- Addition of yeast extract in active components viz., glucose, a few amino acids with or without nucleotides.
- Presence of various metal ions such as Ca^{++}, Mg^{++}, etc.
- Presence of dipicolinic acid.
- Addition of L-alanine.

Growth and Multiplication of Bacteria

Bacteria divide by binary fission and cell divides to form two daughter cells. Nuclear division precedes cell division and therefore, in a growing population, many cells having two nuclear bodies can be seen. Bacterial growth may be considered as two levels, increase in the size of individual cells and increase in number of cells. Growth in numbers can be studied by bacterial counts that of total and viable counts. The total count gives the number of cells either living or not and the viable count measures the number of living cells that are capable of multiplication.

Bacterial Growth Curve

When bacteria is grown in a suitable liquid medium and incubated its growth follows a definite process. If bacterial counts are carried out at intervals after innoculation and plotted in relation to time, a growth curve is obtained. The curve shows the following phase

Lag Phase

Immediately following innoculation there is no appreciable increase in number, though there may be an increase in the size of the cells. This initial period is the time required for adaptation to the new environment and this lag phase varies with species, nature of culture medium and temperature.

Log or Exponential Phase

Following the lag phase, the cell starts dividing and their numbers increase exponentially with time.

Stationary Phase

After a period of exponential growth, cell division stops due to depletion of nutrient and accumulation of toxic products. The viable count remains stationary as an equilibrium exists between the dying cells and the newly formed cells.

Phase of Decline

This is the phase when the population decreased due to cell death.

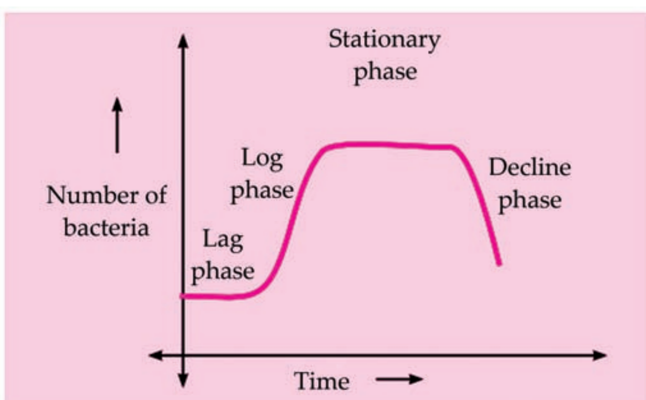


Figure 12: The growth curve of bacteria showing different phases.

The various stages of bacterial growth curve are associated with morphological and physiological alterations of the cells. The maximum cell size is obtained towards the end of the lag phase. In the log phase, cells are smaller and stained uniformily. In the stationary phase, cells are frequently gram variable and show irregular staining due to the presence of intracellular storage granules. Sporulation occurs at this stage. Also, many bacteria produce secondary metabolic products such as exotoxins and antibiotics. Involution forms are common in the phase of decline.

Factors That Affect the Growth of Bacteria

Many factors affect the generation time of the organism like temperature, oxygen, carbon dioxide, light, pH, moisture, salt concentration.

Nutrition

The principal constituents of the cells are water, proteins, polysaccharides, lipids, nucleic acid and mucopeptides. For growth and multiplication of bacteria, the minimum nutritional requirement is water, a source of carbon, nitrogen and some inorganic salts.

Bacteria can be classified nutritionally, based on their energy requirement and on their ability to synthesize essential metabolites. Bacteria which derive their energy from sunlight are called phototrophs, those who obtain energy from chemical reactions are called chemotrophs. Bacteria which can synthesize all their organic compounds are called autotrophs and those that are unable to synthesize their own metabolites are heterotrophs.

Some bacteria require certain organic compounds in minute quantities. These are know as growth factors or bacterial vitamins. Growth factors are called essential when growth does not occur in their absence, or they are necessary for it.

Oxygen

Depending on the influence of oxygen on growth and viability, bacteria are divided into aerobes and anaerobes. Aerobic bacteria require oxygen for growth. They may be obligate aerobes like cholera, vibrio, which will grow only in the presence of oxygen or facultative anaerobes which are ordinarily aerobic but can grow in the absence of oxygen. Most bacterial of medical importance are facultative anaerobes. Anaerobic bacteria, such as clostridia, grow in the absence of oxygen and the obligate anaerobes may even die on exposure to oxygen. Microaerophilic bacteria are those that grow best in the presence of low oxygen tension.

Carbon Dioxide

All bacteria require small amounts of carbon dioxide for growth. This requirement is usually met by the carbon dioxide present in the atmosphere. Some bacteria like Brucella abortus require much higher levels of carbon dioxide.

Temperature

Bacteria vary in their requirement of temperature for growth. The temperature at which growth occurs best is known as the optimum temperature. Bacteria which grow best at temperatures of 25-40°C are called mesophilic. Psychrophilic bacteria are those that grow best at temperatures below 20°C. Another group of non-pathogenic bacteria, thermophiles, grow best at high temperatures, 55-80°C. The lowest temperature that kills a bacterium under standard conditions in a given time is known as thermal death point.

Moisture and Drying

Water is an essential ingredient of bacterial protoplasm and hence drying is lethal to cells. The effect of drying varies in different species.

Light

Bacteria except phototrophic species grow well in the dark. They are sensitive to ultraviolet light and other radiations. Cultures die if exposed to light.

H-ion concentration

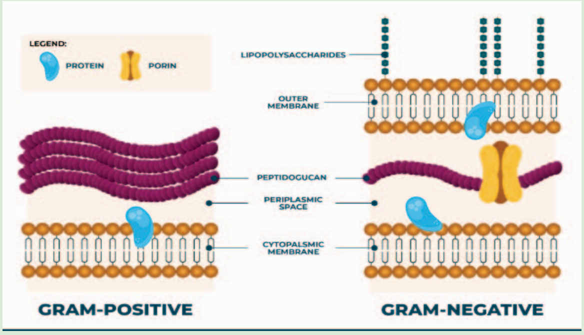
Bacteria are sensitive to variations in pH. Each species has a pH range, above or below which it cannot survive and an optimum pH at which it grows best. Majority of pathogenic bacteria grow best at neutral or slightly alkaline pH (7.2 – 7.6)

Osmotic Effect

Bacteria are more tolerant to osmotic variation than most other cells due to the mechanical strength of their cell wall. Sudden exposure to hypertonic solutions may cause osmotic withdrawal of water and shrinkage of protoplasm called plasmolysis.

CONCEPT OF COMPONENTS OF BACTERIAL CELL WALL

The peptidoglycan that makes up the rigid portion of the cell wall is called murein and is made up of N-acetyl glucosamine and N-acetylmuramic acid molecules that alternate in chains and are cross-connected by peptide subunits. Gram-positive bacteria have simpler chemical structures in their cell walls than Gram-negative bacteria.



Gram-negative Cell Wall

The gram-negative cell wall has the following components:

- Thin peptidoglycan layer – Peptidoglycan is a polymer of sugar and amino acids and is often single-layered in the gram-negative cell wall.
- Outer membrane – It is a protein layer that acts as a target site for antibiotics and phages.
- Lipoprotein layer – Links the peptidoglycan to the outer cell membrane.
- Lipopolysaccharide – It constitutes the endotoxicity of gram-negative bacteria. The toxicity is associated with the lipid A molecule of the lipopolysaccharide.
- Periplasmic space – It is the space between the outer and inner membranes.

Gram-positive Cell Wall

The gram-positive cell wall has the following components:

- Thick or multilayered peptidoglycan. It provides structural strength and mechanical rigidity.
- Teichoic acid – It is a water-soluble polymer that constitutes the major surface antigens of gram-positive bacteria.

It lacks an outer membrane and lipopolysaccharide but has other components such as polysaccharides and proteins.

SUMMARY

- The term morphology comes from the Greek for *form* and means the form and structure of living organisms. The morphology of bacteria describes the external appearance of bacterial cells including shape, arrangement, and size.
- The shape of bacterial cells is inherited and subject to the cell wall and cytoskeleton structure. The bacterial shape plays a role in how well bacteria perform life functions such as obtaining nutrition and avoiding predators. In pathogenic bacteria, the shape contributes to the ability to cause disease. The most common shapes of bacteria are cocci, bacilli, and spiral.
- Bacteria multiply via binary fission, a reproductive cell division that results in two identical daughter cells. These daughter cells usually separate completely and sometimes remain attached. The type of attachment after binary fission is a morphological feature called arrangement. There are many types of arrangement in bacterial cells, each characterized by the number of cells and pattern in the grouping. The most common types of arrangement are single, diplo arrangement, tetrad, sarcina, staphylo- arrangement, and strepto-arrangement.
- Bacteria are single-celled microorganisms with prokaryotic cells, which are single cells that do not have organelles or a true nucleus and are less complex than eukaryotic cells. Bacteria with a capital B refers to the domain Bacteria, one of the three domains of life. The other two domains of life are Archaea, members of which are also single-celled organisms with prokaryotic cells, and Eukaryota.
- The morphological study of bacteria requires the use of microscopes. Microscopy has come a long way since Leeuwenhoek first observed bacteria using hand-ground lenses.
- Bacteria are pokaryotic in their organization. The bacterial cell is surrounded by a capsule which encloses the nuclear structure or incipient nucleus. They do not have membrane-bounded organelles or a well-organized nucleus and their cell wall composition is entirely different.
- Cell wall is very rigid and plays a vital role in bacterial biology. It consists of lipids, carbohydrates, proteins, phosphorus and other inorganic substances.
- The capsule is not essential for the survival of bacteria under favorable growth conditions. However, it protects the bacteria against adverse conditions. The capsule may also function in the storage of food substances, and as a storage site for the disposal of waste material.
- The plasma embrane, also called the cytoplasmic membrane, is the most dynamic structure of a procaryotic cell. Its main function is as a selective permeability barrier that regulates the passage of substances into and out of the cell.

MULTIPLE CHOICE QUESTIONS

1. **How do bacteria reproduce?**
   a. Sexual reproduction
   b. Horizontal gene transfer
   c. Binary fission
   d. Mitosis
2. **Which is not one of the three main shapes of bacteria?**
   a. Coccus
   b. Bacillus
   c. Spiral
   d. Star
3. **When did bacteria first begin to exist on Earth?**
   a. 4 billion years ago
   b. 2 billion years ago
   c. 1.6 billion years ago
   d. 1 billion years ago
4. **Bacterial cells grown in a medium exposed to high osmotic pressure, changes shape from rod-shaped to __________ shaped.**
   a. elongated
   b. irregular
   c. rod shaped
   d. spherical
5. **Bacteria having clusters of flagella at both poles of cells are known as?**
   a. Amphitrichous
   b. Monotrichous
   c. Peritrichous
   d. Lophotrichous

Answers

1. (c) 2. (d) 3. (a) 4. (d) 5. (a)

REVIEW QUESTIONS

1. Discuss about bacterial morphology.
2. Explain the different types of bacteria.
3. Describe the structure of cytoplasmic membrane. Write its function.
4. Describe the components of cytoplasm of bacteria with a neat diagram.
5. Classify bacteria based on shaped and arrangement with examples.
6. What is the function of an endospore?
7. Explain the factors affecting the growth of the bacteria.
8. Describe the structure of cell wall.

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CHAPTER 7

MEDICAL MICROBIOLOGY

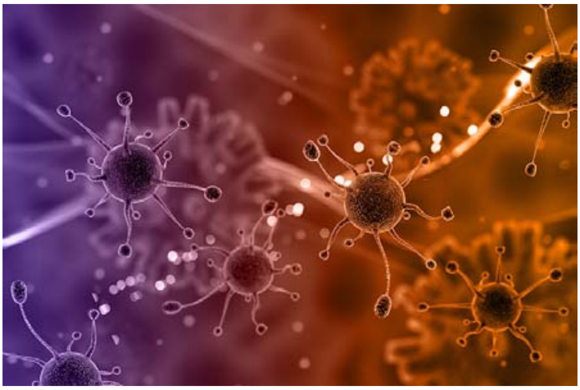
Medical microbiology, also known as clinical microbiology, is a subdiscipline of microbiology dealing with the study of microorganisms (parasites, fungi, bacteria, viruses, and prions) capable of infecting and causing diseases in human.

After studying this chapter, you will understand the core concepts of:

- Human microbial diseases
- Control of microorganisms

INTRODUCTION

Medical microbiology, the large subset of microbiology that is applied to medicine, is a branch of medical science concerned with the prevention, diagnosis and treatment of infectious diseases. In addition, this field of science studies various clinical applications of microbes for the improvement of health. There are four kinds of microorganisms that cause infectious disease: bacteria, fungi, parasites and viruses, and one type of infectious protein called prion.



A medical microbiologist studies the characteristics of pathogens, their modes of transmission, mechanisms of infection and growth. The academic qualification as a clinical/Medical Microbiologist in a hospital or medical research center generally requires a Bachelors degree while in some countries a Masters in Microbiology along with Ph.D. in any of the life-sciences (Biochem, Micro, Biotech, Genetics, etc.). Medical microbiologists often serve as consultants for physicians, providing identification of pathogens and suggesting treatment options. Using this information, a treatment can be devised. Other tasks may include the identification of potential health risks to the community or monitoring the evolution of potentially virulent or resistant strains of microbes, educating the community and assisting in the design of health practices. They may also assist in preventing or controlling epidemics and outbreaks of disease. Not all medical microbiologists study microbial pathology; some study common, non-pathogenic species to determine whether their properties can be used to develop antibiotics or other treatment methods.

CORE CONCEPT: HUMAN MICROBIAL DISEASES

Infectious diseases are illnesses caused by harmful agents (pathogens) that get into your body. The most common causes are viruses, bacteria, fungi and parasites. Infectious diseases usually spread from person to person, through contaminated food or water and through bug bites. Some infectious diseases are minor and some are very serious.

In the decades either side of 1900, advances in microbiology meant that a causative organism could be assigned to most of the major infectious diseases. This section is devoted to human microbial diseases, although it must be appreciated that only a relatively superficial coverage is possible in an introductory text such as this. We shall consider some of the mechanisms used by microorganisms to cause disease, together with modes of disease transmission, before describing some representative examples of diseases caused by different types of microorganisms.

As we have seen, only a small proportion of the vast numbers of microbial species give rise to infections; these are termed pathogens. An infectious disease can be defined as a change in the state of an individual's health brought about by the invasion and colonization of the body by pathogens, which may be bacteria, viruses, fungi or protists. It is worth making clear the distinction between infection and disease in this context. An infection is the colonization, multiplication, invasion or persistence of a pathogen in or on a host, whereas the term infectious disease is only used when this results in significant harm.

A primary infection is one where there is a clear invasion and multiplication of the microorganisms leading to local tissue damage. A secondary infection occurs when there is an invasion of microorganisms subsequent to a primary infection, such as bacterial pneumonia following a viral lung infection. An opportunistic infection is one that occurs only when the host's defense systems have been compromised in some way; the infection of a tissue injury by normal host microflora would be an example of this, or the infection of an AIDS patient by normally harmless microorganisms due to the person's suppressed immune system. Accidental infections are caused by organisms that are not normally associated with humans but that in unusual circumstances can cause disease. For example, the anaerobe Clostridium tetani is widespread in soils and only gains access to the body as a result of being introduced via a deep puncture or scratch, where it can cause the potentially fatal condition of tetanus.

The ability of an organism to cause disease is called its pathogenicity, and the degree of its pathogenicity is its virulence. Many factors affect the virulence of an organism, including the numbers of organisms involved, the route of entry, the state of competence of the host's defenses and virulence factors of the pathogen. Virulence can be quantified by determining the number of bacteria required to kill 50% of a group of experimental animals within a defined time period, the lethal dose 50 (LD_{50}).

Tuberculosis is an infectious disease that most often affects the lungs and is caused by a type of bacteria.

The virulence of a pathogen may be closely related to factors such as its mode of transmission and ability to live outside the host. If a pathogen depends on direct contact for transmission, it would not be to its benefit to damage its host so extensively that it was unable to contact others and continue transmission of the disease. The common cold, for example, has a low virulence; sufferers are able to carry on with their lives, and in doing so continue to come into contact with other people and pass the virus on to them. When a pathogen is not dependent on host mobility, the effect on the host can be more drastic; despite the death of the host, the pathogen may still flourish if it can be transmitted effectively by other means, such as by water or an insect vector. Similarly, those pathogens that are unable to survive outside their hosts and therefore depend on direct contact transmission will again be dependent on the survival of their host and tend to be less virulent. Those that can readily survive in the environment can afford to cause serious damage or death to their host because they are not dependent on it for their survival. The causative agents of diseases such as **tuberculosis** and diphtheria can survive for months without entering a human host. It is also important to realize that the virulence of a given organism is not a constant, but may change under the influence of a range of environmental factors.

To be able to induce an infectious disease, a pathogen must be able to:

- travel/be carried to the host;
- attach to, colonies or invade the host;
- resist host defenses and multiply or complete its life cycle within the host;
- cause damage to the host by chemical or mechanical means.

We now considering each of these in turn.

Transmission

Transmission of a pathogen may be direct (involving person-to-person contact) or indirect.

Direct transmission involves person-to-person transmission without the involvement of any intermediate, such as in the case of the common cold, influenza and hepatitis A, as well as sexually transmitted diseases. Pathogens transmitted in this way are often very sensitive to desiccation and other environmental factors and cannot survive away from the body for even a short time, hence the need for direct transmission. Also classed as directly transmitted are skin pathogens such as staphylococci that cause boils and pimples, and fungi that give rise to ringworm, for example. These, however, are more resistant to environmental factors, hence contact literally from person to person is not always necessary. Organisms that are spread by droplets of saliva or mucus as a result of coughing, sneezing or even laughing are also regarded as being directly transmitted. Droplets generated in this way travel less than a meter through the air and the organisms are not regarded as being truly airborne. A special form of direct transmission is the passing of a pathogen from mother to unborn or newborn child (vertical transmission). AIDS, toxoplasmosis and congenital syphilis can all be transmitted in this way.

Indirect transmission involves an intermediate of some sort. Fomites are inanimate objects on which pathogens can remain viable until contacted by a second individual. Common examples include towels, razors, syringe needles and children's toys. Athlete's foot and tetanus are both transmitted by fomites. Microbial diseases can also be transmitted by living intermediates called vectors, such as insects or cats and dogs. Sometimes the pathogen simply hitches a ride on the surface of the vector, which thus plays an entirely passive role; for example, flies can carry gastrointestinal pathogens of humans such as Shigella on their feet and bodies after visiting a source of faeces. The rat flea, on the other hand, which transmits Yersinia pestis, the causative agent of plague, does so by actually ingesting the bacteria with a blood meal and then passing them on when feeding on another host. Both of these are examples of mechanical transmission. In biological transmission, the pathogen undergoes some part of its life cycle inside the vector; an example of this is the malaria protozoan Plasmodium, which multiplies and accumulates inside the mosquito that transmits it.

Norovirus is a very contagious virus that causes vomiting and diarrhea.

Common vehicle transmission refers to those pathogens that are transmitted by means of an inanimate reservoir of disease used by both the pathogen and its prospective host. Contaminated water (e.g. with cholera) is the most obvious example, but air, blood transfusions, food and medications are others. To be regarded as truly airborne transmission, the droplet or dust particle carrying the pathogen needs to travel over a meter from the disease source to the host. Some microorganisms can be carried in small droplets of mucus called droplet nuclei. These result from the evaporation of droplets from sneezes and the like and are so light that they can stay airborne over long distances. Tuberculosis and measles are largely airborne transmitted diseases. Finally, microbial infections may be transmitted through food that is contaminated with a pathogen such as Salmonella or **Norovirus,** which causes diarrhoea and food poisoning.

Attachment and Colonization

Having reached the host, the first thing a potential invader of the body will come across is an epithelial surface of some description – the skin, or the lining of the various entrances to the body, that is, the alimentary, respiratory and genitourinary tracts. These represent a formidable series of physical and chemical barriers that must be overcome before the invader can gain entry to the body.

Skin

Skin forms an effective barrier to invading organisms for a number of reasons:

- Few organisms can penetrate the intact skin because the outer layer is made up of thick, closely packed keratinized cells, resistant to enzymatic attack.
- Any organisms that do manage to adhere will soon be removed as the process of desquamation continually removes the outer cells.
- The skin is mildly acidic (pH 5–6) due to the production of fatty acids such as oleic acid from the breakdown of lipids; this inhibits the growth of many microorganisms. The low pH of sweat has a similar effect.
- Lysozyme, which breaks the NAG–NAM bond in peptidoglycan, is to be found in secretions such as tears and saliva.
- The relative dryness of the skin slows down microbial growth but it flourishes in damper areas such as the groin, under arms and between the toes.
- Normal, harmless skin microflora compete for nutrients and attachment sites and will in many cases act antagonistically against invading pathogens.

Mucous Membranes

The alimentary, respiratory and genitourinary tracts have mucous membranes whose stratified epithelium and secretions of mucus provide a protective covering that resists penetration. In the respiratory tract, particles greater than 10 µm in size are trapped by the hairs and cilia of the nasal lining, whereas smaller particles are deposited on the mucous surface; particles such as bacteria that are trapped in this way are then removed by mechanisms such as coughing and ciliary movement. Mucus is made up of long, branched acidic carbohydrates called mucins, which are good at trapping and sticking to bacteria and preventing them from attaching to receptors on the epithelial cells. In the alimentary tract, saliva washes bacteria down to the acidic stomach, which destroys most bacteria and their toxins. Toxins of Staphylococcus and Clostridium are exceptions to this, and cysts of protozoans such as Cryptosporidium can also survive. Microorganisms that have survived the acid environment of the stomach, for example by being protected by a layer of food, pass to the small intestine, where bile, pancreatic enzymes and enzymes from intestinal secretions may destroy them. The process of peristalsis and the normal loss of columnar epithelium lining the small intestine help to remove invading organisms, and in addition, specific immune proteins in the alimentary tract prevent bacterial attachment. In extreme cases of microbial ingestion and toxin production, a vomiting reflex may be triggered. In the large intestine there is a large resident microflora, which is important in preventing invaders from getting established. It does this by competing for attachment sites and nutrients and by producing inhibitory metabolic products. A layer of mucus here also performs a protective function.

The bacterial count of a healthy stomach is as few as 10 cells per ml!

The kidneys, ureters and bladders of mammals are normally sterile, as is the urine itself. Some bacteria are killed because of the relatively acid pH of the urine, and also because of the presence of urea and other metabolic products such as uric and hippuric acids. In the kidney itself, conditions are so hypertonic that few microorganisms are able to survive. Potential pathogens are flushed from the urinary tract some 4–10 times daily by urination. In the female genitourinary tract, acid-tolerant lactobacilli degrade glycogen from the vaginal epithelium to form lactic acid, producing a pH of 3–5, which is unfavorable to most organisms. A final, less obvious opening to the body is represented by the eye. The conjunctiva is a specialized mucus-secreting epithelial membrane that lines the inner surface of each eyelid and the exposed surface of the eye itself. This is protected by the secretions in tears, which, as well as keeping the conjunctiva moist, are rich in lysozyme and other antimicrobial substances.

In order to give rise to an infection, pathogens need either to colonies the cell surface or to penetrate within it. To do this, they need to overcome the host's defense mechanisms described above. The routes by which they achieve this are called portals of entry. Many pathogens have a preferred portal of entry, and are only able to cause an infection if that route is used (Table 1). Streptococci, for example, can cause pneumonia if they invade the respiratory tract, but if they are swallowed there are generally no harmful effects. Other pathogens, such as Mycobacterium tuberculosis can gain entry by a variety of routes.

Table 1. Portals of entry for human pathogens

Mucous membranes:
respiratory tract
gastrointestinal tract
genitourinary tract
eyes
Breaks in host barriers, mainly skin (parenteral route)
Placenta

The virulence of an organism depends on a combination of its own properties and the ability of the host to resist it. Pathogenic bacteria often possess virulence factors – structural or physiological features that enhance their disease-causing capabilities by helping with attachment to the host or evasion of its defenses. Virulence factors may also take the form of toxins that actually cause the adverse effects.

Certain pathogenic strains of E. coli such as O157:H7 are able to synthesize a protein that acts as a receptor for their own adhesin. The protein, known as Tir (translocated intimin receptor), is secreted into the membrane of the cells lining the small intestine, where it enables the adhesin intimin, which is expressed on the bacterial surface, to bind.

Many bacteria and viruses gain access to the body by penetrating mucous membranes lining the respiratory, gastrointestinal and genitourinary tracts. The most commonly invaded of these is the respiratory mucosa by, for example, the causative agents of the common cold, influenza, pneumonia, or whooping cough. Microorganisms ingested with food or drink or from dirty fingers enter the gastrointestinal tract, but most are destroyed by enzymatic secretions and the low pH of the stomach. Those that survive, either by being protected inside a food bolus or through sheer weight of numbers, can cause gastroenteritis, cholera, typhoid and dysentery. They are eliminated in the faeces and passed to the next host via contaminated water, poor food hygiene, etc. The route of entry for most urinary infections and sexually transmitted diseases is via the genitourinary tract. Microorganisms may also be introduced through a breakage of the skin or mucous membranes, from cuts, bites or burns.

How Do Pathogens Penetrate the Mucosa?

Some bacteria release extracellular enzymes, sometimes known as 'spreading factors' that facilitate entry to host cells. Hyaluronidase breaks down hyaluronic acid, a key component of the matrix that holds cells together, by digesting it; bacteria such as Streptococcus pyogenes and Staphylococcus aureus are able to penetrate more deeply into a tissue by passing through the intercellular spaces. Collagenase is another enzyme that breaks down connective tissue, in this case collagen.

Coagulase, produced by Staphylococcus aureus, works by accelerating the clotting of blood. Plasma leaking out of vessels into tissues will be clotted; this can have the effect of coating the pathogens with fibrin and protecting them from the host's immune cells. This explains the isolated nature of S. aureus infections in boils and pimples. Another bacterial enzyme, streptokinase (produced by Streptococcus pyogenes), has the opposite effect; it dissolves blood clots, releasing trapped bacteria and allowing them to spread to other tissues. It does this by activating the production of plasmin, which digests fibrin. Once inside a host, most bacteria are able to attach themselves to the host's epithelia, frequently with a high degree of specificity, both for host and type of tissue. This attachment takes place by means of cell surface molecules called adhesins. These are proteins or glycoproteins found on surface structures such as fimbriae, pili or capsules. The term is sometimes extended to describe the structures themselves.

Most adhesins only allow the pathogen to attach to certain cells or tissues, because these carry specific receptors for them. Campylobacter spp. and Vibrio cholerae, for example, both have adhesins on their flagella that bind to specific receptors on the intestinal lining, whereas Streptococcus mutans, which causes tooth decay, attaches via its capsule to tooth enamel. The possession of fimbriae and capsules also makes the bacterium more difficult for the host's phagocytic cells to deal with. Viruses attach by binding to specific receptors on the surface of their host cells.

Host macrophages form part of the front line of defense against pathogens. They function by internalizing the invaders by phagocytosis, and then causing their destruction by fusing with lysosomes, membrane-bound packets of potent destructive enzymes. Some pathogens, however, have developed ways of avoiding lysis, so the process of internalization actually aids their invasion of the host.

Bacterial Toxins

Disease is rarely caused simply by the presence of microorganisms, even in large numbers, although there are exceptions to this, such as when a crucial vessel or heart valve becomes blocked due to sheer weight of bacterial numbers.

In most cases, the pathogens exert their effect by producing toxins. A toxin is a substance, generally a metabolic product of the pathogen, which causes damage to the host. Bacterial toxins can be conveniently divided into two types, exotoxins and endotoxins (Table 2).

Exotoxins are soluble proteins released from living bacteria into host tissues; they may be transported away from the site of infection to produce damage at distant sites.

Endotoxins form part of the bacterial cell surface and are mainly released when the cell is lysed. They are lipopolysaccharides and are limited to Gram-negative bacteria.

Table 2. Features of endotoxins and exotoxins

Exotoxins	Endotoxins
Proteins	Lipopolysaccharides
Secreted from bacterial cell	Integral part of bacterial outer membrane
Denatured by temperatures >60°C	Stable to autoclaving
Usually site-specific	Systemic
Toxic at very low concentrations	Only toxic at much higher concentrations
Not fever-inducing	Fever-inducing
Strongly immunogenic	Weakly immunogenic

Exotoxins

Exotoxins are among the most potent toxic substances known, per unit weight. They are mostly produced by Gram-positive bacteria, together with a few Gram-negative. They have no known function for the bacteria, and have been shown to be non-essential for survival by deleting the genes coding for them without ill effects.

The site of action of most exotoxins is restricted to certain cells or cell receptors; we can categorize exotoxins on the basis of this site specificity, and will now consider some examples in more detail.

'Botox' injections, much in vogue in certain circles as a cosmetic treatment, involve low doses of C. botulinum exotoxin. By acting as a muscle relaxant, they are intended to reduce the facial wrinkles that develop with the passing of time.

Tetanus and botulinum toxins are both produced by members of the genus Clostridium (C. tetani and C. botulinum), and both are examples of neurotoxins. They are also both A-B toxins, that is, they comprise two subunits joined by covalent bonding, the B component binding to a cell surface receptor, allowing the toxic A subunit to be transferred into the host cell where it exerts its effect. The A subunit acts as a proteolytic enzyme, preventing neurotransmitter release.

The botulinum toxin is one of the most toxic substances known; it has been calculated that a milligram of pure botulinum toxin would be enough to kill over 10 000 people. It acts by binding to presynaptic membranes of stimulatory motor neurons, preventing the release of the neurotransmitter acetylcholine. This means that muscles do not receive an excitatory signal and therefore cannot contract, leading to flaccid paralysis. Without adequate treatment around 30% of patients can die within 2–3 days from respiratory failure or cardiac arrest. Botulinum toxin is used clinically to relieve the effects of conditions in which there is uncontrolled muscular contraction. One such condition is blepharospasm, in which uncontrolled contractions of the muscles around the eyes may render the patient functionally blind. Injection of minute amounts of botulinum toxin every few month's results in the blocking of nerve impulses to the muscles, allowing the eyes to be opened.

Tetanus toxin is another A-B toxin that affects the nervous system. It is transported back through the motor neurons to the spinal cord, where it binds specifically to the surface of inhibitory neurons. These normally work by releasing glycine, which acts as an inhibitory neurotransmitter, preventing the release of acetylcholine and therefore inhibiting muscle contraction. If the glycine release is inhibited, there will be a continual release of acetylcholine and uncontrolled muscle contraction. This is what the tetanus toxin tetanospasmin does. One manifestation of this is lockjaw, a prolonged spasm restricting movement of the mouth, and if respiratory muscles are involved, death can result from asphyxiation.

Thus although the two toxins both have serious and even fatal outcomes, the mechanisms by which they work are in a sense the exact opposite of each other. An important difference between them at another level is that while C. tetani grows within its host and secretes its toxin, botulism is caused by the ingestion of food contaminated with toxin, rather than of the organism itself. For this reason, antibiotic therapy for foodborne botulism would be ineffective and treatment focuses instead on neutralizing the exotoxin. Botulism is therefore not strictly speaking an infectious disease, but an intoxication.

Enterotoxins cause a huge secretion of water and electrolytes from the intestinal epithelium into the lumen of the intestine, resulting in the symptoms of diarrhoea. They are produced by organisms such as Salmonella enteroviridis, Clostridium perfringens and Vibrio cholerae.

A third main group of exotoxins are the cytotoxins. They are a heterogeneous group of soluble proteins that are secreted by a range of bacteria, and share in common the feature of either killing host cells or affecting their functions. An example is the diphtheria toxin, which has the distinction of being the first exotoxin to be discovered. It is yet another A-B toxin, with the B subunit binding to the host cell surface, and the A subunit entering to have its toxic effect as before. The mode of action of the A subunit is different again, however. It acts by blocking transfer of amino acids from tRNA to the growing polypeptide chain, and so disrupting protein synthesis. The actual effect is on a protein called elongation factor 2. The bacterium in question, Corynebacterium diphtheriae, only produces the diphtheria toxin if it has been infected by a bacteriophage known as phage β. This carries the tox gene responsible for encoding the diphtheria toxin. Some cytotoxins work by disrupting host membranes, causing cell lysis. They do this enzymatically, or by the insertion of protein channels into the host cytoplasmic membrane. Leukocidins are cytotoxins produced by certain streptococci and staphylococci that are capable of killing host white blood cells, specifically macrophages and neutrophils. Since these are phagocytic cells and form an important part of the host's immune system, this affects the host's ability to defend itself against microbial attack. Leukocidins act by inserting into the host cell membrane to form a pore. Haemolysins lyse red blood cells by a similar mechanism. Their presence can be demonstrated in the laboratory by areas of clearing around the colonies of growth when the bacteria are cultured on blood agar. α-Haemolysis results in a greenish halo around the bacteria due to lysis of the red blood cells and partial breakdown of haemoglobin, whereas β-haemolysis involves complete breakdown of haemoglobin, giving a clear area. Destruction of red cells is not on a scale to have any harmful effects for the host, but the breakdown of haemoglobin provides the bacteria with a source of iron, essential for bacterial growth. Red cells are not the only cells attacked by haemolysins, but provide an easy method of detection. Streptococci and staphylococci both produce haemolysins.

Unlike endotoxins, exotoxins are strongly immunogenic, that is, they stimulate the immune system to produce antibodies (antitoxins) against them, providing the host with immunity against future infections by the same organism. Most individuals who suffer with recurrent Clostridium difficile-associated diarrhoea are only able to produce low levels of antibodies against the C. difficile exotoxins TcdA (an enterotoxin) and TcdB (a cytotoxin).

Toxoids are toxins that have been altered so that they have lost their toxicity but retained their immunogenicity, and are administered in order to develop immunity against the toxin. Tetanus injections contain the tetanus toxoid.

Endotoxins

The cell wall of Gram-negative bacteria has a so-called outer membrane, comprising a bilayer of phospholipid and lipopolysaccharide, and that the lipid A component of the latter can act as an endotoxin. Many Gram-negative bacteria including Salmonella and Yersinia pestis owe their toxic properties to their endotoxin. It is the lipid A component that contributes nearly all the toxic properties of the lipopolysaccharide. Other Gram-negative bacteria may contain a version of lipopolysaccharide lacking specific parts of the lipid A and as a result are not toxic. Endotoxins are produced when Gram-negative bacteria such as E. coli, Shigella and Salmonella die and their cell walls are fragmented. Endotoxins produce the same symptoms when released into the bloodstream, regardless of the specificity of the O-polysaccharide component, although they may differ in their degree. Macrophages are stimulated to produce inflammatory **cytokines** known collectively as endogenous pyrogens, and the net effect is to induce inflammation, intravascular coagulation, haemorrhage and shock (Gram-negative septicaemia or septic shock). In serious cases this can lead to multiple organ failure and death. As mentioned above, endotoxins are poorly immunogenic compared with exotoxins; their immunogenicity is due to the O-polysaccharide fraction. Antibiotics taken to combat Gram-negative infections work by lysing the bacterial cells; in so doing, they may release endotoxins and thereby cause a temporary worsening of the condition.

Cytokines are small proteins released from cells of the immune system that are involved in cellular communication and regulation.

It is important that when we receive medical treatment in the form of antibiotics and other drugs, they are free of endotoxins, otherwise some of the adverse effects just described could result. To this end rigorous standards are observed and sensitive tests have been developed for the detection of small amounts of endotoxin contamination. One of these is based on the fact that trace amounts of endotoxin will react with the clotting protein from the amoebocytes of Limulus, the horseshoe crab. In the commercial assay, the presence of endotoxin causes the clotting protein to precipitate, resulting in changes in turbidity, which can be measured spectrophotometrically. The Limulus amoebocyte lysate (LAL) assay is sensitive down to picogram quantities of lipolysaccharide.

Superantigens

Superantigens are bacterial proteins that produce a set of symptoms very similar to those of septic shock, but by a different mechanism. They activate a large proportion of the immune system's T-lymphocytes in a non-specific way and elicit a massive inflammatory reaction due to an overproduction of cytokines. This may result in the same symptoms of diarrhoea and vomiting, fever or potentially fatal systemic shock. Toxic shock syndrome (Staphylococcus aureus) and scarlet fever (Streptococcus pyogenes) are both caused by bacterial superantigens.

Much of the iron in the blood is not available to bacteria because it is bound up as either haemoglobin (in red cells) or transferrin (in plasma). Similarly iron in other body fluids is tied up as lactoferrin.

Siderophores

The final class of virulence factors we're going to consider are siderophores. Both bacteria and their hosts require iron for growth and metabolism, and animals have developed mechanisms of withholding it from their tissue fluids, making conditions unfavorable for invading bacteria. Siderophores are structurally diverse molecules that share the common feature of high affinity for ferric ions. An environment deficient in iron stimulates the transcription of bacterial genes that encode enzymes involved in the synthesis of siderophores. In some cases the affinity of siderophores may be so high as to solubilize iron already tied up as lactoferrin or transferrin and transfer it to the bacteria. Once the iron-siderophore complex reaches the bacterial surface it binds to a specific receptor protein, whose synthesis is also stimulated by the lack of environmental iron. The iron is then released into the bacterial cell and reduced to the Fe^{2+} form.

Bacterial Diseases in Humans

We start with bacterial diseases, with an emphasis on the different ways they can be transmitted. Table 3 summarizes the principal bacterial diseases of humans.

Table 3. Some bacterial diseases of humans.

	Genus	**Disease**
Gram-positive	*Staphylococcus*	Impetigo, food poisoning, endocarditis, bronchitis, toxic shock syndrome
	Streptococcus	Pneumonia, pharyngitis, meningitis, scarlet fever, dental caries
	Enterococcus	Enteritis
	Bacillus	Anthrax
	Clostridium	Tetanus, botulism, gangrene
	Corynebacterium	Diphtheria
	Listeria	Listeriosis
	Mycobacterium	Leprosy, tuberculosis
	Propionibacterium	Acne
	Mycoplasma	Pneumonia, vaginosis

Gram-negative	*Salmonella*	Salmonellosis
	Escherichia	Gastroenteritis
	Shigella	Dysentery
	Neisseria	Gonorrhoea, meningitis
	Bordetella	Whooping cough
	Legionella	Legionellosis
	Pseudomonas	Infections of burns
	Vibrio	Cholera
	Campylobacter	Gastroenteritis
	Helicobacter	Peptic ulcers
	Haemophilus	Bronchitis, pneumonia
	Treponema	Syphilis
	Chlamydia	Pneumonia, urethritis, trachoma

Waterborne Transmission: Cholera

Cholera is most commonly contracted as a result of ingesting contaminated water containing faecal material. It is a form of gastroenteritis caused by a toxin released by Vibrio cholerae. The bacteria attach by means of adhesins to the intestinal mucosa, where, without actually penetrating the cells, they release the cholera exotoxin. This comprises an ‘A’ and several ‘B’ subunits; the former is the active component, while the latter attach to epithelial cells by binding to a specific glycolipid in the membrane. This allows the passage of the ‘A’ subunit into the cell, where it causes the activation of an enzyme called adenylate cyclase (Figure 1). This results in uncontrolled production of cyclic AMP, causing active secretion of Cl^- and HCO_3^- ions into the intestinal lumen, followed by large volumes of water in an effort to maintain ionic balance.

Many of the diseases listed may be caused only by a particular species within the genus

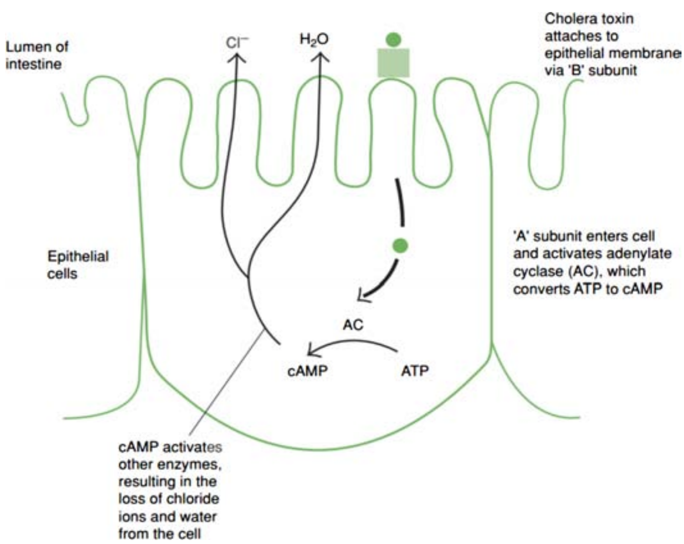


Figure 1. The action of the cholera exotoxin.

The outcome of this is huge fluid loss (10 liters or more per day) through profuse and debilitating diarrhoea. In the young, old and sick, death through dehydration and salt depletion can follow within a few hours. If proper liquid and electrolyte replacement therapy is available, recovery rates are very high, but left untreated, mortality is 50% or more. Although it is now very rare in the developed world, cholera remains a major killer in less developed countries, claiming over 100 000 lives each year. It is easily preventable by means of clean water supplies and improved sanitation; however, when these services break down, for example in times of war or after natural disasters, cholera outbreaks quickly follow; indeed, the most severe outbreak of recent times occurred in the wake of the Haiti earthquake of 2010.

Airborne Transmission: 'Strep' Throat

Streptococcal pharyngitis, commonly known as strep throat, is one of the commonest bacterial diseases of humans, being particularly common in children of school age. The primary means of transmission is by the inhalation from coughs and sneezes of respiratory droplets containing Streptococcus pyogenes (β-haemolytic type A streptococci), although other routes (infected handkerchiefs, kissing) are possible. The primary symptoms are a red and raw throat (and/or tonsils), accompanied by headaches and fever. S. pyogenes attaches to the throat mucosa, stimulating an inflammatory response and secreting virulence factors that destroy host blood cells. Although selflimiting within a week or so, strep throat should be treated with penicillin or erythromycin as more serious streptococcal diseases such as scarlet fever and rheumatic fever may follow if it is left untreated.

Contact Transmission: Syphilis

Causative organisms of sexually transmitted diseases such as Neisseria gonorrhoeae (gonorrhoea) and Treponema pallidum (syphilis) are extremely sensitive to the effects of environmental factors such as UV light and desiccation. They are therefore unable to live outside their human host, and rely for transmission on intimate human contact.

The spirochaete T. pallidum enters the body through minor abrasions, generally on the genitalia or mouth, where a characteristic lesion called a chancre develops. The disease may proceed no further than this, but if T. pallidum enters the bloodstream and passes around the body, the more serious secondary stage develops, lasting some weeks. Following a latent period of several years, around half of secondary syphilis cases go on to develop into the tertiary stage of the disease, whose symptoms may include mental retardation, paralysis and blindness. Congenital syphilis is caused by T. pallidum being passed from a mother to her unborn child.

The primary and secondary stages of syphilis are readily treated by penicillin; however, the tertiary stage is much less responsive to such therapy.

Vector-Borne Transmission: Plague

A limited number of bacterial diseases reach their human hosts from their main host, usually another species of mammal, via an insect intermediary.

Plague (bubonic plague, the Black Death) has been responsible for the deaths of untold millions of people in terrible epidemics such as the ones that wiped out as much as one-third of the population of Europe in the Middle Ages. It is caused by the Gram-negative bacterium Yersinia pestis, whose normal host is a rat, but which can be spread to humans by fleas. The bacteria pass to the lymph nodes, where they multiply, causing the swellings known as buboes. Y. pestis produces an exotoxin, which prevents it from being destroyed by the host's macrophages; instead, it is able to multiply inside them. From the lymph nodes, the bacteria spread via the bloodstream to other tissues such as the liver and lungs. Once established in the lungs (pneumonic plague), plague can spread from human to human by airborne transmission in respiratory droplets. Plague is extremely infectious, and left untreated has a high rate of fatality, particularly for the pneumonic form of the disease. Early treatment with streptomycin or tetracycline, however, is largely successful. Improved public health measures and the awareness of the dangers of rats and other rodents have meant that confirmed cases of plague are now relatively few.

Viral Diseases in Humans

Viruses are responsible for some of the most serious infectious diseases to affect humans. Some important examples are listed in Table 4.

Table 4. Some important viruses of humans

Virus	Family	Disease	Genome type
Adenovirus	*Adenoviridae*	Respiratory infections	dsDNA
Ebola virus	*Filoviridae*	Haemorrhagic fever	(−)ssRNA
Epstein–Barr	*Herpesviridae*	Infectious mononucleosis	dsDNA
Hepatovirus A	*Picornaviridae*	Hepatitis A	(+)ssRNA
Herpes simplex Type I	*Herpesviridae*	Cold sores	dsDNA
Herpes simplex Type II	*Herpesviridae*	Genital warts	dsDNA
Human immunodeficiency virus (HIV)	*Retroviridae*	Acquired immunodeficiency syndrome (AIDS)	(+)ssRNA*
Human papillomavirus	*Papovaviridae*	Warts	dsDNA
Influenza virus	*Orthomyxoviridae*	Influenza	(−)ssRNA
Lassa virus	*Arenaviridae*	Lassa fever	(−)ssRNA
Morbillivirus	*Paramyxoviridae*	Measles	(−)ssRNA
Norovirus	*Calciviridae*	Enteritis	(+)ssRNA
Paramyxovirus	*Paramyxoviridae*	Mumps	(−)ssRNA
Polio virus	*Picornaviridae*	Poliomyelitis	(+)ssRNA
Rabies virus	*Rhabdoviridae*	Rabies	(−)ssRNA
Rhinovirus	*Picornaviridae*	Common cold	(+)ssRNA
Rotavirus	*Reoviridae*	Enteritis	dsRNA
Rubella virus	*Togaviridae*	German measles	(+)ssRNA
Smallpox virus	*Poxviridae*	Smallpox	dsDNA
Varicella-zoster	*Herpesviridae*	Chicken pox, shingles	dsDNA
Yellow fever virus	*Flaviviridae*	Yellow fever	(+)ssRNA

*The genome of HIV, like that of other retroviruses, also has a DNA phase.

Airborne Transmission: Influenza

Influenza is a disease of the respiratory tract caused by members of the Orthomyxoviridae. Transmission occurs as a result of inhaling airborne respiratory droplets from an infected individual. Infection by the influenza virus results in the destruction of epithelial cells of the respiratory tract, leaving the host open to secondary infections from bacteria such as Haemophilus influenzae and Staphylococcus aureus. It is these secondary infections that are responsible for the great majority of fatalities due to influenza. Generally, sufferers from influenza recover completely within 10–14 days, but some people, notably the elderly and those with chronic health problems, may develop complications such as pneumonia.

The influenza virus has an envelope, and a segmented (−)sense ssRNA genome (Figure 2). The envelope contains two types of protein spike, each of which plays a crucial role in the virus's infectivity:

- *Neuraminidase* is an enzyme that hydrolyses sialic acid, thereby assisting in the release of viral particles.
- *Haemagglutinin* enables the virus to attach to host cells by binding to epithelial sialic acid residues. It also helps in the fusion of the viral envelope with the cell membrane.

Both types of spike act as antigens, proteins that stimulate the production of antibodies in a host. One of the reasons that influenza is such a successful virus is that the 'N' and 'H' antigens are prone to undergoing changes (antigenic shift) so that the antigenic 'signature' of the virus becomes altered, and host immunity is evaded. Different strains of the influenza virus are given a code denoting which variants of the antigens they carry; the subtype N_1H_1, for example, caused both the 1918 'Spanish flu' and the 2009 'swine flu' pandemics, while H_5N_1 was responsible for the 'bird flu' outbreak in SE Asia in 2003–4.

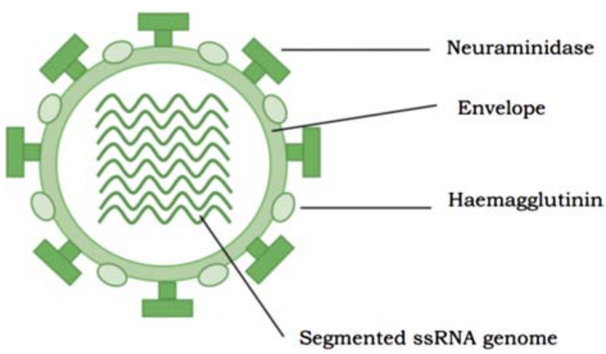


Figure 2. The influenza virus.

Transmission by Water or Food: Viral Gastroenteritis

Most people will be only too familiar with the symptoms of gastroenteritis – sickness, diarrhoea, headaches and fever. The cause of this gastroenteritis may be bacterial (e.g. Salmonella) or viral. The major cause of the viral form is the human rotavirus, which, together with the norovirus, is responsible for the majority of reported cases.

The viruses damage the villi in the upper part of the intestinal tract, affecting normal ion transport, and resulting in the characteristic water loss. Transmission of gastroenteritis is via the faecal-oral route, that is, by the ingestion of faecally contaminated food or water. Poor hygiene practice or contaminated water supplies are usually to blame for the perpetuation of the cycle. Normally, the condition is self-limiting, lasting only a couple of days; the normal treatment is fluid replacement therapy. In areas where clean water supplies are not available, however, the outcome can be much more serious. In less developed countries, the condition is a major killer; it is the principal cause of infant mortality, and the cause of some five to ten million deaths per year.

Arboviruses are a group of viruses that are transmitted by insects to humans.

Vector-Borne Transmission

Some human viral diseases are spread by means of arthropods such as ticks and mosquitoes that feed on human blood. The viruses are known as **arboviruses** (arthropod-borne viruses), and are mostly enveloped RNA viruses. Note, however, that the arboviruses do not form a discrete taxonomic grouping, but simply share a means of transmission. Diseases caused by arboviruses include dengue fever, yellow fever, and forms of viral encephalitis such as West Nile encephalitis and St Louis encephalitis.

Latent and Slow (Persistent) Viral Infections

After an infection has passed, a virus may sometimes remain in the body for long periods, causing no harm. It may be reactivated, however, by stress or some change in the individual’s health, and initiate a disease state. Well-known examples of latent viral infections are cold sores and shingles, both caused by members of the herpesvirus family. A virus of this sort will remain with an individual throughout their lifetime.

Whereas latent virus infections are characterized by a sudden increase in virus production, in persistent (slow) infections the increase is more gradual, building up over several years. Such infections have a serious effect on the target cells, and are generally fatal. An example is the measles virus, which can re-manifest itself after many years in a rare condition called subacute sclerosing panencephalitis (SSPE).

Viruses and Cancer

A number of chemical and physical agents are known to trigger the uncontrolled proliferation of cells that characterize cancers, but in the last two decades it has become clear that at least six types of human cancer can be virally induced. How do cells lose control of their division, and how are viruses able to bring this about? It is now known that cells contain genes called proto-oncogenes, involved in normal cell replication. They are normally under the control of other, tumor-suppressor genes, but these can be blocked by proteins encoded by certain DNA viruses. When this happens, the proto-oncogene functions as an **oncogene**, and cell division is allowed to proceed uncontrolled. Retroviruses have a different mechanism; they carry their own, altered, version of the cellular oncogene, which becomes integrated into the host's genome and leads to uncontrolled cell growth. Retrovirus oncogenes are thought to have been acquired originally from human (or animal) genomes, with the RNA transcript becoming incorporated into the retrovirus particle.

Oncogene is a gene associated with the conversion of a cell to a cancerous form.

Emerging and Re-emerging Viral Diseases

As a result of changes in the pathogen or in the host population, completely new infectious diseases may arise, or we may experience the reappearance of diseases previously considered to be under control. These are known as, respectively, emerging and re-emerging infections. Social, economic and climatic changes and changed patterns of human population movement may all create the conditions that are often responsible for the development of such infections. Frequently, emerging virus infections are zoonotic in origin, that is, they are transferred to humans from animal reservoirs. The human immunodeficiency virus, for example, is thought to have developed from a similar virus found in certain types of monkey.

Virus Vaccines

Smallpox, once the scourge of millions, was in 1979 the first infectious disease to be declared successfully eradicated. This followed a worldwide campaign of vaccination by the World Health Organization over the previous decade, and was made feasible by the fact that humans are the only reservoir for the virus. Vaccination is a preventive strategy that aims to stimulate the host immune system, by exposing it to the infectious agent in question in an inactivated or incomplete form. There are four main classes of virus vaccines.

Attenuated (= 'weakened') vaccines contain 'live' viruses, but ones whose pathogenicity has been greatly reduced. The aim is to mimic an infection in order to stimulate an immune response, but without bringing about the disease itself. A famous example of this type of vaccine is the polio vaccine developed by Albert Sabin in the 1960s. The cowpox virus used by Edward Jenner in his pioneering vaccination work in the late eighteenth century was a naturally occurring attenuated version of the smallpox virus.

Inactivated vaccines contain viruses that have been exposed to a denaturing agent such as formalin. This has the effect of rendering them noninfectious, while at the same time retaining their ability to stimulate an immune response. Vaccines directed against influenza are of this type.

Subunit vaccines depend on the stimulation of an immune response by just a part of the virus. Since the complete virus is not introduced, there is no chance of infection, so vaccines of this type have the attraction of being very safe. Subunit vaccines are often made using recombinant DNA technology; the first example to be approved for human use was the hepatitis B vaccine, which consists of part of the protein coat of the virus produced in specially engineered yeast cells.

DNA vaccines are also the product of modern molecular biology techniques. DNA encoding virus antigens is directly injected into the host, where it is expressed and triggers a response by the immune system. Whilst showing great potential, vaccines of this type have not so far been approved for use in humans.

Protists and Disease

When microbial diseases are discussed, it is generally bacteria and viruses that come to mind; however, a significant number are caused by eukaryotic pathogens, chiefly protozoans. Some important protozoan diseases such as cryptosporidiosis and toxoplasmosis are particularly prevalent in immuno-compromised individuals such as those with AIDS. In some parts of the world, notably the tropics and subtropics, infections by protozoans are endemic; it is estimated that over two million people die each year from malaria alone. With international travel now commonplace, however, cases of travelers from Western countries returning home with these diseases are on the increase; around 2000 cases of malaria are now reported annually in the United Kingdom.

Each of the four categories of protozoans contains representatives capable of causing disease in humans (Table 5). Some of the better known examples are illustrating the range of infectious cycles that exist among protozoan parasites. Some parasitic protozoans complete their life cycle in a single host, whilst others require more than one host to complete the different stages of their life cycle. The host in which the sexual stage of the life cycle is completed is called the definitive host, and the one with the asexual stage the intermediate host.

Table 5. Some major protozoal diseases of humans

Protozoan	Disease	Mode of infection
Mastigophora		
Trypanosoma brucei	African trypanosomiasis	Insect vector (tsetse fly)
Trypanosoma cruzi	Chagas' disease	Insect vector (kissing bug)
Giardia lamblia	Giardiasis	Faecally contaminated water
Leishmania	Leishmaniasis	Insect vector (sandfly)
Trichomonas vaginalis	Vaginitis	Sexual contact
Ciliophora		
Balantidium coli	Balantidial dysentery	Faecally contaminated water
Apicomplexa		
Cryptosporidium	Cryptosporidiosis	Contaminated water
Cyclospora cayetanensis	Cyclosporosis	Contaminated water
Plasmodium spp.	Malaria	Insect vector (mosquito)
Toxoplasma gondii	Toxoplasmosis	Ingestion of oocysts
Sarcodina		
Entamoeba histolytica	Amoebic dysentery	Faecally contaminated water

Fungal Diseases in Humans

A limited number of fungi are pathogenic to humans (Table 6). Mycoses (sing. mycosis) in humans may be cutaneous or systemic; in the latter, spores generally enter the body by inhalation, but subsequently spread to other organ systems via the blood, causing serious, even fatal disease.

Table 6. Some fungal diseases of humans

Disease	Fungus
Histoplasmosis	*Histoplasma capsulatum*
Blastoplasmosis	*Blastomyces dermatitidis*
Cryptococcosis	*Cryptococcus neoformans*
Cutaneous mycoses	*Trichophyton* spp.
Pneumocystis pneumonia	*Pneumocystis jirovecii*
Candidiasis ('thrush')	*Candida albicans*
Aspergillosis	*Aspergillus fumigatus*

Cutaneous mycoses are the most common fungal infections found in humans, and are caused by fungi known as dermatophytes, which are able to utilize the keratin of skin, hair or nails by secreting the enzyme keratinase. Popular names for such infections include ringworm and athletes' foot. They are highly contagious, but not usually serious conditions.

Systemic mycoses can be much more serious, and include conditions such as histoplasmosis and blastomycosis. The former is caused by Histoplasma capsulatum, and is associated with areas where there is contamination by bat or bird excrement. It is thought that the number of people displaying clinical symptoms of histoplasmosis represents only a small proportion of the total number infected. If confined to the lungs, the condition is generally selflimiting, but if disseminated to other parts of the body such as the heart or central nervous system, it can be fatal. The causative agents of both diseases exhibit dimorphism; they exist in the environment as mycelia but convert to yeasts at the higher temperature of their human host.

A number of fungi act as opportunistic pathogens, including Aspergillus fumigatus (aspergillosis), Candida albicans (candidiasis or 'thrush') and Pneumocystis jirovecii (previously named P carinii) (pneumonia). The latter is found in a high percentage of AIDS patients, whose immune defenses have been compromised. The causative organism was previously considered to be a protozoan, and has only been classed as a fungus relatively recently, as a result of DNA/RNA sequence evidence. It lives as a commensal in a variety of mammals, and is probably transmitted to humans through contact with dogs.

The incidence of opportunistic mycoses has increased greatly since the introduction of antibiotics, immunosuppressants and cytotoxic drugs. Each of these either suppresses the individual's natural defenses, or eliminates harmless microbial competitors, allowing the fungal species to flourish. Microsporidiosis is an opportunistic infection, mainly of the intestine, caused by members of the Microsporidia. It is mainly found in immunocompromised individuals, especially those with AIDS but also recipients of organ transplants. Many fungi produce natural mycotoxins; these are secondary metabolites, which, if consumed by humans, can cause food poisoning, which can sometimes be fatal. Certain species of mushrooms ('toadstools'), including the genus Amanita, contain substances that are highly poisonous to humans. Other examples of mycotoxin illnesses include ergotism and aflatoxin poisoning. Aflatoxins are carcinogenic toxins produced by Aspergillus flavus, which grows on stored peanuts. In the early 1960s, the turkey industry in the UK was almost crippled by 'turkey X disease', caused by the consumption of feed contaminated by A. flavus.

It is thought likely that all animals are parasitized by one fungus or another. Extraordinary though it may seem, there are even fungi that act as predators on small soil animals such as nematode worms, producing constrictive hyphal loops that tighten, immobilizing the prey.

Algal Diseases of Humans

Shellfish can accumulate toxins produced by dinoflagellates, causing disease when consumed by humans. A range of such toxins is produced by different dinoflagellates, which act on specific cellular targets and lead to different symptoms, including gastrointestinal and neurological symptoms. Paralytic shellfish poisoning (PSP) is caused by toxins called saxitoxins, produced principally by dinoflagellates of the genus Alexandrium. They act by blocking sodium channels in nerve cells, leading to tingling, numbness and paralysis. If symptoms extend to respiratory paralysis, the condition can be fatal if not treated in time.

Ciguatera is a form of food poisoning caused by consuming fish from tropical regions that have accumulated potent neurotoxins called ciguatoxins from dinoflagellates, notably Gambierdiscus toxicus. The toxin accumulates as it passes up the food chain until it reaches fish such as snapper and barracuda. Ciguatoxins also affect the opening of sodium channels, but in a different way, thus leading to different clinical symptoms. Like saxitoxins they are heatstable, and so are not destroyed by cooking.

CORE CONCEPT: CONTROL OF MICROORGANISMS

Control of microorganisms is essential in order to prevent the transmission of diseases and infection, stop decomposition and spoilage, and prevent unwanted microbial contamination. Microorganisms are controlled by means of physical agents and chemical agents. Physical agents include such methods of control as high or low temperature, desiccation, osmotic pressure, radiation, and filtration. Control by chemical agents refers to the use of disinfectants, antiseptics, antibiotics, and chemotherapeutic antimicrobial chemicals.

Prior to Lister's pioneering work with **antiseptics**, around four out of every ten patients undergoing an amputation failed to survive the experience, such was the prevalence of infections associated with operative procedures, and yet, only 40 years later, this figure had fallen to just 3 in 100. This is a dramatic demonstration of the fact that in some situations it is necessary for us to destroy, or at least limit, microbial growth, because of the undesirable consequences of the presence of microorganisms, or their products.

Antiseptic is a chemical agent of disinfection that is mild enough to be used on human skin or tissues.

Control of microorganisms can be achieved by a variety of chemical and physical methods. Sterilization is generally achieved by using physical means such as heat, radiation and filtration. Agents that destroy bacteria are said to be bactericidal. Chemical methods, whilst effective at disinfection, are generally not reliable for achieving total sterility. Agents that inhibit the growth and reproduction of bacteria without bringing about their total destruction are described as bacteriostatic.

Sterilization

One of the oldest forms of antimicrobial treatment is that of heating, and in most cases this remains the preferred means of sterilization, provided that it does not cause damage to the material in question. The benefits of boiling drinking water have been known at least since the 4th century BC, when Aristotle is said to have advised Alexander the Great to order his troops to take this precaution. This of course was many centuries before the existence of microorganisms had been demonstrated or perhaps even suspected.

Sterilization by Heat

Boiling at 100°C for 10 minutes is usually enough to achieve sterility, provided that organisms are not present in high concentrations; in fact most bacteria are killed at about 70°C. If, however, endospores of certain bacteria (notably Bacillus and Clostridium) are present, they can resist boiling, sometimes for several hours. The causative agents of some particularly nasty conditions, such as botulism and tetanus, are members of this group. In order to destroy the heat-resistant endospores, heating beyond 100°C is required, and this can be achieved by heating under pressure in a closed vessel (Table 7). A typical laboratory treatment is 15 minutes at a pressure of 103 kPa (15 lb/in^2 (psi)), raising the temperature of steam to 121°C. This is carried out in an autoclave, which is, to all intents and purposes, a large-scale pressure cooker (Figure 3). Air is driven out of the system so that the atmosphere is made up entirely of steam; the desired temperature will not be reached if this is not achieved (Figure 4). Large loads, large volumes of liquids, and loads that contain trapped air, such as blankets or bedding, may need a longer treatment time in order for the heat to penetrate throughout. The problem of trapped air can be overcome with a prevacuum autoclave, which removes air from the chamber prior to sterilization, thereby reducing the time needed for the instrument to reach its operating temperature. Modern autoclaves include probes designed to assess the temperature within a load rather than that of the atmosphere. It is essential that the intended temperatures are reached, and most autoclaves produce a printout of the temperatures and pressures for each run. Sometimes, spores of Geobacillus stearothermophilus are introduced into a system along with the material to be sterilized; if subsequent testing shows that the spores have all been destroyed, it is reasonable to assume that the system has also destroyed any other biological entity present. Special tape that changes color if the necessary temperature is reached can act as a more convenient but less reliable indicator.

Table 7. Temperature of steam at different pressures

Pressure		Temperature (°C)
Pounds/in^2 (psi)	Kilopascals (kPa)	
0	0	100
10	68.9	115
15	103.4	121
20	137.9	126
25	172.4	130

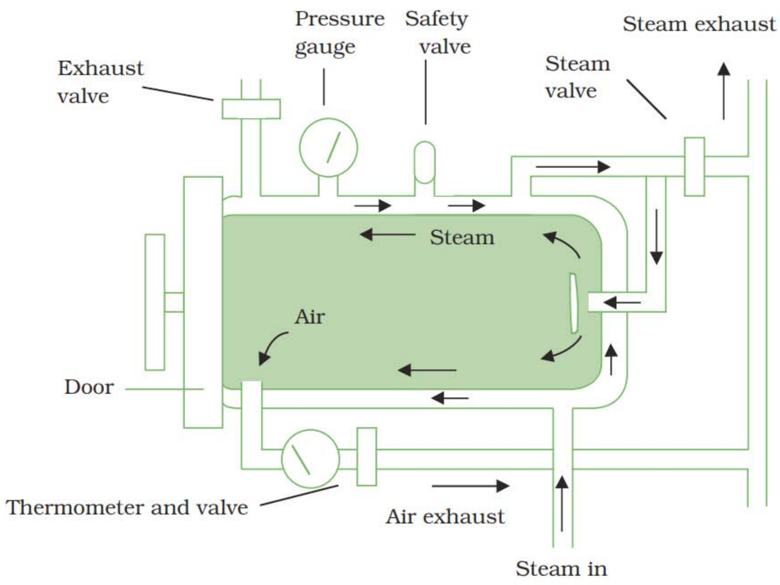


Figure 3. The autoclave. Steam enters the chamber, driving air out. As the pressure of the steam reaches 103 kPa (15 psi) the temperature reaches 121°C, sufficient to kill resistant endospores as well as vegetative cells.

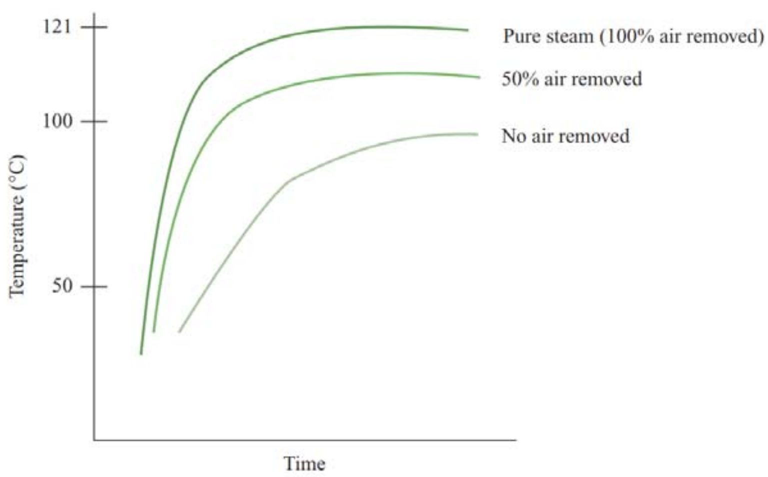


Figure 4. Temperatures achieved in an autoclave at 103 kPa (15 psi) in different atmospheres.

An effect similar to that achieved by autoclaving can be obtained by a method called intermittent steaming, or tyndallization (after the Irish physicist, John Tyndall, who was one of the first to demonstrate the existence of heat-resistant microbial forms). This is used for those substances or materials that might be damaged by the high temperatures used in autoclaving. The material is heated to between 90 and 100°C for about 30 minutes on each of three successive days, and left at 37°C in the intervening periods. Vegetative cells are killed off during the heating period, and during the 37°C incubation any endospores that have survived will germinate. Once these have grown into more vegetative cells, they too are killed in the next round of

Sterilization is the process by which all microorganisms present on or in an object are destroyed or removed.

steam treatment. Clearly this is quite a long-winded procedure, and it is therefore reserved for those materials that might be harmed by steam **sterilization**.

High temperatures can cause damage to the taste, texture and nutritional value of many food substances, and in such instances it is sufficient to destroy vegetative cells by a process of pasteurization (among his many other achievements, Pasteur demonstrated that the microbial spoilage of wines could be prevented by short periods of heating). Milk was traditionally pasteurized by heating large volumes at 63°C for 30 minutes, but the method employed nowadays is to pass it over a heat exchanger at 72°C for 15 seconds (HTST – high temperature, short time). This is not sterilization as such, but it ensures the destruction of disease-causing organisms such as Brucella abortus and Mycobacterium tuberculosis, which at one time were frequently found in milk, as well as significantly reducing the organisms that cause food spoilage, thus prolonging the time the milk can be kept. Some protocols exceed these minimum values in order to reduce further the microbial content of the milk. One type of milk on sale in the shops is subjected to more extreme heating regimes; this is ‘UHT’ milk, which can be kept for several weeks without refrigeration, though many find that this is at some cost to its palatability! It is heated to ‘ultra-high temperatures’ (150°C) for a couple of seconds using superheated steam. The product is often referred to as being ‘sterilized’, but this is not true in the strictest sense. Milk is not the only foodstuff to be pasteurized; others include beer, fruit juices and ice cream, and each has its own time/temperature combination.

All the above methods employ a combination of heat and moisture to achieve their effect; the denaturation of proteins, upon which these methods depend, is enhanced in the presence of water. Heat is more readily transferred through water than through air, and the main reason that endospores are so resistant is because of their very low water content. In some situations, however, it is possible to employ dry heat, using an oven to sterilize metal instruments or glassware, for example. It is a more convenient procedure, but a higher temperature (160–170°C) and longer exposure time (2 hours) are required. Dry heat works by oxidizing (‘burning’) the cell’s components.

Microorganisms are quite literally burnt to destruction by the most extreme form of dry heat treatment, incineration. Soiled medical dressings and swabs, for example, are potentially hazardous, and are destroyed in this way at many hundreds of degrees celsius. Sterilizing the loops and needles used to manipulate microorganisms by means of flaming is a routine part of aseptic procedures.

The world's biggest UV wastewater treatment plant was opened in Manukau, New Zealand in 2001. Its 8000 UV lamps are able to treat up to 50 000 cubic meters of water per hour.

Sterilization by Irradiation

Certain types of irradiation are used to control the growth of microorganisms. These include both ionizing and non-ionizing radiation.

The most widely used form of non-ionizing radiation is ultraviolet (UV) light. Wavelengths around 260 nm are used because these are absorbed by the purine and pyrimidine components of nucleic acids, as well as certain aromatic amino acids in proteins. The absorbed energy causes a rupture of the chemical bonds, so that normal cellular function is impaired. UV light causes the formation of thymine dimers, where adjacent thymine nucleotides on the same strand are linked together, and inhibiting DNA replication. Although many bacteria are capable of repairing this damage by enzyme-mediated photoreactivation, viruses are much more susceptible. UV lamps are commonly found in food preparation areas, operating theatres and specialist areas such as tissue culture facilities, where it is important to prevent contamination. Because they are also harmful to humans (particularly the skin and eyes), UV lamps can only be operated in such areas when people are not present. UV radiation has very poor penetrating powers – a thin layer of glass, paper or fabric can impede the passage of the rays. The chief application is therefore in the sterilization of work surfaces and the surrounding air, although it is increasingly finding an application in the treatment of water supplies, where its ability to kill protozoan pathogens such as Giardia and Cryptosporidium gives it an advantage over conventional chlorine treatment. The use of UV irradiation in this context offers the additional advantage of degrading chemical pollutants such as pesticides.

Ionizing radiations have a shorter wavelength and much higher energy, giving them greater penetrating powers. The effect of ionizing radiations is due to the production of highly reactive free radicals, which disrupt the structure of macromolecules such as DNA and proteins. Surgical supplies such as syringes, catheters and rubber gloves are commonly sterilized employing gamma (γ) rays from the isotope cobalt-60 (^{60}Co).

Gamma radiation has been approved for use in over 50 countries for the preservation of food, which it does not only by killing pathogens and spoilage organisms but also by inhibiting processes that lead to sprouting and ripening. The practice has aroused a lot of controversy, largely due to concerns about health and safety, although the first patent applications for its use date back over a hundred years! Although the irradiated product does not become radioactive, there is a general suspicion on the part of the public about anything to do with radiation, which has led to its use on food being only very gradually accepted by consumers. Tight European Union regulations mean that the use of food irradiation here remains limited to a single product; however, in the United States, a positive attitude towards irradiation of food both by professional

Antibiotic is a metabolic product produced by one microorganism that inhibits or kills other microorganisms.

bodies and the media during the last 20 years has led to a more widespread acceptance of the technology. The Radura symbol is used internationally to indicate that a food product has been irradiated. Gamma radiation is used in situations where heat sterilization would be inappropriate, because of undesirable effects on the texture, taste or appearance of the product. This mainly relates to fresh produce such as meat, poultry, fruit and vegetables. Irradiation is not suitable for some foodstuffs, such as those with a high fat content, where unpleasant tastes and odors result. Ionizing radiations have the great advantage over other methods of sterilization that they can penetrate packaging.

Filtration

Many liquids such as solutions of **antibiotics** or certain components of culture media become chemically altered at high temperatures, so the use of any of the heat regimes described above is not appropriate. Rather than killing the microorganisms, an alternative approach is simply to remove them. This can be done for liquids and gases by passing them through filters of an appropriate pore size. Filters used to be made from materials such as asbestos and sintered glass, but these have been largely replaced by membrane filters, commonly made of nitrocellulose or polycarbonate (Figure 5). These can be purchased ready-sterilized and the liquid passed through by means of pressure or suction. Supplies of air or other gases can also be filter-sterilized in this way. A pore size of 0.22 μm is commonly used; this will remove bacteria plus, of course, anything bigger, such as yeasts; however, Mycoplasma spp. and viruses are able to pass through pores of this size. With a pore size one-tenth this size, only the smallest of viruses can pass through, so it is important that an appropriate pore size is chosen for any given task.

A drawback with all filters, but especially those of a small pore size, is that they can become clogged easily. Filters in general are relatively expensive, and are not the preferred choice if alternative methods are available.

High efficiency particulate air (HEPA) filters are used to create clean atmospheres in areas such as operating theatres and laboratory biological safety cabinets.

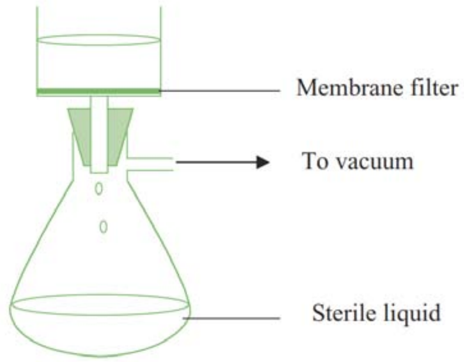


Figure 5. Membrane filtration. Membrane filters are used to sterilize heat-labile substances. They are available in a variety of pore sizes, according to the specific application.

Disinfection is the elimination of microorganisms, but not necessarily endospores, from inanimate objects or surfaces.

Sterilization Using Ethylene Oxide

Generally, chemical methods achieve only **disinfection**; the use of the gas ethylene oxide, however, is effective against bacteria, bacterial spores and viruses. It is used for sterilizing large items of medical equipment, and materials such as plastics that would be damaged by heat treatment. Ethylene oxide is particularly effective in sterilizing items such as dressings and mattresses, due to its great powers of penetration. In the food industry, it is used as an antifungal fumigant, for the treatment of dried fruit, nuts and spices. The materials to be treated are placed in a special chamber, which is sealed and filled with the gas in a humid atmosphere at 40–50°C for several hours. Ethylene oxide is highly explosive, so must be used with great caution; its use is rendered safer by administering it in admixture (10%) with a nonflammable gas such as carbon dioxide. It is also highly toxic, so all items must be thoroughly flushed with sterile air following treatment to remove any trace of it. Ethylene oxide is an alkylating agent; it denatures proteins by replacing labile hydrogens such as those on sulphydryl groups with a hydroxyl ethyl radical (Figure 6).

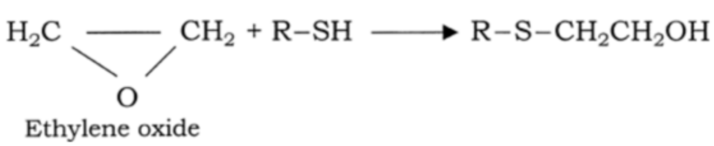


Figure 6. Ethylene oxide is an alkylating agent.

Disinfection

Decontamination is the treatment of an object or inanimate surface to make it safe to handle.

We saw at the beginning of this section how sterilization is an absolute term, implying the total destruction of all microbial life. Disinfection, by comparison, allows the possibility that some organisms may survive, with the potential to resume growth when conditions become more favorable. A disinfectant is a chemical agent used to disinfect inanimate objects such as work surfaces and floors. In the food and catering industry, especially in the United States, the term sanitization is used to describe a combination of cleaning and disinfection. Disinfectants are incapable of killing spores within a reasonable time period, and are generally effective against a narrower range of organisms than physical means. **Decontamination** is a term sometimes used interchangeably with disinfection, but its scope is wider, encompassing the removal or inactivation of microbial products such as toxins as well as the organisms themselves.

The lethal action of disinfectants is mainly due to their ability to react with microbial proteins, and therefore enzymes. Consequently, any chemical agent that can coagulate, or in any other way denature proteins will act as a disinfectant, and compounds belonging to a number of groups are able to do this.

Alcohols

The antimicrobial properties of ethanol have been known for over a century. It was soon realized that it worked more effectively as a disinfectant at less than 100% concentration, that is, when there was some water present. This is because denaturation of proteins proceeds much more effectively in the presence of water (recall that moist heat is more effective than dry heat for the same reason). It is important, however, not to overdo the dilution, as at low percentages some organisms can actually utilize ethanol as a nutrient! Ethanol and isopropanol are most commonly used at a concentration of 70%. As well as denaturing proteins, alcohols may act by dissolving lipids, and thus have a disruptive effect on membranes, and on the envelope of certain viruses. Both bacteria and fungi are killed by alcohol treatment, but spores are often resistant because of problems in rehydrating them; there are records of anthrax spores surviving in ethanol for 20 years! The use of alcohols is further limited to those materials that can withstand their solvent action.

Alcohols may also serve as solvents for certain other chemical disinfectants. The effectiveness of iodine, for example, can be enhanced by the iodine being dissolved in ethanol. Many commercially available hand **sanitizers** are ethanol-based.

Sanitizer is an agent that reduces microbial numbers to a safe level.

Halogens

Chlorine is an effective disinfectant as a free gas, and as a component of chlorine-releasing compounds such as hypochlorite and chloramines. Chlorine gas, in compressed form, is used in the disinfection of municipal water supplies, swimming pools and the dairy industry. Sodium hypochlorite (household bleach) oxidizes sulphydryl (–SH) and disulphide (S–S) bonds in proteins. Like chlorine, hypochlorite is inactivated by the presence of organic material. Chloramines are more stable than hypochlorite or free chlorine, and are less affected by organic matter. They are also less toxic and have the additional benefit of releasing their chlorine slowly over a period of time, giving them a prolonged bactericidal effect.

Iodine acts by combining with the tyrosine residues on proteins; its effect is enhanced by being dissolved in ethanol (1% I_2 in 70% ethanol) as tincture of iodine, an effective skin **disinfectant**. This is being superseded by iodophores (e.g. Betadine, Isodine), compounds in which iodine is combined with a solubilizing agent, usually a detergent. Iodophores can combat bacteria, viruses and fungi, and are widely used in the food and drink industry.

Disinfectant is an agents used to disinfect inanimate objects but generally to toxic to use on human tissues.

Phenolics

The germicidal properties of phenol (carbolic acid) were first demonstrated by Lister in the middle of the nineteenth century. Since phenol is highly toxic, its use in the disinfection of wounds has long since been discontinued, but derivatives such as cresols and xylenols continue to be used as disinfectants and antiseptics. These are both less toxic to humans and more effective against bacteria than the parent compound. Phenol is still used, however, as a benchmark against which the effectiveness of related disinfectants can be measured. The phenol coefficient compares the dilution at which the derivative is effective against a test organism with the dilution at which phenol achieves the same result. A phenol coefficient of more than one means that the new compound is more effective than phenol against the organism tested, whereas a value of less than one means that it is not as effective as phenol.

Phenol derivatives act by combining with and denaturing proteins, as well as disrupting cell membranes. Their advantages include the retention of activity in the presence of organic substances and detergents, and their ability to remain active for some time after application; hence their effect increases with repeated use. Familiar disinfectants such as Dettol, Lysol and Hibitane, are all phenol derivatives. Hexachlorophene (Figure 7) is very effective against Gram-positive bacteria such as staphylococci and streptococci, and used to be a component of certain soaps, surgical scrubs, shampoos and deodorants. Its use is now confined to specialist applications in hospitals since the finding that in some cases, prolonged application can lead to brain damage.

A surfactant reduces the tension between two molecules at an interface.

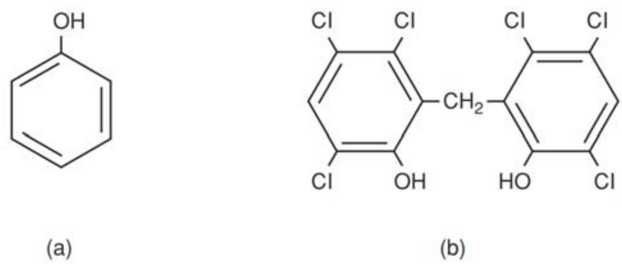


Figure 7. Phenolics. The structure of (a) phenol and (b) hexachlorophene.

Surfactants

Surface active agents, or surfactants, such as soaps and detergents, have the property of orienting themselves between two interfaces to bring them into closer contact (Figure 8).

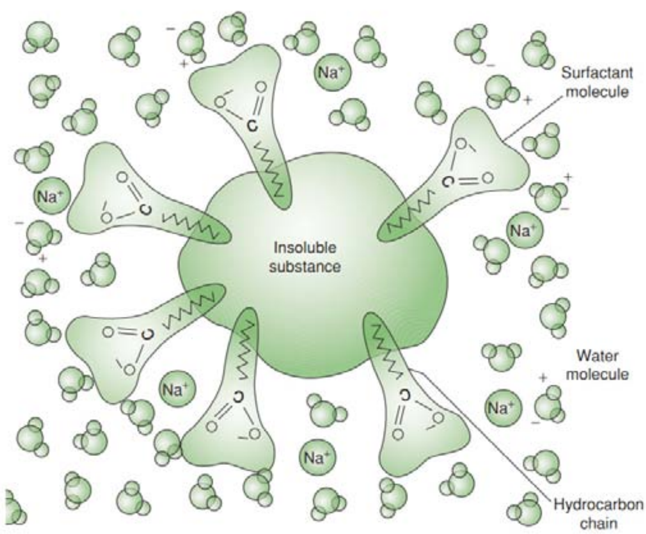


Figure 8. The action of surfactants.

The value of soap has less to do with its disinfectant properties than with its ability to facilitate the mechanical removal of dirt and microorganisms. It does this by emulsifying oil secretions, allowing the debris to be rinsed away. Detergents may be anionic (negatively charged), cationic (positively charged) or nonionic. Cationic detergents such as quaternary ammonium compounds (ammonium chloride with each chlorine replaced by an organic group, Figure 9) act by combining with phospholipids to disrupt cell membranes and affect cellular permeability.

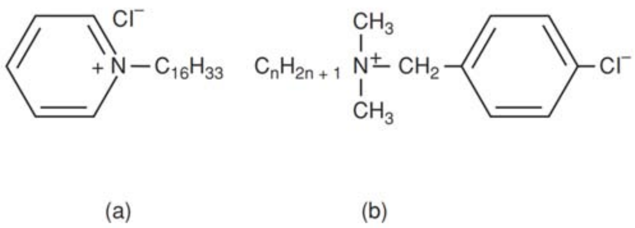
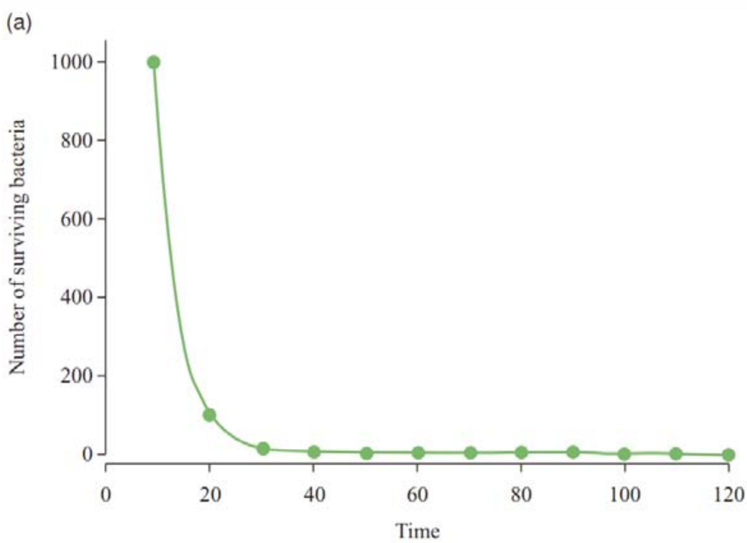


Figure 9. Quaternary ammonium compounds. (a) Cetylpyridinium chloride and (b) benzalkonium chloride. Although nontoxic and effective against a wide range of bacteria, quaternary ammonium compounds are easily inactivated by soap.

The Kinetics of Cell Death

When microorganisms are exposed to any of the treatments outlined in the preceding pages, they aren't all killed instantaneously. During a given time period, only a certain proportion of them will die. Suppose we had 1000 cells (an unrealistically small number, but it keeps the arithmetic simple) and that 10% were killed each minute. After one minute, 900 cells would remain, and after the second minute 10% of these would die, leaving us with 900 − 90 = 810 survivors.



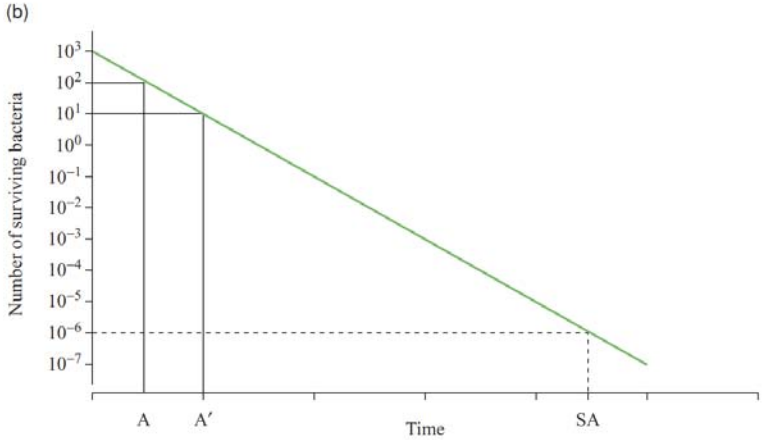


Figure 10. The kinetics of cell death. (a) During heat sterilization, the number of living cells decreases by the same proportion per unit time, giving an exponential curve. (b) When plotted on a logarithmic scale, the decrease in numbers is seen as a straight line, whose slope is a reflection of the rate of killing.

After a further minute, another 10% of the survivors would be killed, so 810 − 81 = 729 would be left. A plot of the surviving cells against time of exposure gives a graph such as Figure 10a. The curve is exponential; theoretically, there will never be zero survivors, but after a while we're going to have less than one cell, let's say one-tenth of a cell, which clearly can't happen. What this really means is that in a given unit volume, there will be a 1 in 10 chance of there being a cell present. Sterility is generally assumed when this figure falls as low as 1 in a million (see Figure 10b). Since only a proportion of the surviving population is killed per unit time, it follows that the more cells you have initially, the longer it will take to eliminate them (Figure 11).

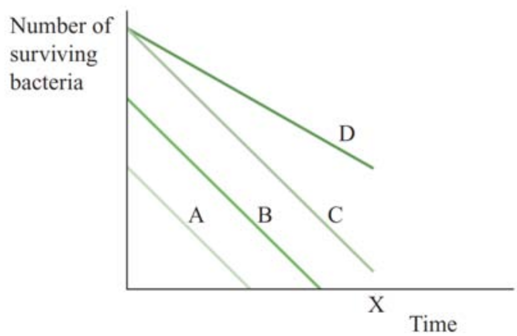


Figure 11. Sterilization time is dependent on starting population. Populations A, B and C all have the same decimal reduction rate, but different starting populations, therefore after a given time they have different numbers of survivors. Population D has the same starting numbers as C, but because a smaller proportion are killed per minute, i.e. the slope is less steep, a greater number survive after time X.

The steepness of the slope in Figure 10b is an indication of the effectiveness of heat sterilization. The decimal reduction time, or D value, is the time needed to reduce the population by a factor of 10 (i.e. to kill 90% of the population) using a particular heat treatment. The D value is a valuable parameter in settings such as the canning industry. It applies to a particular temperature; at higher temperatures, the rate of killing is enhanced, and so the D value is reduced (Figure 12). The increase in temperature required to reduce D by a factor of 10 is the Z value.

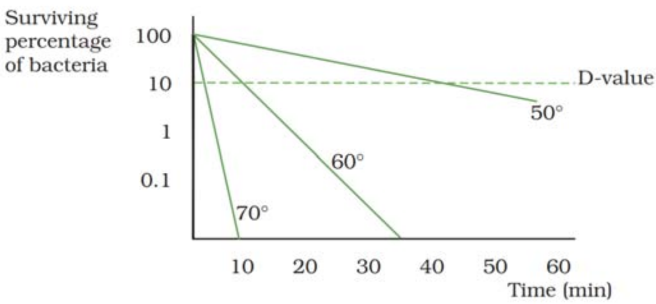


Figure 12. The D value is reduced at higher temperatures. The survival rate of a mesophilic bacterium at three different temperatures. The D value is the time taken to reduce the population to one-tenth of its starting value: note how it falls from 42 min (50°C) to 12 min (60°C) to 3 min (70°C).

Since in real life, the microbial population is certain to be a mixed one, the critical factor is the death rate of the most resistant species, that is, the one with the highest percentage of survivors per minute. Sterilization protocols should therefore be based on the rate of destruction of endospores. It should be stressed that an effective treatment regime for one organism may be wholly inappropriate for another, since organisms differ in their susceptibility to different agents. Organism A may resist heat treatment better than organism B, but be more sensitive to a particular chemical treatment.

Killing by Irradiation

Microorganisms as a group are much more resistant to the effects of ionizing radiation than are higher organisms, and some are more resistant than others. A plot of log surviving numbers against time of irradiation is shown in Figure 13. The decimal reduction value (D10) is analogous to the D value used for heat sterilization.

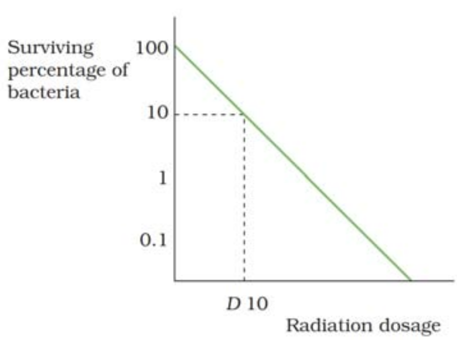
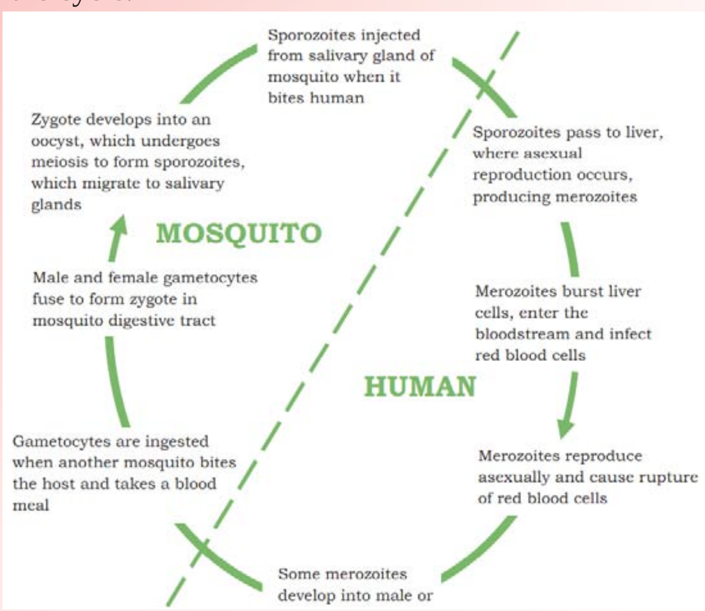


Figure 13. Killing by irradiation. The decimal reduction value (D10) is the dose of radiation necessary to reduce the population of microorganisms to one-tenth of its value. The value for D10 differs greatly between organisms.

CONCEPT OF MALARIA

Malaria, toxoplasmosis and cryptosporidiosis are all caused by members of the Apicomplexa group of protozoans. Their life cycles vary considerably and are described below; however, all depend on being able to penetrate host cells by means of their apical complex of specialized organelles. Four species of Plasmodium cause malaria in humans – P. falciparum, P. vivax, P. malariae and P. ovale – with the first two of these being responsible for around 80% of cases. In addition, a fifth species, P. knowlesi, which infects macaque monkeys, has also been shown to infect humans. The life cycle of Plasmodium species involves both asexual and sexual reproduction, which take place respectively in the mammalian host and mosquitoes of the genus Anopheles. Within an hour of being introduced into the human bloodstream by a mosquito taking a blood meal, sporozoites invade liver cells, and initiate the infection. The subsequent invasion and bursting of red blood cells every two or three days causes the classic malaria symptoms of periodic chills and fever. Some of the released merozoites colonise other red blood cells, perpetuating the so-called erythrocytic cycle. Some develop into gametocytes, which are ingested by another mosquito when it bites the infected person. From here the sexual stage of the life cycle proceeds, resulting in a zygote that develops into an oocyst, the only diploid stages of Plasmodium. Meiosis produces large numbers of sporozoites, which travel to the mosquito's salivary gland, ready to recommence the cycle.



SUMMARY

- Medical microbiology, the large subset of microbiology that is applied to medicine, is a branch of medical science concerned with the prevention, diagnosis and treatment of infectious diseases.
- An infectious disease can be defined as a change in the state of an individual's health brought about by the invasion and colonization of the body by pathogens, which may be bacteria, viruses, fungi or protists.
- Disease is rarely caused simply by the presence of microorganisms, even in large numbers, although there are exceptions to this, such as when a crucial vessel or heart valve becomes blocked due to sheer weight of bacterial numbers.
- Exotoxins are soluble proteins released from living bacteria into host tissues; they may be transported away from the site of infection to produce damage at distant sites.
- Endotoxins form part of the bacterial cell surface and are mainly released when the cell is lysed. They are lipopolysaccharides and are limited to Gram-negative bacteria.
- Control of microorganisms can be achieved by a variety of chemical and physical methods. Sterilization is generally achieved by using physical means such as heat, radiation and filtration. Agents that destroy bacteria are said to be bactericidal. Chemical methods, whilst effective at disinfection, are generally not reliable for achieving total sterility. Agents that inhibit the growth and reproduction of bacteria without bringing about their total destruction are described as bacteriostatic.

MULTIPLE CHOICE QUESTIONS

1. Which of the following pathogens causes cholera in humans?
   a. Fungi
   b. Bacteria
   c. Virus
   d. Protozoan

2. Which of the following is a set of bacterial diseases?
   a. Malaria, poliomyelitis, mumps
   b. Mumps, cholera, typhoid
   c. Plague, Leprosy, Diphtheria
   d. Measles, Tuberculosis, Tetanus

3. Which of the following pathogens causes Tuberculosis __________.
   a. Virus
   b. Bacterium
   c. Protozoan
   d. Malnutrition

4. Which of the following types of medical items requires sterilization?
   a. needles
   b. bed linens
   c. respiratory masks
   d. blood pressure cuffs

5. Which of the following is suitable for use on tissues for microbial control to prevent infection?
   a. disinfectant
   b. antiseptic
   c. sterilant
   d. water

Answers

1. (b) 2. (c) 3. (b) 4. (a) 5. (b)

REVIEW QUESTIONS

1. What is medical microbiology?
2. Name the two types of disease transmission.
3. How do pathogens penetrate the mucosa?
4. What are the methods of physical control of microorganisms?
5. What is chlorine?

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Index

Core Concepts in Biology: Microbiology

2nd Edition

Microbes are found everywhere in significant quantities. Their presence extends beyond our familiar earthly realm, even reaching into the vastness of space. These tiny organisms play a crucial role in the natural cycle of life by efficiently recycling organic matter. Living organisms like bacteria and fungi have the remarkable ability to decompose all naturally occurring substances since they are biodegradable. While many of these microbes positively impact plants, animals, and humans in various ways, it is important to acknowledge that some can also pose a threat. Microorganisms play a significant role in the natural world by actively participating in various biogeochemical cycling processes and assisting in the decomposition of organic matter in both soil and aquatic ecosystems. These tiny lifeforms engage in symbiotic relationships with plants, leading to mutual benefits for both parties. On the other hand, certain microorganisms can also act as pathogens, causing diseases in plants and animals. This diverse range of microbial activities greatly influences the overall balance and functioning of ecosystems.
Core Concepts in Microbiology is a comprehensive and detailed guide that investigates the captivating realm of microorganisms. This extensive book provides a vast array of information, covering various essential topics, such as the remarkable diversity of microorganisms, their intricate metabolic processes, their interactions with hosts, and their crucial role in both environmental and industrial processes. Whether you are a student pursuing a study in microbiology or simply someone with a curious mind seeking to explore this fascinating field, Core Concepts in Microbiology will undoubtedly enrich your understanding. This invaluable resource offers unparalleled insights and clarity, serving as an indispensable tool for gaining a comprehensive understanding of the captivating world of microbiology.

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Kan Hung Cheng is an assistant professor in the Department of Microbiology and Molecular Genetics. He has keen interest in biochemistry, physiology, cell biology, ecology, evolution, and clinical aspects of microorganisms, including host response to these pathogens. He has authored more than 10 research papers, contributed chapters to 2 books, and published 2 books. He has participated in several conference proceedings and workshops.

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